START HERE

CFA Level I: Economics

Every Learning Module of the 2026 Level I Economics syllabus as swipeable cards: a short story, a diagram, the formula, a worked example, and exam-style questions at the end of each chapter.

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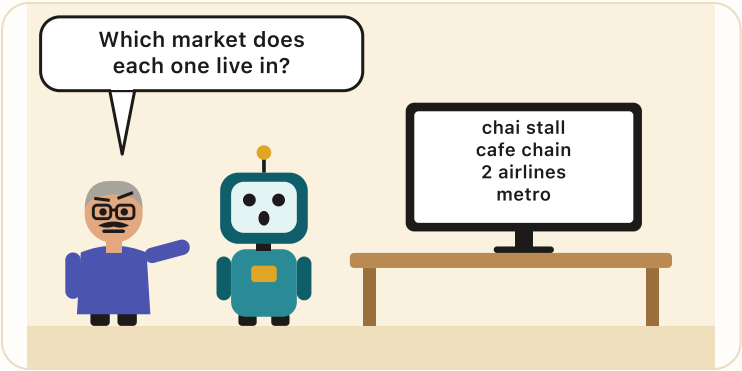
BEFORE YOU START

The cast, and words this chapter uses

A Mumbai investment firm. Sharma ji trusts nothing without data ("Data dikhao"); Pinky is sure of everything, especially when wrong.

- **Price elasticity |E|:** % change in quantity demanded / % change in price, sign dropped. Above 1: elastic.
- **Opportunity cost:** the value of the best alternative given up.
- **Consumer surplus:** what buyers would pay minus what they do pay.
- **Lakh, crore:** 1 lakh = 100,000; 1 crore = 10 million.

So: **You can start this chapter cold. Everything it needs is on this card.**



STORY

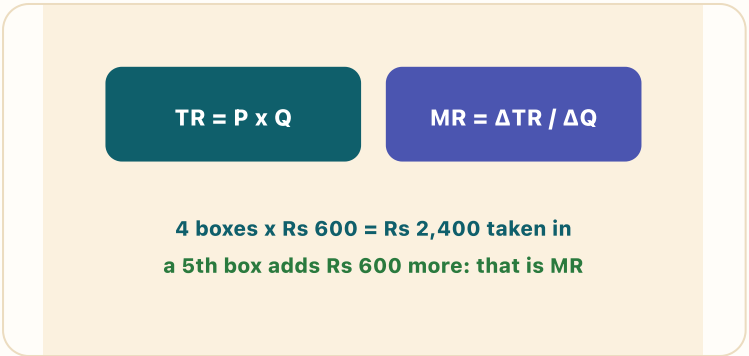
Four firms, one question

Monday, Nariman Point. Chintu must rate four firms: a chai stall at Churchgate station, a Bandra cafe chain, one of two airlines on the Mumbai to Goa route, and the city's only metro operator.

Pinky: "Easy. Buy the one with the highest margin." Sharma ji: "Margin today, yes. Margin in five years? Tell me what market each one lives in. Data dikhao."

The chai stall cannot raise prices; the metro can. That difference is **market structure**, and it decides whether profits survive.

So: **Market structure decides pricing power, and pricing power decides whether today's profit lasts.**



FORMULA

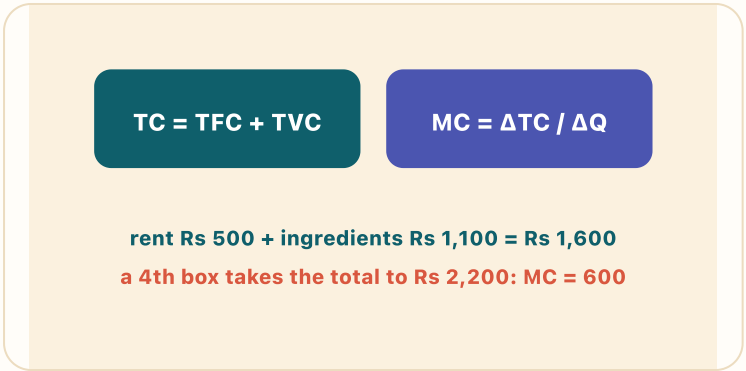
Revenue words: TR, P, Q and MR

$$TR = P \times Q$$
$$MR = \Delta TR \div \Delta Q$$

Symbol	In plain words	Example
TR	total revenue: all the money customers pay you	4 x 600 = Rs 2,400
P	price of one unit	Rs 600 a kulfi box
Q	quantity: how many units you sell	4 boxes
Δ	"delta": change in. New value minus old value	ΔQ = 5 - 4 = 1
MR	marginal revenue: extra money from selling one more unit	box 5 adds Rs 600

Marginal just means "the next one". MR answers: if I sell one more, how much more money comes in?

So: **Total revenue is all the money in; marginal revenue is the money the next unit adds.**



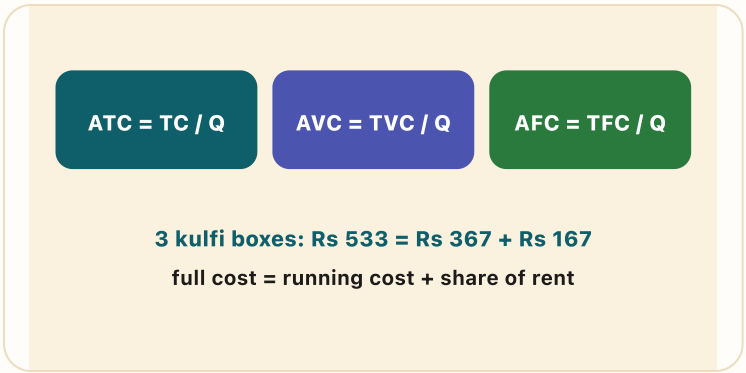
FORMULA

Cost words, part 1: totals and MC

```
TC = TFC + TVC
MC = ΔTC ÷ ΔQ
```

Symbol	In plain words	Example
TFC	total fixed cost: paid even if you make nothing (rent, licence)	freezer rent Rs 500
TVC	total variable cost: grows with output (milk, sugar, gas)	Rs 1,100 for 3 boxes
TC	total cost = TFC + TVC	500 + 1,100 = 1,600
Δ	"delta": change in. New value minus old value	ΔQ = 4 - 3 = 1
MC	marginal cost: extra cost of one more unit	2,200 - 1,600 = Rs 600

So: Total cost = fixed + variable; MC is what the next unit adds to it.



FORMULA

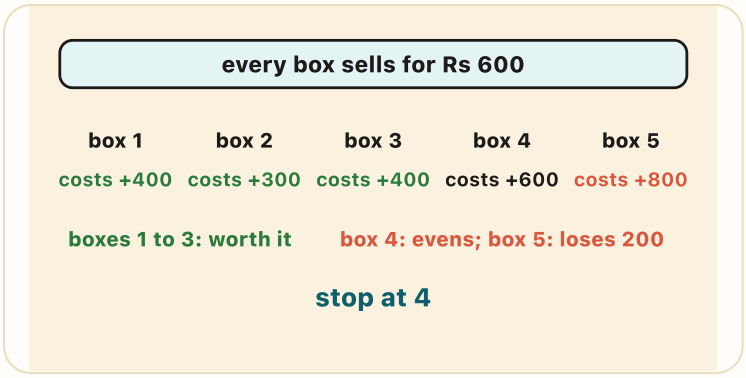
Cost words, part 2: per-unit costs

$$ATC = TC \div Q$$
$$AVC = TVC \div Q$$
$$ATC = AVC + AFC$$

Symbol	In plain words	Example
Q	units made	3 boxes
ATC	average total cost: full cost per unit	1,600 ÷ 3 = Rs 533
AVC	average variable cost: running cost per unit	1,100 ÷ 3 = Rs 367
AFC	average fixed cost: rent per unit, TFC ÷ Q	500 ÷ 3 = Rs 167

Price against ATC says profit or loss. Price against AVC says stay open or shut today.

So: **Divide each total by Q to get the cost per unit.**



STORY

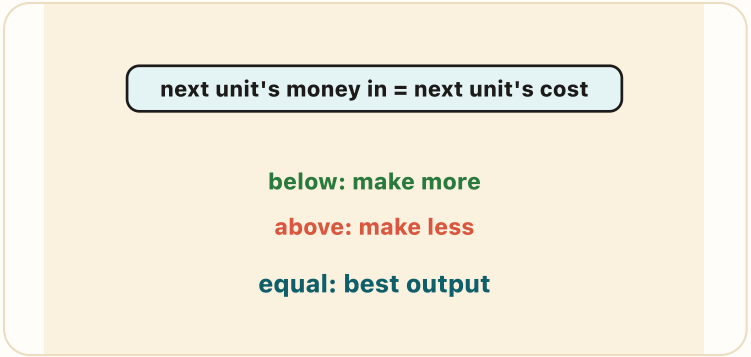
One more box? The kulfi seller's rule

Ramesh sells kulfi at Juhu beach. Every box sells for **Rs 600**, whatever he does: 40 other sellers keep the price there.

Each extra box needs more milk, sugar and gas. Boxes 1 to 3 add only Rs 300 to 400 each to his costs, less than Rs 600, so each one is worth making. Box 4 adds Rs 600: it just pays for itself. Box 5 adds Rs 800, because his one freezer is overfull and his helper wants overtime. Rs 600 in, Rs 800 out: a Rs 200 loss.

His rule: keep making boxes while the next box brings in at least what it costs. Economists write this as **MR = MC**.

So: **Make one more only if it brings in at least what it costs to make.**



FORMULA

The profit rule: MR = MC

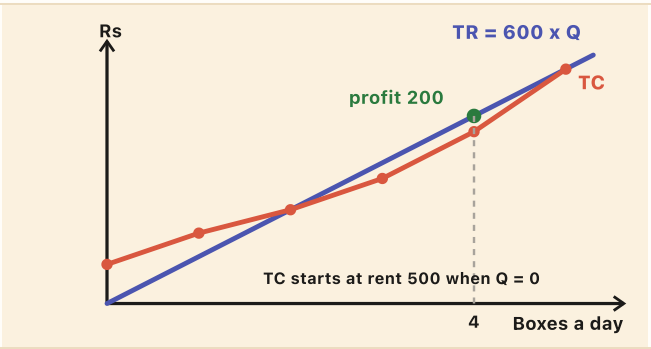
```
Best output: where MR = MC
(with MC rising)
```

Symbol	In plain words	Example
MR	marginal revenue: extra money from one more unit	Rs 600 a box
MC	marginal cost: extra cost of one more unit	box 5: Rs 800
Profit	money in minus all costs, rent included	Rs 200 at 4 boxes

MR = price here: Ramesh sells all he wants at Rs 600, so each extra box brings exactly Rs 600 (a **price taker**).

"MC rising": MC can cross MR twice; pick the crossing where MC is climbing, since every unit after it loses money.

So: **Keep adding units until the last one brings in just what it costs.**



WORKED EXAMPLE

Kulfi in numbers: fill in the table

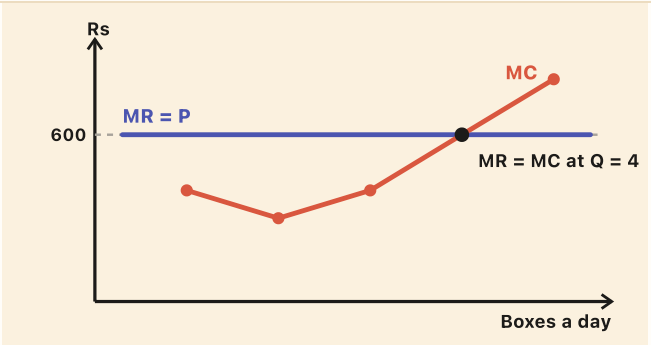
Each box sells for Rs 600 (P). Freezer rent (TFC) is Rs 500 a day, paid even at 0 boxes. Q = boxes; TC = total cost, rent included; TR = money in.

Q boxes	TC Rs	MC Rs	TR Rs	Profit Rs
1	900	400	600	-300
2	1200	300	1200	0
3	1600	400	1800	200
4	2200	600	2400	200
5	3000	800	3000	0

1. MC of box 4 = TC at 4 - TC at 3 = 2,200 - 1,600 = 600
   Extra cost of one more box: new total cost minus old total cost.
2. Profit at 4 = TR - TC = 600 x 4 - 2,200 = 200
   Money in (boxes x Rs 600) minus every cost, rent included.

MC column: 400, 300, 400, 600, 800.

So: MC comes from the change in TC; profit comes from TR minus TC.



WORKED EXAMPLE

MR = MC: how many kulfi boxes?

What we know

- **MR** = Rs 600: extra money from one more box (the price)
- **MC** = 400, 300, 400, 600, 800: extra cost of boxes 1 to 5

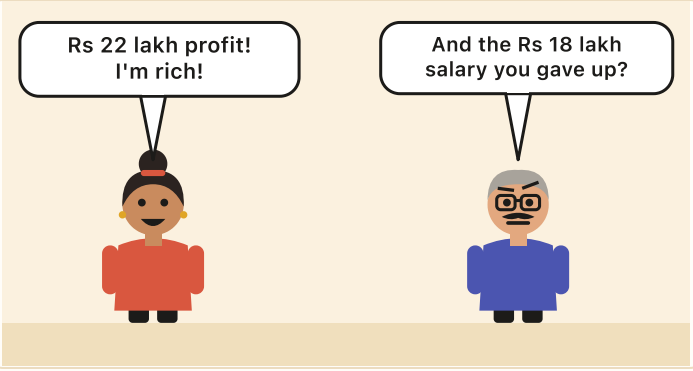
What we want: how many boxes a day give the most profit?

1. Boxes 1 to 3: MC 400, 300, 400, all below MR 600
   Each brings in more than it costs, so make it.
2. Box 4: MC 600 = MR 600
   It just pays for itself: the stopping point.
3. Box 5: MC 800 > MR 600
   It would cost Rs 200 more than it earns, so do not make it.

Best output: 4 boxes, profit Rs 200 a day.

Sense check: profit at 3 boxes is also Rs 200 (1,800 - 1,600): box 4 adds exactly Rs 0, which is what MR = MC means.

So: Produce where MR = MC and MC is rising.



STORY

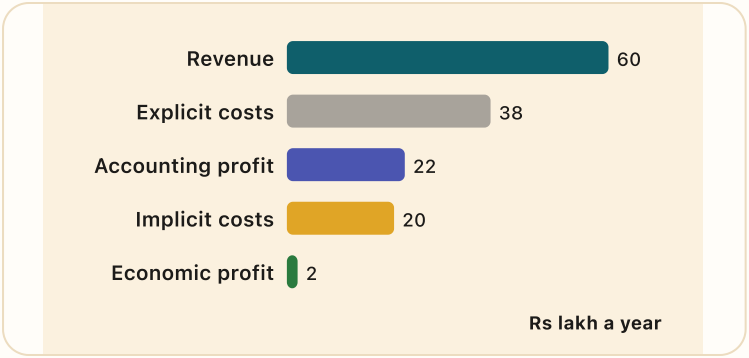
Pinky's chai chain: is it really making money?

Pinky left a job paying Rs 18 lakh a year to run a chai chain. She also put in Rs 20 lakh of her savings, which a bank FD would pay 10% on: Rs 2 lakh a year.

The chain takes in Rs 60 lakh. Rent, wages and milk cost Rs 38 lakh. Her accountant says: profit Rs 22 lakh.

Sharma ji asks what she **gave up**: the Rs 18 lakh salary and the Rs 2 lakh interest, Rs 20 lakh in all. No bill ever arrives for these, but they are real. So the chain beats her old life by only Rs 22 - 20 = **Rs 2 lakh**.

So: Real profit counts what you gave up, not only the bills you paid.



WORKED EXAMPLE

Accounting profit, economic profit, normal profit

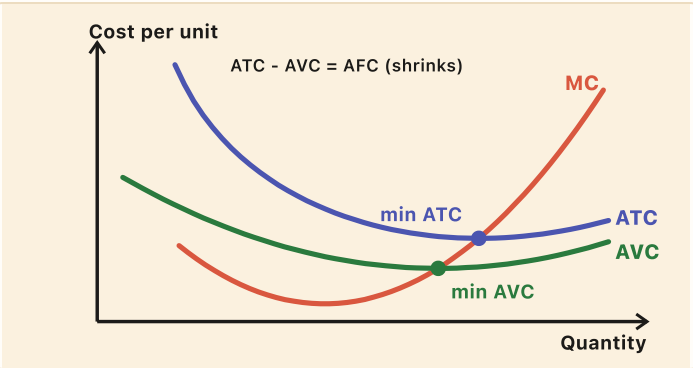
What we know

- **Revenue** = 60: Rs lakh a year (1 lakh = 100,000)
- **Explicit costs** = 38: bills paid: rent, wages, milk
- **Salary given up** = 18: her old job's pay
- **Savings put in** = 20: could earn 10% elsewhere

1. Accounting profit = 60 - 38 = 22
   Money in minus bills paid.
2. Implicit costs = 18 + 10% x 20 = 20
   What she gives up: salary plus lost interest (opportunity cost).
3. Economic profit = 22 - 20 = 2
   Above 0: she beats her next best option.

Economic profit Rs 2 lakh. Normal profit (accounting profit that just covers implicit costs) = 20.

So: Zero economic profit is not failure: it is a normal profit, exactly what the owner gave up.



PICTURE IT

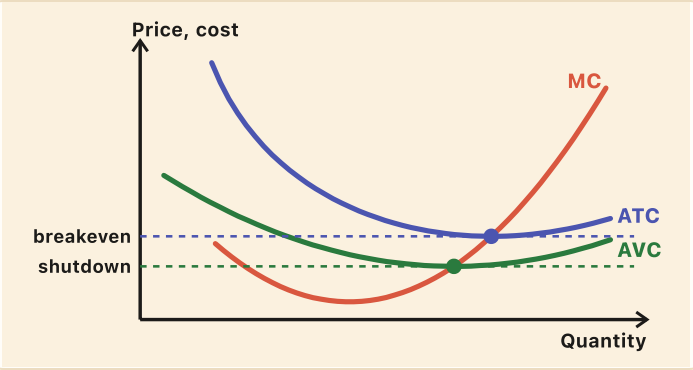
MC cuts AVC and ATC at their lowest points

MC first falls (specialisation), then rises (diminishing marginal returns: a third chaiwala at one stove gets in the way).

Averages follow the marginal like a class average follows the newest marks. While MC is below an average, it pulls the average down. Once MC is above, it pulls it up. So MC crosses **AVC** and **ATC** exactly at their minimums.

ATC sits above AVC by AFC, which shrinks as fixed cost spreads over more units. So the two curves converge.

So: MC passes through the bottom of AVC first, then the bottom of ATC, which is at a higher output.



PICTURE IT

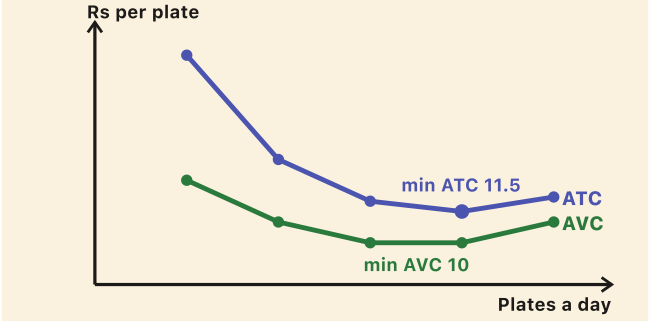
Breakeven and shutdown points

Price (= MR)	Short run	Long run
P ≥ ATC (TR ≥ TC)	Operate	Stay
AVC ≤ P < ATC (TVC ≤ TR < TC)	Operate at a loss	Exit
P < AVC (TR < TVC)	Shut down	Exit

P = price. ATC = full cost per unit; AVC = running cost per unit (rent left out). TR, TC, TVC = total revenue, total cost, total running cost.

Breakeven: P = min ATC: TR = TC, economic profit zero. **Shutdown point:** P = min AVC. Below it, every unit sold loses cash on top of the fixed cost.

So: **Short run: compare price with AVC. Long run: compare it with ATC.**



WORKED EXAMPLE

The vada pav cost table

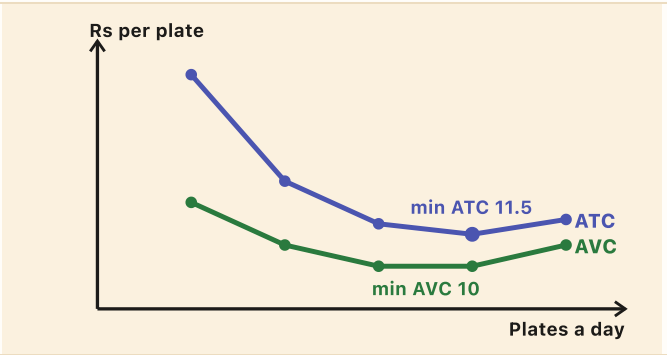
Dadar vada pav stall. Q = plates a day; TFC = rent Rs 600, paid even with no sales; TVC = potatoes, oil, pav, gas.

Q	TVC	TC	AVC	ATC
100	1300	1900	13	19
200	2200	2800	11	14
300	3000	3600	10	12
400	4000	4600	10	11.5
500	5500	6100	11	12.2

1. TC = TVC + TFC = 4,000 + 600 = 4,600
   At 400 plates: making cost plus rent.
2. AVC = TVC ÷ Q = 4,000 ÷ 400 = Rs 10; ATC = TC ÷ Q = 4,600 ÷ 400 = Rs 11.5
   Per plate: running cost only (AVC), then full cost with rent (ATC).

Lowest AVC: Rs 10. Lowest ATC: Rs 11.5.

So: **Find min ATC and min AVC first; price questions then become comparisons.**



WORKED EXAMPLE

Vada pav: breakeven and shutdown prices

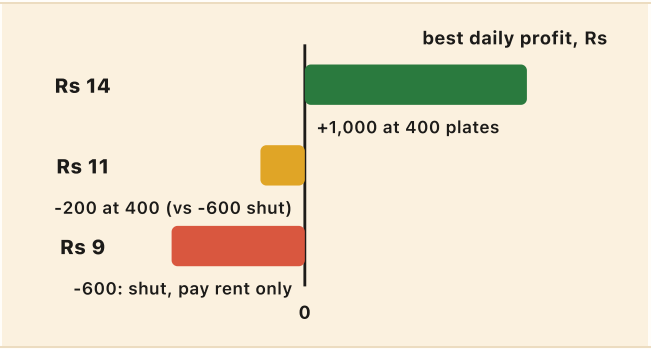
What we know

- **min ATC** = Rs 11.5: lowest full cost per plate, rent included
- **min AVC** = Rs 10: lowest running cost per plate, rent left out
- **TFC** = Rs 600: daily stall rent

1. Breakeven price = min ATC = Rs 11.5
   400 plates x 11.5 = 4,600: every cost, rent included, is paid.
2. Shutdown price = min AVC = Rs 10
   Sales just pay potatoes, oil and gas; nothing left for rent.
3. Above 11.5: profit. 10 to 11.5: open at a loss. Below 10: close
   Below Rs 10, each plate adds to the loss, so closing loses less.

Breakeven price Rs 11.5; shutdown price Rs 10.

So: **Breakeven = lowest full cost per unit; shutdown = lowest running cost per unit.**



WORKED EXAMPLE

Three prices, three decisions

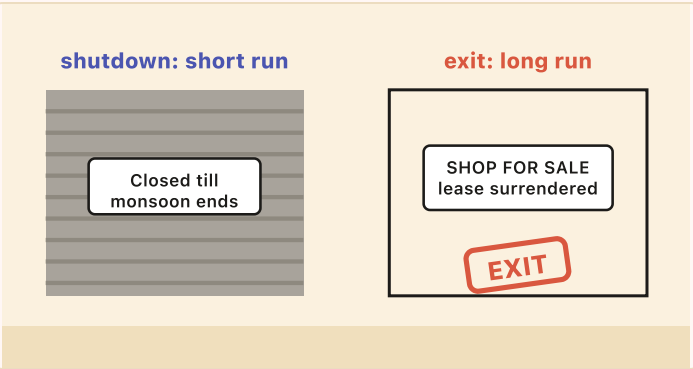
What we know

- **TFC** = Rs 600: stall rent, paid even when closed
- **TC** = Rs 4,600; Rs 3,600: total cost at 400 plates; at 300
- **min ATC, min AVC** = Rs 11.5, Rs 10: lowest full and running cost per plate

1. **Rs 14:** 14 x 400 = 5,600 in; minus 4,600 = +1,000
   Price beats full cost per plate (11.5), so she makes a profit.
2. **Rs 11:** 11 x 400 = 4,400 in; minus 4,600 = -200
   A loss, but closing loses the whole 600 rent. Sales pay the 4,000 making cost and 400 of rent.
3. **Rs 9:** best is 300 plates: 2,700 in; minus 3,600 = -900
   Even the making cost is not covered. Closing loses only the 600 rent.

Rs 14: profit. Rs 11: operate at a loss, exit later if it lasts. Rs 9: shut down now.

So: Between AVC and ATC, keep running: each sale still pays something towards the fixed cost.

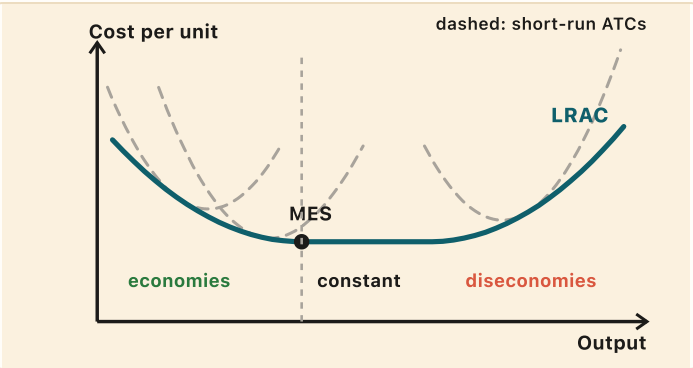


EXAM TRAP

Shutdown is not exit

- **Shutdown** (short run): produce zero, but fixed costs remain. The loss equals TFC.
- **Exit** (long run): leave the industry; no fixed costs remain because all costs are variable in the long run.
- Trap: "P < ATC, so shut down." Wrong. In the short run the test is P against **AVC**.
- Trap: "The firm makes zero economic profit, so it should leave." Wrong. Zero economic profit is a normal profit.
- Breakeven is P = min ATC, not P = MC.

So: Short run: P < AVC means shut down. Long run: TR < TC means exit.



PICTURE IT

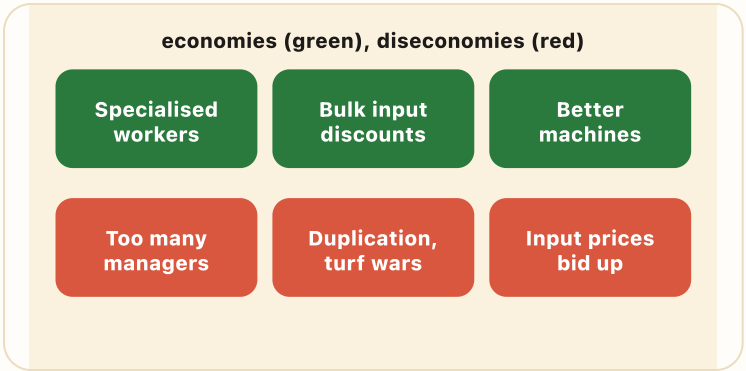
Long run: economies and diseconomies of scale

Short run: at least one input is fixed, usually the plant. **Long run:** every input can change, so the firm picks its plant size.

The **LRAC** curve wraps around the short-run ATC curves of every possible plant. Where LRAC falls, there are **economies of scale**; where it rises, **diseconomies**.

Minimum efficient scale (MES): the smallest output at which LRAC reaches its minimum. In competitive markets, firms are pushed to operate at or beyond it.

So: Economies of scale mean bigger is cheaper per unit; MES is the smallest size that gets the lowest LRAC.



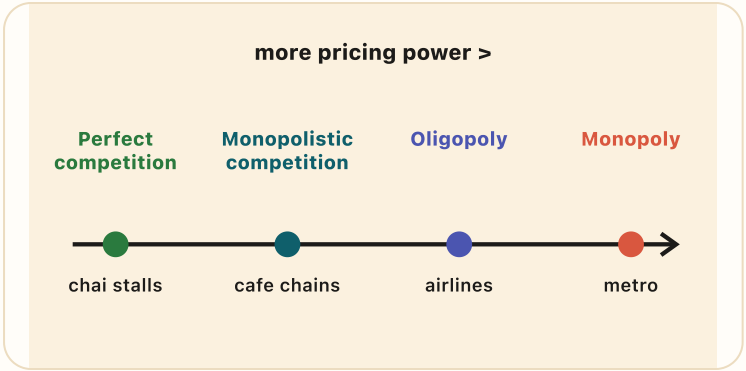
AT A GLANCE

Why big gets cheaper, then dearer

Economies of scale	Diseconomies of scale
Division of labour, specialised staff	Bureaucracy, slow decisions
Bulk buying discounts	Duplicated work across units
Large, efficient machines and technology	Big purchases push input prices up
Fixed costs (R&D, ads) spread over more units	Workers feel lost; motivation falls

A Diwali sweets factory buys sugar by the truck, cheaper per kg. Grow to 40 factories and approvals need six signatures.

So: Economies come from specialisation and spreading fixed costs; diseconomies from coordination problems.



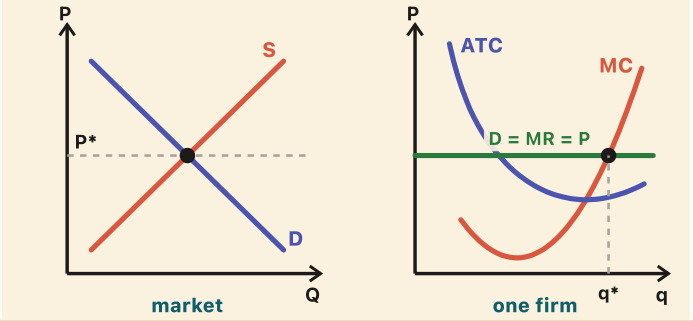
AT A GLANCE

The four market structures

	Perfect comp.	Monop. comp.	Oligopoly	Mono-poly
Sellers	Many	Many	Few	One
Product	Identical	Different	Same or different	Unique
Barriers	Very low	Low	High	Very high
Pricing power	None	Some	Some to a lot	A lot
Non-price rivalry	None	Ads, brand	Ads, brand	Ads, PR

A monopoly sells a product with no close substitute; its pricing power is often regulated.

So: **Count sellers, check differentiation and barriers: those three tell you the structure.**



PICTURE IT

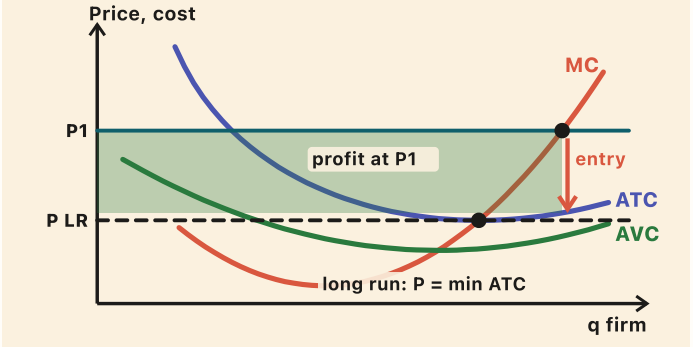
Perfect competition: market sets, firm takes

Thousands of identical sellers, like the onion sellers at Lasalgaon mandi. The **market** (left) sets the price where industry supply meets demand.

Each **firm** (right) is a price taker. It can sell all it wants at P*, and nothing above it. So its demand curve is flat, and **P = MR = AR**.

It produces q* where P = MC. Its short-run supply curve is its MC curve above min AVC.

So: For a perfectly competitive firm, demand is horizontal and price equals marginal revenue.



PICTURE IT

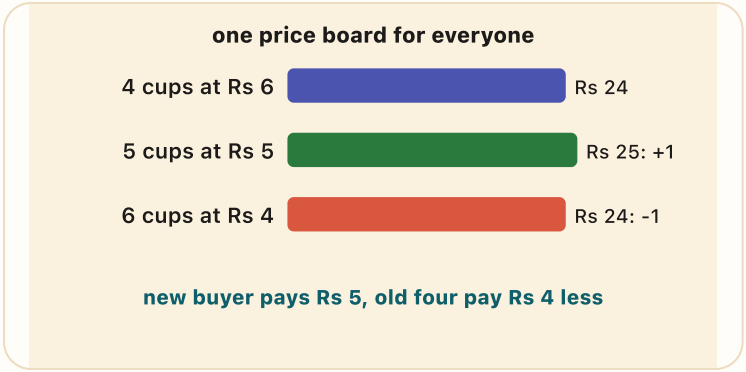
Perfect competition: profits do not last

Short run: if price is above ATC (P1), firms earn economic profit (shaded). If price is between AVC and ATC, they make losses but keep producing.

Long run: profit invites entry. Market supply shifts right and price falls until it reaches min ATC. Losses trigger exit, which pushes price back up.

The end point: **P = MR = MC = min ATC**. Zero economic profit, lowest possible cost per unit, no excess capacity.

So: Long-run perfect competition: zero economic profit, produced at the bottom of ATC.



STORY

Why the 5th cup adds only Rs 1

A tea kiosk at a bus stop has one price board for everyone. At **Rs 6** a cup, 4 people buy: Rs 24.

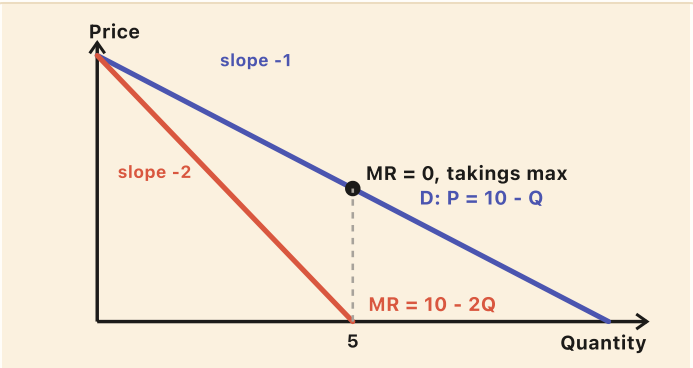
To win a 5th buyer she must drop the board to **Rs 5**. The new buyer pays Rs 5. But the 4 regulars now pay Rs 1 less each: Rs 4 lost.

Net gain from the 5th cup: 5 - 4 = **Rs 1**. Check: 5 x 5 = Rs 25, just Rs 1 more than 24.

A 6th cup at Rs 4 brings in Rs 4 but loses Rs 1 on each of the first five: net **-Rs 1**.

This extra money from one more sale is **marginal revenue (MR)**. When the price must fall for everyone, MR is below the price.

So: **Selling one more means a lower price for all, so the extra money is less than the price.**



FORMULA

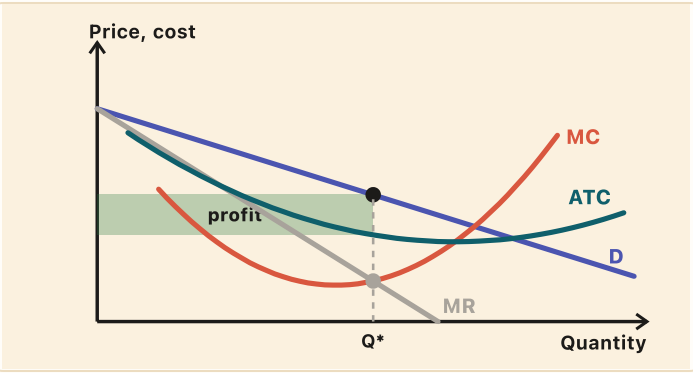
Straight-line demand: MR falls twice as fast

Demand: P = a − bQ
MR = a − 2bQ

Symbol	In plain words	Example
P, Q	price; cups sold	Rs 6; 4
a	price at which nobody buys (top of the line)	Rs 10
b	price cut needed to sell one more	Rs 1
MR	extra money from one more sale; 5th cup read at Q = 4.5	10 - 9 = 1

Why 2b: one more sale cuts the price by b for the new buyer *and* all earlier buyers. MR = 0 at Q = a ÷ 2b = 5: takings peak.

So: **Linear demand: MR has the same starting price a and twice the slope b.**



PICTURE IT

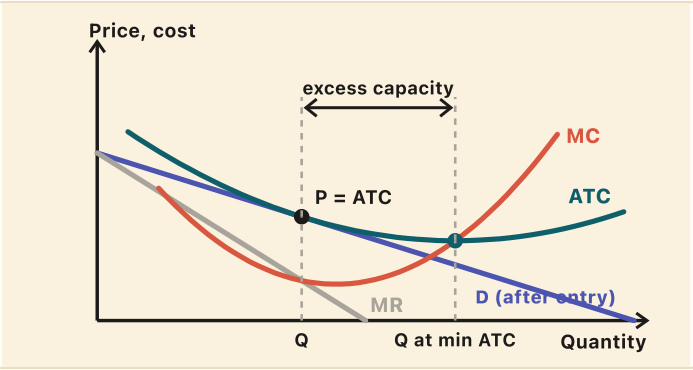
Monopolistic competition: the short run

Many cafes in Bandra, each slightly different: decor, playlist, the cold coffee people queue for. Each faces a **downward-sloping** demand curve, so **MR is below demand**.

The cafe picks Q* where MR = MC and charges the highest price demand allows at Q*. If that price is above ATC, it earns economic profit (shaded).

Firms compete through **advertising, brand and product quality**, not only price. A loyal brand makes demand less elastic.

So: Short run: MR = MC, price from the demand curve, profit possible.



PICTURE IT

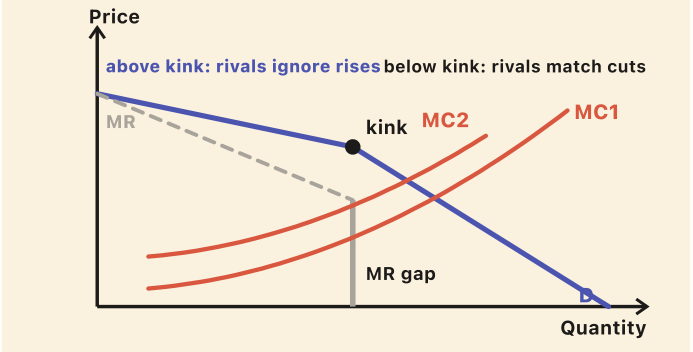
Monopolistic competition: the long run

Profits attract new cafes because entry barriers are low. Each new cafe takes customers, so each firm's demand curve shifts left until it just touches ATC.

Long-run result: **P = ATC**, zero economic profit. But the touching point is on the falling part of ATC, so output is below the min ATC output. That gap is **excess capacity**. Price is also above MC.

The cost of variety: slightly higher prices. Customers get choice in return.

So: Long run: zero economic profit, but P > MC and the firm produces left of min ATC.



PICTURE IT

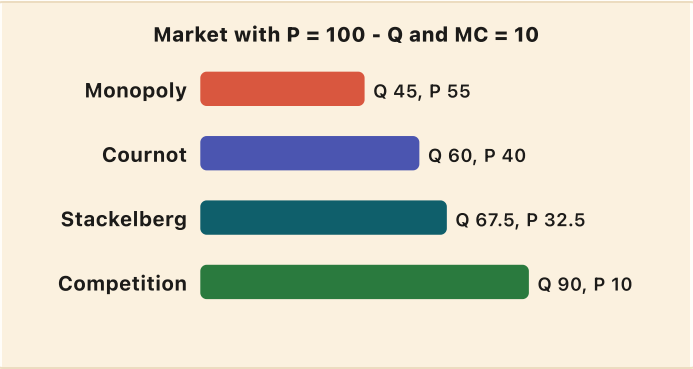
Kinked demand: why oligopoly prices stick

Two cement makers in one district. If one **raises** price, the rival does not follow and steals customers: demand above the kink is elastic. If one **cuts** price, the rival matches at once: demand below is inelastic.

The kink creates a **gap in MR**. MC can move anywhere inside the gap (MC1 to MC2) and the best price does not change. So prices stay sticky.

Weakness: the model explains why price stays put, but not how the kink price was set in the first place.

So: Kinked demand: rivals match cuts, ignore rises, so price is rigid over a range of MC.



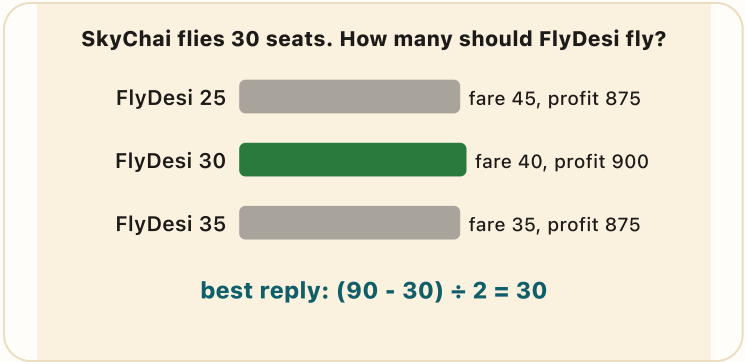
IDEA

Cournot and Stackelberg: guessing the rival's output

Cournot: two firms choose *quantities* at the same time. Each takes the rival's output as given and picks its own best response. Equilibrium output and price land **between monopoly and perfect competition**. Add more firms and the result moves towards competition.

Stackelberg: the same, but one firm moves first. The **leader** commits to a big output; the follower then responds. The first mover wins a larger share and higher profit. Total output is higher and price lower than under Cournot.

So: Cournot is simultaneous and lands between monopoly and competition; Stackelberg rewards the first mover.



FORMULA

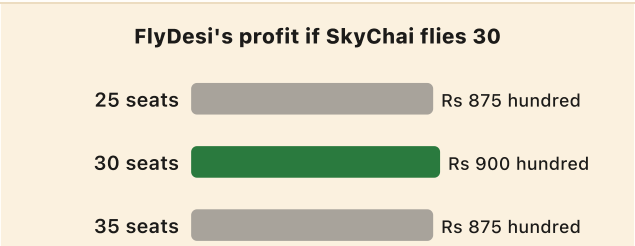
Cournot: one airline's best reply

$$q_A = (90 - q_B) \div 2$$

Symbol	In plain words	Example
P	fare, in Rs hundred, set by total seats: P = 100 - Q	40 = Rs 4,000
Q	total seats both airlines fly: $q_A + q_B$	60
q_A	seats FlyDesi chooses	30
q_B	seats the rival, SkyChai, flies	30
90	100 - MC: top fare minus the cost of a seat (MC = 10)	100 - 10

Why: FlyDesi is a monopolist on the leftover demand, $P = (100 - q_B) - q_A$. Its $MR = 100 - q_B - 2q_A$; set it equal to MC = 10.

So: Each Cournot firm picks its best output for the rival's output, like a monopolist on the leftover demand.



WORKED EXAMPLE

Best reply, played out in money

What we know

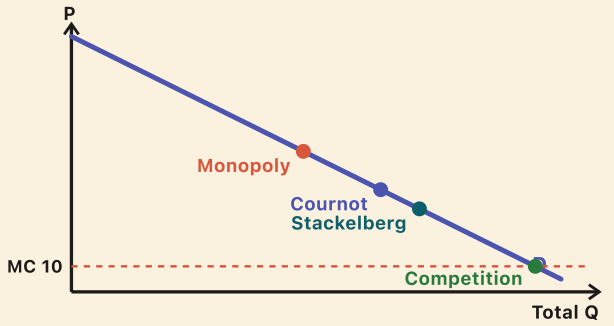
- **Demand** = P = 100 - Q: fare P (Rs hundred) for Q total seats
- **MC** = 10: cost of one more seat
- q_B = 30: seats SkyChai flies

What we want: FlyDesi's profit at 25, 30 and 35 seats. Profit = (fare - MC) x its seats.

1. 25 seats: Q = 55, fare 45. Profit = (45 - 10) x 25 = 875
   Fewer seats keep the fare high, but there are few seats to sell.
2. 30 seats: Q = 60, fare 40. Profit = (40 - 10) x 30 = 900
   The sweet spot.
3. 35 seats: Q = 65, fare 35. Profit = (35 - 10) x 35 = 875
   More seats, but the fare falls on every one of them.

Best reply to 30 is 30 seats, exactly what (90 - 30) ÷ 2 gives.

So: **The best-reply rule just finds the most profitable output against the rival's choice.**



WORKED EXAMPLE

Cournot duopoly in numbers

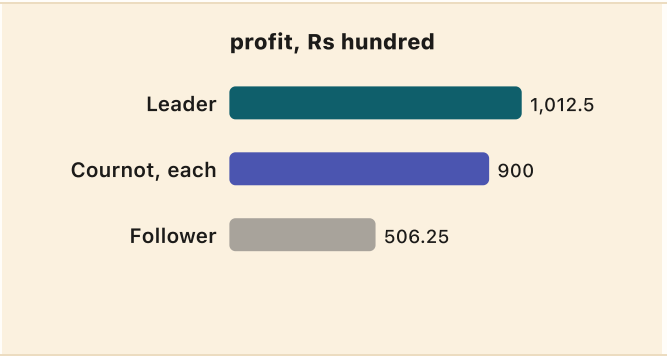
What we know

- **Demand** = P = 100 - Q: fare P in Rs hundred for Q total seats
- **MC** = 10: cost of one more seat; no fixed cost

1. Monopoly: MR = 100 - 2Q = 10, so Q = 45, P = 55; profit 45 x 45 = 2,025
   One airline: MR keeps the start, doubles the slope. Profit = (fare - MC) x seats.
2. Cournot: q = (90 - q) ÷ 2, so q = 30 each; Q = 60, P = 40
   Each airline flies q seats, the best reply to the other's q.
3. Cournot profit = (40 - 10) x 30 = 900 each
   1,800 together, below 2,025: rivalry costs them.

Cournot price 40 sits between monopoly 55 and competition (P = MC = 10, Q = 90).

So: **Ranking of price: monopoly, Cournot, Stackelberg, competition. Output ranks the other way.**



WORKED EXAMPLE

Stackelberg: the airline that moves first

What we know

- **Demand, MC** = P = 100 - Q; 10: fare P (Rs hundred) for Q seats; cost per seat
- **q_L, q_F** = ?: seats of the leader (moves first) and follower

1. Follower will fly $q_F = (90 - q_L) \div 2$

   The leader knows the follower's best-reply rule.

2. $P = 100 - q_L - (90 - q_L) \div 2 = 55 - 0.5\, q_L$

   So the fare depends only on the leader's seats.

3. $MR = 55 - q_L = 10$, so $q_L = 45$

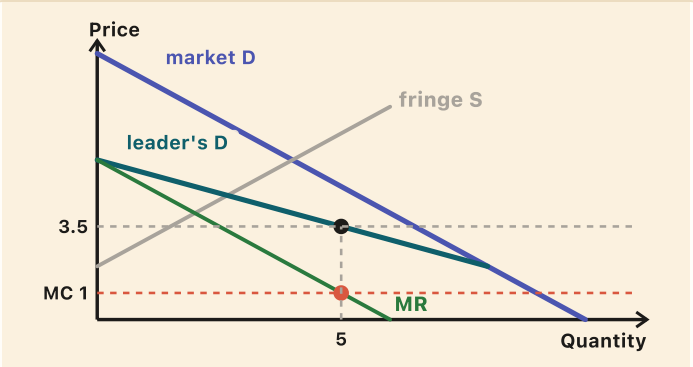
   Leader's MR = MC.

4. $q_F = (90 - 45) \div 2 = 22.5$; $Q = 67.5$; $P = 32.5$

   Follower replies; fare = 100 - 67.5.

Profit = margin 22.5 x seats: leader 1,012.5 (beats Cournot's 900), follower 506.25.

So: Moving first and committing to big output pays.



PICTURE IT

The dominant firm model

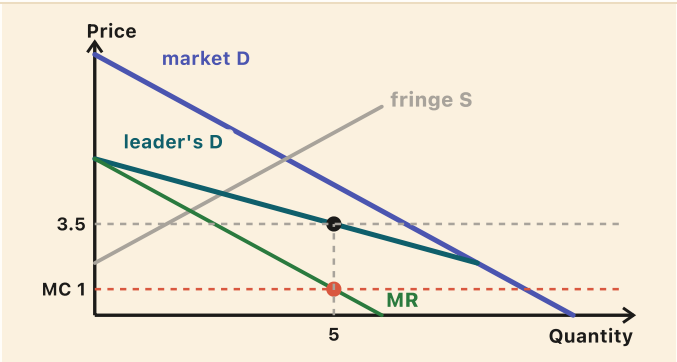
One big, low-cost firm (often 40% share or more) plus a fringe of small price takers.

The leader's demand = market demand minus what the fringe supplies at each price. It sets MR = MC on that leftover demand and picks the price. The fringe accepts it.

The chart: the leader's MC (cost of one more unit) is Rs 1. Where its MR meets MC it sells 5, at a price of Rs 3.5 read off its own demand line. The worked numbers are on the next card.

If the leader's costs rise or the fringe grows, its share shrinks.

So: Dominant firm: price leader on residual demand, small firms take its price.



WORKED EXAMPLE

Dominant firm in numbers

What we know

- **Market demand** = Q = 10 - P: units buyers want at price P
- **Fringe supply** = Q = P - 2: units small firms sell at P
- **MC** = Rs 1: leader's cost of one more unit

1. Leader's demand = (10 - P) - (P - 2) = 12 - 2P
   What buyers want minus what the small firms sell at that price.
2. Flip it: P = 6 - 0.5Q, so MR = 6 - Q
   Price in terms of the leader's sales; MR has the same start, double the slope.
3. 6 - Q = 1, so Q = 5; P = 6 - 2.5 = 3.5
   MR = MC sets the leader's sales; its demand line sets the price.

Leader sells 5 at Rs 3.5. Fringe: 3.5 - 2 = 1.5. Market: 10 - 3.5 = 6.5 = 5 + 1.5.

So: Leader's demand = market demand minus fringe supply; then MR = MC as usual.

profit Rs crore (FlyDesi, SkyChai)

	SkyChai: high fare	SkyChai: low fare
FlyDesi: high	50, 50 best together	10, 70
FlyDesi: low	70, 10	25, 25 Nash

PICTURE IT

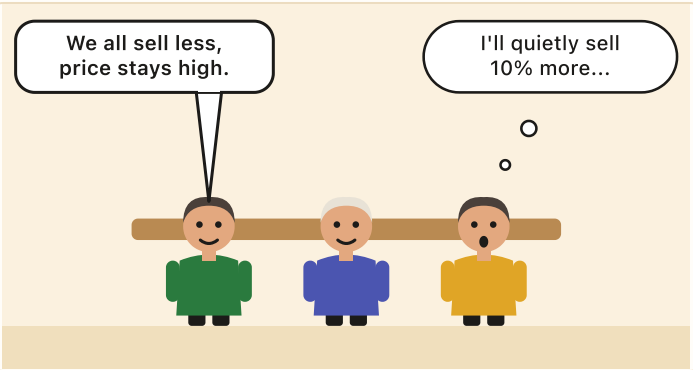
Nash equilibrium and the prisoner's dilemma

Nash equilibrium: each player's choice is its best response to what the other is doing, so nobody gains by changing alone.

Check FlyDesi. If SkyChai goes high, low pays 70 against 50. If SkyChai goes low, low pays 25 against 10. Low wins either way: a **dominant strategy**. SkyChai faces the same numbers.

So both go low and earn 25 each, though both high would earn 50 each. That is the **prisoner's dilemma**: self-interest beats the joint best.

So: In a prisoner's dilemma the Nash equilibrium is worse for both players than cooperating.



STORY

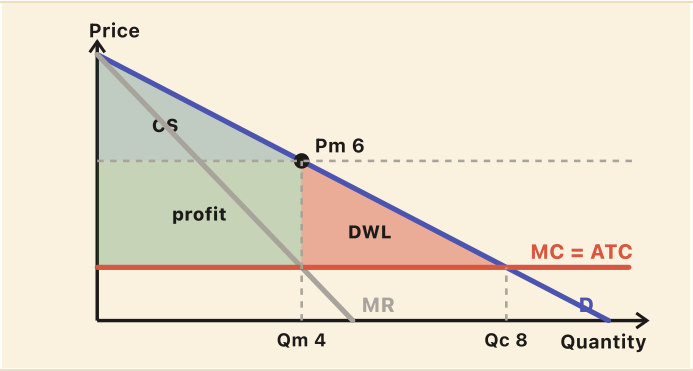
Cartels: a promise everyone wants to break

A **cartel** is a group of firms that agree to restrict output and raise price together, like OPEC in oil. Run well, it behaves like one monopolist.

It breaks because each member gains by **cheating**: at the cartel price, P is far above its MC, so one secret extra truckload is pure profit. If everyone thinks so, output rises and price collapses.

Collusion is more likely to last with **few firms**, similar products and costs, small frequent orders, easy detection, a credible threat to punish cheats, and little outside competition. It is illegal in many countries.

So: Cartels are unstable: each member's best private move is to cheat.



PICTURE IT

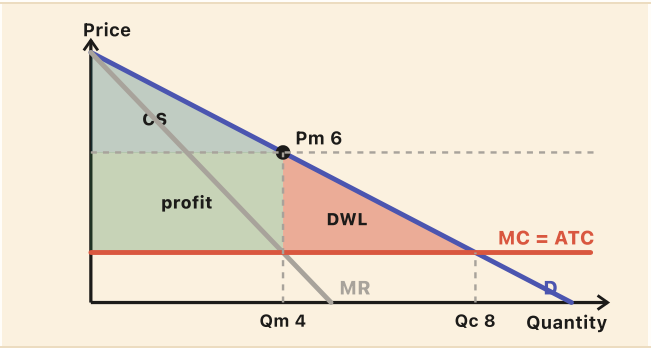
Monopoly: MR = MC and the deadweight loss

The city's only metro faces the whole demand curve D. It runs trains up to where **MR** (extra fare money from one more trip) equals **MC** (extra cost of one more trip), then charges the highest fare riders will pay for that many trips.

Result: fewer trips (**Qm**, monopoly quantity) at a higher fare (**Pm**) than a competitive market would give (**Qc**, where fare = MC).

The trips between Qm and Qc were worth more to riders than they cost, yet never happen. That lost value, the red triangle, is the **deadweight loss (DWL)**. **CS**, consumer surplus, is what riders would pay minus what they do pay.

So: Monopoly: less output, higher price, a deadweight loss triangle between Qm and Qc.



WORKED EXAMPLE

The metro's fare, worked out

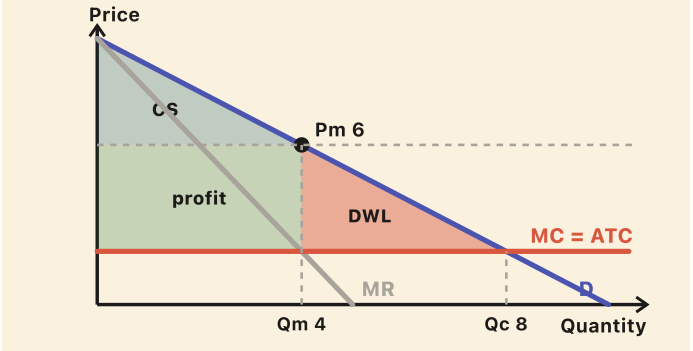
What we know

- **Demand** = P = 10 - Q: the fare in Rs (P) at which riders take Q lakh trips a day
- **MC = ATC** = Rs 2: extra cost, and average cost, per trip

1. MR = 10 - 2Q
   For a straight demand line, MR keeps the start (10) and doubles the slope (1 to 2).
2. 10 - 2Q = 2, so Q = 4
   Run trips until extra fare money equals extra cost (MR = MC).
3. P = 10 - 4 = 6
   Read the fare off demand, not MR: the most riders pay for 4 lakh trips.
4. Profit = (6 - 2) x 4 = 16
   Rs 4 margin per trip (fare minus cost per trip) x 4 lakh trips.

Fare Rs 6, 4 lakh trips a day, profit Rs 16 lakh a day.

So: Set MR = MC for quantity, then read the price off the demand curve.



WORKED EXAMPLE

The metro's deadweight loss

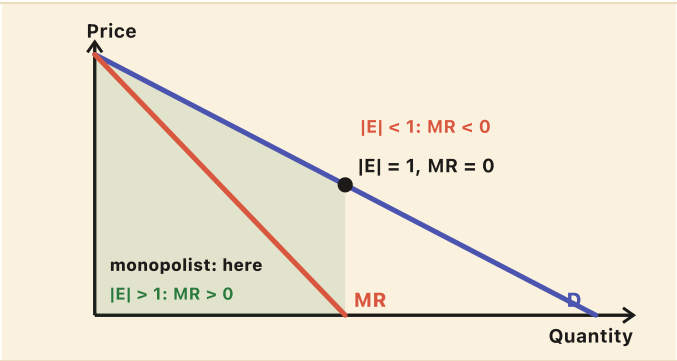
What we know

- **Demand** = P = 10 - Q: fare in Rs (P) for Q lakh trips a day
- **MC** = Rs 2: extra cost of one more trip
- **Qm, Pm** = 4 lakh, Rs 6: monopoly trips and fare

1. Competition: P = MC = 2, so 2 = 10 - Q, Q = 8
   With rivals, fares fall to cost and 8 lakh trips happen.
2. Lost trips = 8 - 4 = 4 lakh
   Trips riders valued above their Rs 2 cost that the monopoly never runs.
3. DWL = ½ x 4 x (6 - 2) = 8
   Triangle area: the first lost trip was worth Rs 4 more than its cost, the last Rs 0; on average Rs 2, x 4 lakh trips.

Deadweight loss Rs 8 lakh a day: value nobody gets, not riders and not the metro.

So: DWL is the triangle between monopoly output and competitive output.



FORMULA

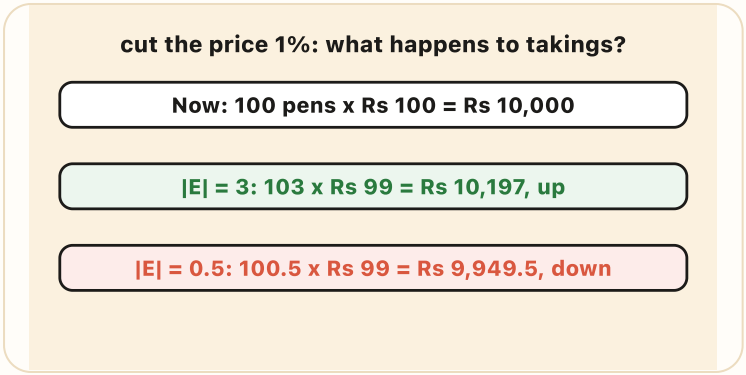
MR and elasticity: MR = P(1 - 1/|E|)

```
MR = P x (1 – 1 / |E|)
```

Symbol	In plain words	Example
MR	extra money from selling one more unit	Rs 66.7
P	price now	Rs 100
E	price elasticity: % change in quantity ÷ % change in price	-3
|E|	E with the minus sign dropped ("absolute value")	3
1 / |E|	per extra unit sold, the share of its price given away as a discount to earlier buyers	1 ÷ 3 = 0.33

|E| > 1: MR positive. |E| = 1: MR = 0. |E| < 1: MR negative. A monopolist sets MR = MC, and MC is positive, so it **always sits on the elastic part**.

So: A monopolist never chooses the inelastic part of the demand curve.



PLAYED OUT IN MONEY

MR and elasticity, in rupees

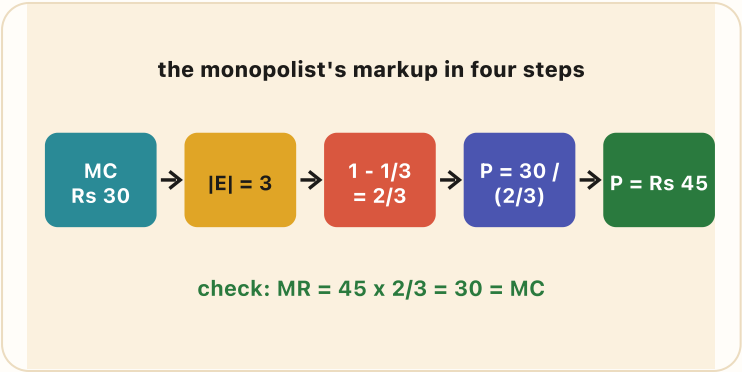
A pen maker sells **100 pens at Rs 100**: Rs 10,000. It cuts the price 1% to Rs 99.

Responsive buyers (|E| = 3): sales rise 3% to 103. Takings: 99 x 103 = Rs 10,197. The 3 extra pens added Rs 197, about **Rs 66 each**. Formula: 100 x (1 - 1/3) = Rs 66.7. A match.

Sticky buyers (|E| = 0.5): sales rise only 0.5% to 100.5. Takings: 99 x 100.5 = Rs 9,949.5. Selling more brought in **less** money: MR is negative. Formula: 100 x (1 - 1/0.5) = **-Rs 100**.

So a monopolist never stops where buyers are sticky: a higher price there brings in more money and means fewer pens to make.

So: Elastic buyers: a price cut raises takings (MR above 0). Inelastic buyers: it lowers them (MR below 0).



WORKED EXAMPLE

Pricing with elasticity

A monopoly makes a patented insulin pen.

What we know

- **MC** = Rs 30: extra cost of one more pen
- **|E|** = 3: elasticity, sign dropped: a 1% price cut lifts sales 3%

What we want: the price P; could |E| be 0.8 there?

1. P x (1 - 1/3) = 30
   Best output: MR = MC. Swap in MR = P(1 - 1/|E|).
2. 1 - 1/3 = 2/3, so P = 30 ÷ (2/3) = Rs 45
   MR is 2/3 of the price, so price = 1.5 x MC.
3. If |E| = 0.8: MR = P x (1 - 1/0.8) = -0.25P
   Negative MR can never equal a positive MC.

P = Rs 45. |E| = 0.8 at the best output is impossible.

So: Plug |E| into MR = P(1 - 1/|E|), set MR = MC, solve for P.



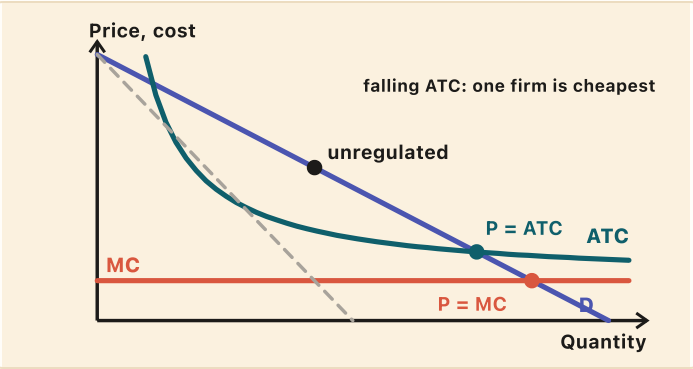
AT A GLANCE

Price discrimination: three degrees

Degree	How	Effect
First (perfect)	Each buyer pays their maximum	Takes all consumer surplus; competitive output, no DWL
Second	Unit price falls with quantity	Buyers sort by volume
Third	Different prices by group	Less elastic group pays more

Same good, different prices, not explained by cost. **Needs:** pricing power, a way to split buyers by willingness to pay, and no resale.

So: **Third degree: charge more to the group with less elastic demand, and block resale.**



PICTURE IT

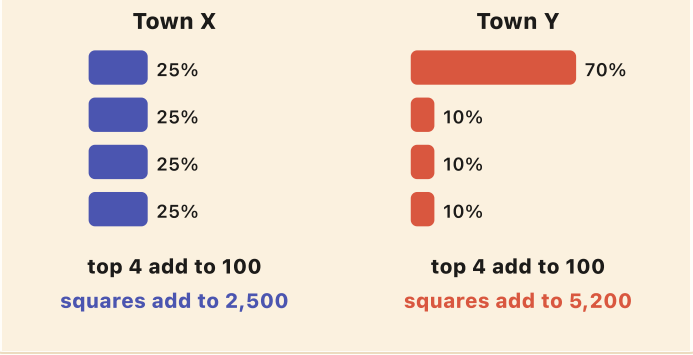
Natural monopoly and how it is regulated

Power grids and water pipes have huge fixed costs and tiny MC. ATC falls over the whole market, so one firm is cheapest: a **natural monopoly**. Regulators then pick the price.

- **Marginal cost pricing** (P = MC): efficient output, but P < ATC, so the firm loses money and needs a subsidy.
- **Average cost pricing** (P = ATC): the firm earns a normal profit with no subsidy. Output is a little lower, so a small DWL remains.

Either beats the unregulated MR = MC outcome on output and price.

So: **Natural monopoly: P = MC is efficient but needs a subsidy; P = ATC covers costs.**



STORY

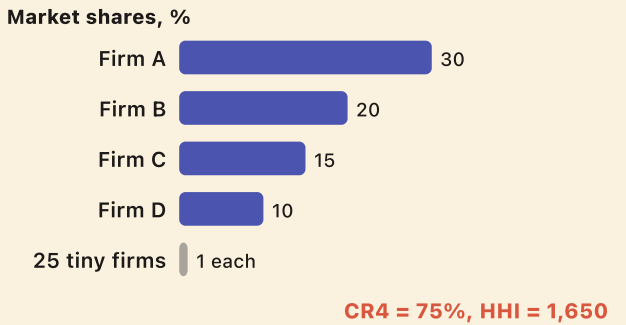
Two towns, four cable operators each

Town X has four cable operators with 25% of homes each. Town Y also has four, but one has 70% and the other three 10% each.

Add up the top four shares: 100% in both towns. This measure says the towns are the same. They are not: in Town Y one operator calls the shots.

Square each share, then add: Town X: 4 x 625 = 2,500. Town Y: 4,900 + 100 + 100 + 100 = 5,200. Squaring makes big numbers much bigger (10 becomes 100, but 70 becomes 4,900), so one giant firm shows up loudly.

So: Adding shares hides a giant; squaring them exposes it.



FORMULA

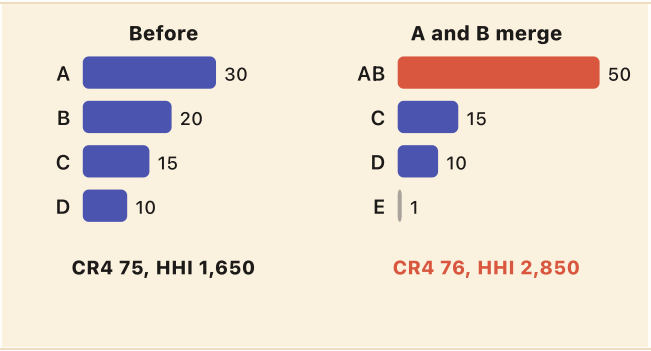
Concentration ratio and HHI

$$CR(N) = s_1 + s_2 + \ldots + s_N$$
$$HHI = s_1^2 + s_2^2 + \ldots \text{ (every firm)}$$

Symbol	In plain words	Example
s_1, s_2	shares (% of industry sales) of the biggest firm, the next, and so on	30, 20
N	how many top firms you count	4
CR(N)	N-firm concentration ratio: the top N shares added	CR4 = 75%
HHI	Herfindahl-Hirschman Index: every share squared, then added	1,650

CR(N) near 0: competitive; 100: monopoly. HHI tops out at 10,000 (one firm: 100^2). With decimal shares it tops out at 1, and 1 ÷ HHI = the number of equal firms it is like (four at 0.25: HHI 0.25, 1 ÷ 0.25 = 4).

So: **HHI squares shares, so it reacts to how unequal the big firms are; CR(N) does not.**



WORKED EXAMPLE

HHI and a merger

What we know

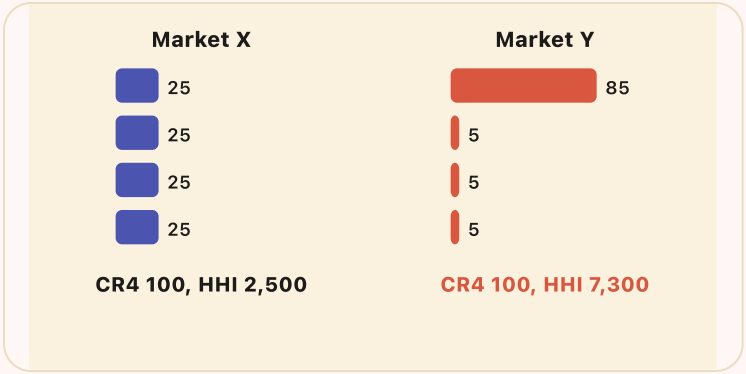
- **Shares** = 30, 20, 15, 10: % of industry sales: A, B, C, D
- **Tiny firms** = 25 of them: firms with 1% of sales each

What we want: CR4 (top four shares added) and HHI (every share squared, then added), before and after A and B merge.

1. Before: CR4 = 30 + 20 + 15 + 10 = 75%
   Add the four biggest shares.
2. HHI = 900 + 400 + 225 + 100 + 25 x 1^2 = 1,650
   Square every firm's share, tiny ones too ($1^2 = 1$), and add.
3. After: CR4 = 50 + 15 + 10 + 1 = 76%
   A and B are now one firm (50), so a 1% firm joins the top four.
4. HHI = 2,500 + 225 + 100 + 25 = 2,850
   Rise 1,200 = 2 x 30 x 20, since $50^2 = 30^2 + 20^2 + 2 \times 30 \times 20$.

CR4 moves 75 to 76; HHI jumps 1,650 to 2,850.

So: A merger between top firms barely moves the concentration ratio but lifts HHI sharply.

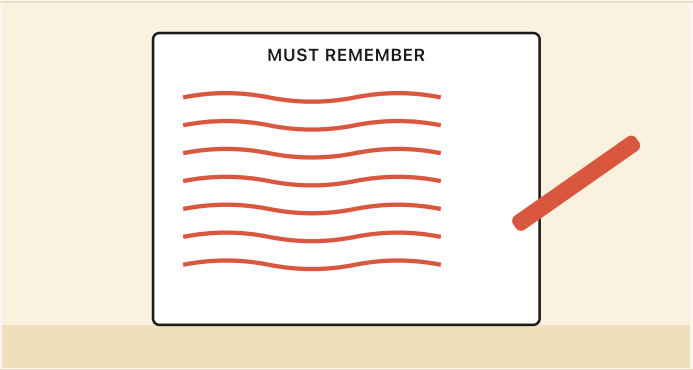


EXAM TRAP

What concentration measures miss

- Same CR4, very different markets: four equal firms (HHI 2,500) versus one giant and three dwarfs (HHI 7,300).
- Neither measure sees **barriers to entry**. A lone firm facing easy entry may price like a competitive one, because potential entrants discipline it.
- Neither captures **demand elasticity** or close substitutes from outside the defined market.
- Results depend on how the market is defined: "Mumbai coffee" or "Indian beverages".
- Trap: "high concentration proves pricing power." It only suggests it.

So: Concentration is a clue to market power, not proof: always ask about entry barriers and elasticity.

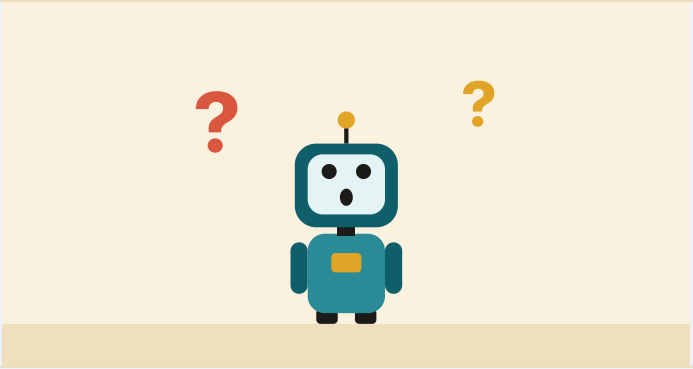


SHARMA JI'S RED PEN

7 things to remember

1. MR = MC, MC rising. Zero economic profit = normal profit.
2. Breakeven P = min ATC. Shutdown P < min AVC. In between: run at a loss, exit later.
3. MC cuts AVC and ATC at their lowest points. Perfect competition, long run: P = MC = min ATC.
4. Monopolistic competition, long run: P = ATC, but left of min ATC.
5. Kinked demand: sticky prices. Cournot: in between. Cartels: cheating pays.
6. Monopoly: MR = P(1 - 1/|E|), elastic part only, DWL.
7. HHI squares shares; a top-firm merger moves HHI, not CR(N).

So: **Scribbled by Sharma ji on the back of a Bandra cafe bill.**



EXAM QUESTION 1 OF 8

A firm in a perfectly competitive market sells at Rs 8. At its best output, AVC is Rs 7 and ATC is Rs 9. In the short run the firm should most likely:

A. shut down, because price is below ATC.

B. keep producing, even though it makes a loss.

C. exit the industry immediately.

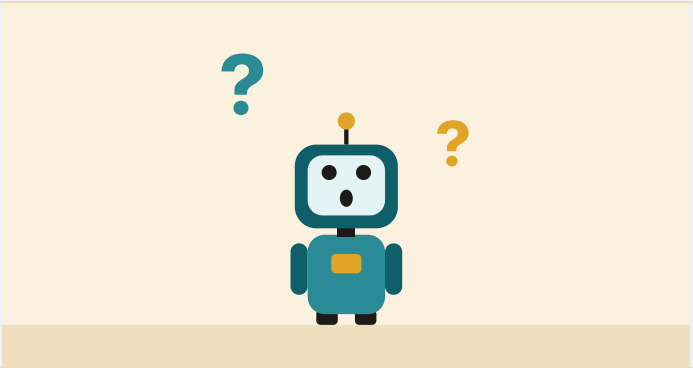
Answer on the next page.

So: **Pick an answer, then turn the page.**

ANSWER

A firm in a perfectly competitive market sells at Rs 8. At its best output, AVC is Rs 7 and ATC is Rs 9. In the short run the firm should most likely:

Answer: B. Price (8) is above AVC (7), so each unit covers its variable cost and adds Re 1 towards fixed costs. The loss is smaller than the fixed cost it would lose by shutting. A is wrong: the short-run shutdown test is P against AVC, not ATC. C is wrong: exit is a long-run decision, taken if TR stays below TC.



EXAM QUESTION 2 OF 8

Four firms hold market shares of 40%, 30%, 20% and 10%. The Herfindahl-Hirschman Index is closest to:

A. 3,000

B. 2,500

C. 100

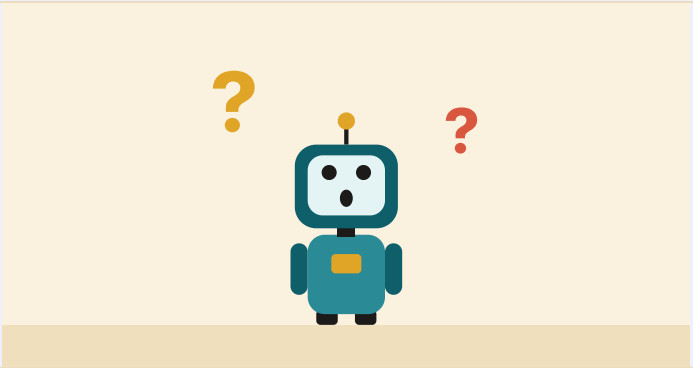
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So: Pick an answer, then turn the page.

ANSWER

Four firms hold market shares of 40%, 30%, 20% and 10%. The Herfindahl-Hirschman Index is closest to:

Answer: A. HHI = $40^2 + 30^2 + 20^2 + 10^2$ = 1,600 + 900 + 400 + 100 = 3,000. B is the HHI of four equal firms at 25% each, which ignores the actual inequality. C is just the sum of the shares, which is the four-firm concentration ratio (100%), not the HHI.



EXAM QUESTION 3 OF 8

A monopolist's marginal cost is 20. At its profit-maximising output, the absolute price elasticity of demand is 2. Its price is closest to:

A. 20

B. 30

C. 40

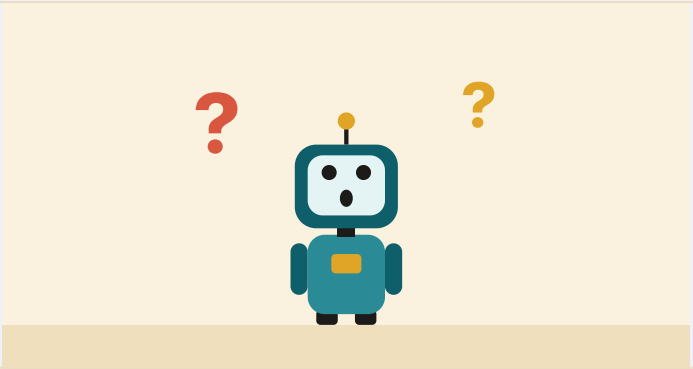
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So: Pick an answer, then turn the page.

ANSWER

A monopolist's marginal cost is 20. At its profit-maximising output, the absolute price elasticity of demand is 2. Its price is closest to:

Answer: C. Set MR = MC: P(1 - 1/2) = 20, so P = 20 / 0.5 = 40. A sets price equal to MC, the competitive outcome, which a monopolist does not choose. B adds half of MC as a markup, a guess with no basis in the formula.



EXAM QUESTION 4 OF 8

In long-run equilibrium, a firm in monopolistic competition most likely:

A. earns positive economic profit protected by its brand.

B. produces at the minimum of its average total cost.

C. charges a price equal to average total cost, at an output below the minimum-ATC output.

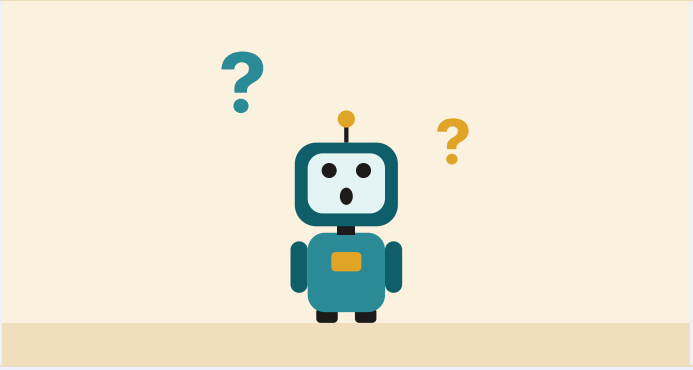
Answer on the next page.

So: **Pick an answer, then turn the page.**

ANSWER

In long-run equilibrium, a firm in monopolistic competition most likely:

Answer: C. Low entry barriers let new firms in until each firm's demand curve just touches ATC: zero economic profit, on the falling part of ATC, so there is excess capacity. A is wrong: entry erodes long-run economic profit. B describes perfect competition, where the demand curve is flat and touches the bottom of ATC.



EXAM QUESTION 5 OF 8

The kinked demand curve model of oligopoly assumes that when one firm changes its price, rivals:

A. match price cuts but ignore price increases.

B. match price increases but ignore price cuts.

C. ignore all price changes.

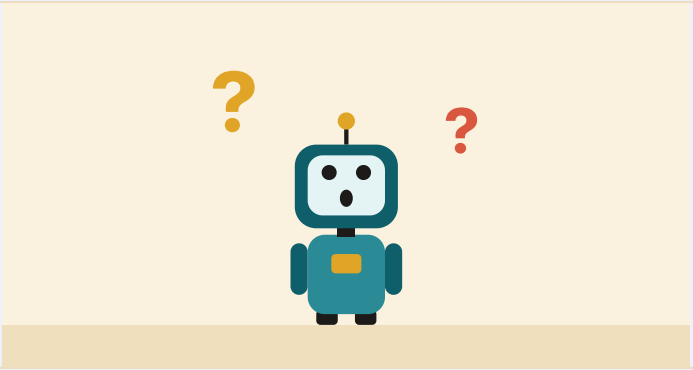
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

The kinked demand curve model of oligopoly assumes that when one firm changes its price, rivals:

Answer: A. Rivals match cuts (so a cut wins few customers: inelastic below the kink) and ignore rises (so a rise loses many: elastic above). That creates the MR gap and sticky prices. B reverses the assumption and would not produce the kink. C would give a single straight demand curve with no kink.



EXAM QUESTION 6 OF 8

Compared with a monopoly and a perfectly competitive market facing the same demand and costs, a Cournot duopoly most likely produces:

A. the same output as a monopoly, because two firms collude.

B. an output between the monopoly and competitive outputs.

C. the competitive output, because each firm sets P = MC.

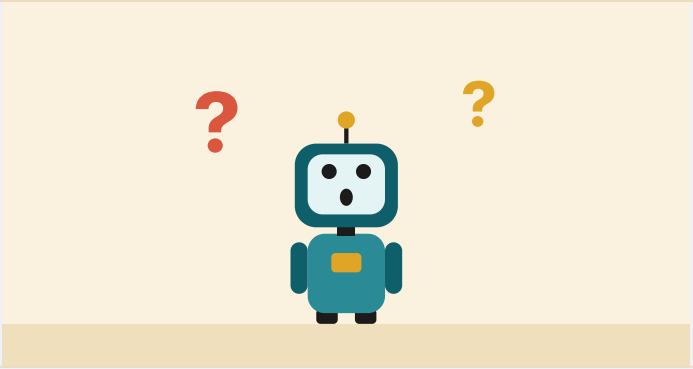
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So: **Pick an answer, then turn the page.**

ANSWER

Compared with a monopoly and a perfectly competitive market facing the same demand and costs, a Cournot duopoly most likely produces:

Answer: B. Each Cournot firm chooses its output taking the rival's as given. Total output ends above the monopoly level but below the competitive level, and price lands between the two. A describes a successful cartel, not Cournot. C is the perfectly competitive outcome; Cournot firms keep P above MC.



EXAM QUESTION 7 OF 8

A cinema charges students less than other adults for the same seat. For this pricing to raise profit, the cinema most likely needs students to have:

A. more elastic demand than other adults.

B. less elastic demand than other adults.

C. the same elasticity of demand as other adults.

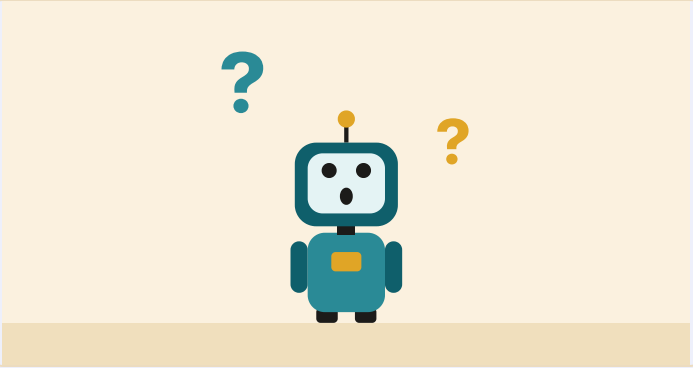
Answer on the next page.

So: **Pick an answer, then turn the page.**

ANSWER

A cinema charges students less than other adults for the same seat. For this pricing to raise profit, the cinema most likely needs students to have:

Answer: A. Third-degree price discrimination charges the lower price to the more price-sensitive (more elastic) group. Students react strongly to price, so a discount brings many more of them in. B is backwards: the less elastic group pays more. C leaves no reason to charge different prices.



EXAM QUESTION 8 OF 8

A market has shares of 30%, 20%, 15% and 10%, with the rest spread across many firms of 1% each. If the two largest firms merge, which is most likely?

A. Both the four-firm concentration ratio and the HHI rise sharply.

B. The four-firm ratio rises sharply while the HHI barely changes.

C. The HHI rises sharply while the four-firm ratio barely changes.

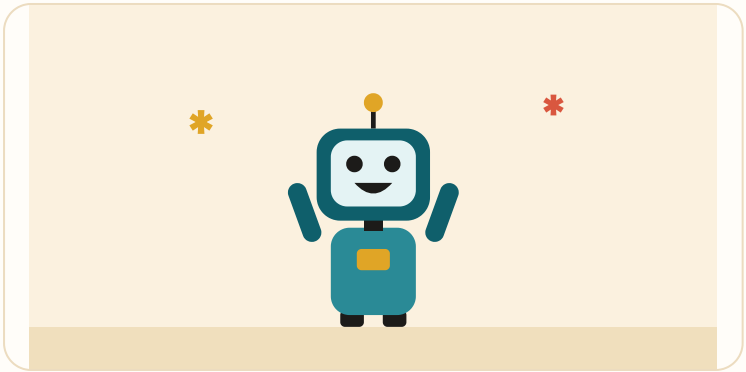
Answer on the next page.

So: **Pick an answer, then turn the page.**

ANSWER

A market has shares of 30%, 20%, 15% and 10%, with the rest spread across many firms of 1% each. If the two largest firms merge, which is most likely?

Answer: C. The merged firm's 50% was already inside the top four, so CR4 only gains the next firm's 1% (75 to 76). HHI rises by 2 x 30 x 20 = 1,200 (1,650 to 2,850) because squaring rewards size. A and B miss that the concentration ratio is blind to a merger among top firms.



CHAPTER 1 DONE

You can now name any market, and price it

Chintu's verdict for Sharma ji: the chai stall earns normal profit at best, the cafe chain earns profit only until rivals copy it, the airline duopoly lives on rivals' reactions, and the metro needs a regulator watching its price. Sharma ji nodded once. From him, that is applause.

So: **Swipe on, or jump to another chapter from the Menu.**



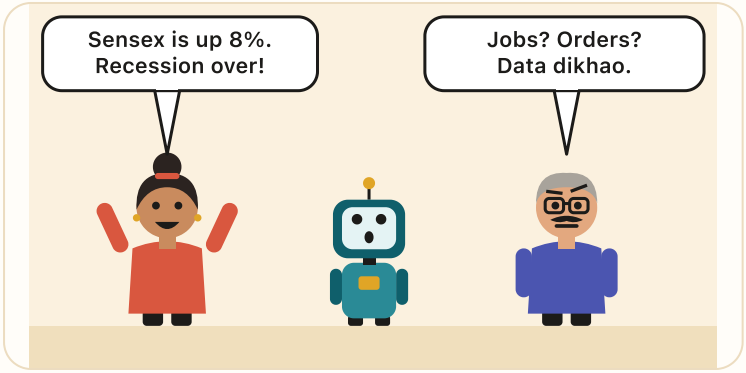
BEFORE YOU START

The cast, and words this chapter uses

A Mumbai investment firm. Sharma ji trusts nothing without data ("Data dikhao"); Pinky is sure of everything, especially when wrong.

- **Real GDP:** an economy's total output, adjusted for inflation.
- **Potential GDP:** the output it can sustain without pushing inflation up.
- **Monetary, fiscal policy:** central bank rates and money; government spending and taxes.
- **Collateral:** an asset pledged to a lender, like a flat behind a home loan.
- **Lakh crore:** 1 trillion.

So: **You can start this chapter cold. Everything it needs is on this card.**



STORY

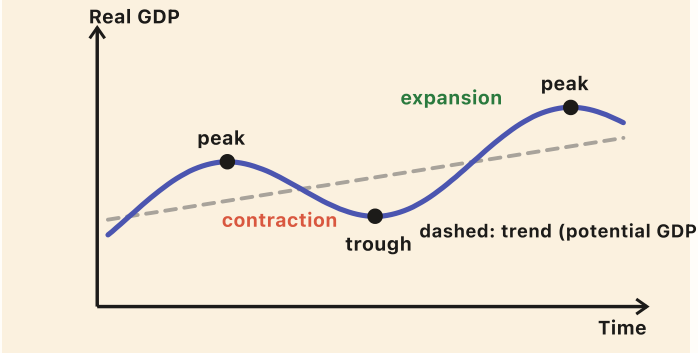
Is the recession over?

Pinky storms into the Monday meeting. "Sensex up 8% in a month. The slowdown is over. Buy everything."

Sharma ji does not look up. "Factories are running at 68% of capacity. Jobless claims are still rising. Car sales fell before Diwali. Data dikhao, Pinky."

Chintu's job this week: read where the economy sits in its cycle, using the right data for each question. Some numbers move first, some move together, some arrive late to the party.

So: Different data turn at different times; knowing which is which tells you where the cycle is.



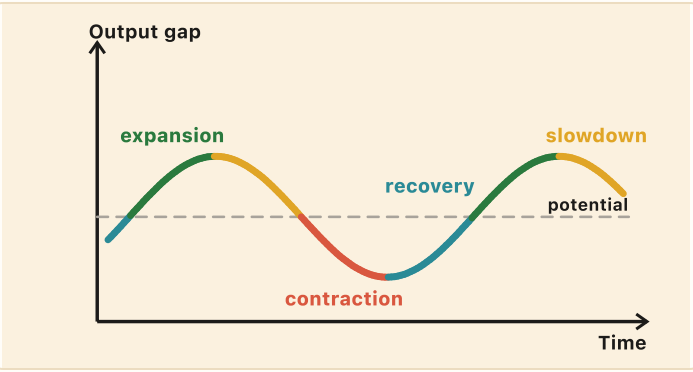
PICTURE IT

The business cycle: up, down, up again

A **business cycle** is the swing of real GDP and other activity around its long-run trend. Cycles are recurring but irregular: no two last the same time.

- **Peak:** activity tops out. **Trough:** it bottoms out.
- **Expansion:** trough to peak. **Contraction:** peak to trough.
- **Recession:** a significant contraction. A common rule of thumb is two quarters in a row of falling real GDP; official dating bodies, like the NBER in the US, look at many indicators.

So: Peak and trough are turning points; expansion and contraction are the stretches between them.



AT A GLANCE

Four phases, read from the output gap

The **output gap** is actual GDP minus potential GDP. The CFA curriculum splits the cycle into four phases using its sign and direction:

Phase	Output gap	Note
Recovery	Negative, closing	Starts at the trough
Expansion	Positive, widening	Growth above trend
Slowdown	Positive, narrowing	Starts at the peak
Contraction	Negative, widening	Ends at the trough

So: **Below potential and rising: recovery. Above and rising: expansion. Above and falling: slowdown. Below and falling: contraction.**



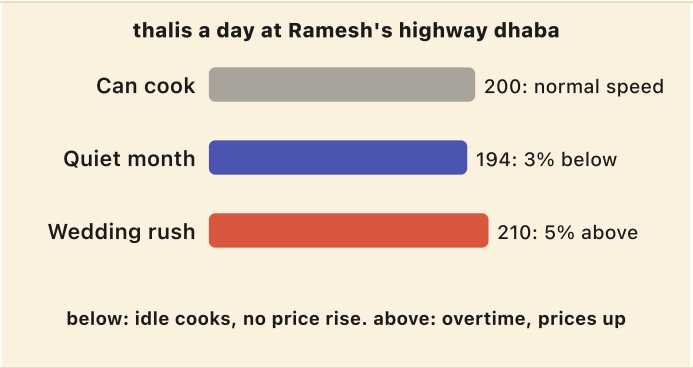
AT A GLANCE

What GDP, jobs, prices and stocks of goods do

Phase	GDP	Jobs	Prices	Stocks
Recovery	Turns up	Still weak	Low	Rebuilt
Expansion	Above trend	Hiring	Rising	Built up
Slowdown	Slows	Hiring slows	Peaks	Pile up
Contrac-tion	Falls	Layoffs	Eases late	Cut

Stocks = inventories. Unemployment stays high early in a recovery; inflation stays high into the slowdown.

So: Jobs and inflation turn after GDP: unemployment is still high in recovery, inflation still high in the slowdown.



STORY

Output gap: a dhaba below its best

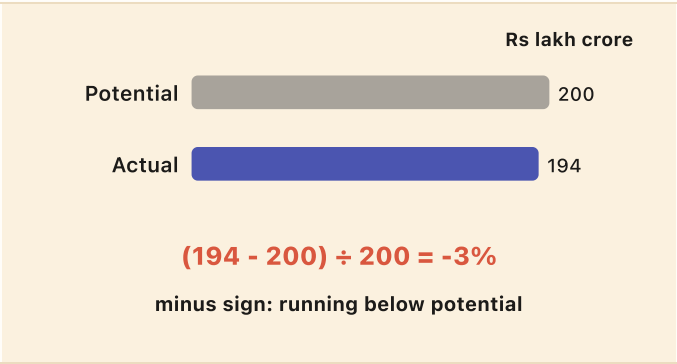
Ramesh's highway dhaba can cook **200 thalis a day** with its normal cooks and stoves, no overtime. That 200 is its **potential**.

Quiet month: only 194 are ordered. Cooks stand around and gas cylinders sit unused. Ramesh will not raise prices; he may offer a free lassi. He is 6 thalis short of 200, which is 6 ÷ 200 = **3% below** potential.

Wedding season: 210 are ordered. Cooks do overtime at extra pay, so Ramesh raises prices. He is 10 ÷ 200 = **5% above** potential.

Swap the dhaba for a whole country and thalis for real GDP: that % is the **output gap**.

So: Below potential: idle capacity and soft prices. Above potential: overtime and rising prices.



FORMULA

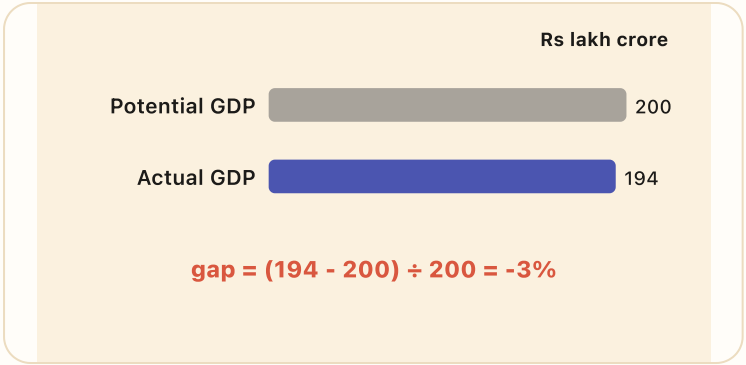
The output gap

Output gap = (Actual GDP − Potential GDP) ÷ Potential GDP

Symbol	In plain words	Example
Actual GDP	real output the economy really produced this year	Rs 194 lakh crore
Potential GDP	output at normal use of workers and machines, without pushing inflation up	Rs 200 lakh crore
Output gap	how far actual sits above (+) or below (-) potential, as a % of potential	-3%

Why divide by potential: it turns the shortfall into a %, so economies of any size compare.

So: **Output gap = (actual - potential) ÷ potential; negative means slack, positive means overheating.**



WORKED EXAMPLE

Output gap in numbers

What we know

- **Potential** = Rs 200 lakh crore: real GDP at normal speed
- **Actual** = Rs 194 lakh crore: real GDP really made

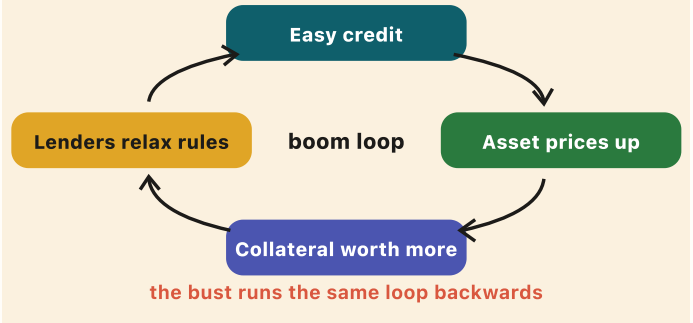
What we want: the output gap: is the economy cold (below potential) or hot?

1. 194 - 200 = -6
   Actual minus potential: Rs 6 lakh crore less than it could make.
2. -6 ÷ 200 = -0.03 = -3%
   Divide by potential to turn the shortfall into a %.

Output gap = -3%: the economy runs 3% below potential.

Sense check: 194 is below 200, so the gap must be negative, like a dhaba cooking 194 of a possible 200 thalis.

So: **Negative gap: slack and low inflation pressure. Positive gap: overheating and rising inflation risk.**



PICTURE IT

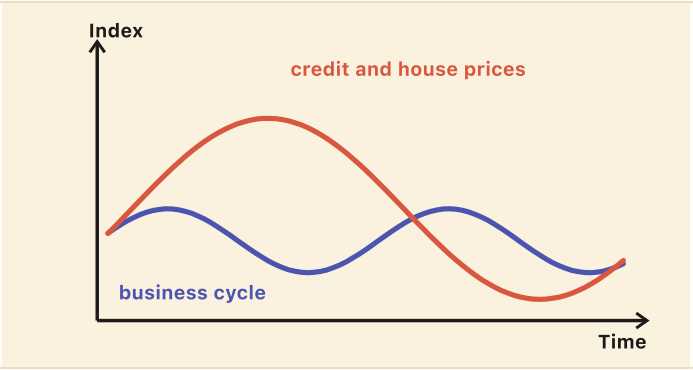
Credit cycles: lending that feeds itself

A **credit cycle** is the rise and fall of how easily, and how cheaply, people and firms can borrow.

Expansion: profits and asset prices rise, defaults are rare, so banks lend more, on looser terms. More borrowing pushes asset prices higher, which makes collateral worth more, which supports more lending.

Contraction: defaults rise, asset prices fall, collateral shrinks, banks tighten. Less credit means less spending, so prices fall further.

So: Credit is procyclical: it is easiest to get when the economy needs it least.



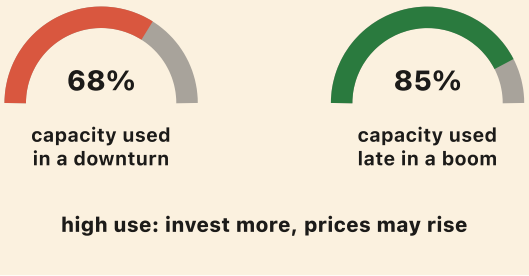
PICTURE IT

Credit, housing booms and deeper busts

Thane, 2030 (made up). Banks compete on home loans, flat prices jump 40%, owners borrow against their flats, builders borrow to build more. Then rates rise. Buyers vanish, prices slide, loans sour, banks stop lending, and construction jobs go.

Credit cycles tend to last longer than business cycles, and housing is where they bite hardest. When a recession coincides with a credit crunch, it is usually **deeper and longer**. Many financial crises begin at a credit peak.

So: Credit cycles amplify business cycles; watch lending and house prices, not only GDP.



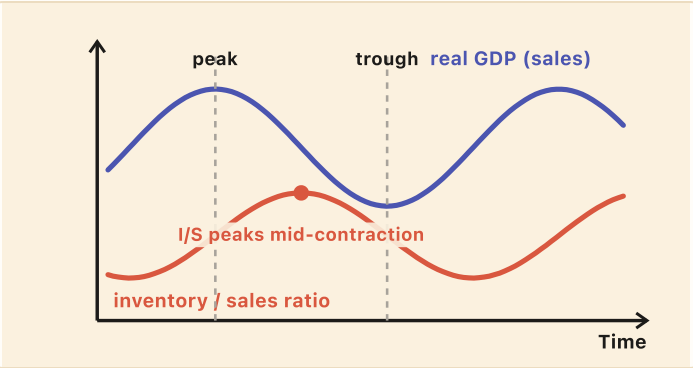
IDEA

Resource use: machines and people first

Capacity utilisation is the share of potential output factories actually use. Low use means slack; firms will not buy new machines. High use pushes firms to invest, and can push prices up.

Labour hoarding: early in a downturn, firms cut hours and overtime but keep skilled staff, because firing and rehiring is costly. Output per worker falls. Early in a recovery, they add hours before hiring.

So: Firms adjust hours and capacity first and headcount last, so employment lags output.



PICTURE IT

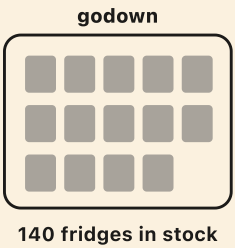
The inventory cycle

The **inventory/sales ratio** = stock of unsold goods / sales.

- **Late expansion, slowdown:** sales slow unexpectedly, goods pile up, the ratio rises.
- **Contraction:** firms slash production to clear stock, so output falls by more than sales. The ratio peaks mid-way, then falls.
- **Recovery:** sales pick up from lean stocks, the ratio is low, and firms raise output to rebuild inventory, adding to growth.

Because the ratio turns after the cycle does, it is a **lagging** indicator.

So: Inventories swing output more than sales; the I/S ratio is high in contractions and low in recoveries.



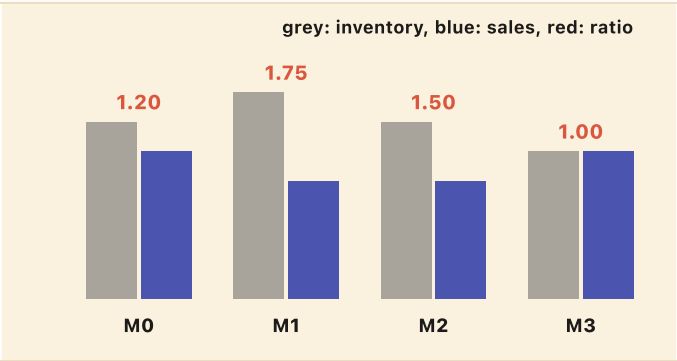
FORMULA

Inventory/sales ratio and closing stock

```
I/S ratio = Inventory ÷ Sales
Closing stock = Opening stock +
Production − Sales
```

Symbol	In plain words	Example
I/S ratio	months of sales sitting unsold	140 ÷ 80 = 1.75
Inventory	unsold goods at month end	140
Sales	units sold in the month	80
Opening stock	units at the start of the month	120
Production	units made in the month	100
Closing stock	units left at month end	140

So: I/S of 1.75 = 1.75 months of sales sitting unsold. Above normal: goods piling up. Below: shelves thin.



WORKED EXAMPLE

Inventory/sales ratio in numbers

What we know

- **Stock** = 120: a Pune fridge maker's godown
- **Sales** = 100, falling to 80 when Diwali flops: fridges sold a month (I/S starts at 1.2)

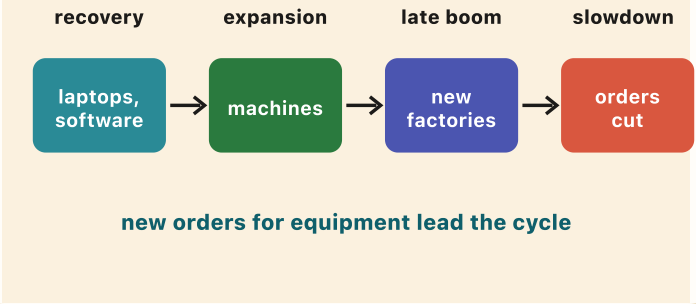
What we want: I/S each month. Closing stock = opening + made - sold.

1. M1: 120 + 100 - 80 = 140; 140 ÷ 80 = 1.75
   Still makes 100, sells only 80: 20 fridges pile up.
2. M2: 140 + 60 - 80 = 120; 120 ÷ 80 = 1.5
   Cuts production to 60, below sales, to clear the pile.
3. M3: 120 + 80 - 100 = 100; 100 ÷ 100 = 1.0
   Sales back to 100, shelves thin: time to make more.

Ratio path: 1.2, 1.75, 1.5, 1.0.

Sense check: Sales fell 20% (100 to 80) but production fell 40% (100 to 60).

So: Output swings more than sales because firms also correct their stock.



IDEA

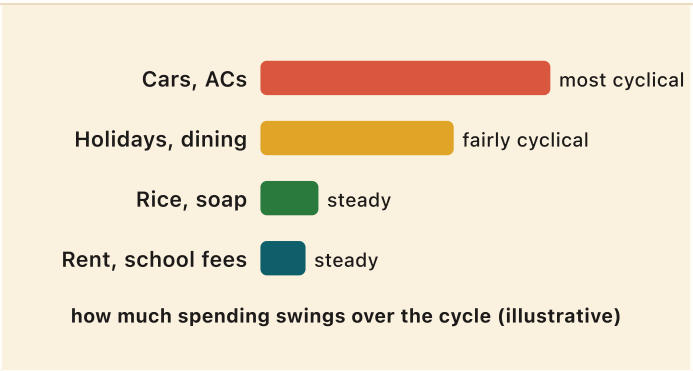
Business capital spending: light first, heavy later

Capital spending (capex) is firms' spending on equipment, software and buildings. It is one of the most cyclical parts of GDP.

- **Recovery:** spare capacity, so firms first buy quick-payback items: computers, software.
- **Expansion:** capacity tightens, so they order heavy machinery, then new plants.
- **Slowdown and contraction:** orders are cut fast, though projects already started continue.

New orders for capital goods move ahead of actual spending, so analysts track them as a leading signal.

So: Capex is volatile and shifts from light to heavy as the expansion matures.



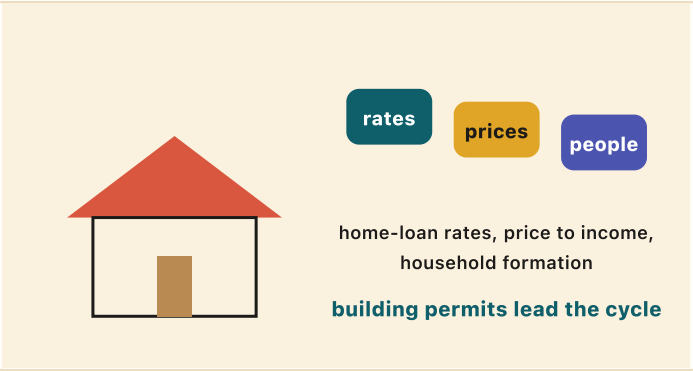
AT A GLANCE

Consumer spending: cars swing, groceries do not

Type	Examples	Over the cycle
Durables	Cars, fridges, ACs	Most cyclical: purchases can be postponed
Non-durables	Food, toiletries, fuel	Steady: bought every week
Services	Rent, school fees; travel, dining	Basic ones steady; discretionary ones cyclical

Households smooth spending to their expected long-run (permanent) income, so total consumption swings less than income. Retail sales and consumer confidence surveys track it.

So: Durables are the first thing households cut and the first they buy again.



IDEA

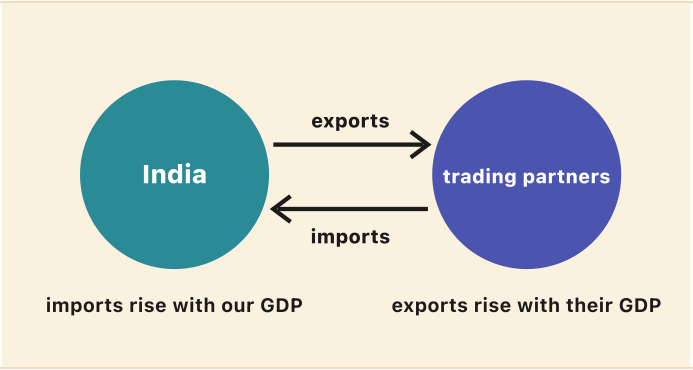
Housing: the interest-rate sensitive sector

Housing is small relative to GDP but swings hard. Its drivers:

- **Mortgage rates:** a 1% rise in home-loan rates adds thousands to an EMI (monthly loan payment) and kills marginal buyers.
- **Affordability:** house prices relative to incomes. Very high ratios eventually stall sales.
- **Demographics:** new households forming, young people moving to cities.

Because it feeds on credit, housing often leads the cycle and can overshoot in booms and busts.

So: Housing responds early to interest rates, so permits and starts are leading indicators.



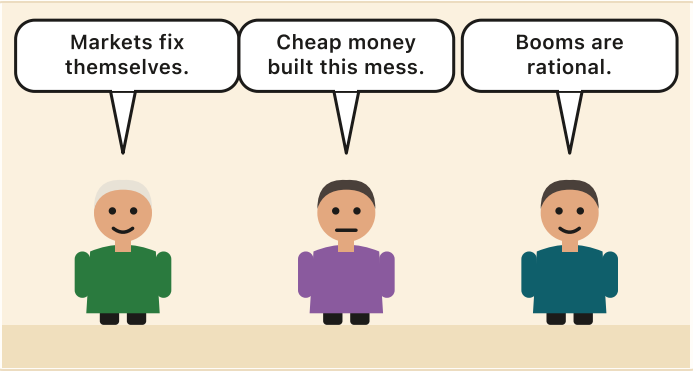
IDEA

Trade: imports follow us, exports follow them

- **Imports** rise with *domestic* GDP: a boom in India sucks in more oil, electronics and gold.
- **Exports** depend on *foreign* GDP: a US or Euro area slowdown hits Tiruppur garment orders.
- **Exchange rates** matter with a lag: a stronger rupee makes exports dearer abroad and imports cheaper at home.

So in a strong domestic expansion the trade balance tends to worsen, and in a domestic recession it tends to improve. Net exports often move against the domestic cycle.

So: **Net exports depend on relative growth and the currency, and often lean against the home cycle.**

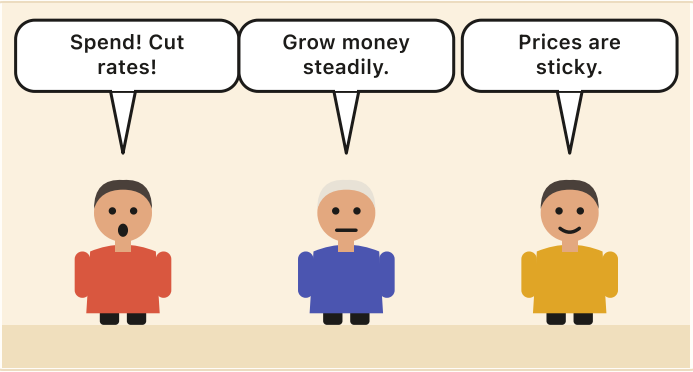


AT A GLANCE

Theories, part 1: leave the market alone

School	What causes cycles	Policy advice
Neoclassical	Technology shocks; markets self-correct (Say's law)	None needed
Austrian	Rates held too low: credit boom, bad investment, bust	Keep rates natural; let the bust clear
New classical (RBC)	Real shocks (technology, energy); cycles are efficient	Do not intervene

So: **These schools see cycles as self-correcting or efficient, so they advise against active stabilisation.**

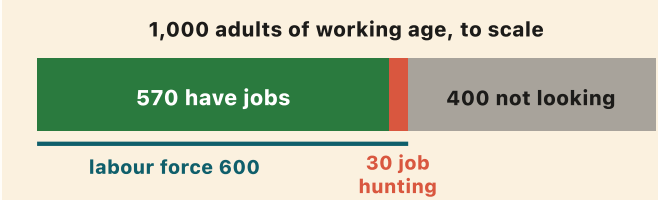


AT A GLANCE

Theories, part 2: policy can help

School	What causes cycles	Policy advice
Keynesian	Demand swings ("animal spirits"); sticky wages	Fiscal and monetary stimulus
Monetarist	Erratic money supply growth	Steady money growth rule; no fine-tuning
New Keynesian	Shocks plus sticky prices (menu costs)	Intervention can help

So: **Keynesians and new Keynesians favour active policy; monetarists want a steady money rule, not discretion.**



STORY

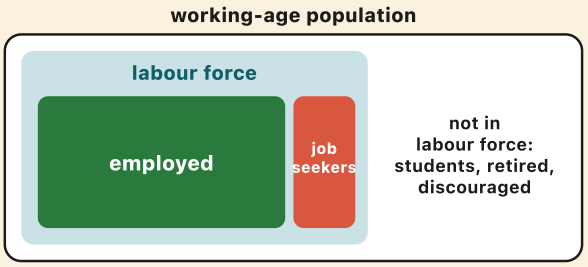
Counting jobs in one housing society

Green Park society in Noida has **1,000 adults** of working age.

- **570** go to work every day.
- **30** have no job and are sending CVs and attending interviews.
- **400** are not looking: college students, homemakers, early retirees, and some who gave up after months of rejections.

How many are "jobless"? Only the 30. The 400 are not counted as unemployed because they are not in the job market. So the jobless share is 30 out of the 600 who are working or looking: **5%**. And 600 of 1,000, or **60%**, take part in the job market at all.

So: Unemployment is measured against people in the job market (working or looking), not against everyone.



FORMULA

Unemployment rate and participation rate

```
LF = Employed + Unemployed
UR = Unemployed ÷ LF
PR = LF ÷ Working-age population
```

Symbol	In plain words	Example
Working-age population	everyone old enough to work	1,000
Employed	has a paid job	570
Unemployed	no job, actively looking	30
LF	labour force: working or looking	600
UR	unemployment rate	5%
PR	participation rate	60%

So: UR divides by the labour force, not by everyone: students and retirees are outside it, not unemployed.



WORKED EXAMPLE

When unemployment falls for a bad reason

What we know

- **Adults** = 1,000 of working age: 570 employed, 30 unemployed and looking
- **Discouraged** = 10 of the 30: jobless who give up looking

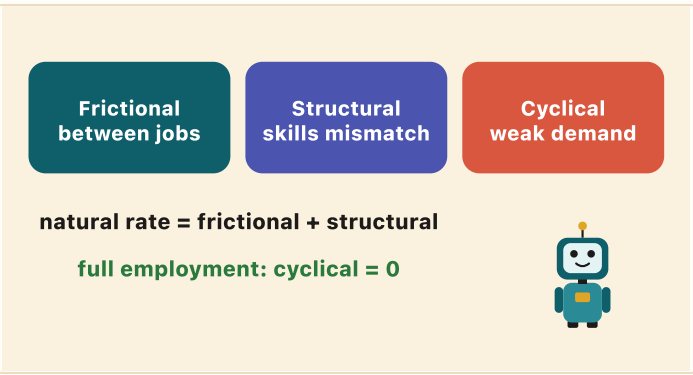
What we want: UR (unemployed ÷ labour force, LF) and PR (LF ÷ adults), before and after.

1. Before: 30 ÷ 600 = 5.0%; 600 ÷ 1,000 = 60%
   LF = 570 working + 30 looking = 600.
2. After: 20 ÷ 590 = 3.4%; 590 ÷ 1,000 = 59%
   The 10 were not hired; they stopped looking and left the LF.

UR falls from 5.0% to 3.4%, with zero new jobs.

Sense check: Still 570 employed, yet the rate fell. Falling participation gives it away.

So: **Discouraged workers make the unemployment rate understate weakness; check participation too.**



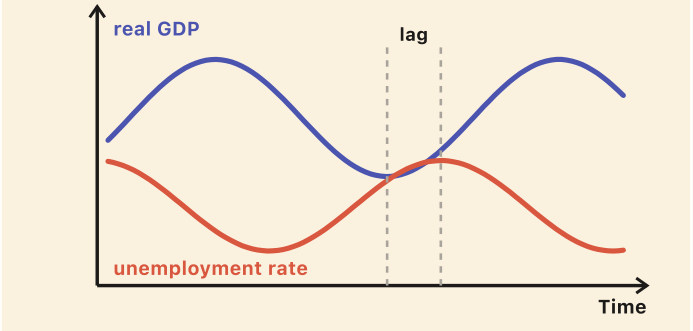
AT A GLANCE

Frictional, structural, cyclical

Type	Cause	Example
Frictional	Time spent searching and matching	An analyst serving notice before a new firm
Structural	Skills or location no longer match jobs	Typists after word processors
Cyclical	Too little demand in a downturn	Factory staff laid off in a recession

Full employment means no cyclical unemployment, not zero unemployment. The rate that remains is the **natural rate**.

So: **Natural rate = frictional + structural; only cyclical unemployment comes and goes with the cycle.**



PICTURE IT

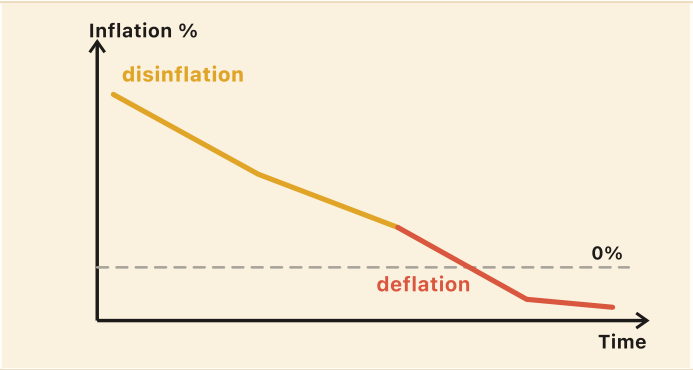
Why unemployment turns late

After the trough, unemployment often keeps rising for a while:

- Firms first add hours to hoarded staff and wait to be sure the recovery is real before hiring.
- **Discouraged workers** (who had stopped looking) return to the search and count as unemployed again.

Underemployed: people with a job below their skills, or part time when they want full time. They count as employed, so the headline rate hides them.

So: The unemployment rate is a lagging indicator; it peaks after output has already turned up.

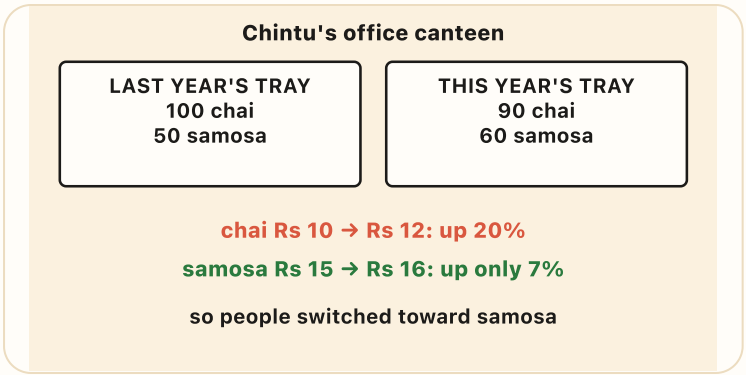


AT A GLANCE

Inflation, disinflation, deflation, hyperinflation

Term	Meaning	Example
Inflation	Sustained rise in the price level	5% a year
Disinflation	Inflation falls, stays positive	6% to 3%
Deflation	Price level falls (below 0%)	-1%
Hyperinflation	Extreme, accelerating; often defined as over 50% a month	Zimbabwe, 2008

So: **Disinflation is slower price rises; deflation is falling prices. Examiners swap them.**



STORY

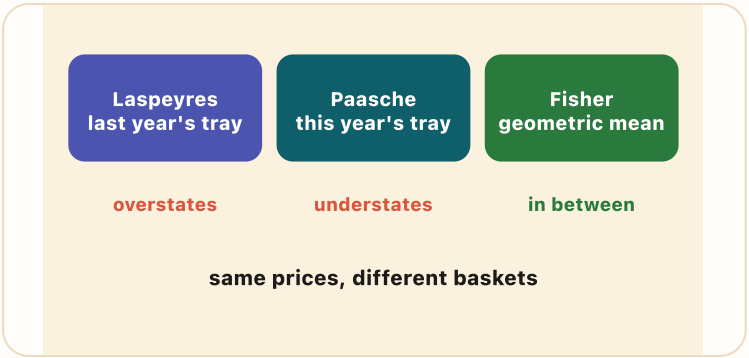
Has lunch got dearer? Depends whose tray

Last year the canteen sold **100 chais at Rs 10** and **50 samosas at Rs 15**. This year chai is Rs 12 (up 20%) and samosa Rs 16 (up about 7%). Chai got relatively dearer, so people switched: **90 chais and 60 samosas**.

To measure inflation you fix one shopping list (a **basket**, here a tray) and price it twice, then and now. But which tray?

- **Laspeyres:** price *last year's* tray at both prices.
- **Paasche:** price *this year's* tray at both prices.
- **Fisher:** split the difference between the two answers.

So: A price index prices one fixed basket at old and new prices; the three indexes differ only in which basket they fix.



FORMULA

Price indexes: Laspeyres, Paasche, Fisher

$$\text{Laspeyres} = \Sigma(P_1 \times Q_0) \div \Sigma(P_0 \times Q_0) \times 100$$

$$\text{Paasche} = \Sigma(P_1 \times Q_1) \div \Sigma(P_0 \times Q_1) \times 100$$

$$\text{Fisher} = \sqrt{(\text{Laspeyres} \times \text{Paasche})}$$

Symbol	In plain words	Example
P_0, P_1	price then, price now	chai 10, 12
Q_0, Q_1	quantity bought then, now	chai 100, 90
Σ	Greek sigma: add up over all items	chai + samosa
$\sqrt{}$	square root	$\sqrt{9} = 3$

Same quantities top and bottom, so only prices differ: one fixed tray, priced now vs then.

So: Laspeyres: old quantities. Paasche: new quantities. Fisher: square root of their product.



then: Rs 1,750

now: Rs 2,000

2,000 ÷ 1,750

= 114.29

same tray, 14.29% dearer

WORKED EXAMPLE

Laspeyres: price last year's tray

What we know

- **Prices** = chai 10 → 12, samosa 15 → 16: then → now
- Q_0 = 100 chai, 50 samosa: last year's tray, kept fixed

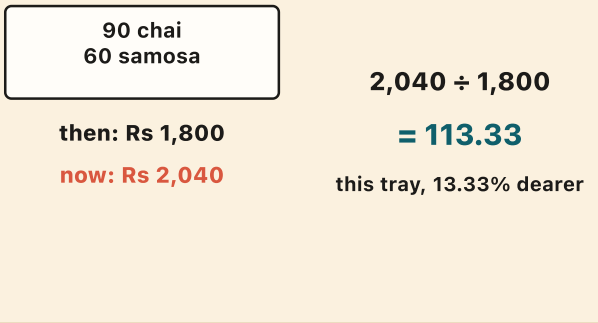
What we want: how much dearer last year's tray is now, x 100.

1. Then: 10 x 100 + 15 x 50 = 1,750
   Old tray at old prices.
2. Now: 12 x 100 + 16 x 50 = 2,000
   Same old tray at today's prices.
3. 2,000 ÷ 1,750 x 100 = **114.29**
   The old tray costs 14.29% more.

Laspeyres index = 114.29.

Sense check: Between chai's 20% and samosa's 7%, nearer chai: chai was 1,000 of the 1,750.

So: Laspeyres fixes the base-year basket: old tray at new prices ÷ old tray at old prices.



WORKED EXAMPLE

Paasche: price this year's tray

What we know

- **Prices** = chai 10 → 12, samosa 15 → 16: then → now
- **Q_1** = 90 chai, 60 samosa: this year's tray, kept fixed

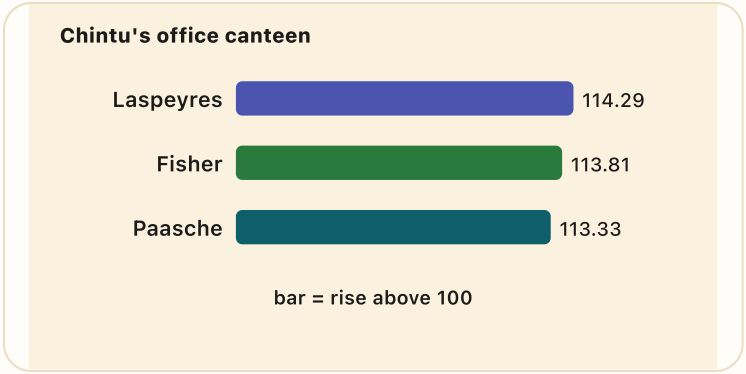
What we want: how much dearer this year's tray is than at last year's prices, x 100.

1. Then: 10 x 90 + 15 x 60 = 1,800
   New tray at last year's prices.
2. Now: 12 x 90 + 16 x 60 = 2,040
   New tray at today's prices: what people pay.
3. 2,040 ÷ 1,800 x 100 = **113.33**
   The new tray costs 13.33% more.

Paasche index = 113.33.

Sense check: Below Laspeyres (114.29): the new tray has more samosa, which rose less.

So: **Paasche fixes the current basket: new tray at new prices ÷ new tray at old prices.**



WORKED EXAMPLE

Fisher: meet in the middle

What we know

- **Laspeyres** = 114.29: old tray
- **Paasche** = 113.33: new tray

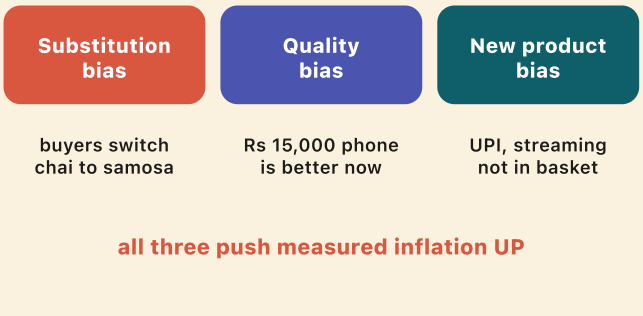
What we want: Fisher = $\sqrt{}$(Laspeyres x Paasche), and the ranking.

1. $\sqrt{}$(114.29 x 113.33) = $\sqrt{}$12,952.5 = **113.81**
   Multiply the two, then take the square root ($\sqrt{}$).
2. 114.29 > 113.81 > 113.33
   Laspeyres ignores the switch to samosa, so it is highest.

Laspeyres 114.29 > Fisher 113.81 > Paasche 113.33.

Sense check: Fisher must land between the two, and 113.81 does.

So: **Buyers switch to cheaper goods: Laspeyres overstates, Paasche understates.**

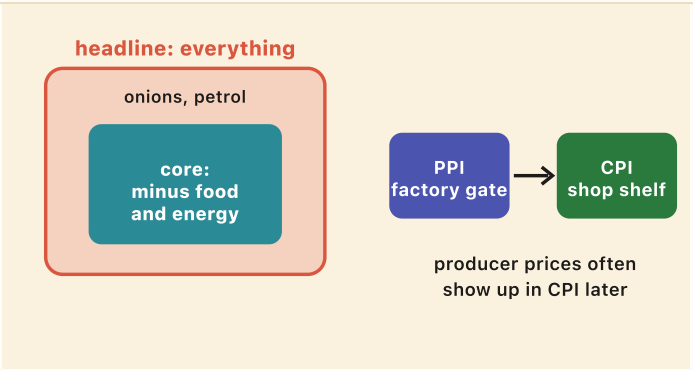


AT A GLANCE

Why a fixed-basket CPI overstates inflation

Bias	What goes wrong	Fix
Substitution	Old basket ignores switching to cheaper goods	Chained index, update weights often
Quality	A price rise for a better product counts as pure inflation	Hedonic pricing: adjust for quality
New product	New, often cheaper-to-use goods enter the basket late	Add new items sooner

So: A Laspeyres-type CPI tends to overstate inflation through substitution, quality and new product biases.



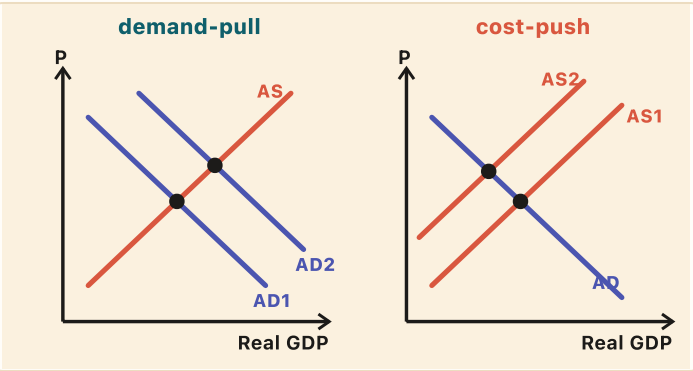
AT A GLANCE

Headline vs core, CPI vs PPI

Measure	What it covers	Use
Headline CPI	All items consumers buy	Cost of living; the RBI's 4% target
Core CPI	Excludes food and energy	Underlying trend, less noise
PPI	Prices producers receive, inputs and outputs	Early sign of cost pressure

Monsoon failure sends onion prices soaring and headline inflation jumps; core may barely move. India also publishes a WPI (wholesale prices), similar in spirit to a PPI.

So: **Headline includes food and energy; core strips them out to show the trend.**

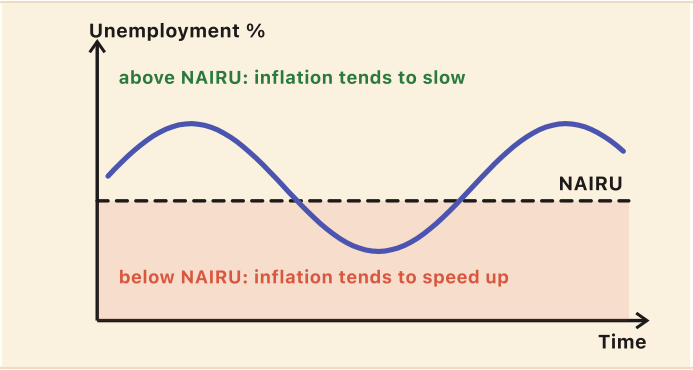


PICTURE IT

Demand-pull versus cost-push inflation

- **Demand-pull:** spending runs ahead of capacity, for example a wedding-season boom on top of easy credit. Prices *and* output rise; the output gap turns positive.
- **Cost-push:** input costs jump, like oil or a failed monsoon. Prices rise while output *falls*.
- **Wage-push:** a cost-push variety. Strong worker bargaining power, often in a tight labour market, lifts wages faster than productivity, so **unit labour costs** rise.

So: Demand-pull: prices and output up together. Cost-push: prices up, output down.



PICTURE IT

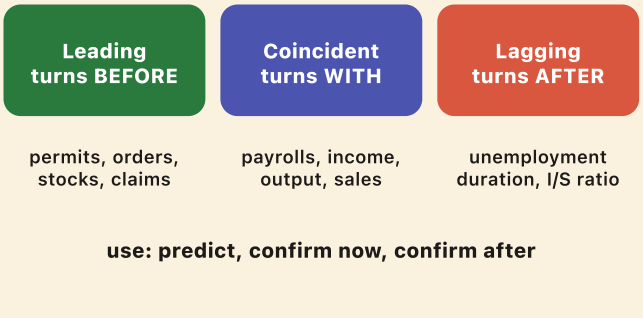
NAIRU: the speed limit for jobs

NAIRU, the non-accelerating inflation rate of unemployment, is the lowest unemployment rate consistent with stable inflation. It is closely linked to the natural rate.

When unemployment drops **below** NAIRU, employers fight for scarce workers, wages outpace productivity, unit labour costs climb, and inflation tends to accelerate. Above NAIRU, slack pulls inflation down.

Analysts watch the output gap and unit labour costs as early signs. NAIRU cannot be observed directly and shifts over time.

So: Unemployment below NAIRU signals rising inflation pressure.

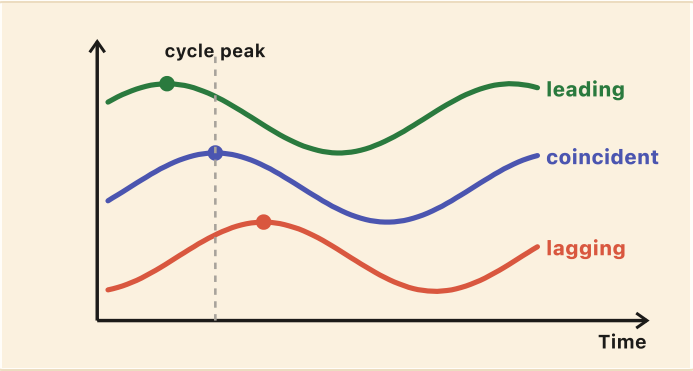


AT A GLANCE

Leading, coincident, lagging indicators

Type	Examples (US Conference Board style)
Leading	Building permits; new orders; stock prices; weekly hours in factories; initial jobless claims (rise = bad); yield curve spread; consumer expectations
Coincident	Payroll employment; real personal income; industrial production; manufacturing and trade sales
Lagging	Average duration of unemployment; inventory/sales ratio; unit labour cost change; bank prime rate; business loans outstanding; services CPI

So: **Leading indicators forecast turns, coincident ones show where we are, lagging ones confirm the turn.**



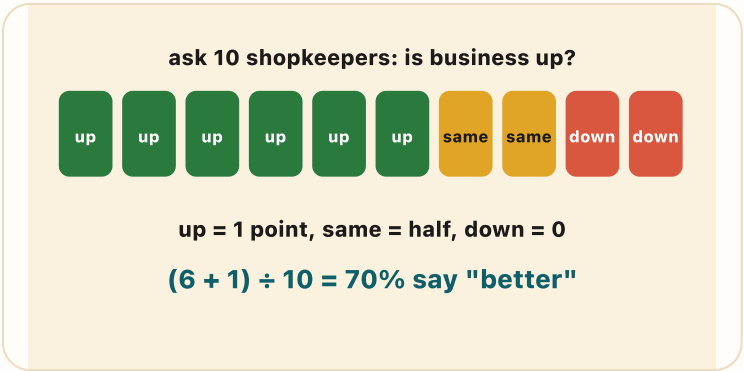
PICTURE IT

Who turns first

Line up three series against the business cycle peak (dashed line). The **leading** one peaks first, the **coincident** one peaks with the cycle, the **lagging** one peaks afterwards.

Single indicators are noisy, so agencies build **composite indexes**, like The Conference Board's Leading Economic Index in the US. Analysts also use surveys such as purchasing managers' indexes (PMIs), where a reading above 50 means more firms report growth than decline.

So: **Leading peaks before the cycle peak; lagging peaks after it.**



FORMULA

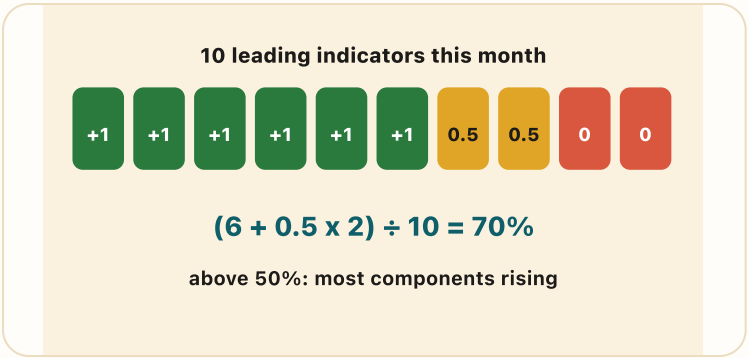
Diffusion index

```
Diffusion index = (Rising + 0.5 x Unchanged) ÷ Total components
```

Symbol	In plain words	Example
Total components	indicators inside the index	10
Rising	went up; 1 point each	6
Unchanged	did not move; 0.5 each	2
Falling	went down; 0 points	2
Diffusion index	share rising, as a %	70%

Like asking 10 Lajpat Nagar shopkeepers if business is up: 6 up, 2 same, 2 down. Half a point per "same": (6 + 1) ÷ 10 = 70%. It shows how **widespread** a move is, not how big.

So: **Diffusion index = (rising + half of unchanged) ÷ total; above 50% means a broad rise.**



WORKED EXAMPLE

Diffusion index: how broad is the move?

What we know

- **Components** = 10: 6 rose, 2 unchanged, 2 fell: indicators in a leading index

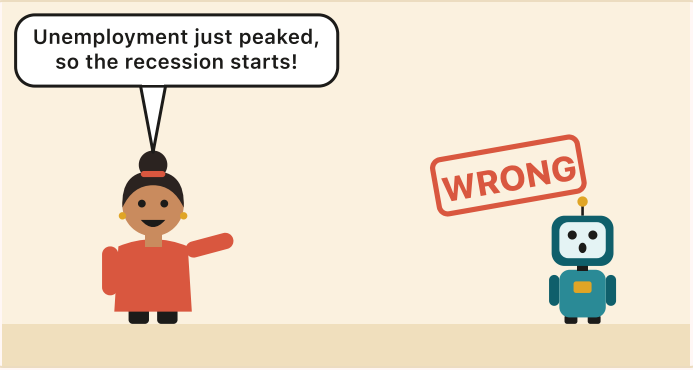
What we want: the diffusion index: the share of components rising.

1. 6 x 1 + 2 x 0.5 + 2 x 0 = 7 points
   Rise: 1 point. Unchanged: half. Fall: nothing.
2. 7 ÷ 10 = 70%
   Divide by all 10 components.
3. 70% is above 50%
   The move is broad, not one or two series.

Diffusion index = 70%.

Sense check: 6 of 10 rose outright (60%); half credit for 2 lifts it a bit above.

So: A diffusion index shows how widespread a change is across an index's components.

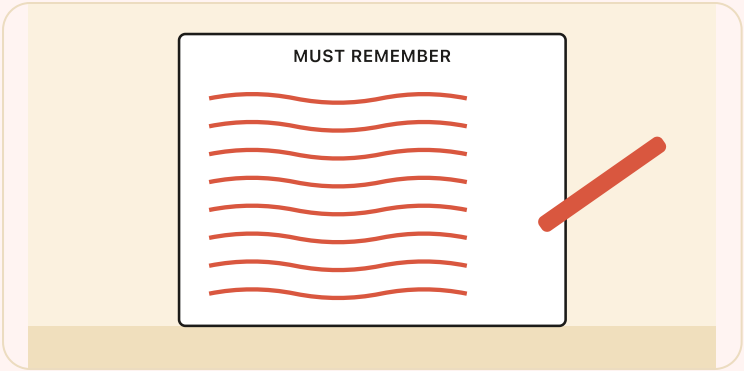


EXAM TRAP

Indicator traps examiners love

- The **unemployment rate** lags, and its average **duration** is a lagging index component. **Initial jobless claims** lead.
- **Stock prices** lead; **industrial production** is coincident.
- The **inventory/sales ratio** lags; it peaks mid-contraction.
- A fall in a leading index does not prove a recession is coming: leading indicators give false signals.
- Disinflation is not deflation. Core excludes food and energy, not housing.

So: **Know each indicator's timing; the exam swaps leading and lagging.**

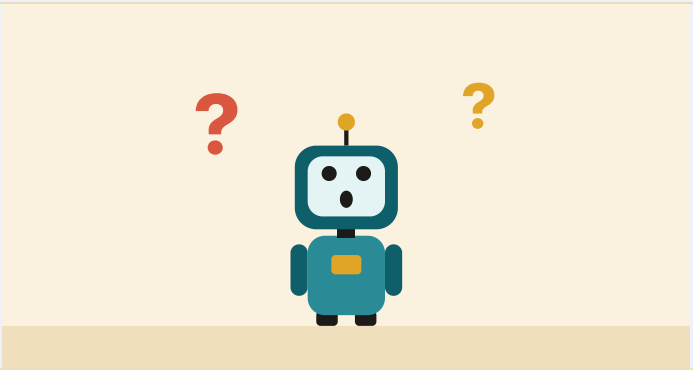


SHARMA JI'S RED PEN

8 things to remember

1. Expansion: trough to peak. Contraction: peak to trough. Four phases: recovery, expansion, slowdown, contraction.
2. Output gap = (actual - potential) / potential.
3. Credit cycles are procyclical, longer, and deepen recessions.
4. Durables and capex swing most; the I/S ratio peaks mid-contraction.
5. Unemployment rate = unemployed / labour force; it lags.
6. Laspeyres > Fisher > Paasche when buyers substitute.
7. Below NAIRU: inflation speeds up.
8. Leading: permits, orders, claims, stocks. Lagging: duration, I/S, prime rate.

So: Scribbled by Sharma ji on the back of a monsoon-soaked GDP release.



EXAM QUESTION 1 OF 8

Sales growth is slowing, inventories are piling up unexpectedly, unemployment is near its cycle low and inflation is at its highest. The economy is most likely in:

A. contraction.

B. slowdown.

C. recovery.

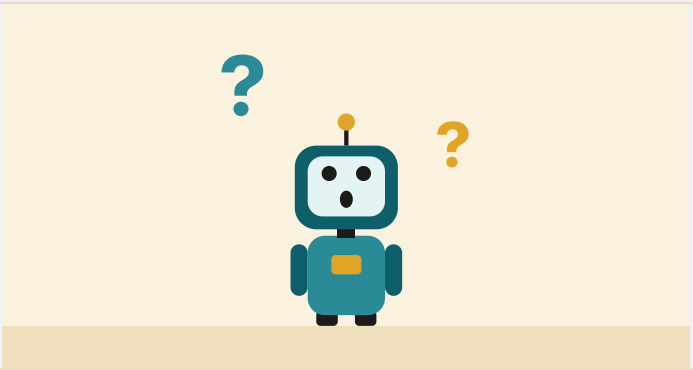
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

Sales growth is slowing, inventories are piling up unexpectedly, unemployment is near its cycle low and inflation is at its highest. The economy is most likely in:

Answer: B. A slowdown follows the peak: output is still above potential but growth is fading, so unsold goods pile up while jobs and inflation have not yet turned. A is wrong: in contraction, GDP falls and layoffs push unemployment up. C is wrong: recovery starts from the trough, with high unemployment and low inflation.



EXAM QUESTION 2 OF 8

A country has a working-age population of 50 million, 30 million employed and 2 million unemployed. Its unemployment rate and participation rate are closest to:

A. 6.25% and 64%.

B. 6.67% and 60%.

C. 4.00% and 64%.

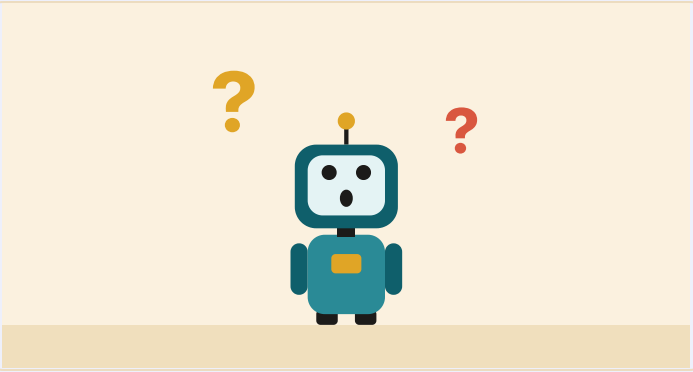
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

A country has a working-age population of 50 million, 30 million employed and 2 million unemployed. Its unemployment rate and participation rate are closest to:

Answer: A. Labour force = 30 + 2 = 32 million. Unemployment rate = 2/32 = 6.25%. Participation = 32/50 = 64%. B divides by employed instead of the labour force (2/30) and leaves the unemployed out of participation (30/50). C divides the unemployed by the whole population (2/50).



EXAM QUESTION 3 OF 8

Base period: good X costs 10, 5 units bought; good Y costs 20, 5 units. Current period: X costs 12, 3 units bought; Y costs 21, 6 units. The Fisher price index is closest to:

A. 108.0

B. 110.0

C. 109.0

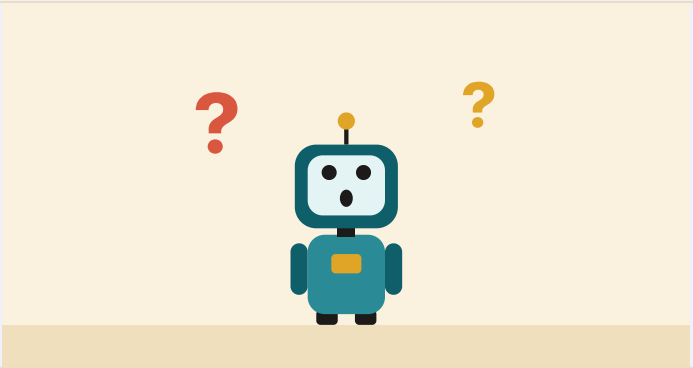
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

Base period: good X costs 10, 5 units bought; good Y costs 20, 5 units. Current period: X costs 12, 3 units bought; Y costs 21, 6 units. The Fisher price index is closest to:

Answer: C. Laspeyres = (12x5 + 21x5) / (10x5 + 20x5) = 165/150 = 110.0. Paasche = (12x3 + 21x6) / (10x3 + 20x6) = 162/150 = 108.0. Fisher = $\sqrt{}$(110 x 108) = 109.0. A is the Paasche index and B is the Laspeyres index, not their geometric mean.



EXAM QUESTION 4 OF 8

Which school of thought blames booms and busts mainly on a central bank holding interest rates below their natural level?

A. Austrian

B. Monetarist

C. Keynesian

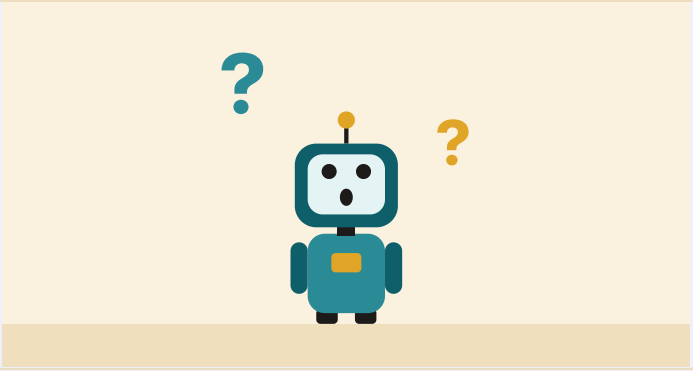
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

Which school of thought blames booms and busts mainly on a central bank holding interest rates below their natural level?

Answer: A. Austrians argue that artificially cheap credit leads to over-investment in the wrong projects, and the bust corrects it. B is wrong: monetarists blame erratic money supply growth and want a steady money rule. C is wrong: Keynesians blame swings in aggregate demand and favour active stimulus.



EXAM QUESTION 5 OF 8

Which of the following is a lagging indicator?

A. Average weekly initial claims for unemployment insurance

B. Industrial production

C. Average duration of unemployment

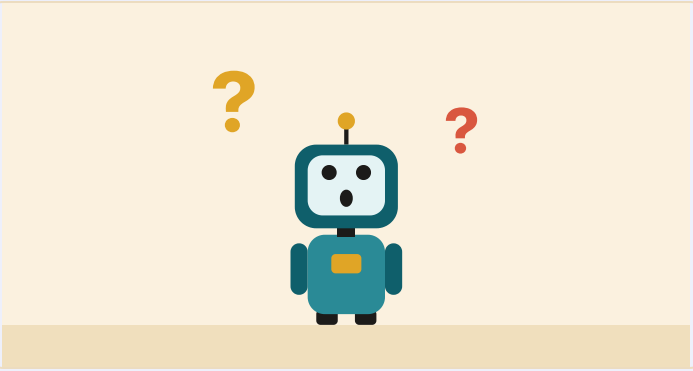
Answer on the next page.

So: **Pick an answer, then turn the page.**

ANSWER

Which of the following is a lagging indicator?

Answer: C. Unemployment spells lengthen well after a downturn begins and stay long into the recovery, so average duration lags. A is a leading indicator: layoffs show up in claims early. B is coincident: output moves with the cycle.



EXAM QUESTION 6 OF 8

How do credit cycles most likely affect business cycles?

A. They have little effect, because credit cycles are shorter.

B. They amplify them, because rising asset prices raise collateral values and encourage more lending.

C. They dampen them, because banks lend more in recessions.

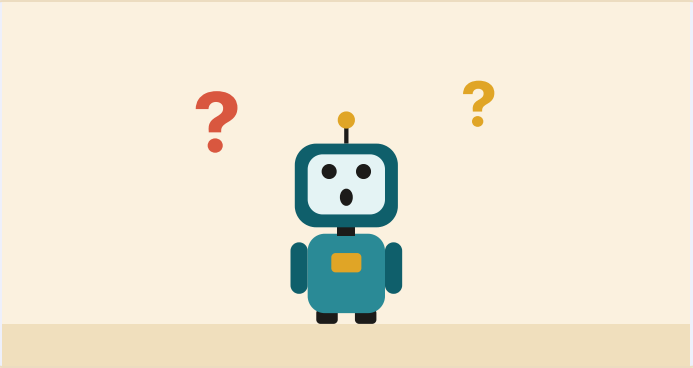
Answer on the next page.

So: **Pick an answer, then turn the page.**

ANSWER

How do credit cycles most likely affect business cycles?

Answer: B. Credit is procyclical: in booms, higher asset prices raise collateral values and lenders loosen standards; in busts, the reverse. That deepens both phases. A is wrong: credit cycles tend to be longer, not shorter, and matter a lot. C is backwards: banks tighten lending in recessions.



EXAM QUESTION 7 OF 8

A country's annual inflation rate falls from 6% to 3%. This is best described as:

A. disinflation.

B. deflation.

C. hyperinflation.

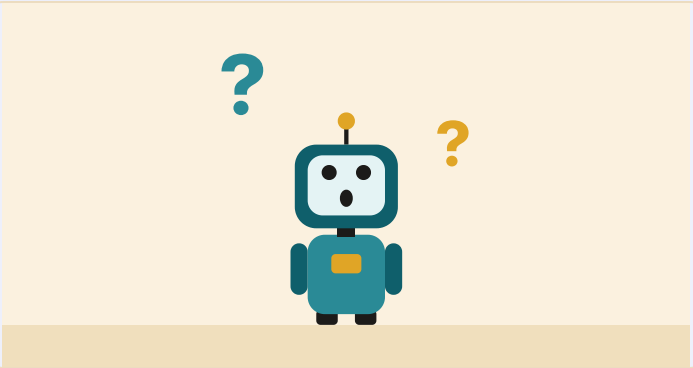
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

A country's annual inflation rate falls from 6% to 3%. This is best described as:

Answer: A. Prices are still rising, just more slowly, which is disinflation. B is wrong: deflation needs a falling price level, meaning inflation below zero. C is wrong: hyperinflation is extremely high, accelerating inflation.



EXAM QUESTION 8 OF 8

Unemployment has fallen well below the NAIRU and unit labour costs are rising quickly. The most likely outcome is that inflation will:

A. turn into deflation.

B. stay stable.

C. accelerate.

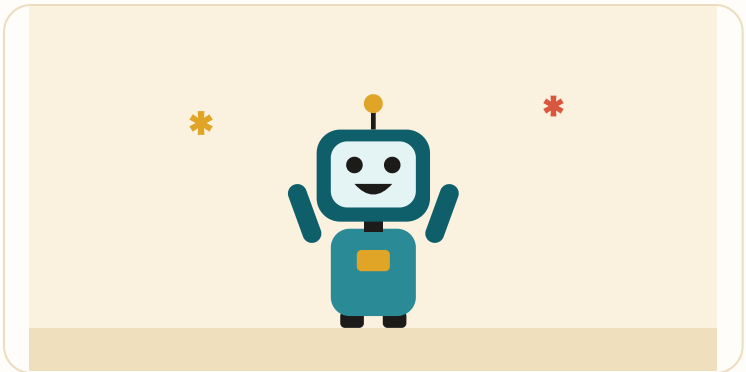
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

Unemployment has fallen well below the NAIRU and unit labour costs are rising quickly. The most likely outcome is that inflation will:

Answer: C. Below the NAIRU, the labour market is tight: wages outpace productivity, unit labour costs rise and firms pass them on, so inflation speeds up. B describes unemployment at the NAIRU. A would need large slack, the opposite of this case.



CHAPTER 2 DONE

You can now read the cycle

Chintu's note to Sharma ji: "Stocks lead, so the Sensex may be early. Payrolls and factory output say we are still near the trough. Unemployment will keep rising for a while; that is normal, it lags. Watch building permits and capital goods orders." Sharma ji forwarded it to Pinky with one word: "Padho." (Read it.)

So: **Swipe on, or jump to another chapter from the Menu.**



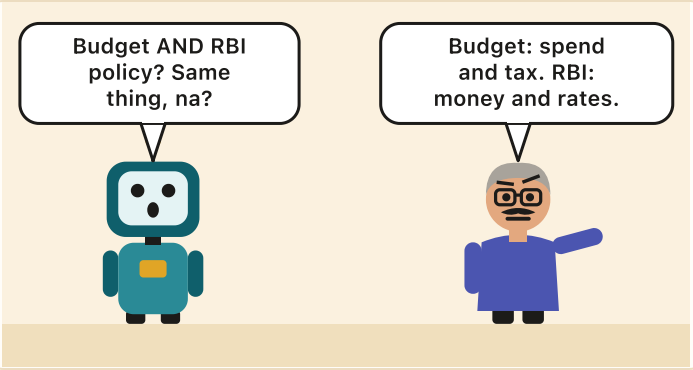
BEFORE YOU START

The cast, and words this chapter uses

A Mumbai investment firm. Sharma ji trusts nothing without data ("Data dikhao"); Pinky is sure of everything, especially when wrong.

- **GDP:** the value of all final output an economy produces in a year.
- **Aggregate demand (AD):** total spending on a country's output.
- **Potential GDP:** output the economy can sustain at full employment.
- **Budget deficit:** government spending minus revenue, when spending is bigger.
- **Bond:** an IOU sold to borrow money; the buyer earns interest.

So: **You can start this chapter cold. Everything it needs is on this card.**



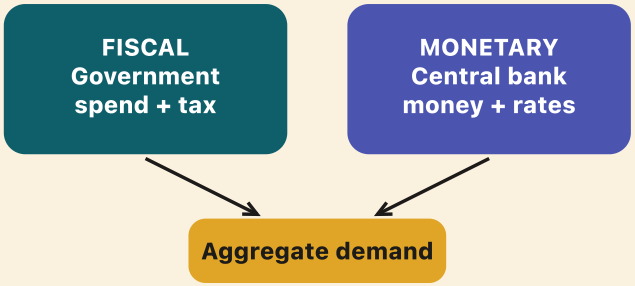
STORY

Budget day or RBI day?

February 1: the Finance Minister reads the Union Budget in Delhi. A few days later the RBI announces its policy rate in Mumbai. Chintu writes one report on both, titled "Government stimulus".

Sharma ji returns it with one red line: "Two different drivers." **Fiscal policy** is the government using spending and taxes to influence the economy. **Monetary policy** is the central bank using the money supply and interest rates. Both push on aggregate demand, through different pedals and at different speeds.

So: Fiscal policy = government spending and taxes. Monetary policy = central bank money and interest rates.

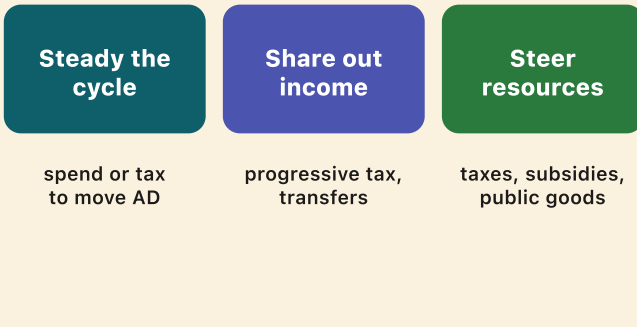


AT A GLANCE

Fiscal vs monetary policy, side by side

	Fiscal	Monetary
Run by	Government and parliament	Central bank (RBI, Fed, ECB)
Tools	Spending, taxes, transfers	Policy rate, open market operations, reserve requirements
Speed to decide	Slow: debate and votes	Fast: a committee meets several times a year
Main targets	Demand, fairness, what gets produced	Price stability, often jobs too

So: **Monetary decisions are quick to make and act broadly; fiscal decisions are slow but can target specific groups.**



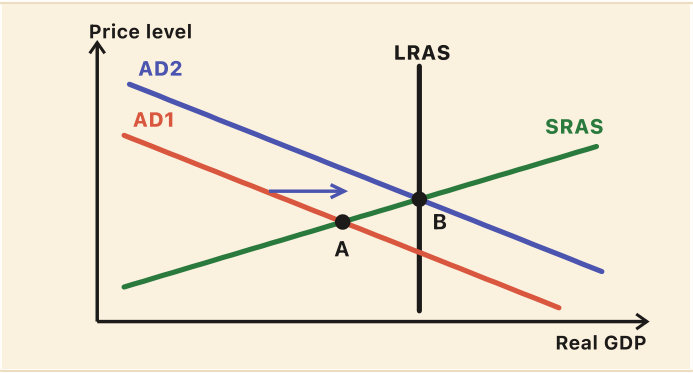
IDEA

The three jobs of fiscal policy

1. **Stabilise the economy** by influencing aggregate demand: spend more or tax less in a slump, the reverse in a boom.
2. **Redistribute income and wealth** through progressive taxes and transfer payments.
3. **Allocate resources**: taxes and subsidies change what gets produced, and the state pays for public goods such as defence that markets underprovide.

Behind all three sit the big goals: steady growth, low unemployment and stable prices.

So: Stabilise, redistribute, allocate: three jobs, one budget.



PICTURE IT

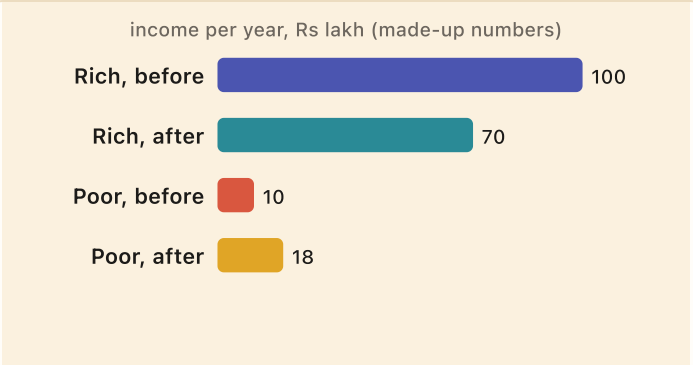
Fiscal stimulus shifts aggregate demand

Start at A, a recession: AD1 meets short-run aggregate supply (SRAS) to the left of the vertical long-run line (LRAS), which marks potential GDP. The shortfall is a **recessionary gap**.

The government spends more or cuts taxes. People and firms buy more, so AD shifts right to AD2 and output climbs to potential at B, with a somewhat higher price level.

In an overheating boom it does the reverse: spend less or tax more, shifting AD left to cool inflation.

So: Expansionary fiscal policy shifts AD right; contractionary fiscal policy shifts it left.



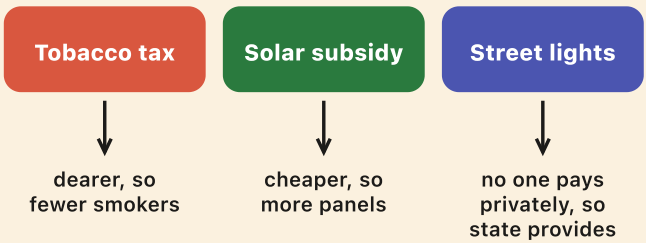
IDEA

Redistribution: squeeze the gap

Mrs Mehta earns Rs 100 lakh a year and pays 30% in income tax, keeping Rs 70 lakh. Raju, a delivery rider, earns Rs 10 lakh, pays no income tax and gets a Rs 8 lakh transfer, so he has Rs 18 lakh.

Before the budget, she had 10 times his income. After it, 3.9 times. That is fiscal policy redistributing income: **progressive taxes** (higher rates on higher incomes) take more from the top, **transfers** add at the bottom.

So: Progressive taxes plus transfer payments are the main fiscal tools for redistributing income and wealth.



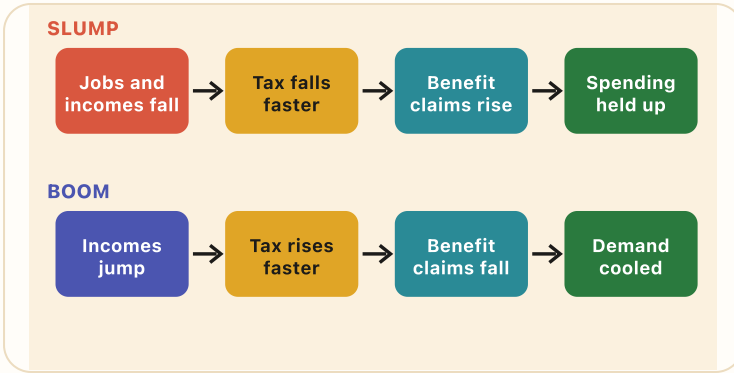
IDEA

Resource allocation: tax it or pay for it

A tax makes something dearer, so people buy less of it. A subsidy makes it cheaper, so they buy more. Governments use this to steer the economy's resources: heavy taxes on cigarettes and alcohol, subsidies for rooftop solar or research.

They also pay for **public goods** such as defence, courts and street lights. Private firms underprovide these because they cannot make each user pay.

So: Taxes and subsidies change what the economy produces; public goods are funded because markets underprovide them.



PICTURE IT

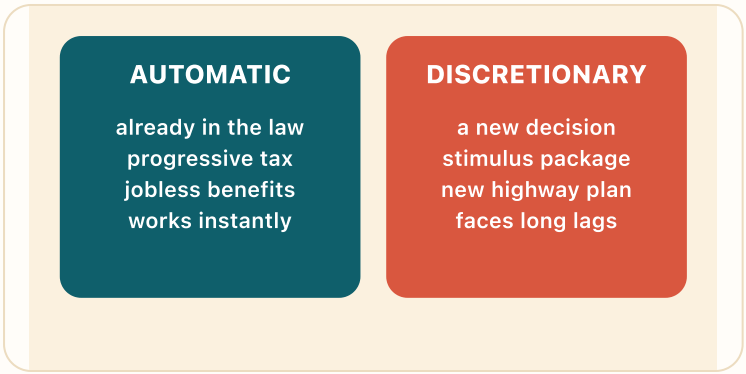
Automatic stabilisers: the budget's shock absorbers

Automatic stabilisers are parts of the budget that lean against the cycle with no new decision.

- **Progressive income tax:** in a slump, falling incomes drop people into lower tax brackets, so tax paid falls faster than income and spending holds up.
- **Unemployment benefits:** more people claim them in a slump, fewer in a boom.

In a boom both run in reverse: tax receipts jump and benefit claims fall, cooling demand.

So: Automatic stabilisers soften the cycle by themselves, and they push the deficit up in slumps and down in booms.



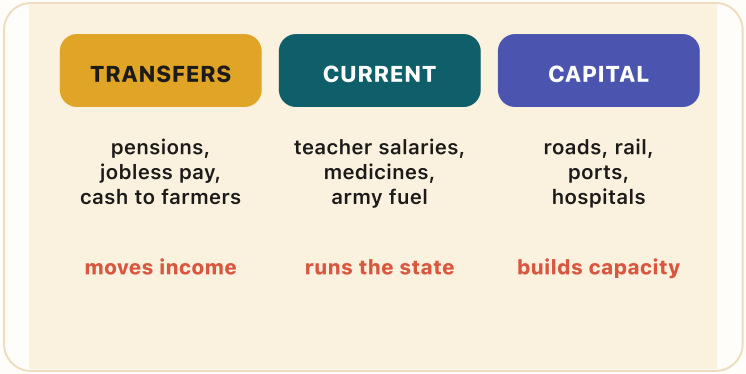
AT A GLANCE

Automatic vs discretionary fiscal policy

Discretionary fiscal policy is a deliberate change in spending or taxes, chosen for the situation.

	Automatic	Discretionary
Trigger	Built into existing rules	A new decision by government
Speed	Works as incomes change	Slow: must be spotted, voted, rolled out
Examples	Progressive income tax, unemployment benefits	Stimulus package, highway programme, temporary GST cut

So: **Discretionary policy needs a decision and suffers every lag; automatic stabilisers kick in on their own.**



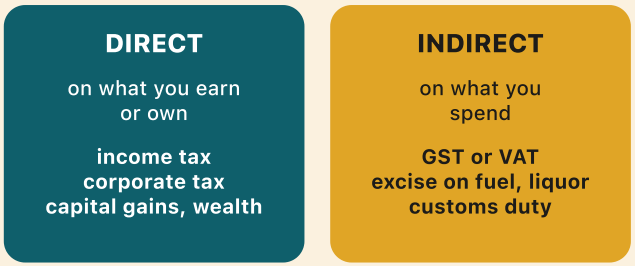
TOOLS

Three ways a government spends

- **Transfer payments:** money paid out with nothing bought in return, such as pensions and unemployment benefits. They redistribute income and are not counted in the G of GDP, because nothing is produced.
- **Current spending:** goods and services used up each year, like salaries and medicines.
- **Capital spending:** long-lived assets such as roads and hospitals, which raise future capacity.

Why spend at all: public goods, infrastructure, help for the poor, research, and steadying the cycle.

So: Transfers move income, current spending runs the state, capital spending builds the economy's future capacity.



TOOLS

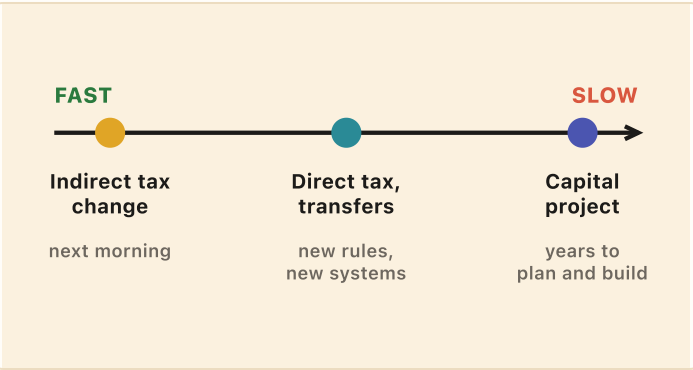
Two ways a government raises money

Direct taxes fall on income and wealth: income tax, corporate tax, capital gains tax, wealth and inheritance taxes, and social security contributions.

Indirect taxes fall on spending: GST or VAT, excise duties on fuel, alcohol and tobacco, and customs duties. You pay them at the till, and the shop passes them to the government.

Chintu's chai costs Rs 20 at the stall. Part of it is GST he never notices. That invisibility is one reason indirect taxes are easy to collect.

So: Direct taxes hit what you earn or own; indirect taxes hit what you buy.

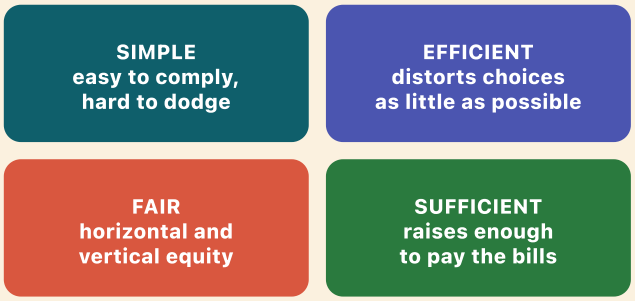


AT A GLANCE

Which tool works fastest?

Tool	Upside	Downside
Indirect taxes	Change almost overnight; hard to evade; can discourage smoking or drinking	Take a bigger share of poorer people's income
Direct taxes and transfers	Good for fairness goals	Slower to change: need new rules and admin systems
Capital spending	Raises long-run growth	Slow to plan and build, so a poor quick boost

So: For speed, indirect taxes; for fairness, direct taxes and transfers; for long-run capacity, capital spending, which is slow.



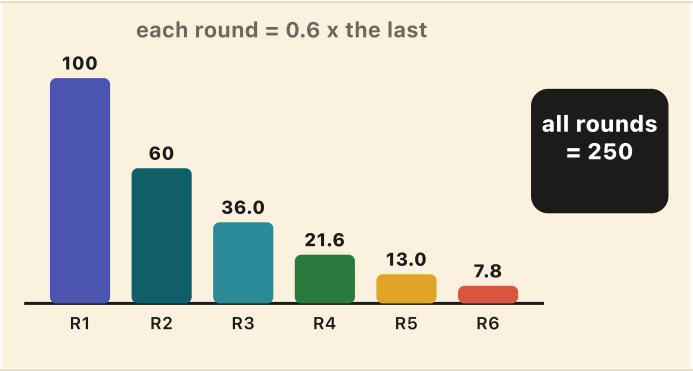
IDEA

What makes a good tax system

- **Simplicity:** easy to comply with and to enforce.
- **Efficiency:** interferes as little as possible with choices about work, saving and spending.
- **Fairness:** *horizontal equity* means people in similar situations pay the same tax. *Vertical equity* means people with more ability to pay pay more.
- **Revenue sufficiency:** raises enough to fund spending.

Two nurses on the same salary paying the same tax: horizontal. A surgeon earning five times more paying a higher rate: vertical.

So: Simple, efficient, fair (horizontal and vertical) and sufficient: the four attributes of good tax policy.



STORY

The multiplier, one round at a time

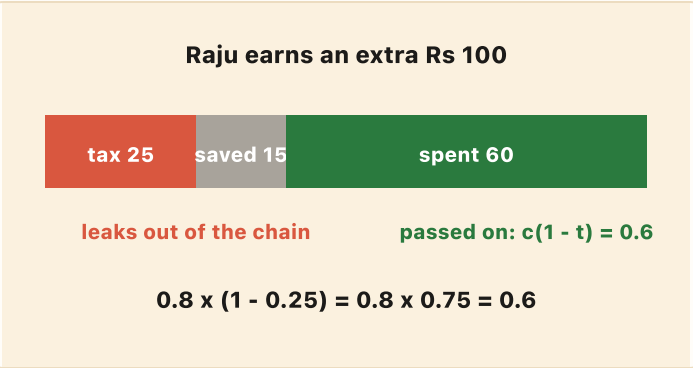
The government pays a contractor Rs 100 crore to build a flyover. Each extra rupee anyone earns is taxed at 25%, and people spend 80% of what is left.

Round 1: workers earn Rs 100 crore, pay Rs 25 crore tax, keep Rs 75 crore and spend 80% of it: Rs 60 crore at shops.

Round 2: shopkeepers earn that Rs 60 crore and, the same way, spend Rs 36 crore. Each round is 0.6 times the last.

Add every round: 100 + 60 + 36 + 21.6 + ... The rounds shrink and the total stops at **Rs 250 crore** of extra GDP from Rs 100 crore of spending.

So: Each rupee of government spending becomes someone's income, which is partly taxed, partly saved and partly spent again.



IDEA

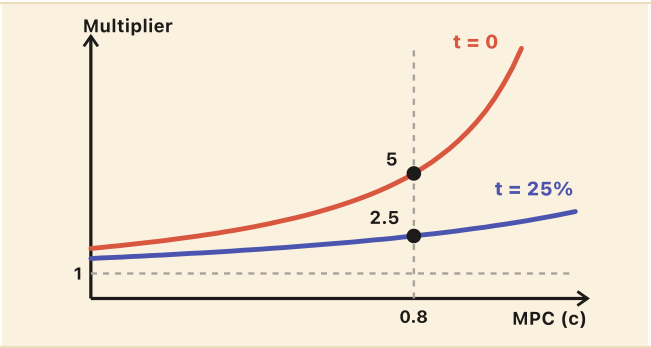
Where each extra rupee goes

Raju gets an extra Rs 100 from the flyover job.

- **Tax rate t = 25%:** Rs 25 goes in tax. He keeps Rs 75, which is (1 - t) = 75% of the 100.
- **MPC c = 0.8:** the marginal propensity to consume is the share of each extra rupee he keeps that he spends. 80% of Rs 75 = Rs 60 spent; Rs 15 saved.
- So Rs 60 of the 100 reaches the next shopkeeper: **c(1 - t)** = 0.8 x 0.75 = 0.6.

Tax and saving are **leaks**: money that drops out of the spending chain each round. Here 40 paise of every rupee leak.

So: c(1 - t) is the share of each extra rupee earned that gets spent again; the rest leaks to tax and saving.



FORMULA

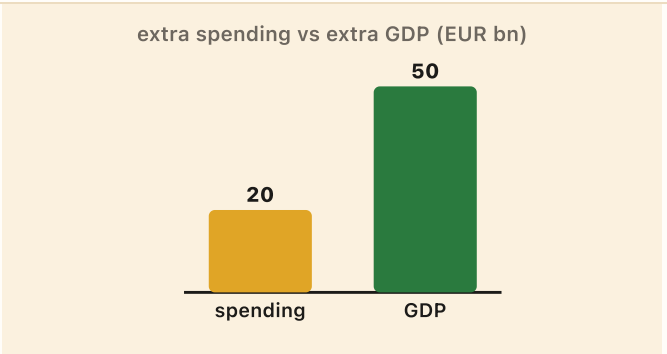
The fiscal multiplier

```
Multiplier = 1 ÷ [1 − c(1 − t)]
Change in GDP = Multiplier x Change in G
```

Symbol	In plain words	Example
c	MPC: share of extra after-tax income spent	0.8
t	tax rate on extra income	0.25
c(1 - t)	share of each extra rupee spent again	0.6
Change in G	extra government spending	Rs 100 cr
Multiplier	rupees of GDP per rupee of spending	2.5

Why 1 ÷ (1 - ...): rounds of 100 + 60 + 36 + ..., each 0.6 times the last, always add up to 100 ÷ (1 - 0.6) = 250.

So: Higher MPC: bigger multiplier. Higher tax rate: smaller one, because tax leaks out of the chain.



WORKED EXAMPLE

Multiplier: a spending boost in euros

What we know

- **c** = 0.8: MPC: share of extra after-tax income spent
- **t** = 25%: tax rate
- **Change in G** = EUR 20 billion: extra road spending

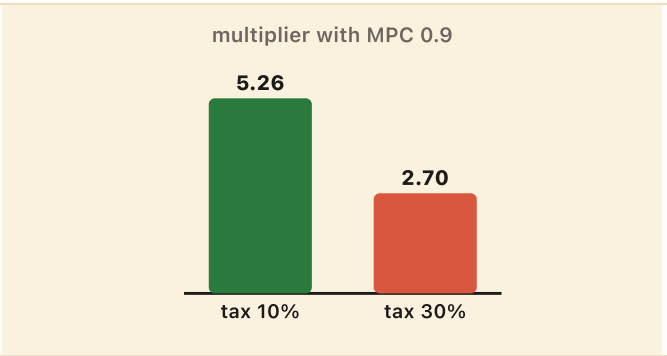
What we want: the change in GDP.

1. c(1 - t) = 0.8 x 0.75 = 0.60
   Of each extra euro earned, 60 cents are spent again.
2. 1 ÷ (1 - 0.60) = 1 ÷ 0.40 = 2.50
   40 cents leak each round, so the chain adds to 1 ÷ 0.40.
3. 2.50 x 20 = EUR 50 billion
   Multiplier times the extra spending.

GDP rises by EUR 50 billion (multiplier 2.5).

Sense check: Rounds go 20, 12, 7.2, ... and creep up toward 50, never past it.

So: Find c(1 - t) first, then 1 divided by one minus it, then multiply by the spending change.



WORKED EXAMPLE

Multiplier: a higher tax rate shrinks it

What we know

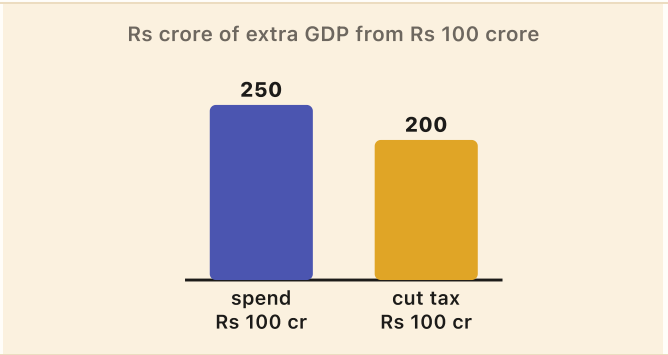
- **c** = 0.9: MPC: share of extra after-tax income spent
- **t** = 10% or 30%: tax rate, two cases
- **Change in G** = USD 10 billion: extra spending

What we want: the multiplier and GDP change under each tax rate.

1. t = 10%: c(1 - t) = 0.9 x 0.9 = 0.81; 1 ÷ 0.19 = 5.26
   Only 19 cents per dollar leak, so the ripple runs long.
2. t = 30%: c(1 - t) = 0.9 x 0.7 = 0.63; 1 ÷ 0.37 = 2.70
   Now 37 cents leak each round, so it dies fast.
3. 5.26 x 10 = 52.6 vs 2.70 x 10 = 27.0
   Same spending, very different punch.

USD 52.6 bn at 10% tax, USD 27.0 bn at 30% tax: about half the punch.

So: **The higher the tax rate, the more each round leaks to the government and the smaller the multiplier.**



WORKED EXAMPLE

Spending increase vs tax cut

What we know

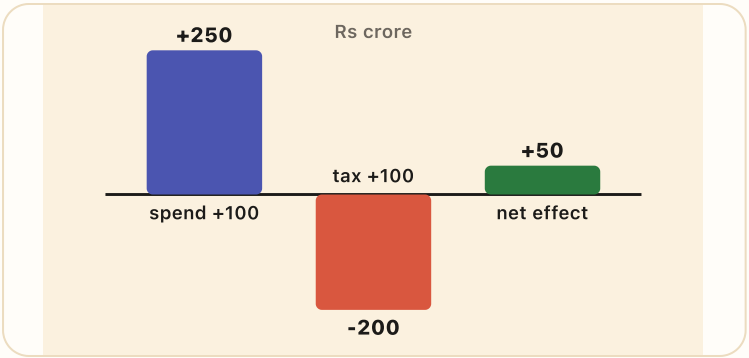
- **c** = 0.8: MPC: share of extra after-tax income spent
- **t** = 25%, so multiplier = 2.5: tax rate
- **Size** = Rs 100 crore either way: spend more, or cut taxes

What we want: which adds more GDP.

1. Spend: round 1 = 100; 2.5 x 100 = Rs 250 crore
   The government spends every rupee on output at once.
2. Tax cut: round 1 = 0.8 x 100 = 80
   Households get 100 back but spend only 80%; they save 20.
3. 2.5 x 80 = Rs 200 crore
   The saved 20 never enters the chain.

Spending: +Rs 250 crore. Tax cut: +Rs 200 crore.

So: A tax cut has a smaller effect than the same amount of spending, because part of the cut is saved before it is ever spent.



WORKED EXAMPLE

The balanced budget multiplier

What we know

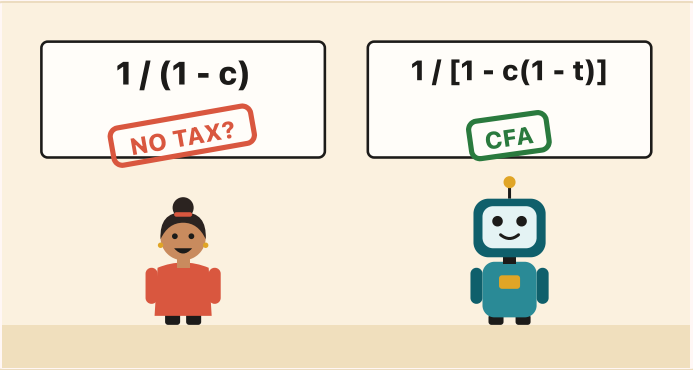
- **c, t** = 0.8, 25%: MPC and tax rate, so multiplier = 2.5
- **Plan** = Rs 100 crore each: spend more, paid by equal new taxes

What we want: the net change in GDP.

1. Spend: 2.5 x 100 = +250
   The government spends all the new tax money.
2. Tax: 2.5 x (0.8 x 100) = -200
   Households pay 100 but spend only 80 less; 20 comes from saving.
3. Net: 250 - 200 = +Rs 50 crore
   Round 1 net is 100 - 80 = 20, times 2.5.

GDP rises by Rs 50 crore though the deficit is unchanged.

So: **The balanced budget multiplier is positive: equal rises in spending and taxes still raise output, because taxes are partly paid out of saving.**



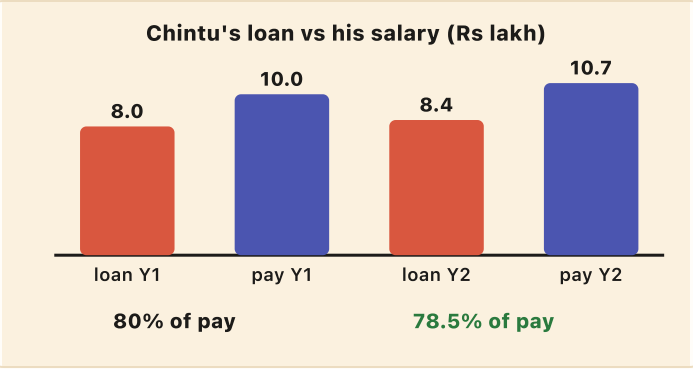
EXAM TRAP

Multiplier mistakes examiners love

c = MPC, the share of extra after-tax income spent; t = tax rate.

- Using 1 / (1 - c) when a tax rate is given. Pinky's favourite: it overstates the multiplier.
- Plugging in t instead of (1 - t): the bracket is the share of income people keep.
- Treating a tax cut like the same amount of spending. A tax cut starts at c x the cut.
- Saying a balanced budget change has zero effect. It is positive.
- Reading "marginal propensity to save" as c. If MPS = 0.2, then c = 0.8.

So: Include the tax rate, use c(1 - t), shrink tax cuts by c; the balanced budget multiplier is positive.



STORY

More loan, yet a lighter burden

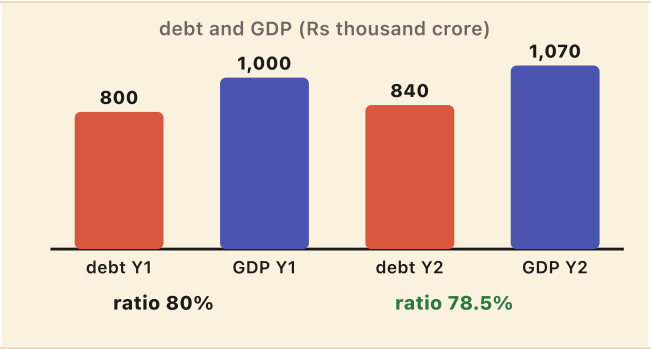
Chintu owes **Rs 8 lakh** and earns **Rs 10 lakh** a year: his loan is 80% of a year's pay.

This year he borrows another **Rs 40,000** for a bike. He also gets a **7% raise**, to Rs 10.7 lakh.

Now he owes Rs 8.4 lakh on Rs 10.7 lakh of pay: 8.4 ÷ 10.7 = **78.5%**. He owes more rupees, yet the loan is a smaller weight on his income.

A government works the same way. Its loan is the **debt**, its pay is **GDP**, and this year's extra borrowing is the **deficit**.

So: What matters is debt relative to income: if income grows faster than the loan, the burden shrinks.



WORKED EXAMPLE

The debt ratio can fall while you borrow

What we know

- **Debt** = 800: what the state owes; this year's deficit (new borrowing) adds 40
- **GDP** = 1,000: a year's output; nominal (in rupees) growth 7%

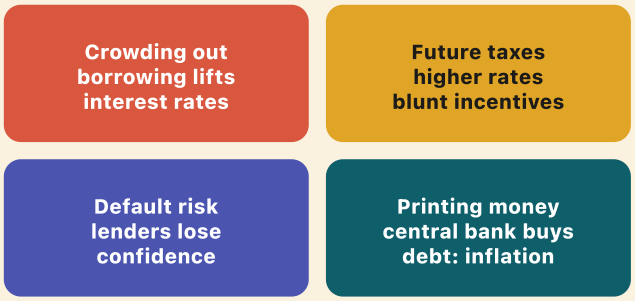
What we want: debt ÷ GDP now and next year.

1. Year 1: 800 ÷ 1,000 = 80%
   Debt is 80% of a year's output.
2. Debt: 800 + 40 = 840. GDP: 1,000 x 1.07 = 1,070
   The deficit adds to debt; GDP grows 7%.
3. Year 2: 840 ÷ 1,070 = 78.5%
   New debt over new GDP.

Ratio falls 80% to 78.5% despite a deficit.

Sense check: Debt grew 5% (40 ÷ 800), GDP 7%: the bottom grew faster.

So: The debt ratio rises only when debt grows faster than nominal GDP.

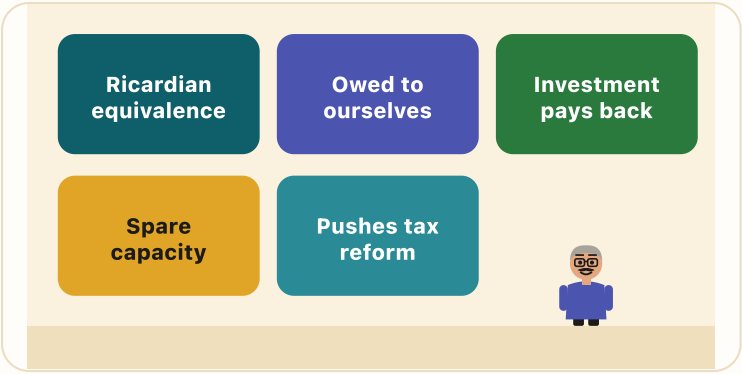


IDEA

Why large deficits and debt worry people

- **Crowding out:** heavy government borrowing pushes up interest rates and squeezes private investment.
- **Higher future taxes:** debt must be serviced, and higher taxes later weaken the incentive to work and invest.
- **Default risk:** if lenders doubt repayment, they demand higher yields or stop lending.
- **Inflation:** a government that cannot borrow may lean on the central bank to create money to fund it (monetisation).

So: Concerns about deficits: crowding out, future tax disincentives, default risk and inflation from monetising debt.

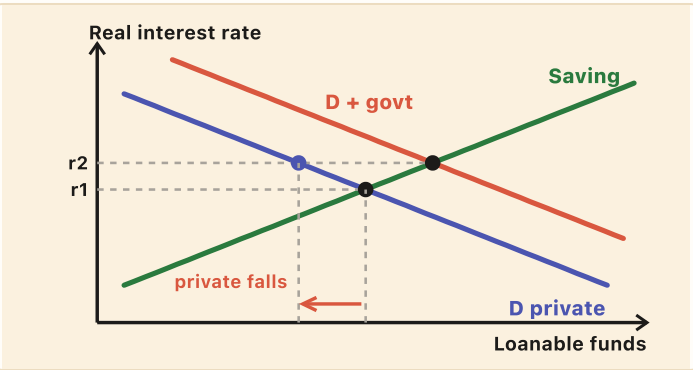


IDEA

Why some economists shrug at deficits

- **Ricardian equivalence:** a tax cut paid for by borrowing makes people expect higher taxes later, so they save the cut. Spending, and the economy, barely change.
- **Owed to ourselves:** debt held by citizens is a transfer within the country.
- **Investment pays back:** borrowing for roads or ports can raise future output and tax receipts.
- **Spare capacity:** with high unemployment, deficits put idle resources to work without crowding out.
- Big deficits can force overdue tax reform.

So: Ricardian equivalence says debt-financed tax cuts are saved, not spent; internal debt and productive investment soften the debt worry.



PICTURE IT

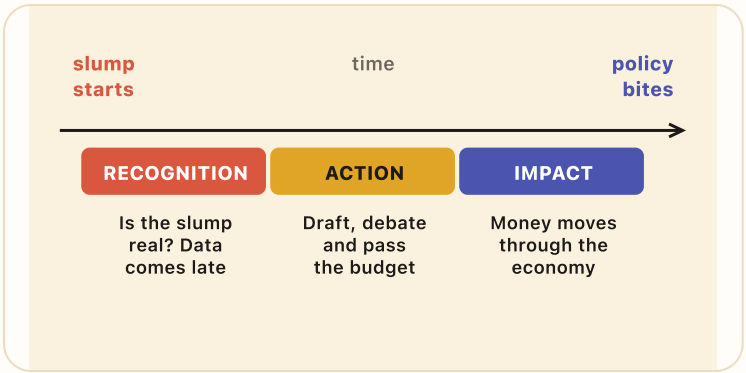
Crowding out on one diagram

Interest rates are set where saving meets the demand for loanable funds. Firms borrow 5.0 units at rate r1.

The government now borrows to fund a deficit. Total demand for funds shifts right, and the rate rises to r2. At that higher rate, firms want only 3.75 units, down from 5.0. Private investment has been **crowded out**.

This is most likely when the economy is near full capacity. With lots of idle saving in a deep slump, rates may barely move.

So: Crowding out: government borrowing raises interest rates and pushes out private investment, which weakens the fiscal multiplier.



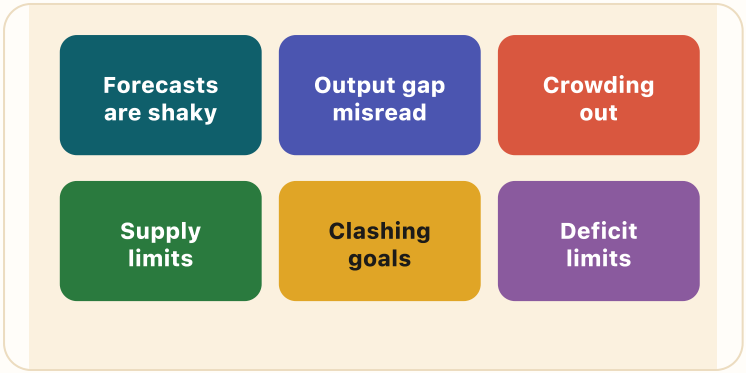
PICTURE IT

Three lags between the slump and the fix

- **Recognition lag:** time to realise the economy has changed. GDP data arrive months late and get revised.
- **Action lag:** time to design the measures and get them through parliament.
- **Impact lag:** time for the spending or tax change to work through the economy.

Add them up and the cure can arrive after the patient has recovered. A stimulus that lands in the next boom adds to inflation instead of fighting the slump.

So: Recognition, action, impact: discretionary fiscal policy can arrive late and push the economy the wrong way.

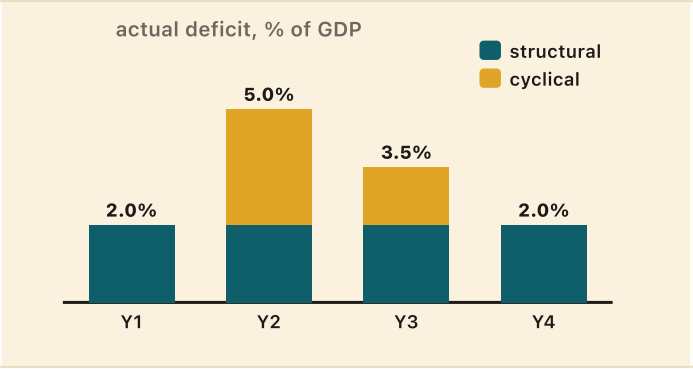


IDEA

More reasons fiscal fine-tuning is hard

- **Shaky forecasts:** macro models are uncertain, so the size of the needed boost is a guess.
- **Misread output gap:** if potential GDP is lower than thought, stimulus brings inflation, not jobs.
- **Crowding out:** borrowing lifts rates and cuts private investment.
- **Supply limits:** if workers lack the right skills, more demand cannot create jobs.
- **Clashing goals:** high unemployment with high inflation needs opposite medicines.
- **Deficit limits:** heavy debt leaves little room to borrow.

So: **Even with the right intent, uncertain forecasts, a misjudged output gap and supply constraints can make fiscal policy miss.**



PICTURE IT

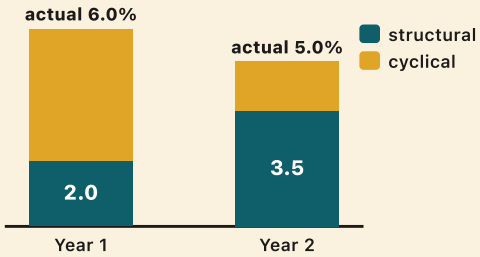
Why the actual deficit misleads

The **actual deficit** moves for two reasons. Policy changes, and the cycle changes. In a slump, automatic stabilisers cut tax receipts and raise benefits, so the deficit grows with no new policy at all.

So economists split it. The **cyclical deficit** is the part caused by the cycle. The **structural (cyclically adjusted) deficit** is what the deficit would be at full employment.

In the chart, policy never changes: the structural deficit stays at 2%. The actual deficit jumps to 5% in the slump of Y2 anyway.

So: Judge the fiscal stance by the structural (cyclically adjusted) deficit, not by the headline deficit.



FORMULA

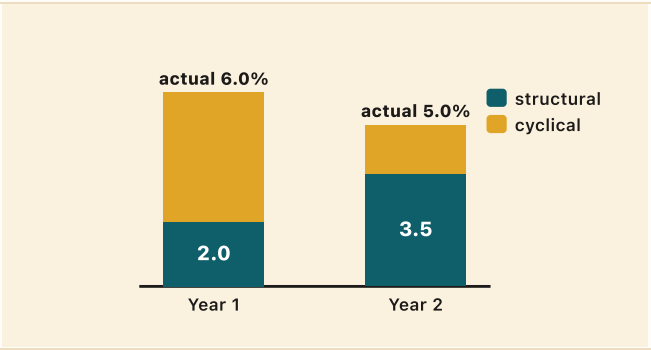
Structural deficit and fiscal stance

Structural deficit = Actual deficit − Cyclical deficit
Fiscal stance = change in the structural deficit

Symbol	In plain words	Example
Actual	spending minus revenue, % of GDP	6.0%
Cyclical	part caused by the slump, no new policy	4.0%
Structural	part the government chose (cyclically adjusted)	2.0%
Stance	is policy adding demand or taking it away?	up = adding

Like a shop's loss in a bad monsoon: remove the monsoon before judging the owner.

So: **Structural up: expansionary. Structural down: contractionary. Unchanged: neutral.**



WORKED EXAMPLE

Reading the fiscal stance

What we know

- **Year 1** = 6.0%, 4.0%: actual deficit, cyclical part (% of GDP)
- **Year 2** = 5.0%, 1.5%: same, as the economy recovers

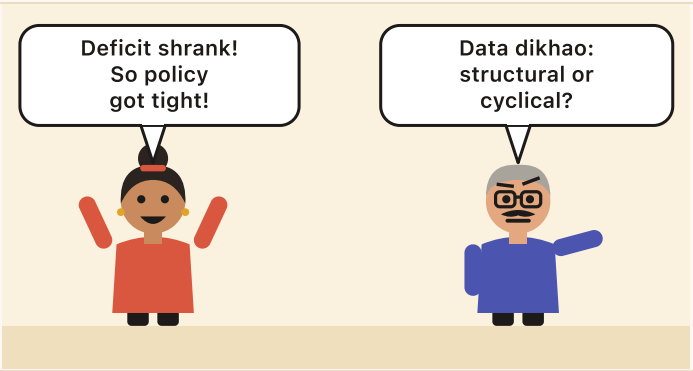
What we want: the structural deficit (actual - cyclical) each year. Pinky says policy tightened: right?

1. Year 1: 6.0 - 4.0 = 2.0%
   Strip the slump's share out of the actual deficit.
2. Year 2: 5.0 - 1.5 = 3.5%
   Recovery shrank the cyclical part.
3. Change: 3.5 - 2.0 = +1.5 points
   The chosen deficit grew, so policy got looser.

Policy turned more expansionary. Pinky is wrong.

Sense check: The actual deficit fell only because recovery shrank the cyclical part.

So: Structural deficit up: expansionary. Down: contractionary. Ignore the headline.

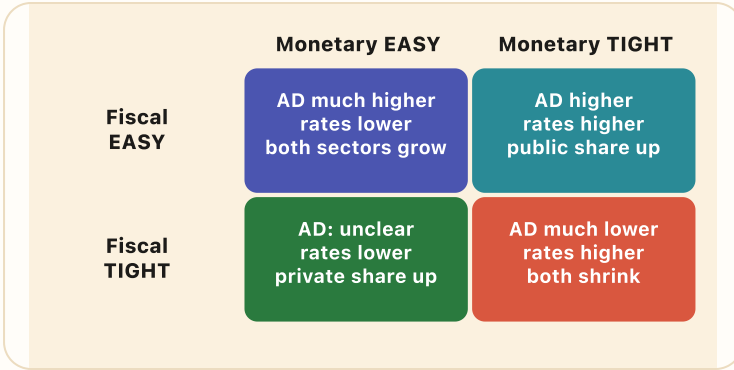


EXAM TRAP

Stance traps

- A bigger deficit in a recession may be pure automatic stabilisers. It is not proof of expansionary policy.
- Stance is about the **change** in the structural deficit, not its level. A structural deficit that falls from 4% to 3% is contractionary, even though 3% is still a deficit.
- A balanced budget is not automatically neutral: the balanced budget multiplier is positive.
- Structural deficits depend on estimates of potential GDP, which are uncertain.

So: Read the change in the cyclically adjusted deficit, and never the headline deficit alone.



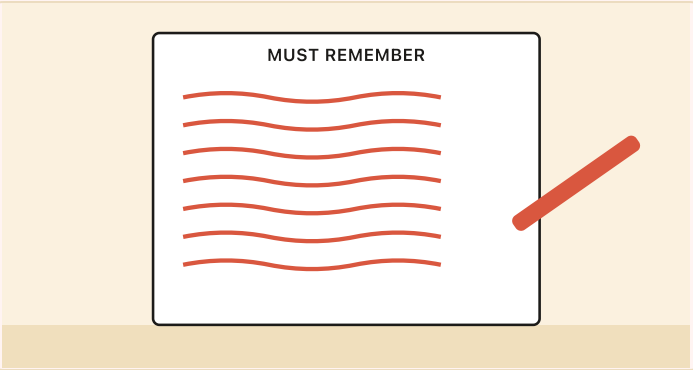
AT A GLANCE

When fiscal and monetary policy meet

	Monetary easy	Monetary tight
Fiscal easy	Strong boost; lower rates; public and private sectors both grow	Higher output; higher rates; public sector's share of GDP rises
Fiscal tight	Lower rates; private sector's share rises, public falls	Strong squeeze; higher rates; both sectors shrink

Interest rates follow the monetary side. The public sector's share follows the fiscal side.

So: Easy fiscal with tight monetary means higher rates and a bigger public sector; tight fiscal with easy monetary means lower rates and a bigger private sector.

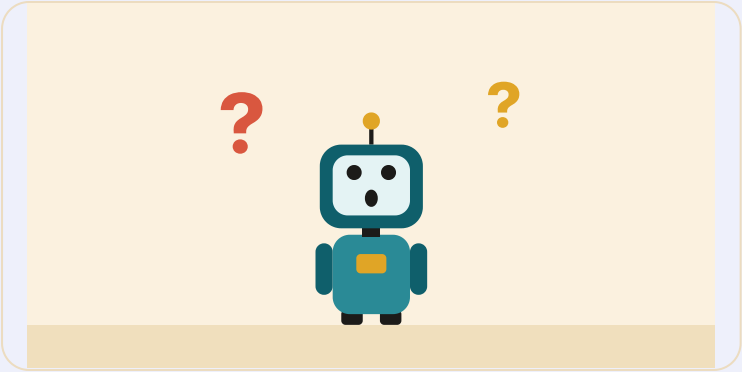


SHARMA JI'S RED PEN

7 things to remember

1. Three jobs: stabilise AD, redistribute, allocate resources.
2. Automatic stabilisers need no decision; discretionary policy faces three lags.
3. Indirect taxes change fastest, capital projects slowest. Transfers are not in G.
4. Good taxes: simple, efficient, fair (horizontal, vertical), sufficient.
5. Multiplier = 1 / [1 - c(1 - t)], c = MPC, t = tax rate. Tax cut: c x the cut. Balanced budget: positive.
6. Debt worries: crowding out, future taxes, default, inflation. Counter: Ricardian equivalence.
7. Stance = change in the structural (cyclically adjusted) deficit.

So: **Scribbled by Sharma ji on the back of the Union Budget speech.**



EXAM QUESTION 1 OF 8

Which of the following is an automatic stabiliser?

A. A progressive income tax system

B. A highway programme approved during a recession

C. A temporary GST cut announced in the budget

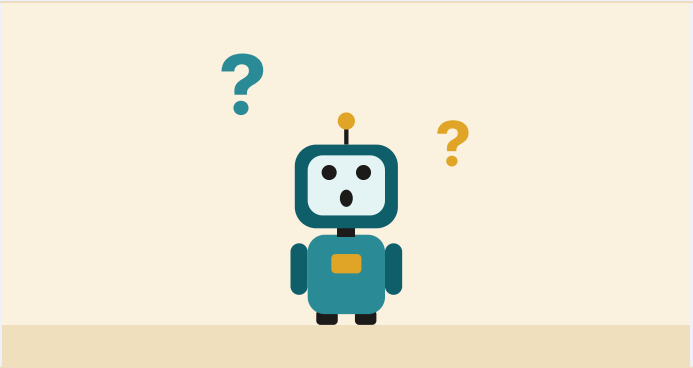
Answer on the next page.

So: **Pick an answer, then turn the page.**

ANSWER

Which of the following is an automatic stabiliser?

Answer: A. A progressive tax takes a smaller share of income when incomes fall and a bigger share when they rise, with no new decision. B and C are discretionary: each needs a new government decision, so each faces recognition and action lags.



EXAM QUESTION 2 OF 8

MPC is 0.75 and the tax rate is 20%. Government spending rises by USD 30 billion. The change in GDP is closest to:

A. USD 35.3 billion

B. USD 120 billion

C. USD 75 billion

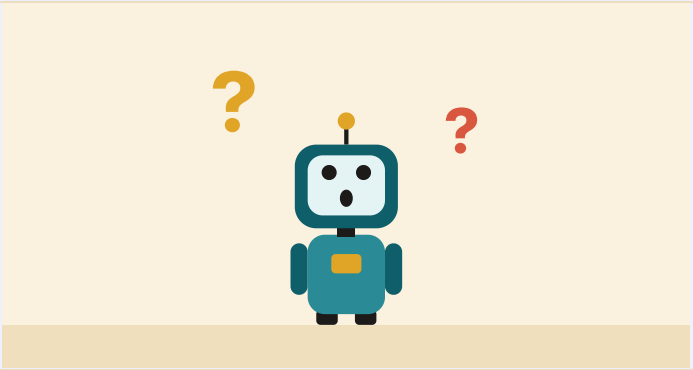
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

MPC is 0.75 and the tax rate is 20%. Government spending rises by USD 30 billion. The change in GDP is closest to:

Answer: C. Multiplier = 1 / [1 - 0.75 x 0.8] = 1 / 0.40 = 2.5, so GDP rises by 2.5 x 30 = USD 75 billion. A wrongly uses c x t instead of c x (1 - t). B ignores tax: 1 / (1 - 0.75) = 4 gives USD 120 billion, which overstates the effect.



EXAM QUESTION 3 OF 8

MPC is 0.8 and the tax rate is 25%. The government raises spending by EUR 60 billion and taxes by EUR 60 billion. GDP will:

A. stay unchanged, because the budget balance does not change

B. rise by about EUR 30 billion

C. rise by about EUR 150 billion

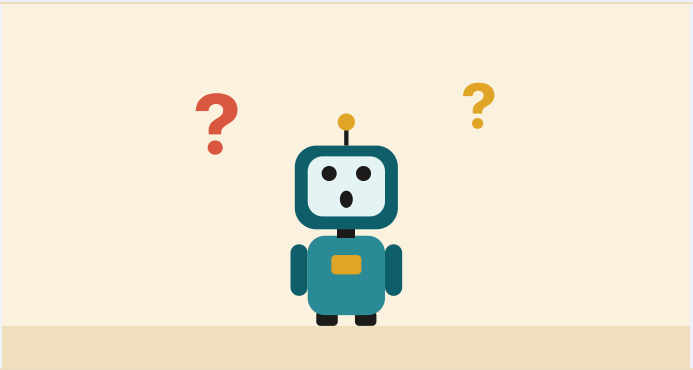
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

MPC is 0.8 and the tax rate is 25%. The government raises spending by EUR 60 billion and taxes by EUR 60 billion. GDP will:

Answer: B. The multiplier is 2.5. Spending adds 2.5 x 60 = 150. The tax rise cuts round-one spending by 0.8 x 60 = 48, so it removes 2.5 x 48 = 120. Net: +30. A ignores that part of the tax is paid from saving, so the balanced budget multiplier is positive. C counts the spending and forgets the tax.



EXAM QUESTION 4 OF 8

Under Ricardian equivalence, a tax cut financed by government borrowing will most likely:

A. have little effect on consumption, because households save to meet expected future taxes

B. raise consumption by the MPC times the tax cut

C. raise inflation, because the central bank must buy the new debt

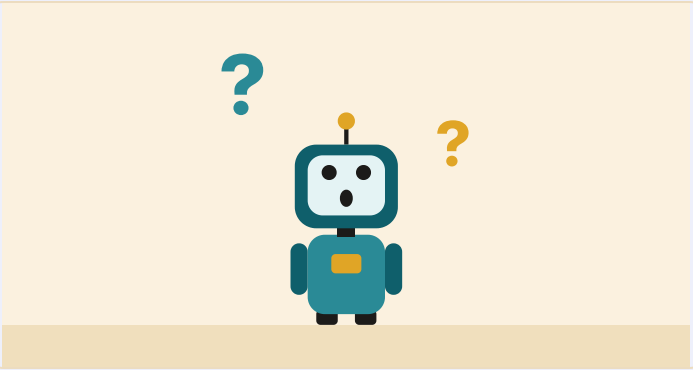
Answer on the next page.

So: **Pick an answer, then turn the page.**

ANSWER

Under Ricardian equivalence, a tax cut financed by government borrowing will most likely:

Answer: A. Ricardian equivalence says households see that today's borrowing means higher taxes later, so they save the cut. B is the standard Keynesian view that Ricardian equivalence rejects. C describes monetisation, which is a separate concern, not part of Ricardian equivalence.



EXAM QUESTION 5 OF 8

Last year the structural deficit was 2.0% of GDP. This year the actual deficit is 5.0% of GDP, of which 3.5% is cyclical. The fiscal stance is:

A. expansionary, because the actual deficit rose

B. neutral, because the deficit was caused by the cycle

C. contractionary

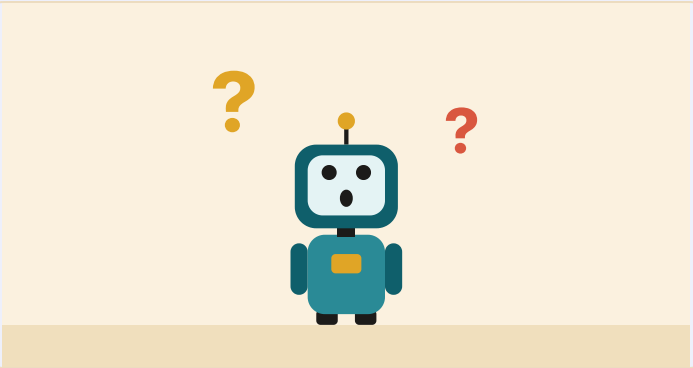
Answer on the next page.

So: **Pick an answer, then turn the page.**

ANSWER

Last year the structural deficit was 2.0% of GDP. This year the actual deficit is 5.0% of GDP, of which 3.5% is cyclical. The fiscal stance is:

Answer: C. Structural deficit this year = 5.0 - 3.5 = 1.5%, down from 2.0%, so policy tightened. A reads the headline deficit, which rose because of automatic stabilisers. B is wrong because the structural part changed, so the stance did too.



EXAM QUESTION 6 OF 8

The months a government needs to draft a stimulus bill and get it through parliament are best described as the:

A. recognition lag

B. action lag

C. impact lag

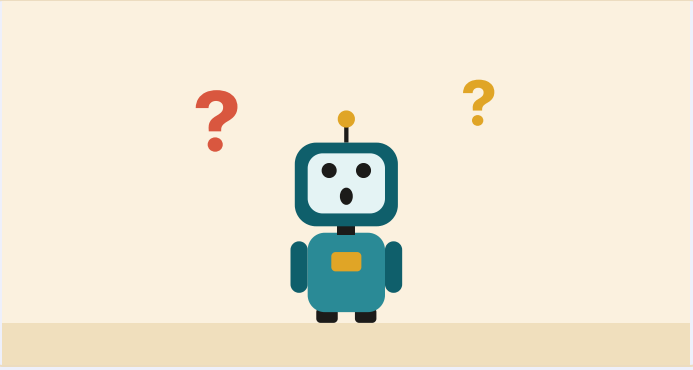
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

The months a government needs to draft a stimulus bill and get it through parliament are best described as the:

Answer: B. The action lag is the time to decide and implement. The recognition lag (A) is the time to realise the economy has changed. The impact lag (C) is the time for the measure, once in place, to affect the economy.



EXAM QUESTION 7 OF 8

Two workers with the same income and circumstances pay the same tax. This tax system shows:

A. vertical equity

B. revenue sufficiency

C. horizontal equity

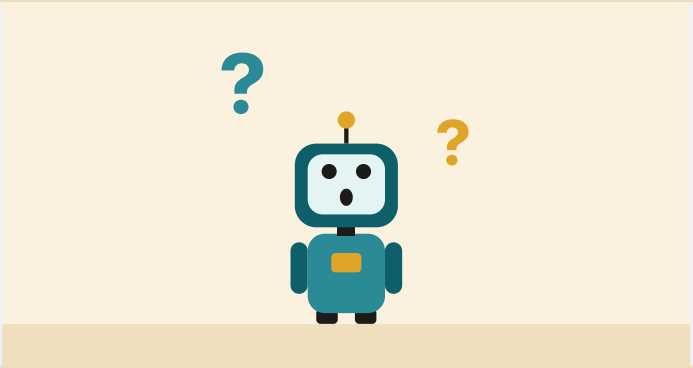
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

Two workers with the same income and circumstances pay the same tax. This tax system shows:

Answer: C. Horizontal equity means people in similar situations pay the same tax. Vertical equity (A) means those with more ability to pay pay more. Revenue sufficiency (B) is about raising enough money to fund spending, not about who pays.



EXAM QUESTION 8 OF 8

A country runs tight fiscal policy and easy monetary policy. The most likely result is:

A. lower interest rates and a larger private sector share of GDP

B. higher interest rates and a larger public sector share of GDP

C. higher interest rates and a fall in both private and public demand

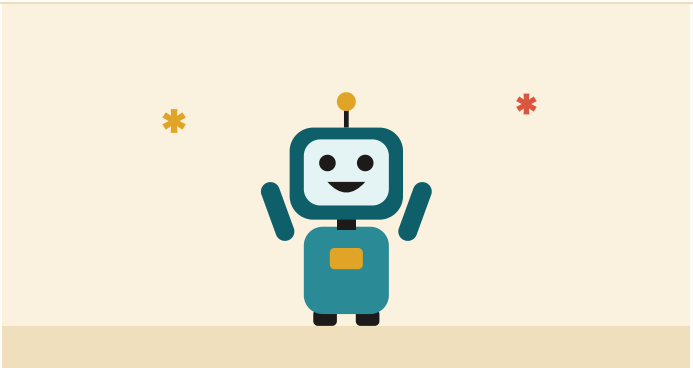
Answer on the next page.

So: **Pick an answer, then turn the page.**

ANSWER

A country runs tight fiscal policy and easy monetary policy. The most likely result is:

Answer: A. Easy money lowers rates, which helps private investment and consumption, while tight fiscal policy shrinks the public sector. B describes easy fiscal with tight monetary policy. C describes tight fiscal with tight monetary policy.



CHAPTER 3 DONE

Fiscal policy, sorted

You can compare fiscal and monetary policy, name the tools and their speeds, compute the fiscal and balanced budget multipliers, argue both sides of the deficit debate, list the lags, and read the stance from the structural deficit.

So: **Swipe on, or jump to another chapter from the Menu.**



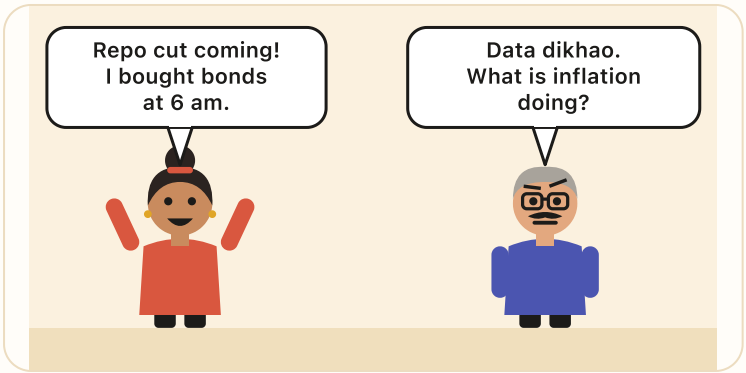
BEFORE YOU START

The cast, and words this chapter uses

A Mumbai investment firm. Sharma ji trusts nothing without data ("Data dikhao"); Pinky is sure of everything, especially when wrong.

- **GDP:** the value of all final output an economy produces in a year.
- **Inflation:** a sustained rise in the general price level.
- **Aggregate demand:** total spending on a country's output.
- **Bond:** an IOU sold to borrow money; its price and yield move oppositely.
- **Exchange rate:** the price of one currency in terms of another.

So: **You can start this chapter cold. Everything it needs is on this card.**



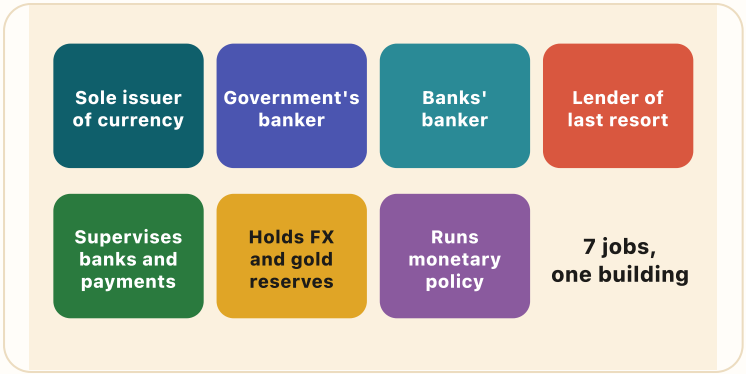
STORY

Policy day on the trading desk

10 am, RBI policy day. Pinky has loaded up on bonds, sure the repo rate will be cut. Chintu asks the obvious question: who decides, and why should one number move every price on her screen?

Sharma ji: "The central bank. **Monetary policy** is what a central bank does to influence the quantity of money and credit, mainly through interest rates. Its main job is to keep inflation low and stable. So before you bet on a cut, check what inflation is doing."

So: Monetary policy is the central bank's control of money and credit, mostly through its policy interest rate, aimed first at stable prices.

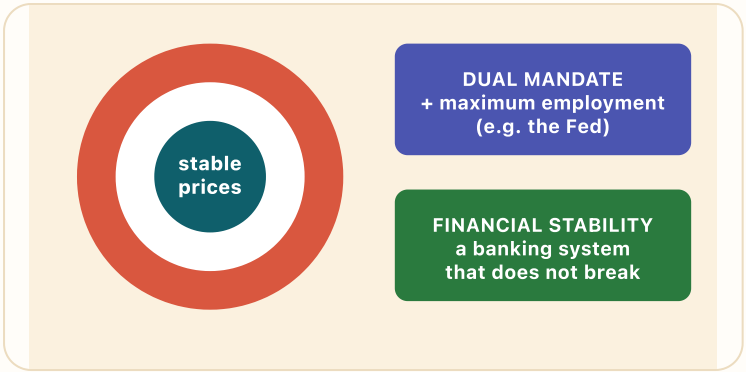


AT A GLANCE

What a central bank actually does

- **Monopoly supplier of currency:** only it issues notes; a Rs 500 note is its promise.
- **Banker to the government** and **to the banks**: holds their accounts, settles payments between banks.
- **Lender of last resort:** lends to a sound bank facing a sudden run, to stop panic spreading.
- **Regulator and supervisor** of banks and the payments system.
- **Manager of foreign exchange and gold reserves.**
- **Conductor of monetary policy.**

So: **Issuer, banker to government and banks, lender of last resort, supervisor, reserve manager and policy maker: six roles to know.**



IDEA

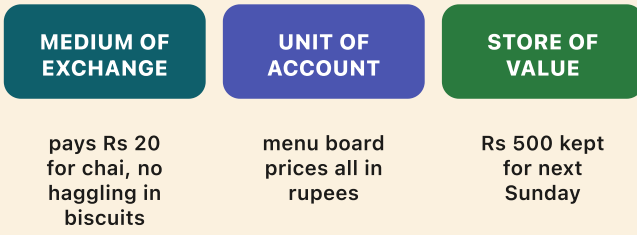
What a central bank is trying to achieve

Price stability is the main goal almost everywhere: low, stable inflation. Why? High or jumpy inflation eats savings, blurs price signals, adds uncertainty that raises interest rates, and hits people on fixed incomes hardest.

Some central banks have a **dual mandate**. The US Fed must pursue maximum employment as well as stable prices.

Most also care about **financial stability**: banks and markets that keep working through a shock.

So: Price stability comes first; some central banks add full employment (a dual mandate), and most also guard financial stability.

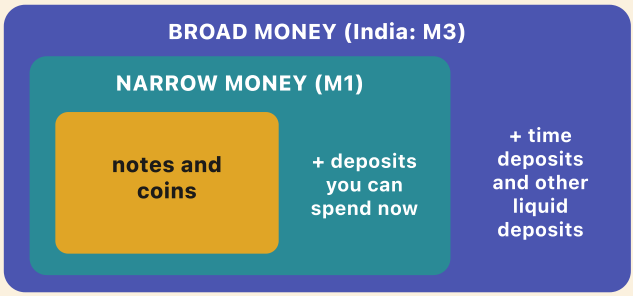


IDEA

Three jobs of money, at one chai stall

- **Medium of exchange:** Chintu pays Rs 20 for chai. Without money he would need a chaiwala who happens to want his spreadsheet skills (barter's "double coincidence of wants").
- **Unit of account:** every price on the board is in rupees, so comparing samosa with vada pav is easy.
- **Store of value:** the Rs 500 note in his wallet will still buy something next Sunday.

So: Money is a medium of exchange, a unit of account and a store of value.



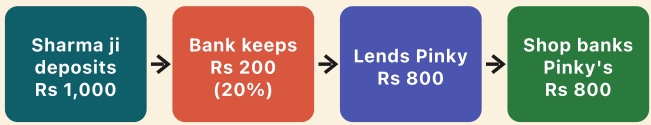
AT A GLANCE

Narrow money vs broad money

	Narrow money	Broad money
Contains	Notes and coins in circulation + deposits you can spend at once (current accounts)	Narrow money + less liquid deposits, such as time deposits
Idea	Money ready for transactions	Money plus near-money

Each country defines its own measures. India's RBI reports M1 (narrow) and M3 (broad); others publish their own versions under similar names.

So: Narrow money is what you can spend right now; broad money adds less liquid deposits.



Deposits now: 1,000 + 800 = Rs 1,800

The extra Rs 800 is new money created by a loan

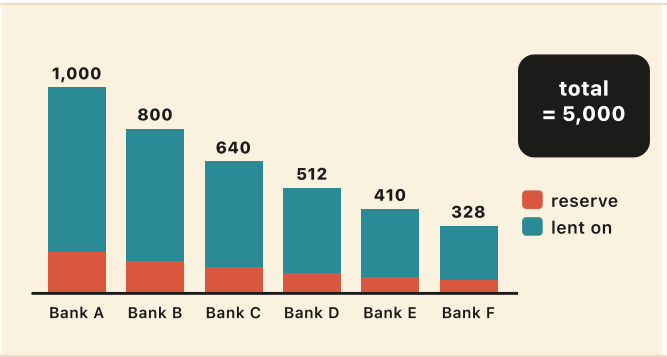
PICTURE IT

How banks create money

In **fractional reserve banking**, a bank keeps only a fraction of each deposit as reserves (vault cash or its account at the central bank) and lends out the rest. The minimum fraction is the **reserve requirement**.

Pinky borrows Rs 800 and pays a shop, which deposits it. Sharma ji still has his Rs 1,000 deposit, and the shop now has Rs 800 too. Deposits are money, so the loan created money.

So: Each bank loan that is spent and redeposited creates a new deposit, so commercial banks create most of an economy's money.



WORKED EXAMPLE

The deposit chain, part 1: bank by bank

What we know

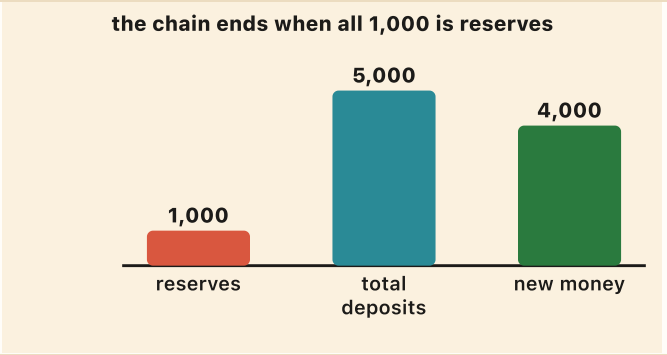
- **r** = 20% = 0.2: reserve requirement: share of each deposit a bank must keep
- **New cash** = Rs 1,000: Sharma ji's deposit; every loan is spent and banked again

What we want: how the one deposit turns into many

1. Bank A: deposit 1,000, keeps 200, lends 800
   The bank must lock away 20% and lends the other 800 to Pinky.
2. Bank B: deposit 800, keeps 160, lends 640
   Pinky pays a shop; the shop banks it. Same rule.
3. Bank C: deposit 640, keeps 128, lends 512
   Each deposit is 80% of the one before.

After six banks, deposits already add to Rs 3,689.

So: Every loan that is spent and banked again becomes a new, slightly smaller deposit.



WORKED EXAMPLE

The deposit chain, part 2: where it stops

What we know

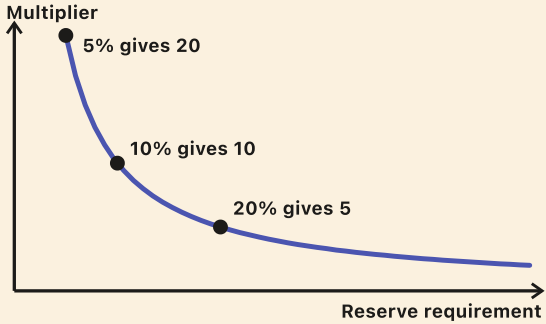
- **r** = 20% = 0.2: reserve requirement: share of each deposit kept as reserves

What we want: the most deposits Rs 1,000 of new cash can build

1. Total reserves can never pass 1,000
   All reserves come out of Sharma ji's 1,000 of cash.
2. Total deposits = 1,000 / 0.2 = 5,000
   Reserves are 20% of deposits, so deposits are 5 times reserves.
3. New money created = 5,000 - 1,000 = 4,000
   The first deposit was already money; lending made the rest.

Total deposits up to Rs 5,000; new money Rs 4,000.

So: Total deposits can grow to new reserves / r, and no further.



FORMULA

The money multiplier

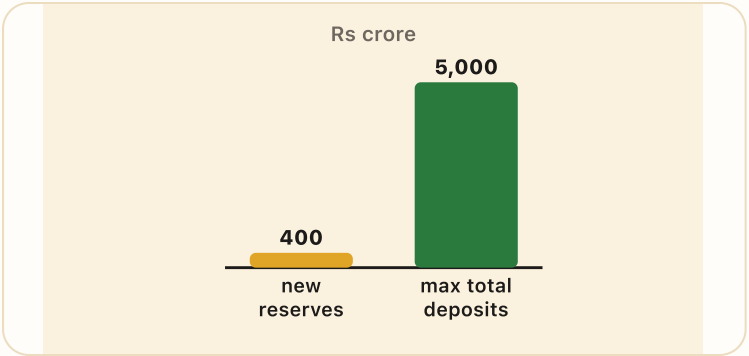
```
Money multiplier = 1 / r
```

Symbol	In plain words	Example
r	reserve requirement: share of deposits banks must keep, as a decimal	10% = 0.10
1 / r	the money multiplier: rupees of deposits each rupee of reserves can back	1 / 0.10 = 10

Use it: most deposits banks can build = multiplier x new reserves. At 10%, Rs 100 of new reserves backs up to Rs 1,000.

Why "1 / r": lending stops when all new cash is locked up as reserves. Reserves = r x deposits, so deposits = reserves / r. It is a **ceiling**: cash kept in wallets makes the real figure smaller.

So: Money multiplier = 1 / reserve requirement: the lower the requirement, the more money each rupee of reserves can support.



WORKED EXAMPLE

Multiplier: a central bank adds reserves

What we know

- **r** = 8% = 0.08: reserve requirement: share of deposits banks must keep
- **New reserves** = Rs 400 crore: paid to banks for bonds the central bank bought

What we want: the multiplier, and the most deposits can rise

1. Multiplier = 1 / 0.08 = 12.5
   One rupee of reserves backs 12.5 of deposits (8% of 12.5 is 1).
2. Maximum increase in deposits = 12.5 x 400 = Rs 5,000 crore
   Scale the new reserves up by the multiplier.

Multiplier 12.5; deposits can rise by up to Rs 5,000 crore.

So: Check: 8% of 5,000 = 400, exactly the new reserves. The answer is a maximum, not a forecast.



Print 10% more notes: Rs 550 x 4 = Rs 2,200

still 1,000 cups, so each now costs Rs 2.20

STORY

A village that only makes chai

Picture a village whose only product is chai. All its notes add up to **Rs 500**. Notes change hands: Chintu pays the chaiwala, the chaiwala pays the milkman, the milkman buys a cup. Over a year each note is spent about **4 times**, so total spending is 500 x 4 = **Rs 2,000**.

The village makes **1,000 cups** a year. Rs 2,000 chasing 1,000 cups means each cup costs **Rs 2**.

Now print 10% more notes: Rs 550. Each is still spent 4 times (Rs 2,200), and there are still only 1,000 cups. So a cup now costs **Rs 2.20**. More money, same chai: only the price moved.

So: Money times how often it is spent equals what all the output costs; extra money with no extra output just raises prices.



money stock x times each unit is spent

price level x real output = nominal GDP

FORMULA

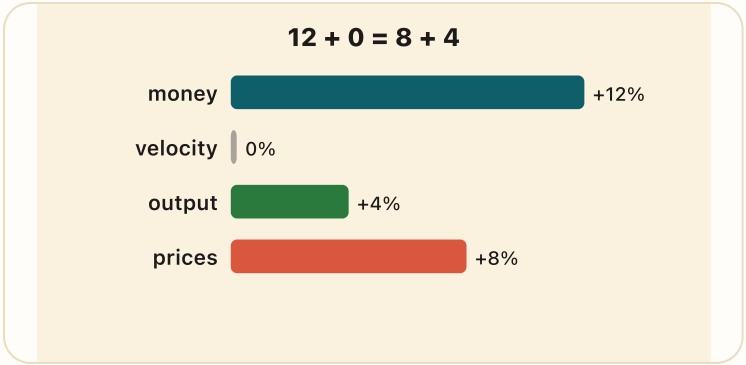
The quantity equation of exchange

$$M \times V = P \times Y$$

Symbol	In plain words	Example
M	money: all the notes and spendable deposits	Rs 500
V	velocity: times each rupee is spent on final goods in a year	4
P	price level: what one unit of output costs on average	Rs 2 a cup
Y	real output: how much is made, in units, not rupees	1,000 cups

Both sides are the same money counted twice: M x V is what buyers spend; P x Y is what sellers receive (nominal GDP). 500 x 4 = 2 x 1,000 = 2,000. The **quantity theory of money** adds a guess: V and Y hardly change, so extra M shows up in P.

So: MV = PY; with velocity and real output fixed, money growth turns into inflation one for one.



FORMULA

The quantity equation in growth rates

`%M + %V = %P + %Y (approximately)`

Symbol	In plain words	Example
%M	growth in the money supply this year	12%
%V	growth in velocity (times each rupee is spent)	0%
%P	growth in the price level: inflation	8%
%Y	growth in real output	4%

Why it works: if spending (M x V) grows 12% and there is only 4% more stuff to buy, the rest of the growth, about 8%, has to show up as higher prices.

It is approximate because growth rates multiply: 1.12 / 1.04 = 1.077, so 7.7% exactly. Exam questions use the simple subtraction.

So: **Inflation is roughly money growth plus velocity growth minus real output growth.**



WORKED EXAMPLE

Quantity theory: print 10% more

What we know

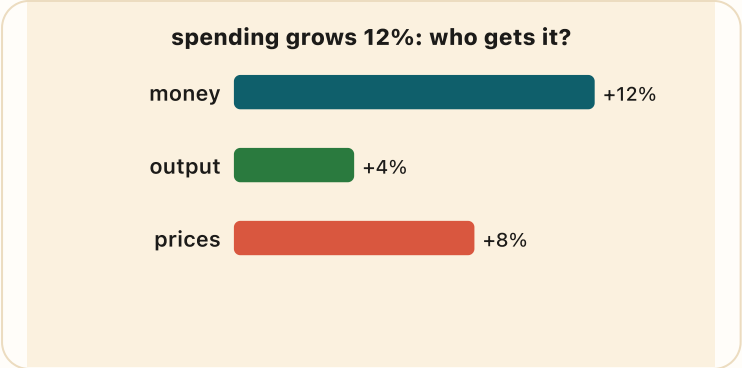
- **M** = 500: money in the economy
- **V** = 4: velocity: times each unit is spent in a year
- **Y** = 1,000: real output, in units

What we want: (a) the price level P; (b) P after M rises 10%, V and Y unchanged

1. (a) P = M x V / Y = 500 x 4 / 1,000 = 2.0
   Total spending is 2,000; spread over 1,000 units, each costs 2.
2. (b) M = 500 x 1.10 = 550, so P = 550 x 4 / 1,000 = 2.2
   Spending rises to 2,200, but output is still 1,000 units.

(a) P = 2.0 (b) P = 2.2, up 10%, the same 10% as money.

So: With V and Y fixed, prices rise in the same proportion as money.



WORKED EXAMPLE

Quantity theory in growth rates

What we know

- **%M** = 12%: money supply growth this year
- **%V** = 0%: velocity growth (velocity is stable)
- **%Y** = 4%: real output growth

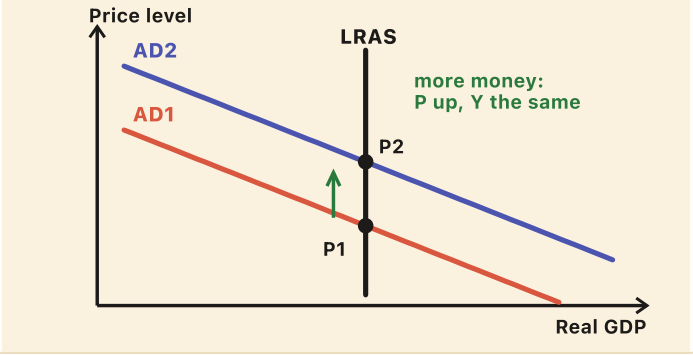
What we want: %P, this year's inflation

1. %M + %V = %P + %Y, so 12 + 0 = %P + 4
   Spending grows 12%; 4 of those points pay for the extra output.
2. %P = 12 - 4 = 8%
   The other 8 points have nothing new to buy, so they push prices up.

Inflation of about 8%.

Sense check: Exact: 1.12 / 1.04 = 1.077, so 7.7%, close to 8%.

So: **With velocity stable, inflation is roughly money growth minus real output growth.**



PICTURE IT

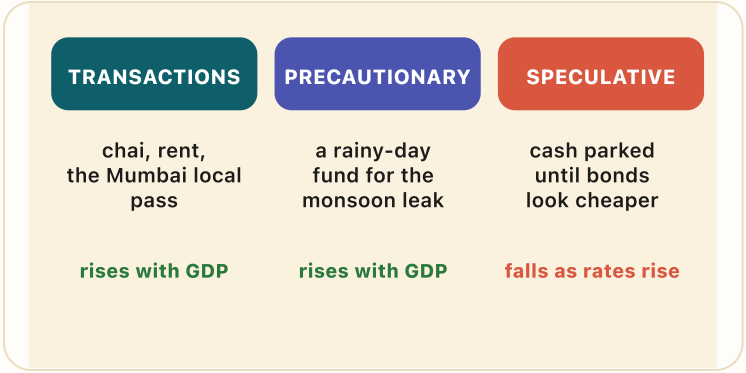
Money neutrality: in the long run, only prices move

Money neutrality says that in the long run, a change in the money supply changes only nominal things, such as the price level and nominal wages. Real things, such as output, jobs and real interest rates, are set by labour, capital and technology.

On the diagram, more money pushes AD up, but long-run supply is vertical, so only prices rise.

In the short run, prices and wages are sticky, so money can move real output. That short-run grip is what monetary policy works with.

So: Money is neutral in the long run (it moves only prices) but not in the short run, which is why monetary policy can steady the cycle.



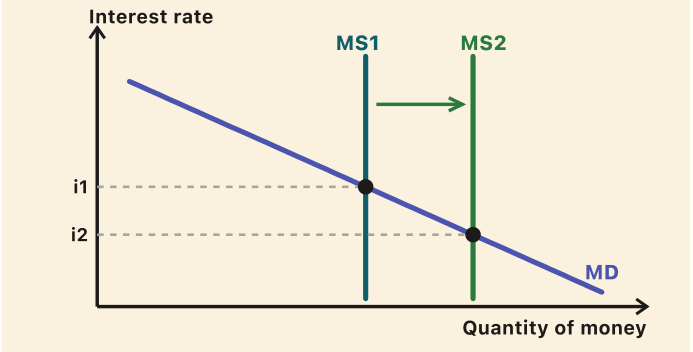
IDEA

Why people want to hold money

- **Transactions demand:** money for everyday buying. It rises with nominal GDP.
- **Precautionary demand:** a cushion for the unexpected. It also rises with income.
- **Speculative (portfolio) demand:** money held instead of bonds or shares. When interest rates are high, holding money costs a lot of lost interest, so people hold less. When rates are low or other assets look risky, they hold more.

Pinky keeps cash when she thinks bond prices are about to fall.

So: Transactions and precautionary demand rise with income; speculative demand falls as interest rates rise.



PICTURE IT

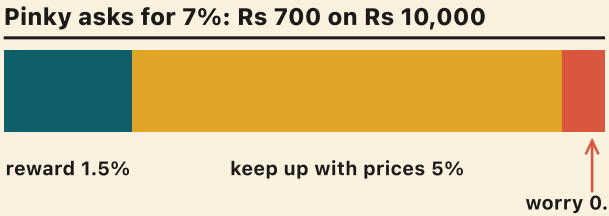
Money supply and demand set the interest rate

The central bank fixes the money supply, so MS is vertical. Money demand slopes down: at lower rates, holding money costs less, so people hold more. They meet at interest rate i1.

If the rate is above i1, people hold more money than they want. They buy bonds, bond prices rise, yields fall, and the rate drops back to i1.

When the central bank adds money (MS1 to MS2), the rate falls to i2.

So: **More money supply means a lower interest rate; less money supply means a higher one.**



STORY

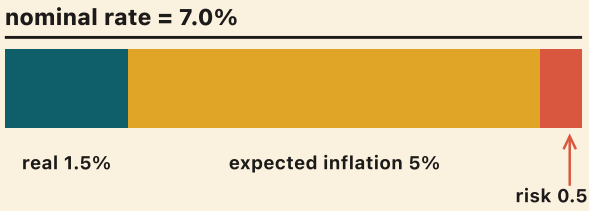
Lending your cousin Rs 10,000 for a year

Pinky's cousin wants Rs 10,000 for a year. Prices are expected to rise **5%**: the Rs 200 thali will cost Rs 210. If Pinky gets back just Rs 10,000, it buys 5% less than today. She has lost by lending.

So she asks for **5%** to keep up with prices, plus **1.5%** as her real reward for waiting, plus **0.5%** in case inflation turns out higher. Total: **7%**, or Rs 700.

If everyone starts expecting 6% inflation, she asks 8%. Her quoted rate moves one for one with the inflation she expects.

So: A quoted interest rate = reward for waiting + expected inflation + a cushion for surprises.



FORMULA

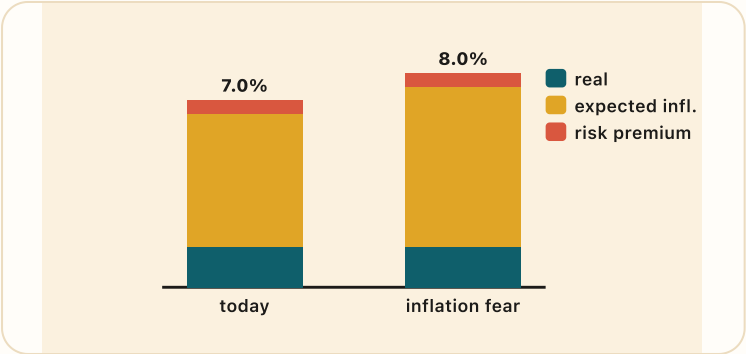
The Fisher effect

$$R_{nominal} = R_{real} + \pi^{e} \text{ (+ risk premium)}$$

Symbol	In plain words	Example
$R_{nominal}$	the quoted rate: an FD rate, a bond yield	7%
R_{real}	the reward for waiting, in what money can buy	1.5%
π^{e}	expected inflation (π "pi" = inflation, e = expected)	5%
Risk premium	extra in case inflation surprises	0.5%

Why "expected": the rate is fixed today, before inflation is known, so lenders build in what they expect. The **Fisher effect**: nominal rates move one for one with expected inflation. Adding is a shortcut; the exact form is (1 + nominal) = (1 + real) x (1 + inflation).

So: **When expected inflation rises, nominal rates rise with it, one for one.**



WORKED EXAMPLE

Fisher: pricing a one-year bond

What we know

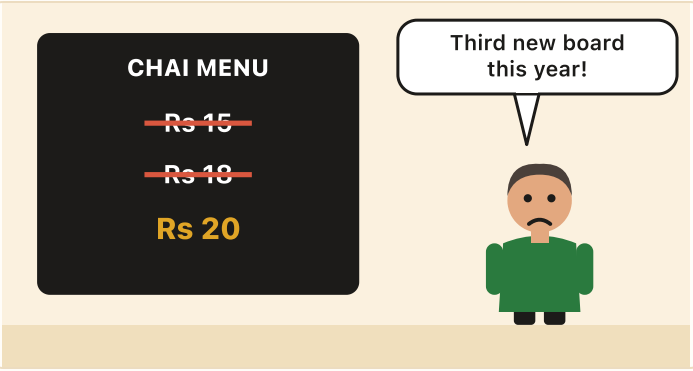
- $\mathbf{R_{real}}$ = 1.5%: real return lenders want
- $\boldsymbol{\pi^e}$ = 5%: expected inflation, then 6% after a scare
- **Risk premium** = 0.5%: extra for inflation surprises

What we want: the quoted rate $R_{nominal}$, (a) now and (b) after the scare

1. (a) $R_{nominal} = 1.5 + 5 + 0.5 = 7.0\%$
   Reward, plus keeping up with prices, plus the cushion.
2. Exact, no premium: 1.015 x 1.05 - 1 = 6.575%, near 6.5%
   Multiplying adds a tiny extra; the exam usually adds.
3. (b) $R_{nominal} = 1.5 + 6 + 0.5 = 8.0\%$
   One more point of expected inflation, one more point of rate.

(a) 7.0% (b) 8.0%, so old 7% bonds fall in price.

So: **Nominal yields track expected inflation one for one.**



AT A GLANCE

The costs of inflation

Type	Cost	What it means
Expected	Menu costs	Reprinting price lists, menus, catalogues
Expected	Shoe-leather costs	Holding less cash, more trips to the bank
Unexpected	Redistribution	Fixed-rate lenders lose, borrowers gain
Unexpected	Uncertainty, muddled signals	Higher risk premiums; firms mistake rising prices for rising demand

So: **Expected inflation brings menu and shoe-leather costs; unexpected inflation is costlier because it redistributes wealth and clouds decisions.**

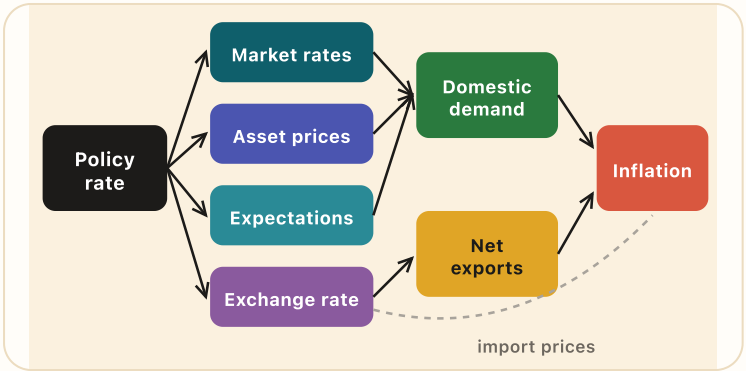
	Open market operations	Policy rate	Reserve requirement
EASE	buy bonds	cut it	lower it
TIGHTEN	sell bonds	raise it	raise it

TOOLS

The three levers of monetary policy

- **Open market operations:** the central bank buys government bonds from banks, paying with new reserves, so banks can lend more. Selling bonds drains reserves.
- **Policy rate:** the rate at which the central bank lends to banks (RBI repo rate, ECB refinancing rate, Bank of England Bank Rate; the Fed targets the federal funds rate). Banks price their own loans off it, so it anchors short-term rates. It is the main tool today.
- **Reserve requirements:** a lower requirement frees reserves to lend; a higher one locks them up.

So: To ease: buy bonds, cut the policy rate, lower reserve requirements. To tighten: do the reverse.



PICTURE IT

How a rate change reaches inflation

Follow a rate **hike**:

1. **Market rates**: banks raise loan and deposit rates.
2. **Asset prices**: bonds, shares and property fall; people feel poorer and spend less.
3. **Expectations**: firms and households expect slower growth and lower inflation.
4. **Exchange rate**: higher rates attract foreign money, the currency rises, exports get dearer and imports cheaper.

Demand cools and import prices fall, so inflation eases, often after more than a year.

So: Policy rate, then market rates, asset prices, expectations and the exchange rate, then spending, then inflation.



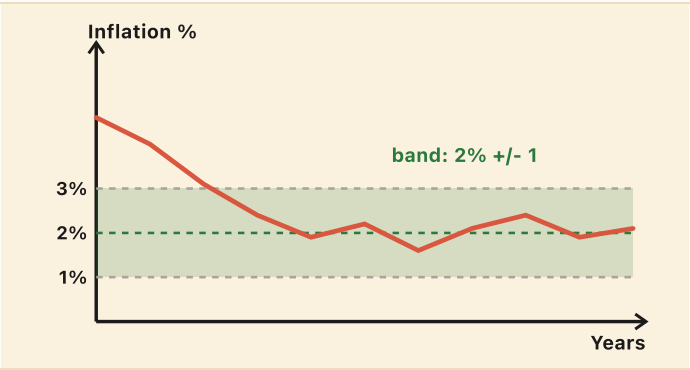
Credibility lets a central bank cool inflation with smaller rate moves: people already expect it to win.

IDEA

What makes a central bank effective

- **Independence:** free from politicians who want cheap money before elections. *Operational independence*: the government sets the target and the bank picks the rate (Bank of England, RBI). *Target independence*: the bank also sets the target (the ECB defines its own).
- **Credibility:** markets and wage setters believe the target, so inflation expectations stay put.
- **Transparency:** published reports, forecasts and minutes, so everyone sees the reasoning.

So: Independence (operational or target), credibility and transparency make a central bank's words and actions work.



PICTURE IT

Inflation targeting, and why 2%

Under **inflation targeting**, the central bank announces a number, often 2% with a band around it, and moves its policy rate to hit it, usually aiming a year or two ahead. India's target is 4% with a 2% to 6% band.

Why not 0%? Measured inflation overstates true inflation, because better quality and new products are hard to capture. A little inflation is also a cushion against deflation, and leaves room to cut rates.

So: Inflation targeting: a public number, usually around 2%, hit with the policy rate; slightly positive to stay clear of deflation.



PICTURE IT

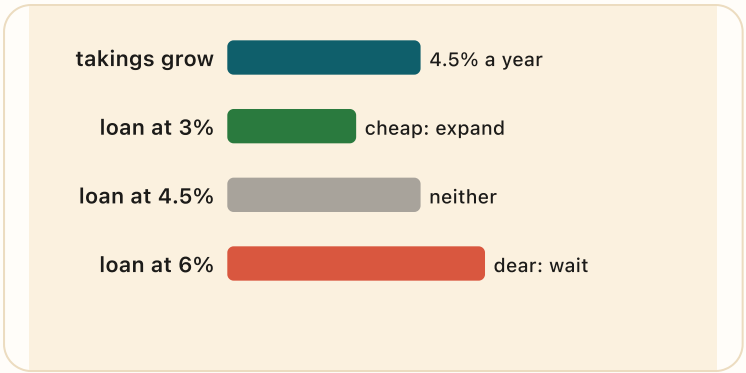
Exchange rate targeting: what you give up

Some central banks target an exchange rate instead of inflation. Hong Kong keeps its dollar between 7.75 and 7.85 per US dollar.

If the local currency weakens, the bank sells foreign reserves and buys its own currency. That shrinks the local money supply and raises local rates, whatever the local economy needs.

So a peg imports the anchor country's monetary policy and, over time, its inflation. Independent monetary policy is the price. The bank also needs large FX reserves.

So: An exchange rate target gives up independent monetary policy: local interest rates must follow the anchor currency.



STORY

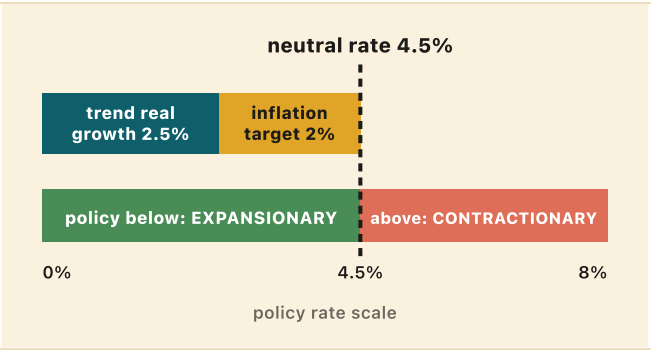
When is a loan neither cheap nor dear?

Chintu's uncle runs a chai stall. Year after year he sells about **2.5%** more cups (real growth), and prices rise about **2%** (inflation). So his rupee takings grow about **4.5%** a year.

A bank offers him a loan for a second counter. At **3%** the loan costs less than his takings grow: he borrows and expands. At **6%** it costs more: he waits. At **4.5%** it neither tempts nor scares him.

Across the whole economy, that 4.5% is the **neutral rate**: the policy rate that neither speeds the economy up nor slows it down.

So: A rate below how fast the economy's rupee income grows pushes spending up; a rate above it holds spending back.



FORMULA

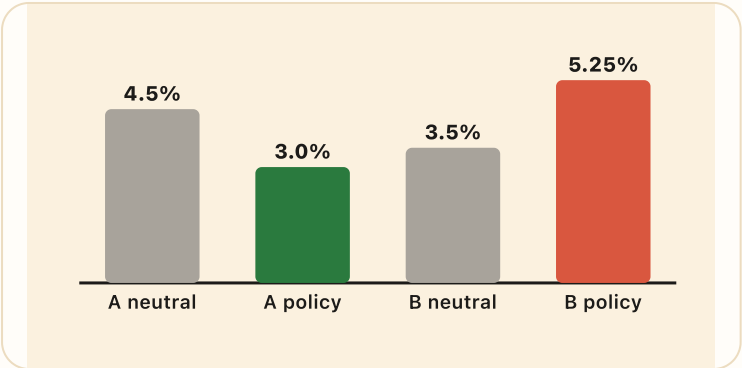
The neutral interest rate

Neutral rate = trend real growth rate + inflation target

Symbol	In plain words	Example
Neutral rate	the rate that neither speeds up nor slows the economy	4.5%
Trend real growth	long-run real GDP growth, not this quarter's	2.5%
Inflation target	the inflation the central bank aims for	2%
Policy rate	the rate the central bank sets (RBI: repo)	3%: easy

Why add them: real growth plus inflation is how fast income grows in rupees; a rate at that pace is neither cheap nor dear. Policy rate below neutral: **expansionary** (easy). Above: **contractionary** (tight).

So: **Neutral = trend real growth + inflation target; compare the policy rate with it to tell easy from tight.**



WORKED EXAMPLE

Neutral rate: easy or tight?

What we know

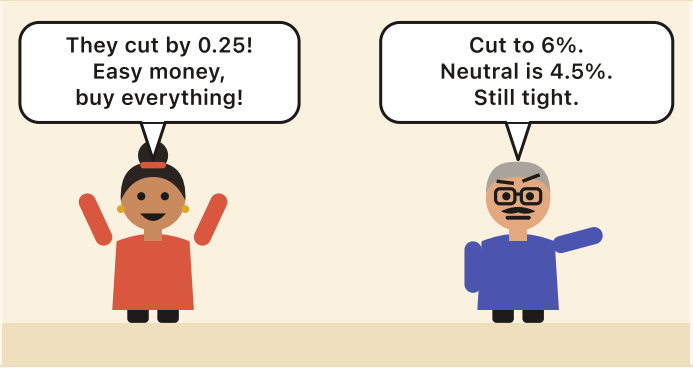
- **Economy A** = 2.5%, 2%, 3.0%: trend real growth, inflation target, policy rate
- **Economy B** = 1.5%, 2%, 5.25%: trend real growth, inflation target, policy rate

What we want: each neutral rate (trend real growth + inflation target), and whether policy is easy or tight

1. A: neutral = 2.5 + 2 = 4.5%. Policy 3.0% is 1.5 points below
   Loans cost less than income grows: borrowing picks up.
2. B: neutral = 1.5 + 2 = 3.5%. Policy 5.25% is 1.75 points above
   Loans cost more than income grows: people hold back.

A is expansionary. B is contractionary.

So: **Below neutral is easy, above neutral is tight, whatever direction the last rate move was.**

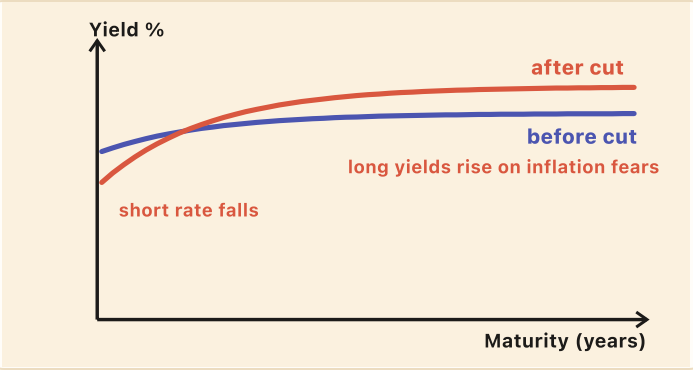


EXAM TRAP

Monetary policy traps

- A rate cut is not automatically expansionary. If the rate is still above neutral, policy is still contractionary, just less so.
- The neutral rate uses **trend** real growth, not this year's GDP growth.
- The money multiplier gives the **maximum** deposits. New money created excludes the original deposit.
- MV = PY (money x times spent = prices x real output): inflation is about money growth minus real growth, if velocity is stable.
- Fisher: rising expected inflation raises nominal rates, not real ones.

So: Compare policy with neutral, use trend growth, and read whether a question asks for total deposits or new money.

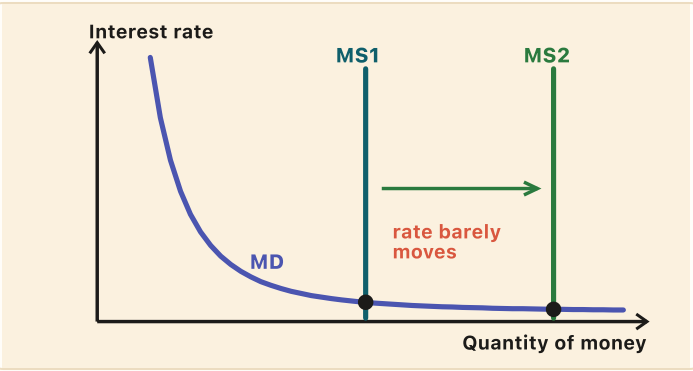


PICTURE IT

When the levers do not grip

- **Weak transmission:** the central bank controls reserves and its own rate, not how much banks lend or how much people borrow. In a gloomy slump, banks may sit on new reserves and firms may not borrow even at low rates.
- **Bond market reaction:** if markets doubt the bank's commitment to low inflation, a rate cut can raise long-term yields, because investors demand compensation for expected inflation. Long-term borrowing then gets dearer, the opposite of the aim.

So: Monetary policy only works if banks lend, people borrow and bond markets believe the central bank.



PICTURE IT

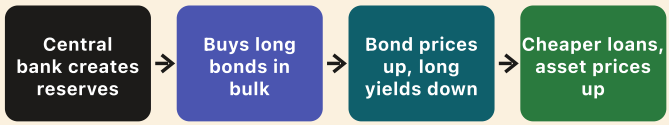
The liquidity trap and deflation

In a **liquidity trap**, interest rates are near zero and people are happy to hold any extra money. Money demand becomes flat, so adding money no longer lowers rates or lifts spending.

Deflation, a falling price level, makes it worse. People delay purchases, the real burden of debt grows, and with nominal rates stuck near zero, the real rate rises: at 0% nominal and -2% inflation, the real rate is +2%.

Japan fought this for much of the 1990s and 2000s.

So: **Near zero rates, extra money just sits idle (a liquidity trap), and deflation pushes real rates up when the central bank cannot cut further.**



Used after 2008 and in 2020 by the Fed, Bank of England, ECB and Bank of Japan

PICTURE IT

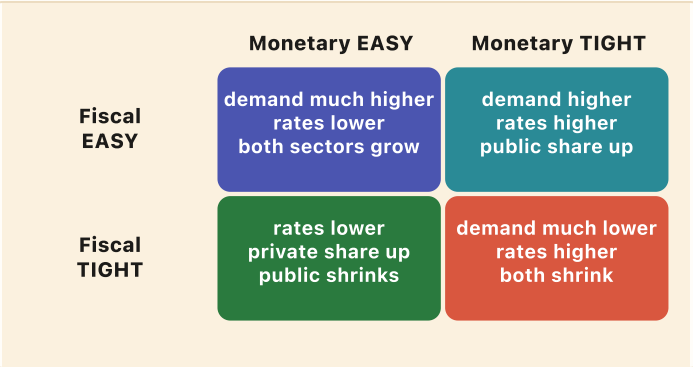
Quantitative easing (QE)

When the policy rate is already near zero, a central bank can still ease. In **quantitative easing**, it creates reserves and uses them to buy large amounts of longer-term government bonds, and sometimes other assets such as corporate bonds.

The aim: push down long-term yields, raise asset prices, and flood banks with reserves so they lend more.

The catch: if banks and borrowers stay cautious, the reserves sit idle. Critics also warn of asset bubbles and future inflation.

So: QE is large-scale asset buying with new reserves, used when the policy rate cannot go lower, to cut long-term yields and support lending.



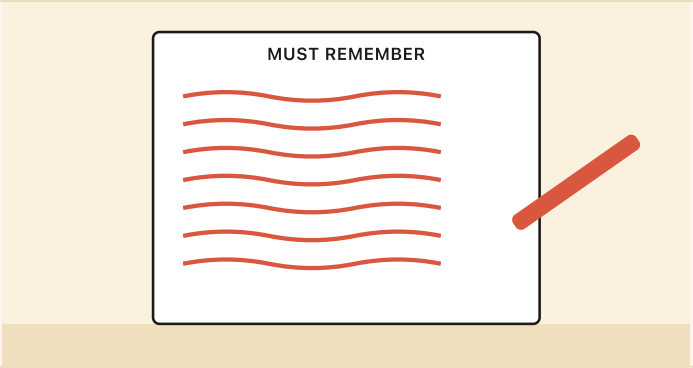
AT A GLANCE

Monetary policy meets fiscal policy

Fiscal policy is the government's spending and taxes. The two can push together or against each other.

- **Both easy:** a strong boost; rates low; public and private sectors grow.
- **Both tight:** a strong squeeze; rates high; both sectors shrink.
- **Easy fiscal, tight money:** rates rise; the public sector's share of GDP grows.
- **Tight fiscal, easy money:** rates fall; the private sector's share grows.

So: Interest rates follow the monetary side; the public sector's share of GDP follows the fiscal side.

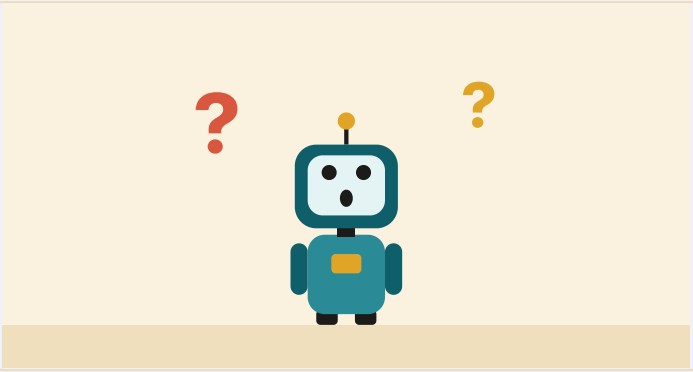


SHARMA JI'S RED PEN

7 things to remember

1. Roles: issuer, bankers' banker, lender of last resort, supervisor, reserves, policy.
2. Money multiplier = 1 / reserve requirement, a maximum.
3. MV = PY; money is neutral in the long run only.
4. Nominal = real + expected inflation (+ risk premium).
5. Ease: buy bonds, cut the policy rate, cut reserve requirements.
6. Neutral = trend real growth + inflation target; below it is easy.
7. Limits: weak transmission, bond doubts, liquidity trap, deflation. QE at zero rates.

So: Scribbled by Sharma ji on the back of the RBI policy statement.



EXAM QUESTION 1 OF 8

A solvent bank faces a sudden run and cannot meet withdrawals. The central bank lends it cash against good collateral. This role is best described as:

A. lender of last resort

B. banker to the government

C. monopoly supplier of currency

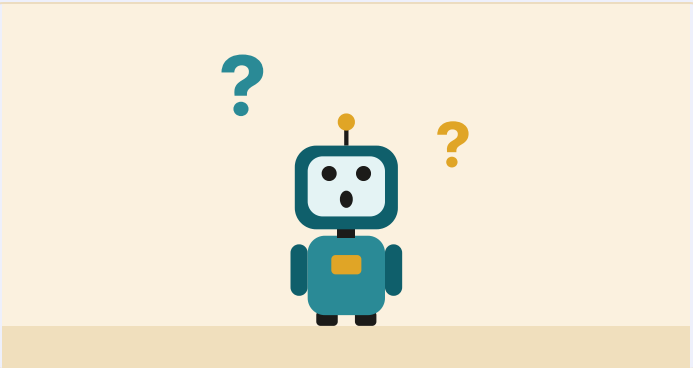
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

A solvent bank faces a sudden run and cannot meet withdrawals. The central bank lends it cash against good collateral. This role is best described as:

Answer: A. Lending to a sound bank in a liquidity crisis, to stop panic spreading, is the lender of last resort role. B is about holding the government's accounts and handling its payments. C is about being the only issuer of notes and coins.



EXAM QUESTION 2 OF 8

The reserve requirement is 5%. A customer deposits 2,000 of new cash. If banks lend out everything they can and all loans are redeposited, total deposits can rise to at most:

A. 2,100

B. 38,000

C. 40,000

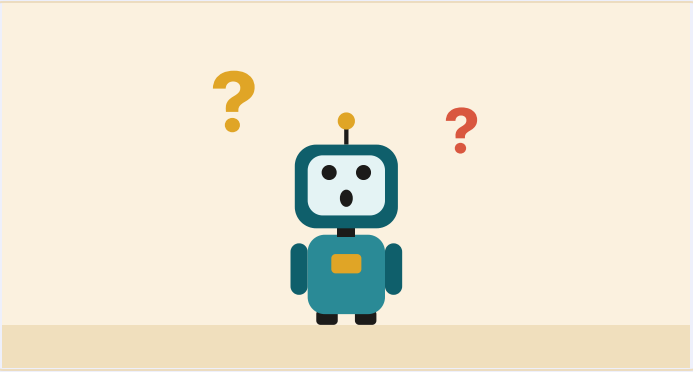
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So: Pick an answer, then turn the page.

ANSWER

The reserve requirement is 5%. A customer deposits 2,000 of new cash. If banks lend out everything they can and all loans are redeposited, total deposits can rise to at most:

Answer: C. Multiplier = 1 / 0.05 = 20, so total deposits = 20 x 2,000 = 40,000. B (38,000) is the new money created by lending, which leaves out the original deposit. A just adds 5% to the deposit and ignores the chain of loans.



EXAM QUESTION 3 OF 8

Investors require a real return of 1.5%, expect inflation of 3%, and want a 0.5% premium for inflation uncertainty. The nominal rate is closest to:

A. 4.5%

B. 5.0%

C. 3.5%

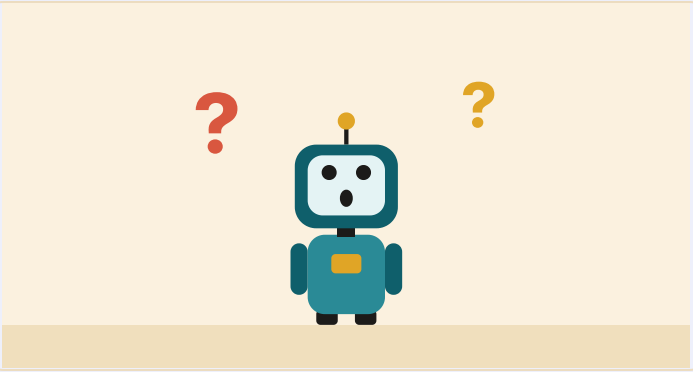
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So: Pick an answer, then turn the page.

ANSWER

Investors require a real return of 1.5%, expect inflation of 3%, and want a 0.5% premium for inflation uncertainty. The nominal rate is closest to:

Answer: B. Nominal = real + expected inflation + risk premium = 1.5 + 3 + 0.5 = 5.0%. A leaves out the risk premium. C leaves out the real return, as if lenders only wanted to stand still.



EXAM QUESTION 4 OF 8

Velocity is constant. The money supply grows 8% and real output grows 3%. Under the quantity theory of money, inflation is closest to:

A. 5%

B. 8%

C. 11%

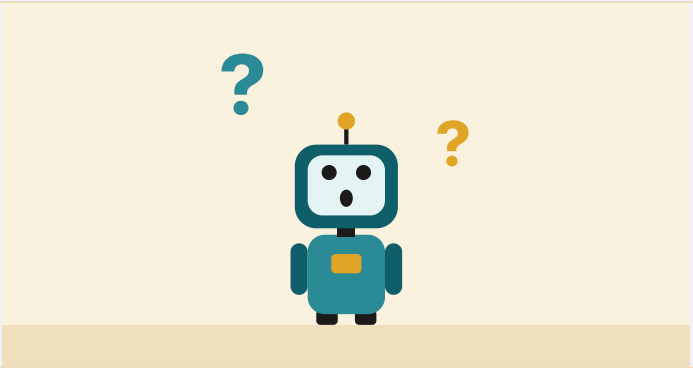
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

Velocity is constant. The money supply grows 8% and real output grows 3%. Under the quantity theory of money, inflation is closest to:

Answer: A. From %M + %V = %P + %Y (money growth + velocity growth = inflation + real output growth): 8 + 0 = %P + 3, so %P = 5%. B assumes real output is fixed. C adds output growth instead of subtracting it.



EXAM QUESTION 5 OF 8

Trend real growth is 2%, the inflation target is 2%, and the policy rate is 3.25%. The monetary policy stance is:

A. contractionary

B. neutral

C. expansionary

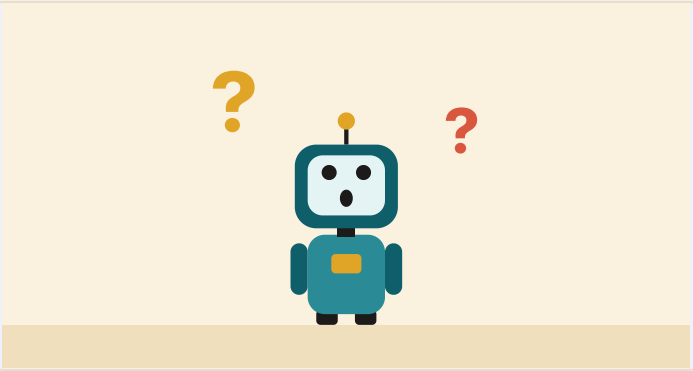
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

Trend real growth is 2%, the inflation target is 2%, and the policy rate is 3.25%. The monetary policy stance is:

Answer: C. Neutral rate = 2 + 2 = 4%. The policy rate of 3.25% is below it, so policy is expansionary. A would need a policy rate above 4%. B would need a policy rate of about 4%.



EXAM QUESTION 6 OF 8

A central bank buys government bonds from commercial banks in the open market. The most likely effect is:

A. bank reserves rise and policy is expansionary

B. bank reserves fall and policy is contractionary

C. no change in reserves, only in bond prices

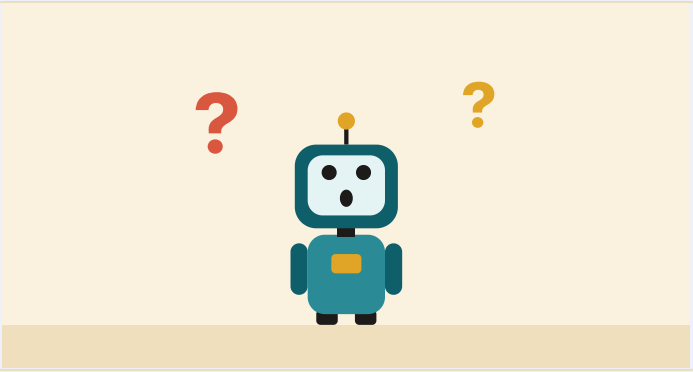
Answer on the next page.

So: **Pick an answer, then turn the page.**

ANSWER

A central bank buys government bonds from commercial banks in the open market. The most likely effect is:

Answer: A. The central bank pays for the bonds by crediting banks' reserve accounts, so reserves rise and banks can lend more: an easing move. B describes selling bonds. C is wrong because paying for the bonds always changes reserves.



EXAM QUESTION 7 OF 8

Interest rates are close to zero and households and banks are holding on to cash. A further increase in the money supply will most likely:

A. lower interest rates sharply

B. leave interest rates and spending largely unchanged

C. raise real interest rates by creating deflation

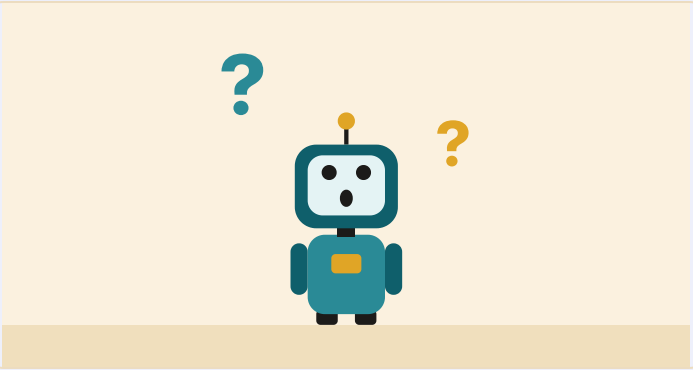
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

Interest rates are close to zero and households and banks are holding on to cash. A further increase in the money supply will most likely:

Answer: B. This is a liquidity trap: money demand is flat near zero rates, so extra money is simply held and rates barely move. A assumes the normal downward-sloping money demand. C is backwards: more money does not create deflation, although deflation itself does raise real rates.



EXAM QUESTION 8 OF 8

A central bank pegs its currency to the US dollar. The main thing it gives up is:

A. the need to hold foreign exchange reserves

B. any role for credibility

C. independent control of domestic interest rates

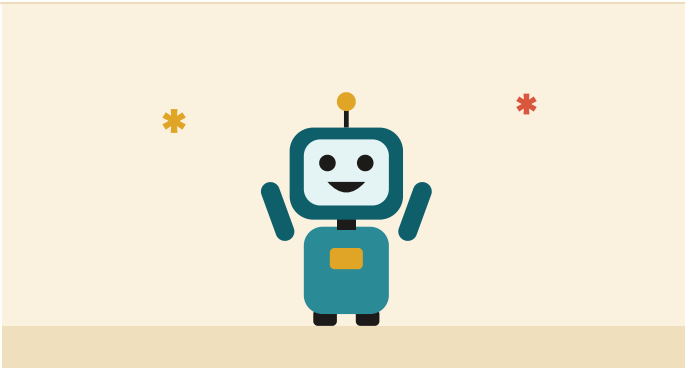
Answer on the next page.

So: **Pick an answer, then turn the page.**

ANSWER

A central bank pegs its currency to the US dollar. The main thing it gives up is:

Answer: C. To defend the peg, local interest rates must follow US rates, so independent monetary policy is lost. A is wrong: a peg needs large reserves to defend it. B is wrong: a peg works only if markets believe it.

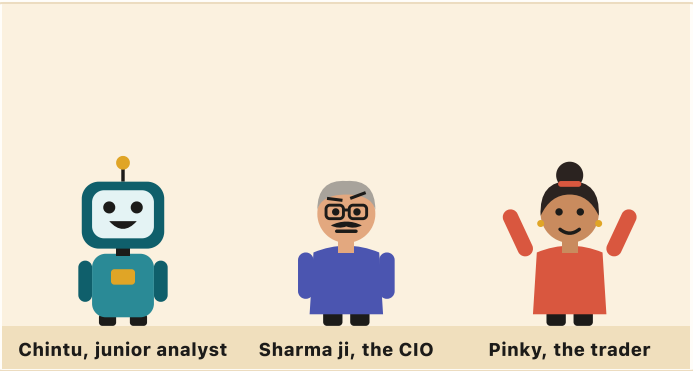


CHAPTER 4 DONE

Monetary policy, sorted

You can list the central bank's roles and goals, work the money multiplier, the quantity equation and the Fisher effect, explain the tools and the transmission chain, judge the stance against the neutral rate, and name the limits from liquidity traps to QE.

So: Swipe on, or jump to another chapter from the Menu.



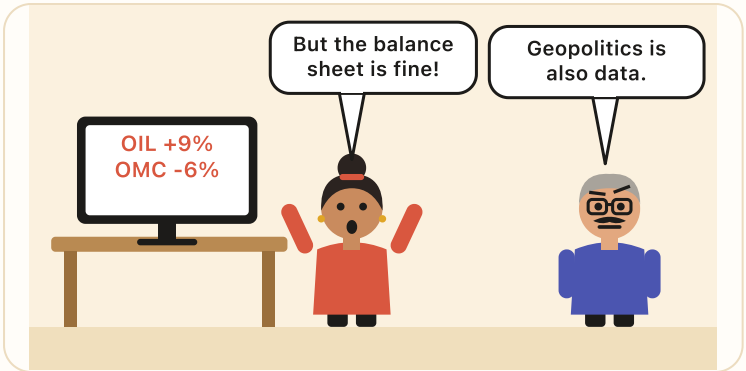
BEFORE YOU START

The cast, and words this chapter uses

A Mumbai investment firm. Sharma ji trusts nothing without data ("Data dikhao"); Pinky is sure of everything, especially when wrong.

- **Sovereignty:** a country's power to decide for itself.
- **Balance of payments problem:** too little foreign currency to pay for imports or debts.
- **Reserve currency:** a currency central banks hold as savings, mostly the US dollar.
- **Risk-off:** investors dump risky assets for safe ones.

So: **You can start this chapter cold. Everything it needs is on this card.**



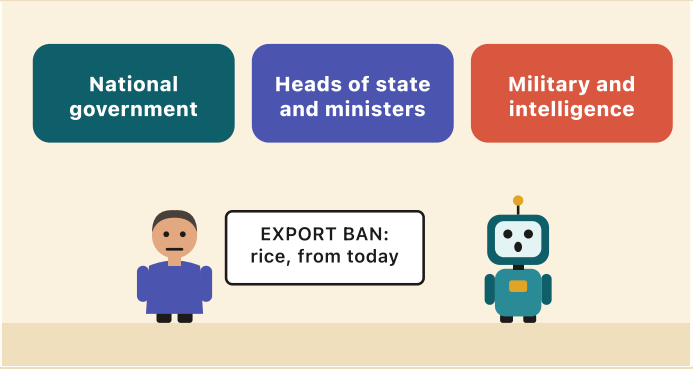
STORY

Monday, 9:02 am: the strait that moved the Nifty

Pinky's oil-marketing stocks fell 6% before her first cutting chai. Nothing in the companies changed. Overnight, two navies faced off near a narrow sea lane that carries much of the world's oil.

Geopolitics is the study of how geography, politics and power shape relations between countries. Elections, trade wars, sanctions and alliances change firms' costs, revenues and supply chains, and the return investors demand for holding risky assets.

So: Geopolitics reaches a valuation through cash flows, costs and the risk premium, so analysts must model it, not ignore it.



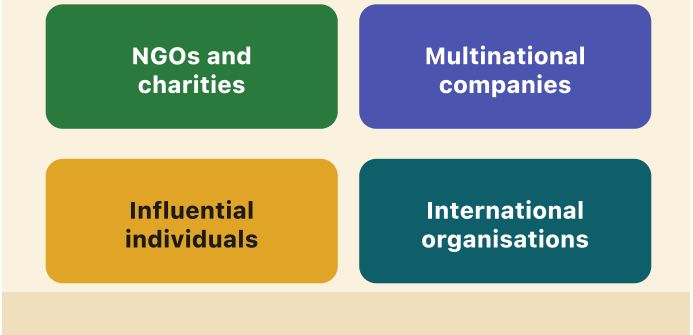
IDEA

State actors: the ones with armies and treasuries

A **political actor** is anyone who shapes how countries deal with each other. **State actors** are those with authority over a country's national security and resources: the national government, heads of state and ministers, and security bodies such as the military and intelligence services.

Example: a government bans rice exports to keep food cheap at home. Within hours, rice importers abroad and rice-trading stocks reprice.

So: State actors control borders, budgets, armies and resources, so their decisions move markets directly.



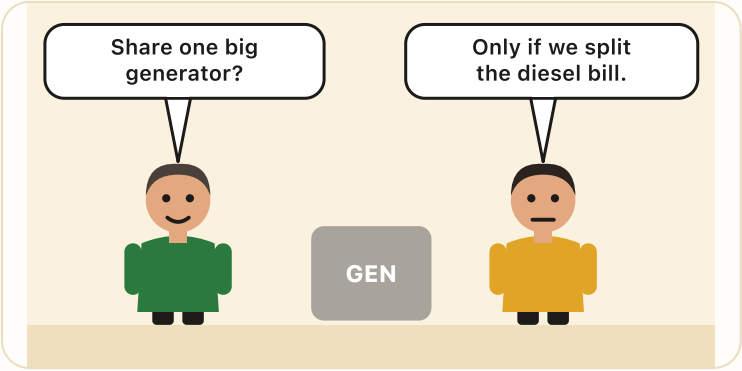
IDEA

Non-state actors: no army, plenty of influence

Non-state actors take part in global politics and economics but do not control a country's army or resources.

- **NGOs and charities:** a campaign against deforestation changes palm-oil rules.
- **Multinational companies:** a chipmaker whose factories every government wants.
- **Influential individuals:** a billionaire, a religious leader, a famous economist.
- **International organisations** such as the IMF, World Bank and WTO: set up by governments, but run by their own rules and staff.

So: Non-state actors cannot declare war, yet they shape rules, flows of money and public opinion across borders.



IDEA

Cooperate or go it alone? A cost-benefit call

A **cooperative** country joins agreements and shared rules with others: trade deals, defence pacts, common standards. A **non-cooperative** country acts alone and changes rules when it suits it.

It is a cost-benefit decision. Benefits: market access, security, cheaper inputs, a voice in writing the rules. Cost: some lost **sovereignty**, because every agreement limits what you may do alone.

It is not all or nothing: a country can cooperate on trade and refuse on defence.

So: Countries cooperate when the expected benefits beat the cost of giving up some freedom of action.

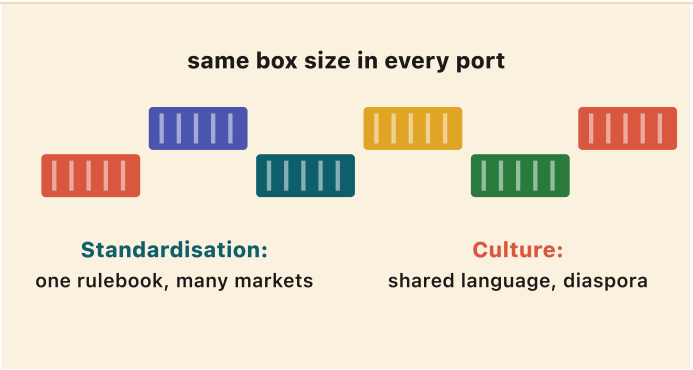


IDEA

Why cooperate? Security, growth and resources

- **National security:** a country with hostile neighbours or a small military seeks allies, shared intelligence and arms deals.
- **Economic growth:** exporters need open markets; a deal that cuts a partner's tariffs lifts sales and jobs at home.
- **Access to resources:** India imports most of its crude oil, so it values friendly suppliers and safe shipping lanes. Resource-rich countries cooperate too, to lock in reliable buyers.

So: What a country lacks, in safety, markets or raw materials, tells you why and with whom it cooperates.



IDEA

Why cooperate? Common rules and shared culture

Standardisation: agreeing on the same rules, regulations and processes makes cross-border business cheaper. Examples: shipping containers of standard sizes, common accounting standards (IFRS), bank capital rules (Basel), aviation safety rules. One rulebook lets a firm sell in many countries.

Cultural considerations: shared history, language, religion or a large diaspora make cooperation easier and more natural. Countries with millions of Indian-origin residents tend to have warm ties with India.

So: Common rules cut the cost of doing business across borders; common culture cuts the cost of trusting each other.



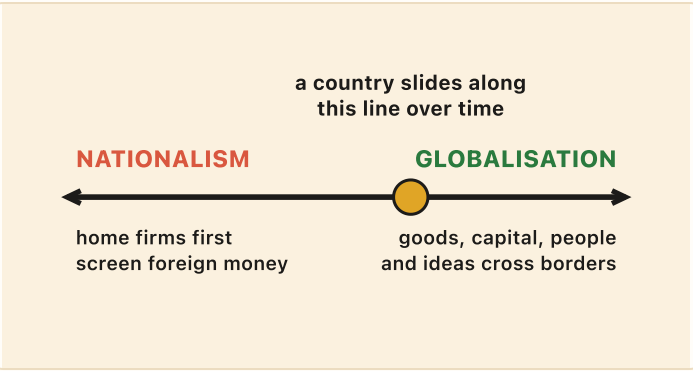
IDEA

Soft power: winning friends without a single tank

Hard power is coercion: military force or economic pressure. **Soft power** is influence through attraction: culture, values, education, media, food.

How it works: people who watch your films, study at your universities, do your yoga and love your biryani build goodwill and familiarity. That goodwill makes their governments more open to your trade deals, your companies and your positions in global forums. Examples: Hollywood and US universities, K-pop and Korean dramas, Bollywood and International Yoga Day.

So: Soft power is influence by attraction, not force; it builds the trust that makes cooperation easier.



IDEA

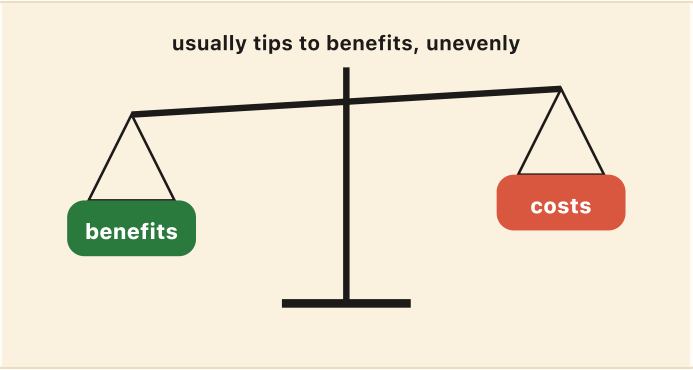
Globalisation versus nationalism

Globalisation is the process of growing interaction and integration between people, firms and governments worldwide: more cross-border flows of goods, services, capital, people and ideas.

Nationalism is the promotion of a country's own economic interests, including the freedom to decide without outside influence.

It is a spectrum, not a switch. Signs of a move towards nationalism: higher tariffs, "buy local" rules, tighter visa and foreign-investment screening.

So: Globalisation and nationalism are two ends of one line, and countries move along it as politics changes.

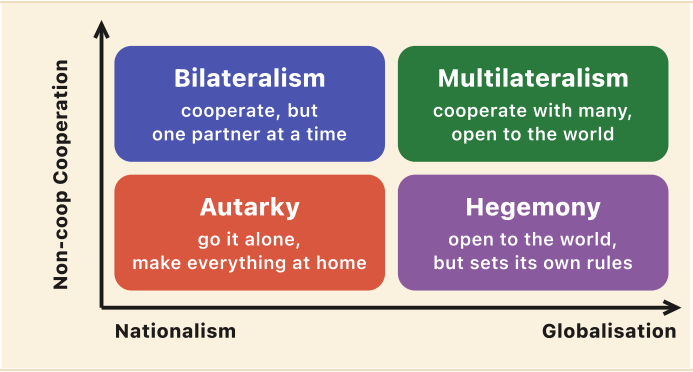


AT A GLANCE

Globalisation: the benefits and the bill

Benefits for a country	Costs for a country
Bigger markets, higher profits for firms	Gains spread unevenly; some regions and workers lose
Cheaper inputs, labour and resources	Dependence: a shock abroad halts production at home
More choice and lower prices for consumers	Local industries and skills may wither
Ideas, technology and cultural exchange	Backlash and political instability

So: Globalisation usually raises total income, but the losers are concentrated and loud, which fuels nationalism.



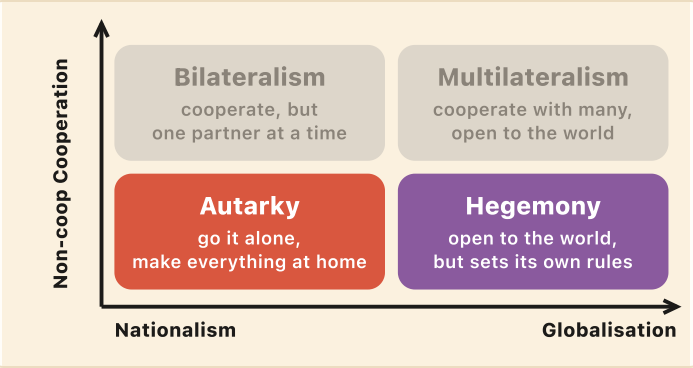
PICTURE IT

The four archetypes on one map

Put the two choices together. Vertical: does the country cooperate with others? Horizontal: is its economy nationalist or globalised?

- **Autarky:** non-cooperative and nationalist.
- **Hegemony:** non-cooperative but globalised.
- **Bilateralism:** cooperative but nationalist leaning.
- **Multilateralism:** cooperative and globalised.

So: **Learn the map as two yes/no questions: cooperate? globalise? Each answer pair gives one archetype.**



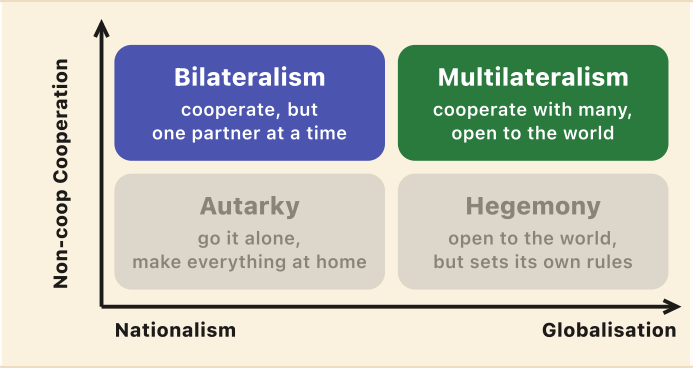
IDEA

Autarky and hegemony, in plain words

Autarky means self-sufficiency: the country tries to make what it needs at home, trades little and keeps the economy under tight state control. Example: North Korea. Kitchen version: the family that grows its own vegetables and never eats outside.

Hegemony: a dominant power is open to global trade and capital but does not cooperate as an equal. It uses its size to set the rules and push others along. Example: Britain in the 19th century. Kitchen version: the mall owner who trades with every shop but writes every lease alone.

So: Autarky shuts the door; hegemony keeps the door wide open but owns the building.



IDEA

Bilateralism, multilateralism and regionalism

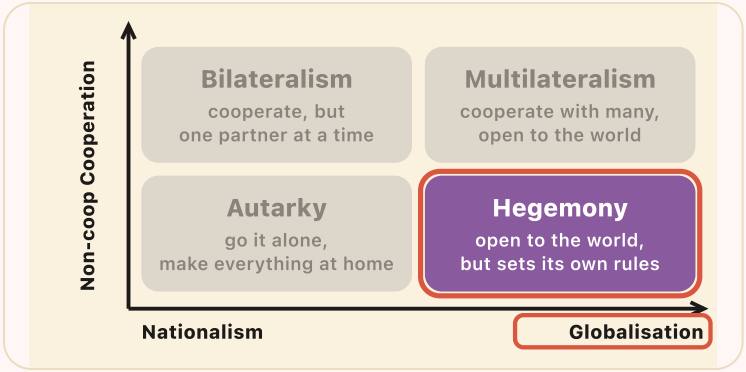
Bilateralism: cooperation between two countries at a time. Example: the 2022 India and UAE trade deal.

Multilateralism: many countries cooperate under shared rules that apply to all. Example: the 160-plus members of the WTO.

Regionalism, a form of multilateralism: a group of neighbours cooperate with each other, not with outsiders. Examples: the EU, ASEAN.

Society version: bilateral is two flats sharing a car pool; multilateral is the whole society agreeing on parking rules.

So: Bilateral is one partner at a time; multilateral is many under one rulebook; regional is multilateral within a club.

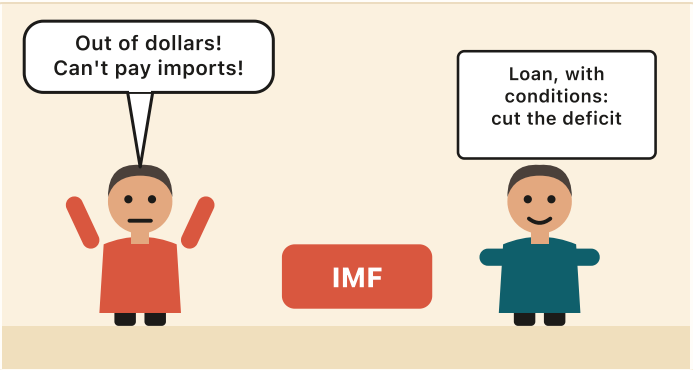


EXAM TRAP

Archetype traps examiners set

- **Hegemony is globalised**, not nationalist. Its "non" is about cooperation, not openness.
- **Bilateralism sits on the nationalist side:** it cooperates, but one partner at a time, on its own terms.
- **Regionalism is not a fifth box.** It is multilateralism inside a group.
- **The axes are separate:** cooperation is policy towards others; globalisation is how open the economy is.
- **Archetypes change.** An election can slide a country from multilateral to bilateral.

So: Ask the two questions separately every time: cooperative or not, globalised or not.



IDEA

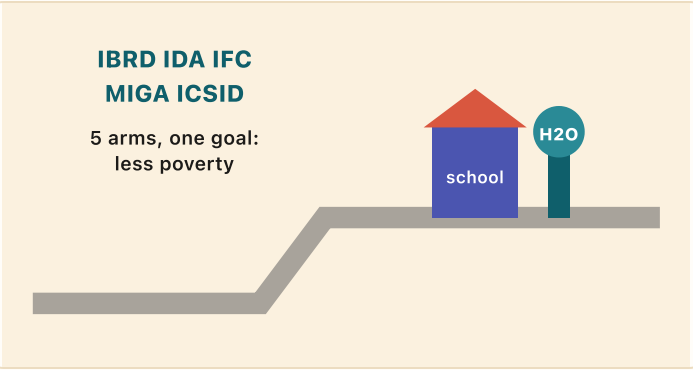
The IMF: the fire brigade for currencies

The **International Monetary Fund** works to keep the international monetary system stable. Three jobs:

- **Surveillance:** it checks every member's economy and policies, usually once a year.
- **Lending:** loans to countries with balance of payments problems, usually with conditions such as cutting the budget deficit.
- **Capacity building:** technical help on tax, central banking and statistics.

Example: India borrowed from the IMF in its 1991 balance of payments crisis.

So: **IMF = short-run stability: watch members, lend when they run out of foreign currency, help them fix policy.**



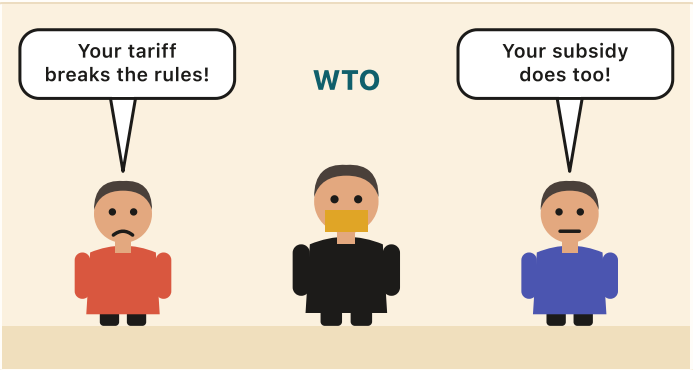
IDEA

The World Bank Group: the long-haul builder

The **World Bank Group** finances long-term development to cut poverty: roads, power, schools, health, water. It has five arms:

- **IBRD:** loans to middle-income governments.
- **IDA:** grants and very cheap loans to the poorest countries.
- **IFC:** invests in private companies.
- **MIGA:** insures foreign investors against political risk.
- **ICSID:** settles disputes between investors and governments.

So: World Bank = long-term development projects; IMF = short-term balance of payments help.



IDEA

The WTO: the referee of world trade

The **World Trade Organization**, set up in 1995 to replace the GATT, runs the global rules of trade between nations. Its jobs:

- a forum to negotiate lower trade barriers;
- settling trade disputes between members;
- reviewing members' trade policies.

Its core principle is **non-discrimination**: a tariff cut offered to one member is offered to all (most-favoured-nation), and imports are treated like home goods once they clear customs (national treatment).

So: WTO = rules and referee for trade between countries; it does not lend money.



EXAM TRAP

IMF, World Bank, WTO: do not mix them up

	IMF	World Bank	WTO
Aim	Financial stability	Cut poverty	Rules-based trade
Horizon	Short run	Long run	Ongoing
Lends?	Yes, for payments crises	Yes, for projects	No

Wording trick: "funds a dam" points to the World Bank; "short of foreign reserves" points to the IMF.

So: **Match the problem to the body: reserves crisis to IMF, poverty project to World Bank, trade dispute to WTO.**



PICTURE IT

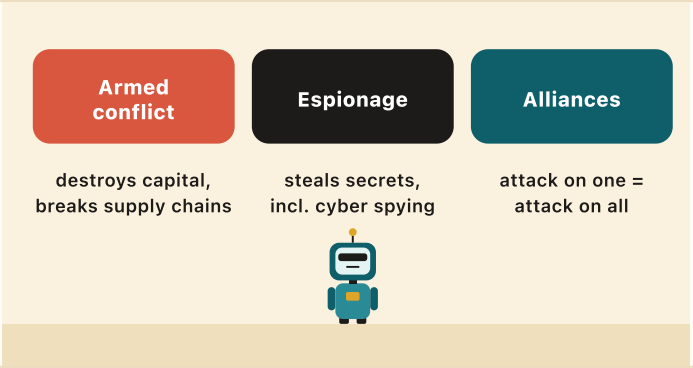
The geopolitical toolbox

Countries chase their aims with three kinds of tools:

- **National security tools:** armed conflict, espionage, alliances. The most extreme, especially conflict.
- **Economic tools:** cooperative ones such as trade agreements, and non-cooperative ones such as tariffs, quotas and nationalisation.
- **Financial tools:** currency reserves and exchange, limits on foreign investment, sanctions, access to payment systems.

Each tool can be used to cooperate with friends or to pressure rivals.

So: Three boxes: security, economic, financial. Know which tool sits in which box.

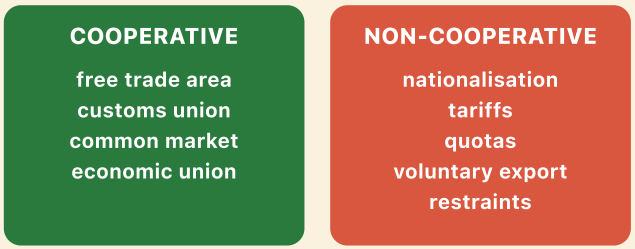


IDEA

National security tools: conflict, spies, alliances

- **Armed conflict:** the most extreme tool. It destroys buildings, factories and lives, displaces people, and breaks supply chains far beyond the war zone, so prices of energy and food can jump worldwide.
- **Espionage:** secretly gathering military, political or business information, today often by cyber attack.
- **Alliances:** military cooperation and mutual defence. NATO members treat an armed attack on one as an attack on all.

So: Conflict is the most damaging tool for markets; alliances are the cooperative face of national security.

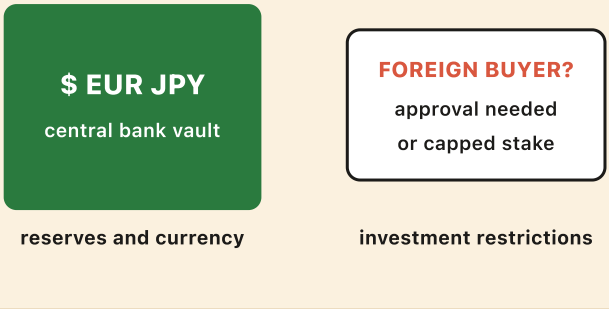


AT A GLANCE

Economic tools: friendly and unfriendly

Tool	What it does
Free trade area, customs union	No tariffs inside; a customs union adds one outside tariff
Common market, economic union	Adds free movement of workers and capital; then shared policy
Nationalisation	State takes over private, often foreign, firms
Tariffs, quotas	Tax imports; cap import volumes
Voluntary export restraint	An exporter "agrees" to cap its exports, under pressure

So: **Cooperative economic tools join markets together; non-cooperative ones protect home firms or punish rivals.**



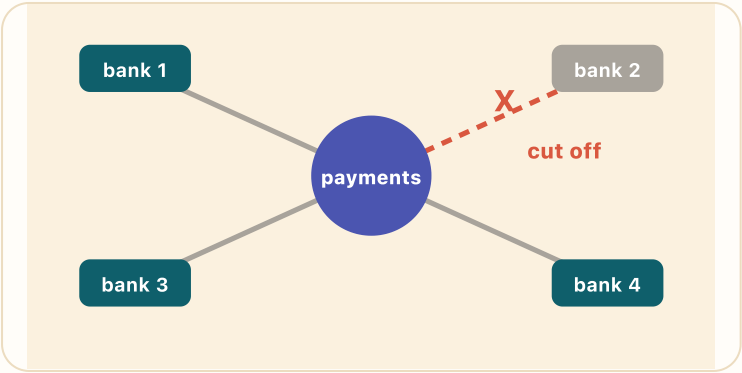
IDEA

Financial tools: reserves, currency and investment limits

Currency reserves and exchange: a country whose currency others hold as reserves, like the US with the dollar, gains influence over global finance. Countries also build reserves, offer currency swap lines to friends, or push trade in their own currency, as India does with rupee settlement.

Investment restrictions: limits on who may buy home companies or assets. Since 2020, India requires government approval for investment from countries that share a land border with it.

So: Whoever controls the money pipes and the purchase approvals holds real geopolitical power.



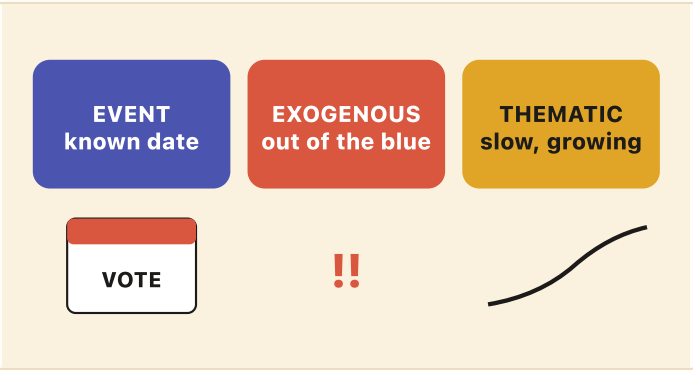
IDEA

Financial tools: sanctions and payment systems

Sanctions restrict dealings with named countries, firms or people: asset freezes, bans on buying their bonds or selling them technology.

Access to payment systems: cross-border payments run through shared networks such as SWIFT, the messaging system banks use to instruct transfers. Cutting a country's banks off makes it hard for them to pay for imports or receive export earnings. Several Russian banks were removed from SWIFT in 2022.

So: Sanctions and payment cut-offs are non-cooperative financial tools that bite without a shot fired.

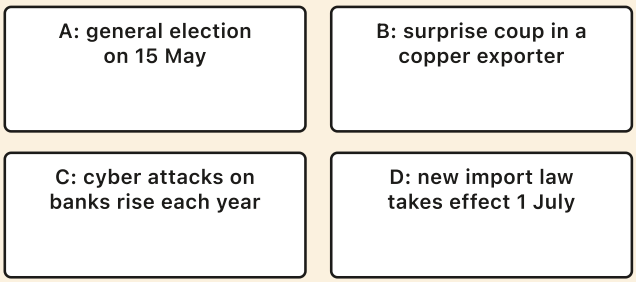


AT A GLANCE

Three types of geopolitical risk

Type	What it is	Examples
Event	Built around known dates, so the timing is known, the outcome is not	Elections, budget day, new laws taking effect, summits
Exogenous	Sudden, unanticipated shocks	Invasion, coup, uprising, attack on infrastructure
Thematic	Known risks that grow and change over time	Climate change, cyber threats, migration, big-power rivalry

So: **Event: you know when. Exogenous: you do not see it coming. Thematic: you see it coming for years.**



WORKED EXAMPLE

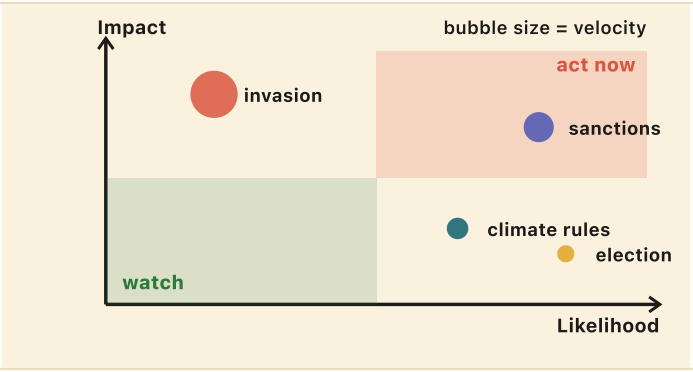
Sort these four risks

Chintu must label four risks for Sharma ji's weekly note.

1. A: the date is known, the result is not. **Event risk.**
2. B: nobody saw it coming. **Exogenous risk.**
3. C: known for years, growing slowly. **Thematic risk.**
4. D: a law with a set start date. **Event risk.**

A and D: event. B: exogenous. C: thematic.

So: **Ask first: is there a date on the calendar? Yes means event; no and sudden means exogenous; no and slow means thematic.**



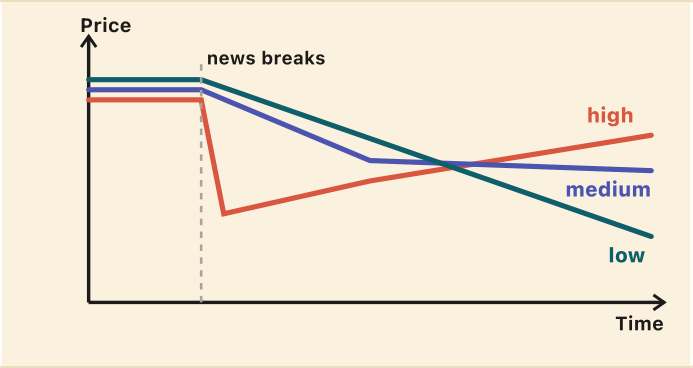
PICTURE IT

Assessing a risk: likelihood, velocity, impact

For each risk, score three things:

- **Likelihood:** how probable is it? Hard for exogenous risks. Clues: are the actors cooperative, is there a history of conflict, are elections near?
- **Velocity:** how fast does it hit markets? High, medium or low.
- **Size and nature of impact:** how big, and is it discrete (one country, sector or firm) or broad (many markets at once)?

So: A risk deserves attention for its impact and speed, not only its probability.

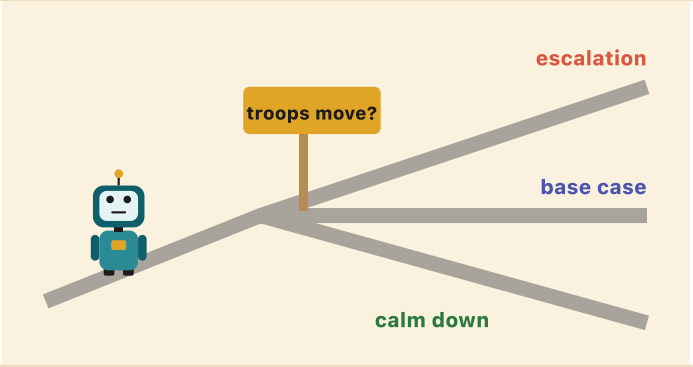


PICTURE IT

Velocity: how fast the risk hits

- **High velocity:** markets react within hours, typically to exogenous shocks. Volatility spikes; prices often recover as the dust settles.
- **Medium velocity:** the damage arrives over weeks or months, through firms' costs and supply chains, for example sanctions that raise input prices.
- **Low velocity:** slow-burn, long-term risks such as climate rules or ageing populations. Easy to ignore today, which is why they can be badly underpriced.

So: Fast risks hit volatility; slow risks hit long-term value, and investors tend to underweight the slow ones.



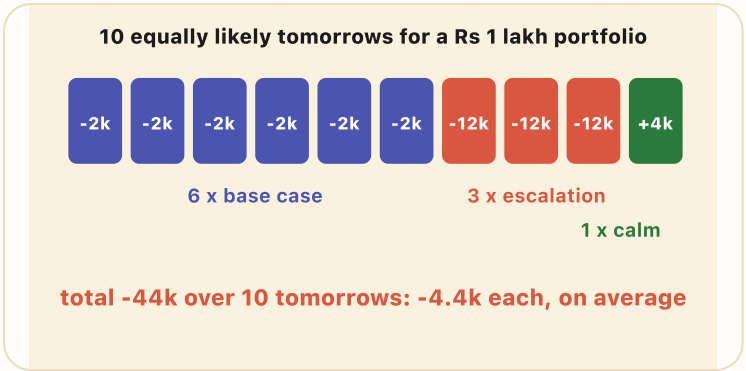
IDEA

Scenario analysis and signposts

Scenario analysis builds a few plausible stories (base case, escalation, calm-down), gives each a probability, and estimates what each does to the portfolio.

Signposts are the indicators you watch to see which story is unfolding: troop movements, sanctions announcements, shipping insurance rates, polls, diplomatic visits. Think of the monsoon reaching Kerala: one signpost that tells farmers and FMCG analysts which rainfall story to believe.

So: Scenarios say what could happen; signposts tell you which one is happening, so you adjust before the crowd.



PLAYED OUT IN MONEY

Ten possible tomorrows

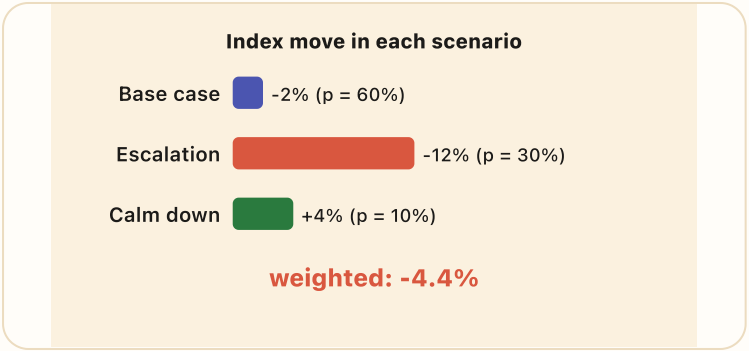
Chintu holds Rs 1 lakh of Indian shares during a border stand-off. Imagine the next month played out **10 times**:

- 6 times the talks drag on (base case): the index falls 2%, he loses Rs 2,000 each time: **-12,000**.
- 3 times it escalates: the index falls 12%, he loses Rs 12,000 each time: **-36,000**.
- 1 time it calms down: the index rises 4%: **+4,000**.

All 10 together: -12,000 - 36,000 + 4,000 = -44,000. Per tomorrow: -44,000 ÷ 10 = **-Rs 4,400**, a 4.4% loss on average.

That average is the **expected** (probability-weighted) move. "6 out of 10" is just a 60% probability.

So: An expected value is the average result if the same situation played out many times.



WORKED EXAMPLE

Probability-weighting three scenarios

What we know

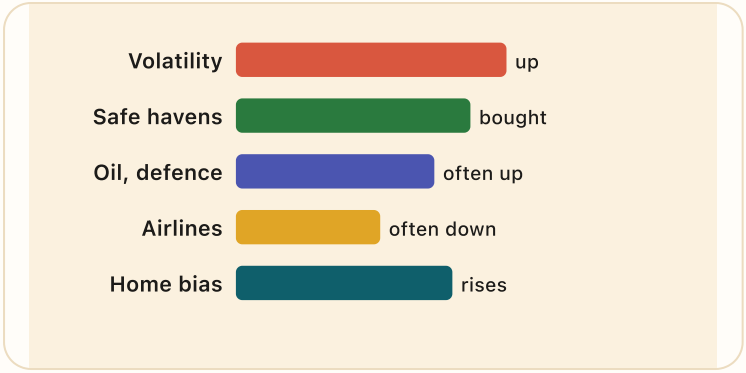
- **p** = 60%, 30%, 10%: how likely each scenario is (they add to 100%)
- **r** = -2%, -12%, +4%: index move if that scenario happens

What we want: the expected index move: add up p x r.

1. Base: 0.60 x (-2%) = -1.2%
   Turn 60% into 0.60, then scale the move by how likely it is.
2. Escalation: 0.30 x (-12%) = -3.6%
   A big fall, but it happens only 3 times in 10.
3. Calm down: 0.10 x (+4%) = +0.4%
   A gain, but rare.

Add them: -1.2 -3.6 +0.4 = -4.4%. Expected index move = -4.4%

So: **The expected loss is small next to the 12% tail, so also check whether the portfolio can survive the bad scenario.**

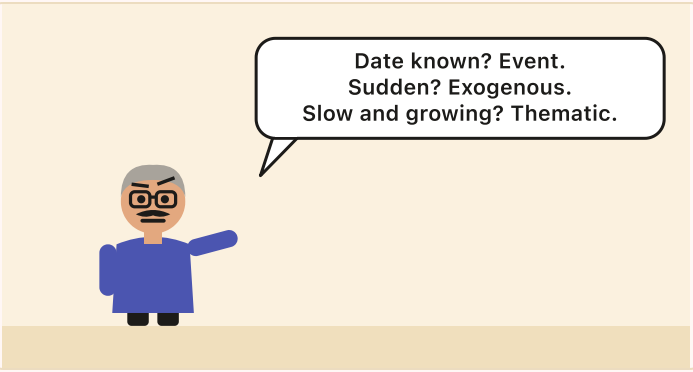


IDEA

What geopolitical risk does to a portfolio

- **Volatility** (price swings) rises, and correlations between risky assets often jump, so diversification helps less exactly when you want it.
- **Asset classes:** risk-off selling of equities; buying of safe havens such as US Treasuries, gold, the Swiss franc and the yen.
- **Sectors:** an oil shock can lift energy and defence stocks and hurt airlines, paint makers and importers.
- **Home bias:** investors retreat to markets they know, which cuts global diversification.

So: Geopolitical risk shows up as higher volatility, flights to safety, sector winners and losers, and more home bias.

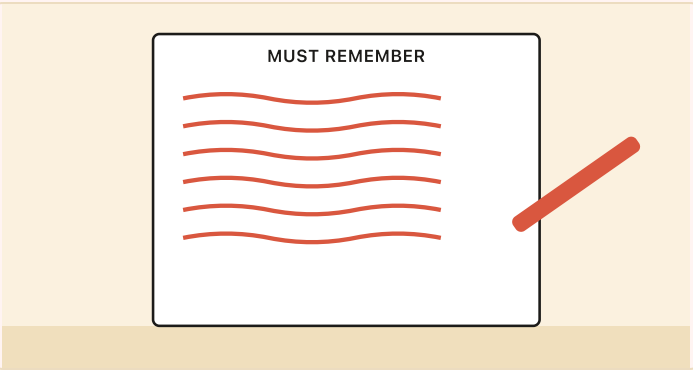


EXAM TRAP

Geopolitical risk traps

- **An election is event risk**, not exogenous: the date is known even if the winner is not.
- **Thematic risks are known**, not unknown. They are dangerous because they creep.
- **Likelihood is not enough.** A low-probability, high-impact risk can matter more than a likely small one.
- **High velocity is often short-lived.** Panicked sell-offs can reverse; low-velocity risks can do more lasting damage.
- **Impact has two parts:** size, and whether it is discrete or broad.

So: **Read the wording: known date, sudden shock, or slow trend. Then score likelihood, velocity and impact separately.**

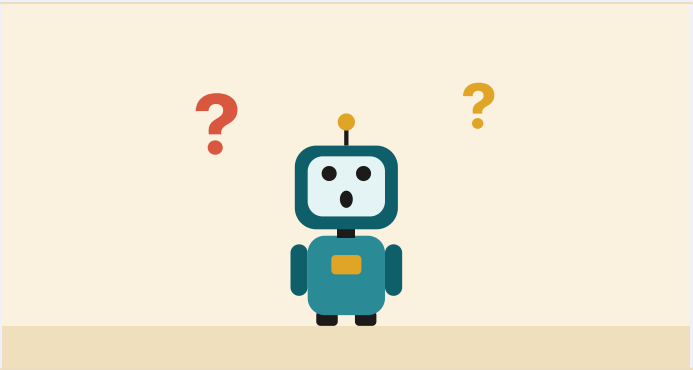


SHARMA JI'S RED PEN

6 things to remember

1. State actors control security and resources; non-state actors only influence.
2. Cooperation motives: security, growth, resources, standards, culture.
3. Autarky, hegemony (non-coop); bilateral, multilateral (coop). Left side nationalist, right side global.
4. IMF: payments crises. World Bank: development. WTO: trade rules, no loans.
5. Risks: event (known date), exogenous (sudden), thematic (slow).
6. Assess likelihood, velocity, impact; track scenarios with signposts.

So: Scribbled by Sharma ji on the geopolitics chapter.



EXAM QUESTION 1 OF 8

A country is deeply open to global trade and capital flows, but it does not cooperate as an equal and uses its economic weight to set the rules for others. This archetype is best described as:

A. autarky

B. hegemony

C. multilateralism

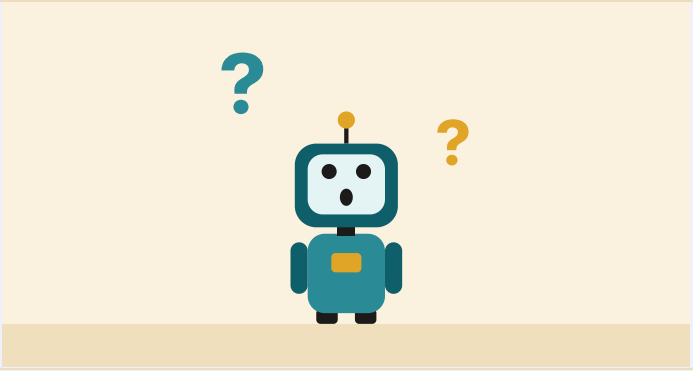
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

A country is deeply open to global trade and capital flows, but it does not cooperate as an equal and uses its economic weight to set the rules for others. This archetype is best described as:

Answer: B. Hegemony combines non-cooperation with globalisation: open to the world, but setting rules on its own terms. Autarky is non-cooperative and nationalist, with little trade, so A fails on openness. Multilateralism is cooperative, so C fails on cooperation.



EXAM QUESTION 2 OF 8

Which of the following is most likely a non-state actor?

A. A country's defence ministry

B. A country's head of government

C. A global charity that campaigns on refugee policy

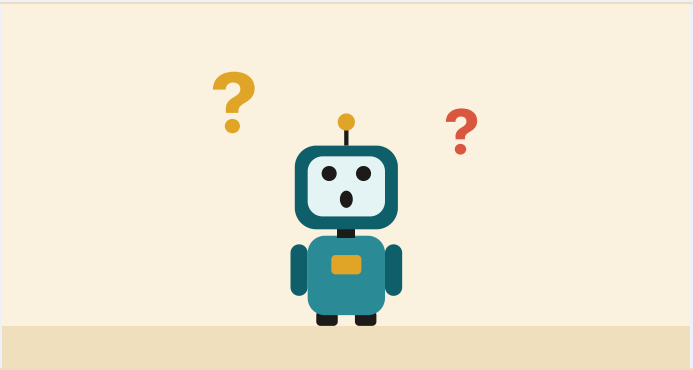
Answer on the next page.

So: **Pick an answer, then turn the page.**

ANSWER

Which of the following is most likely a non-state actor?

Answer: C. Non-state actors influence global affairs without controlling a country's security or resources; a charity fits. A defence ministry and a head of government both have authority over national security and resources, so A and B are state actors.



EXAM QUESTION 3 OF 8

A government is running out of foreign currency and cannot pay for its imports. The international organisation it is most likely to approach for short-term financing is the:

A. International Monetary Fund

B. World Bank

C. World Trade Organization

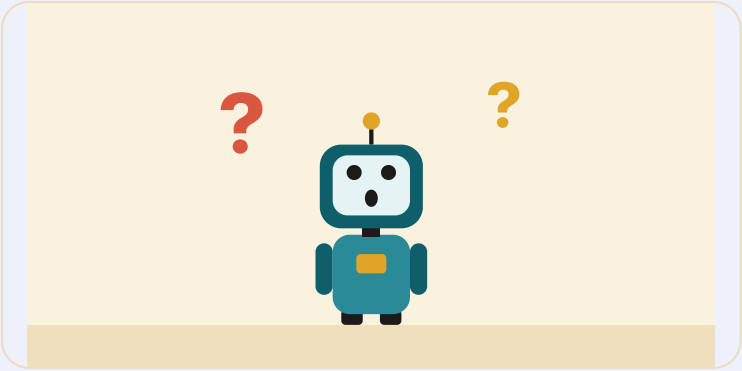
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

A government is running out of foreign currency and cannot pay for its imports. The international organisation it is most likely to approach for short-term financing is the:

Answer: A. The IMF lends to members facing balance of payments problems, usually with policy conditions. The World Bank finances long-term development projects, not short-term reserve shortages, so B is wrong. The WTO sets trade rules and settles disputes; it does not lend, so C is wrong.



EXAM QUESTION 4 OF 8

Without any warning, the army of a large oil exporter seizes power overnight. For an investor, this is best classified as:

A. event risk

B. thematic risk

C. exogenous risk

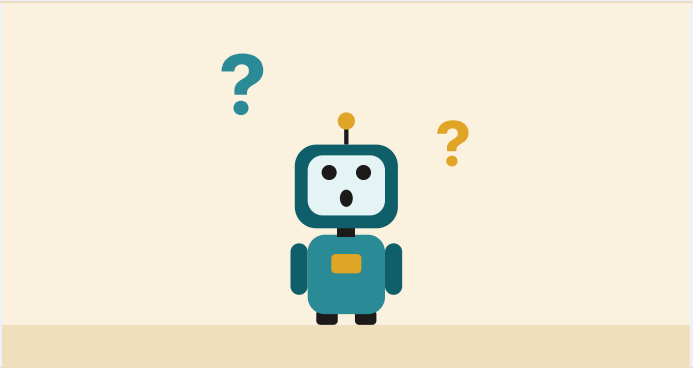
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

Without any warning, the army of a large oil exporter seizes power overnight. For an investor, this is best classified as:

Answer: C. A sudden, unanticipated shock is exogenous risk. Event risk revolves around known dates such as elections, so A is wrong. Thematic risk is a known risk that grows over time, such as climate change, so B is wrong.



EXAM QUESTION 5 OF 8

A country removes a rival's banks from the messaging network banks use for cross-border payments. This is an example of a:

A. national security tool

B. financial tool

C. cooperative economic tool

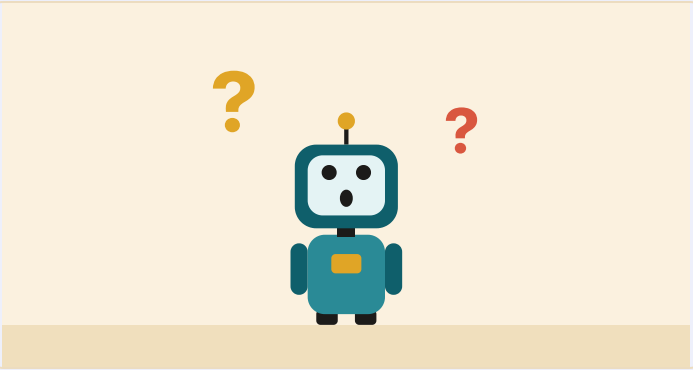
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

A country removes a rival's banks from the messaging network banks use for cross-border payments. This is an example of a:

Answer: B. Restricting access to payment systems is a financial tool, alongside sanctions and investment restrictions. National security tools are armed conflict, espionage and alliances, so A is wrong. Cooperative economic tools join markets, such as free trade areas, so C is wrong.



EXAM QUESTION 6 OF 8

Which risk is most likely described as low velocity?

A. Gradual climate regulation that slowly erodes the value of coal-heavy companies

B. A surprise invasion that triggers an immediate global sell-off

C. New sanctions that push up a manufacturer's input costs over the next quarter

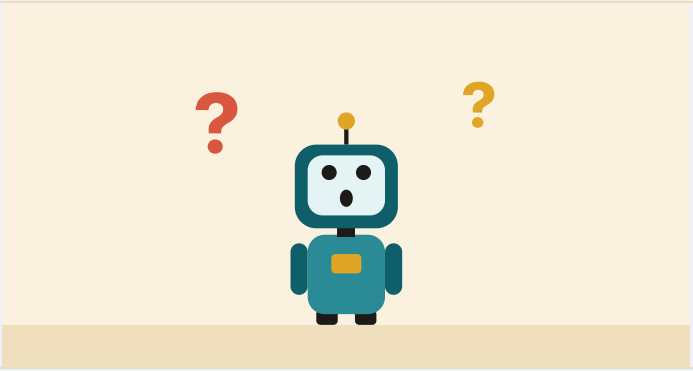
Answer on the next page.

So: **Pick an answer, then turn the page.**

ANSWER

Which risk is most likely described as low velocity?

Answer: A. Low-velocity risks play out over years and affect long-term value; slow climate rules fit. An immediate sell-off is high velocity, so B is wrong. Costs rising over a quarter through supply chains is medium velocity, so C is wrong.



EXAM QUESTION 7 OF 8

An analyst models a trade dispute with three scenarios for an equity index: deal reached, probability 50%, return +1%; escalation, probability 40%, return -10%; full resolution, probability 10%, return +5%. The probability-weighted index return is closest to:

A. -1.3%

B. -4.0%

C. -3.0%

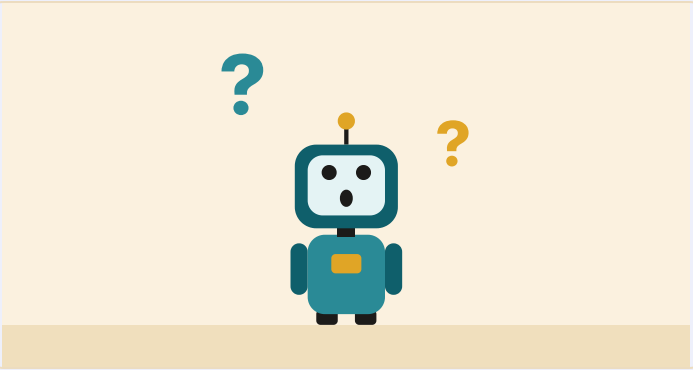
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

An analyst models a trade dispute with three scenarios for an equity index: deal reached, probability 50%, return +1%; escalation, probability 40%, return -10%; full resolution, probability 10%, return +5%. The probability-weighted index return is closest to:

Answer: C. 0.5 x 1% + 0.4 x (-10%) + 0.1 x 5% = 0.5% - 4.0% + 0.5% = -3.0%. A is the simple average of the three returns, which ignores probabilities. B uses only the escalation term and forgets the other two scenarios.



EXAM QUESTION 8 OF 8

Soft power is best described as a country's ability to influence others through:

A. the appeal of its culture, values and institutions

B. trade sanctions and tariffs

C. military alliances and mutual defence

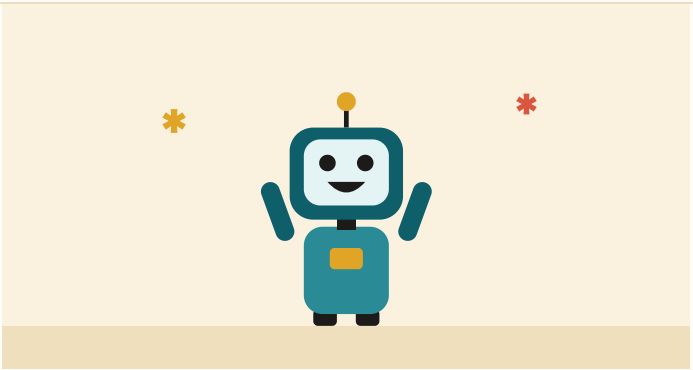
Answer on the next page.

So: **Pick an answer, then turn the page.**

ANSWER

Soft power is best described as a country's ability to influence others through:

Answer: A. Soft power works through attraction: films, universities, food, values. Sanctions and tariffs are coercive economic tools, so B is hard power. Alliances are national security tools, so C is wrong.



CHAPTER 5 DONE

Geopolitics, sorted

You can name the actors, explain why countries cooperate, place any country on the archetype map, match the IMF, World Bank and WTO to their jobs, sort the tools into three boxes, and assess a geopolitical risk by type, likelihood, velocity and impact. Sharma ji's verdict: "Data dikhaya. Accha hai."

So: Swipe on, or jump to another chapter from the Menu.



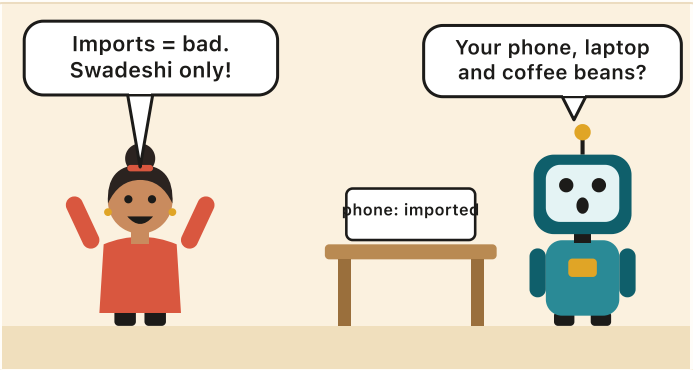
BEFORE YOU START

The cast, and words this chapter uses

A Mumbai investment firm. Sharma ji trusts nothing without data ("Data dikhao"); Pinky is sure of everything, especially when wrong.

- **Consumer surplus:** what buyers would pay minus what they do pay.
- **Producer surplus:** the price sellers get minus their cost to make it.
- **Deadweight loss:** value lost to everyone, gained by no one.
- **World price:** the price a good trades at on world markets.
- **Terms of trade:** export prices relative to import prices.

So: **You can start this chapter cold. Everything it needs is on this card.**



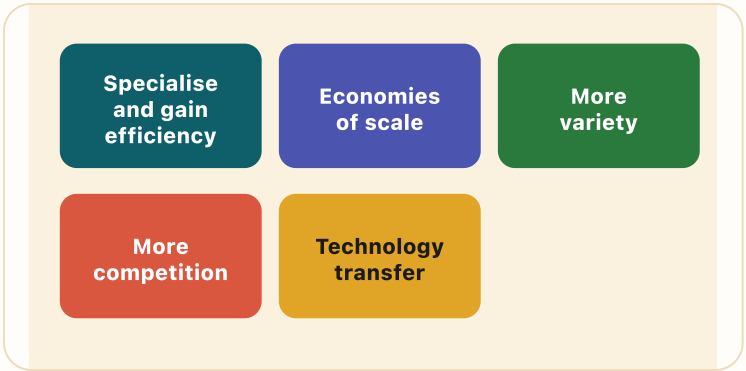
STORY

Pinky's "Buy Indian only" week

Pinky announces a week of buying only Indian goods. By Tuesday she has no phone charger, her laptop needs a part from Taiwan, and her cold coffee is suddenly half the size.

Sharma ji: "Trade is not charity, Pinky. Every country buys what others make cheaper and sells what it makes cheaper. Both sides gain. But not everyone inside a country gains, and that is where the politics starts."

So: Trade raises total income for both countries, but the gains and losses fall on different people.

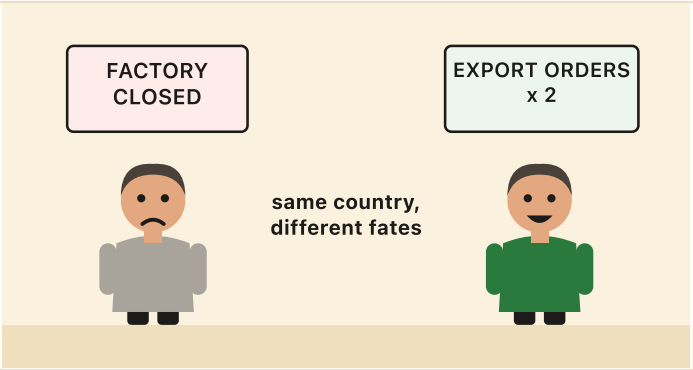


IDEA

Why countries trade: five gains

- **Specialisation and efficiency:** resources move to what each country does relatively best.
- **Economies of scale:** selling to the world, a firm produces more and its cost per unit falls.
- **Variety:** consumers get goods home firms do not make.
- **Competition:** imports keep home firms honest on price and quality.
- **Technology transfer:** foreign firms and machines bring know-how, as in India's phone assembly.

So: Trade lifts output and consumption by moving resources to their most efficient use.



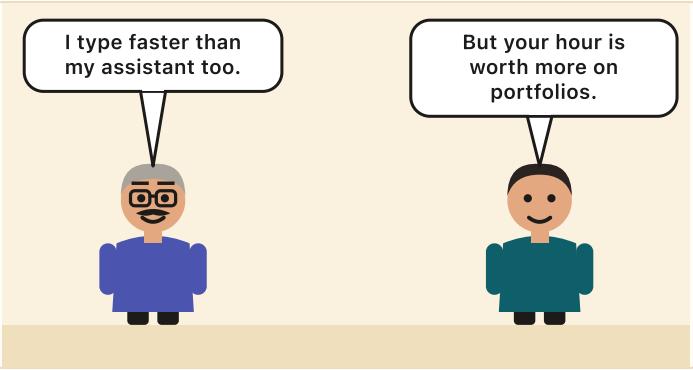
IDEA

Who pays for trade

- **Job losses in import-competing industries:** a toy maker undercut by cheaper imports cuts staff.
- **Inequality:** gains go to exporters, their workers and consumers; losses fall on specific towns and skills, so inequality can rise.
- **Adjustment costs:** moving workers and machines to new industries takes time, retraining and money. A 50-year-old weaver does not become a software tester overnight.

Economists' view: total gains exceed total losses, so winners could compensate losers, but often do not.

So: Gains from trade are broad and spread thin; losses are narrow and concentrated, which drives protectionism.



IDEA

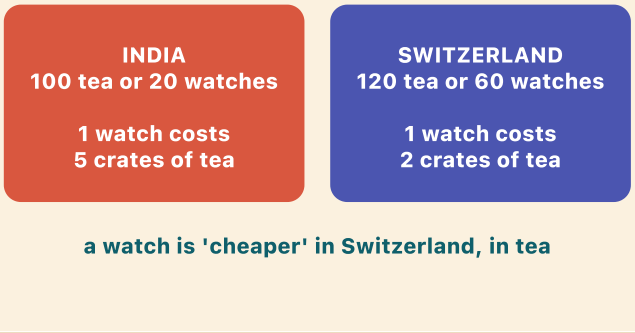
Absolute versus comparative advantage

Absolute advantage: a country makes a good with fewer resources, or more output per worker, than another country.

Comparative advantage: a country makes a good at a lower **opportunity cost**, meaning it gives up less of other goods to make it.

Sharma ji types faster than his assistant (absolute advantage in typing), yet he should still hand over the typing: every hour he types is an hour of portfolio decisions lost.

So: **Trade patterns follow comparative advantage, not absolute advantage. Even a country worse at everything gains from trade.**



STORY

What a watch costs India: tea, not rupees

With all its workers, India can make **100 crates of tea** a day, or **20 watches**, or a mix. Making all 20 watches means giving up all 100 crates, so each watch costs India **5 crates of tea**.

Switzerland can make 120 tea or 60 watches. Each watch costs it only **2 crates**.

That "what you give up" price is the **opportunity cost**. Flip it round: a crate of tea costs India 0.2 of a watch, but costs Switzerland 0.5 of a watch. So Switzerland makes watches more cheaply, and India makes tea more cheaply.

So: The real cost of making something is what you stop making to free up the workers.

$$\text{OC of 1 watch} = \frac{\text{tea given up}}{\text{watches gained}}$$

the lowest OC wins the comparative advantage

FORMULA

Opportunity cost of a good

```
OC of X = Y given up / X gained
```

Symbol	In plain words	Example
X	the good you make more of	watches
Y	the other good, given up to make X	tea
OC of X	opportunity cost: Y lost per extra X	India: 5 tea

Shortcut: moving every worker from tea to watches swaps 100 tea for 20 watches, so each watch costs 100 / 20 = 5 tea. OC of Y = 1 / OC of X: a crate of tea costs India 0.2 watch. The **lower** OC wins the comparative advantage.

So: Compute the opportunity cost of each good in each country; the lower one marks the comparative advantage.



WORKED EXAMPLE

Who should make what?

What we know

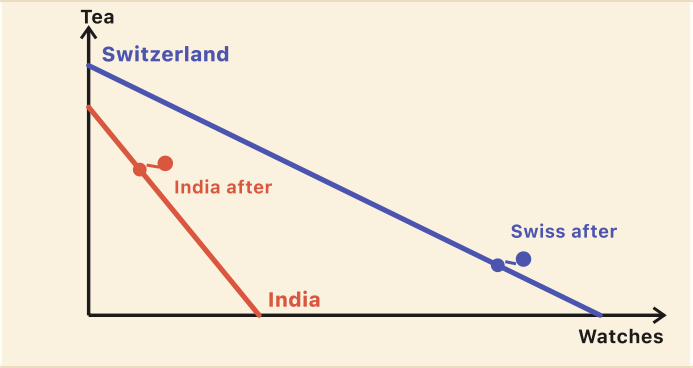
- **India** = 100 tea or 20 watches: a day's max
- **Switzerland** = 120 tea or 60 watches: a day's max

What we want: who wins each good on opportunity cost (OC: what you give up)

1. Absolute: Switzerland makes more of both
   Absolute advantage only asks who makes more.
2. OC of a watch: India 100/20 = 5 tea; Swiss 120/60 = 2 tea
   A Swiss watch costs less tea.
3. OC of a crate: India 20/100 = 0.2 watch; Swiss 60/120 = 0.5 watch
   Indian tea costs fewer watches.

Comparative advantage: India in tea, Switzerland in watches.

So: **Switzerland is better at both, yet India still has the comparative advantage in tea.**



WORKED EXAMPLE

Specialise, and the world gains

What we know

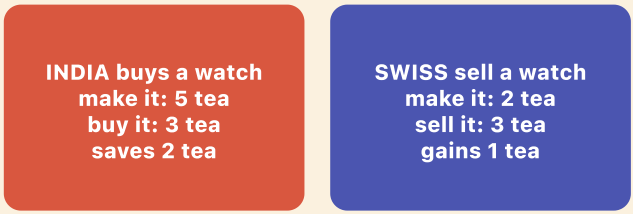
- **India before** = 70 tea, 6 watches: no trade
- **Swiss before** = 24 tea, 48 watches: no trade

What we want: the extra world output if each makes only its cheaper good

1. World before: tea 70 + 24 = 94; watches 6 + 48 = 54
   Each makes some of both for itself.
2. After: India makes only tea, 100; Switzerland only watches, 60
   Each puts all its workers on its cheaper good.
3. Gain: +6 tea and +6 watches
   100 - 94 and 60 - 54: extra output to share out.

+6 tea and +6 watches for the world.

So: Following comparative advantage raises world output without a single extra worker.



STORY

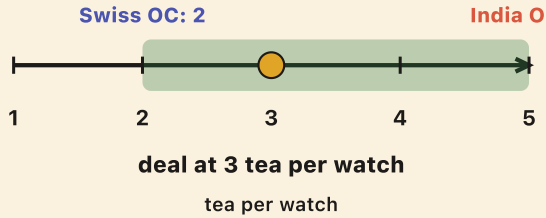
Why 3 tea per watch suits both sides

India and Switzerland agree a swap rate: **3 crates of tea for 1 watch**.

- **India:** making a watch at home costs 5 crates. Buying one costs 3. It saves 2 crates per watch.
- **Switzerland:** a watch costs it 2 crates of tea to make. Selling it brings 3. It gains 1 crate per watch.

Both win, so both sign. At 6 tea per watch India would rather make its own (5). At 1.5, Switzerland would rather grow its own tea (2). Any price **between 2 and 5** works. This swap rate is the **terms of trade**.

So: A trade price between the two opportunity costs gives each side something cheaper than making it at home.



WORKED EXAMPLE

Splitting the gains: the trade price

What we know

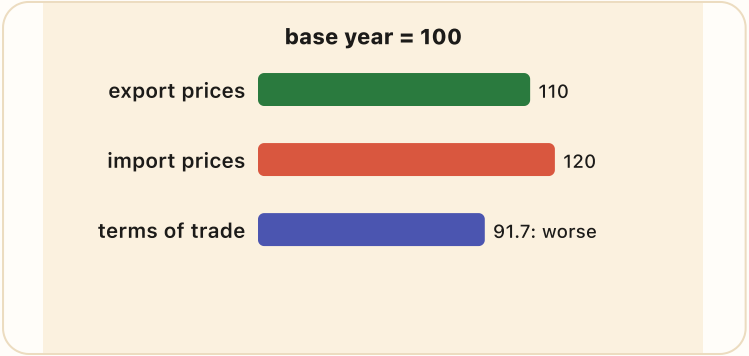
- **Trade price** = 3 tea: tea India pays per watch
- **Tea shipped** = 27: crates India sends to Switzerland

What we want: whether both end up better off than before trade

1. 3 lies between 2 and 5
   Between the two opportunity costs (OC), so both agree.
2. India: 27 / 3 = 9 watches, keeps 100 - 27 = 73 tea
   Was 70 tea, 6 watches: 3 more of each.
3. Swiss: 27 tea, 60 - 9 = 51 watches
   Was 24 tea, 48 watches: 3 more of each.

Both gain 3 tea and 3 watches: both consume beyond what they could make alone.

So: Any trade price between the two countries' opportunity costs leaves both better off.



FORMULA

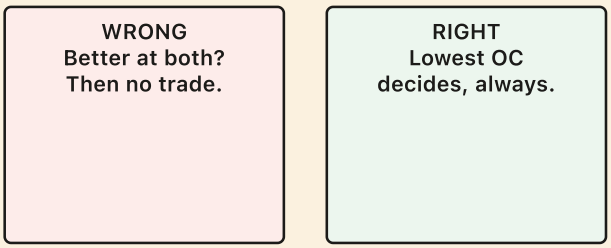
Terms of trade, as an index

```
Terms of trade = (export price index / import price index) x 100
```

Symbol	In plain words	Example
Export price index	average price of what it sells abroad (base year = 100)	110
Import price index	average price of what it buys abroad (base year = 100)	120
Terms of trade	imports one unit of exports buys, as an index	91.7

India sells tea and buys oil. Tea prices rise 10% (index 110), oil 20% (120). Tea now buys less oil: 110 / 120 x 100 = **91.7**, below 100, so India's terms of trade **worsened**.

So: **Above 100: exports buy more imports than in the base year (better). Below 100: they buy fewer (worse).**



EXAM TRAP

Comparative advantage traps

- **Absolute advantage in everything does not kill trade.** Opportunity costs differ, so gains exist.
- **A country cannot have a comparative advantage in both goods** (in a two-good model). Its OC of one is the inverse of the other.
- **Divide the right way:** OC of watches = tea given up / watches gained. Flipping it swaps the answer.
- **The trade price must lie between the two OCs.** Outside that range one country refuses.

So: **Always compute both opportunity costs before you answer; never stop at who makes more.**

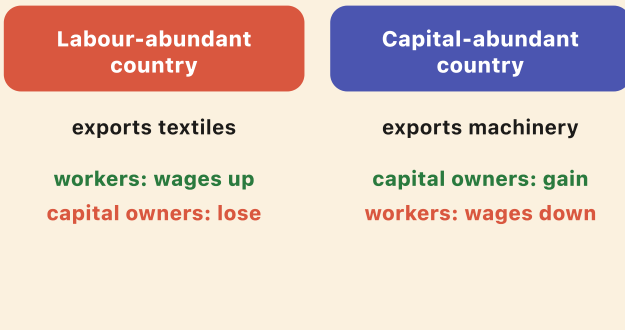


AT A GLANCE

Ricardian versus Heckscher-Ohlin

	Ricardian	Heckscher-Ohlin
Factors	Labour only	Labour and capital
Advantage from	Technology (labour productivity)	Factor endowments; same technology
Exports	What its workers do relatively best	Goods using its abundant factor
Income effect	Not its focus	Abundant factor gains, scarce loses

So: **Ricardian: technology drives trade. Heckscher-Ohlin: factor endowments drive trade, and trade shifts income between factors.**



IDEA

Heckscher-Ohlin: who wins inside each country

Under **Heckscher-Ohlin**, a labour-abundant country (many workers, little capital) exports labour-intensive goods such as garments. A capital-abundant country exports capital-intensive goods such as machinery.

Trade raises demand for each country's abundant factor and lowers demand for its scarce factor. So in the labour-abundant country, wages rise and returns to capital fall. In the capital-abundant country, capital owners gain and workers in import-competing industries lose.

So: **Heckscher-Ohlin: the abundant factor gains from trade and the scarce factor loses, so trade changes income distribution.**

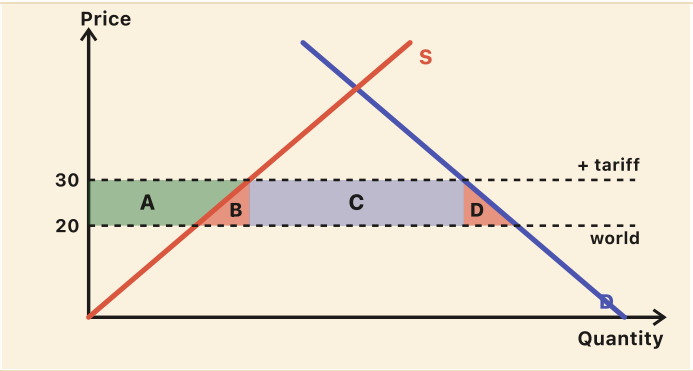


AT A GLANCE

Five ways to restrict trade

Restriction	What it is
Tariff	Tax on imports
Import quota	Cap on import quantity
Export subsidy	Government pays exporters per unit exported
Voluntary export restraint (VER)	Exporting country agrees to cap its exports, usually under pressure
Minimum domestic content	A set share of a product's value must be made at home

So: **Every restriction raises the home price of the protected good; they differ in who pockets the gap.**



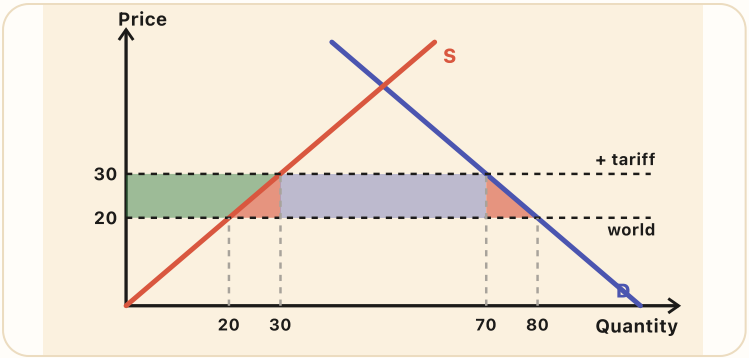
PICTURE IT

A tariff in a small country: four areas

A **small country** cannot move the world price. Without a tariff, the home price equals the world price and imports fill the gap between demand and home supply. A tariff lifts the home price by the full tariff.

- **A:** gain in producer surplus.
- **C:** government tariff revenue.
- **B and D:** deadweight loss. B: costly extra home production. D: consumption that no longer happens.
- Consumer surplus falls by A + B + C + D.

So: Consumers lose A+B+C+D; producers gain A; government gains C; the nation loses B+D.



WORKED EXAMPLE

Tariff, part 1: prices and quantities

What we know

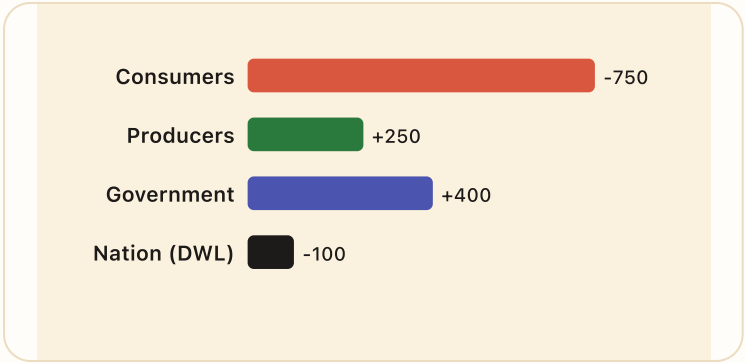
- **Qd = 100 - P** = 80 at $20: units buyers want at price P
- **Qs = P** = 20 at $20: units home firms make at P
- **World price; tariff** = $20; $10: price abroad; tax per import

What we want: home price, output, buying and imports after the tariff

1. Before: Qd = 80; Qs = 20; imports = 60
   At $20: 100 - 20 = 80 wanted, 20 made; imports fill the gap.
2. After: P = 20 + 10 = $30; Qd = 70; Qs = 30; imports = 40
   Imports now cost $30, so home firms charge $30 too. Buyers want 10 fewer; firms make 10 more.

Price $30; home output 30; consumption 70; imports 40, down 20.

So: A tariff cuts imports from both sides: home firms make more and consumers buy less.



WORKED EXAMPLE

Tariff, part 2: who wins, who loses

What we know

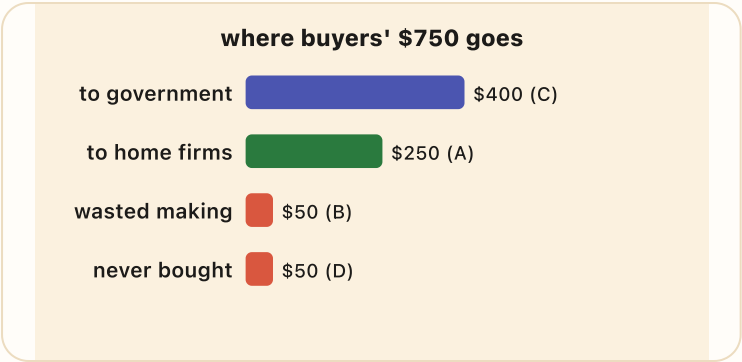
- **Price; bought; made at home** = $20 to $30; 80 to 70; 20 to 30: before to after

What we want: each side's gain or loss, and the deadweight loss

1. Buyers lose (80 + 70) / 2 x 10 = 750
   70 still bought cost $10 more (700); 10 dropped lose $5 each (50).
2. Home firms gain (20 + 30) / 2 x 10 = 250
   20 made anyway earn $10 more (200); 10 extra earn $5 each (50).
3. Tariff revenue = 40 x 10 = 400
   Tax on each unit still imported.
4. Deadweight loss = 750 - 250 - 400 = 100
   Lost by buyers, gained by nobody: 50 + 50.

Deadweight loss = $100

So: **Only the two triangles are truly lost.**



PLAYED OUT IN MONEY

Same tariff, followed dollar by dollar

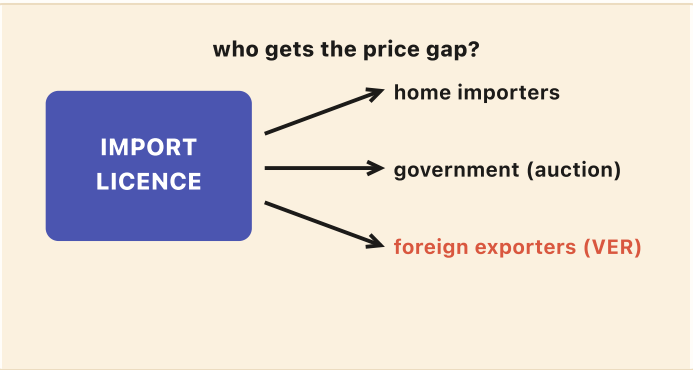
Make it cricket bats. At $20: 80 bought, 20 made at home, 60 imported. A $10 tariff lifts the price to $30.

Buyers pay $10 more on the 70 bats they still buy ($700) and give up 10 bats worth more to them than $20 ($50). Total **$750**. Where it goes:

- **$400** to the government: $10 on 40 imported bats (area C).
- **$250** to home bat makers (area A).
- **$50** wasted: the 10 extra home bats cost $25 each on average to make, versus $20 to import (area B).
- **$50** gone: 10 bats people wanted and no longer buy (area D).

400 + 250 + 50 + 50 = 750. Only B + D, **$100**, is lost to the nation.

So: Most of what buyers lose is moved to the government and home firms; only the triangles B and D vanish.



IDEA

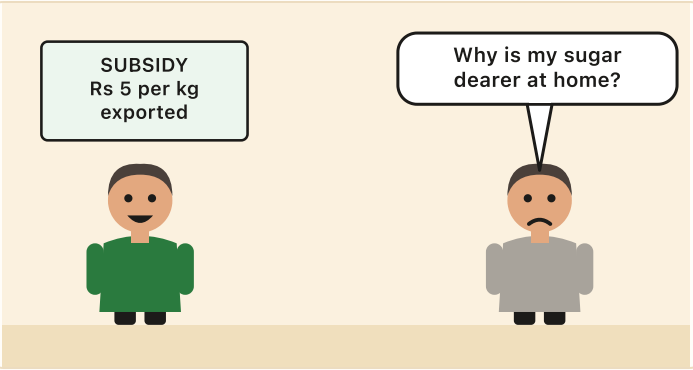
Quotas, VERs and the quota rent

A quota that cuts imports to the same level as a tariff raises the home price by the same amount. Consumers and producers are affected exactly as under the tariff.

The difference is area C, the **quota rent**: buy at the world price, sell at the higher home price. It goes to whoever holds the import licences: home importers, or the government if it auctions them.

Under a **VER**, foreign exporters hold the right to sell, so the rent goes abroad and the importing country loses more.

So: **Same price effect as a tariff; the question is who keeps area C. Under a VER it leaves the country.**



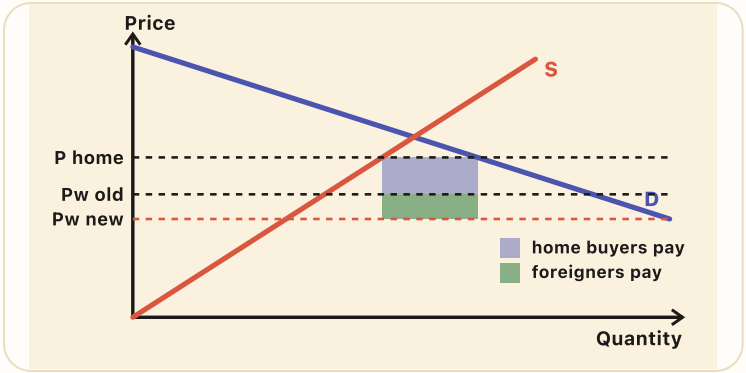
IDEA

Export subsidies and domestic content rules

Export subsidy: exporters earn the world price plus the subsidy abroad, so they will only sell at home for that much. The home price rises. Consumers lose, producers gain, taxpayers pay the subsidy, and the nation suffers a deadweight loss. In a large country, extra exports also push the world price down, worsening its terms of trade.

Minimum domestic content: for example, a rule that 60% of a car's value be made locally. It protects suppliers but raises costs.

So: Export subsidies raise home prices and cost taxpayers; the country as a whole loses.



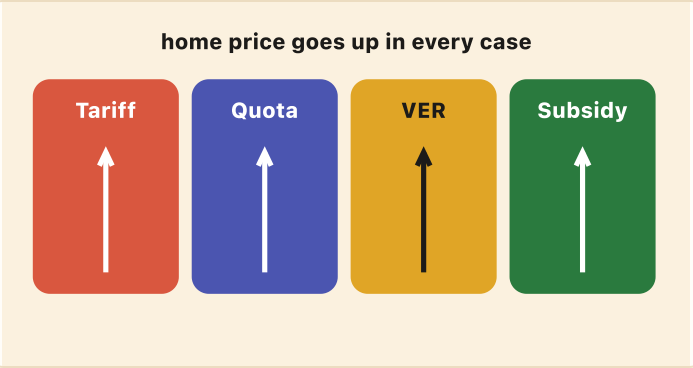
PICTURE IT

A large country's tariff: the terms-of-trade gain

A **large country** buys enough to move the world price. When it imposes a tariff, its imports fall, world demand falls, and the world price drops.

So foreign exporters absorb part of the tariff. The country pays less for its imports: its **terms of trade** improve. If that gain exceeds the deadweight loss, national welfare can even rise, at the exporters' expense. Retaliation can wipe the gain out.

So: **Small country: a tariff always lowers national welfare. Large country: the terms-of-trade gain may offset it.**



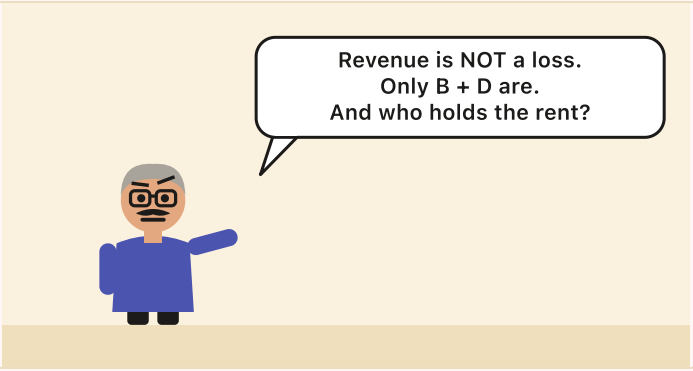
AT A GLANCE

Winners and losers by restriction

In every case, home consumers of the good lose and home producers of it gain. The difference is the government and the nation (small country):

	Government	Nation
Tariff	Gets revenue	Loses the two triangles
Quota	Gets rent only if it sells licences	Loses
VER	Gets nothing; rent goes abroad	Loses most
Export subsidy	Pays the subsidy	Loses

So: **Producers of the protected good always gain; home consumers always lose; the nation loses in a small country.**

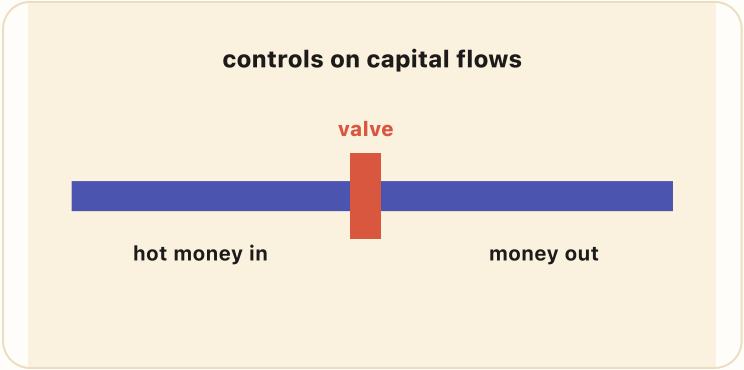


EXAM TRAP

Trade restriction traps

- **Deadweight loss is only the two triangles**, not the whole consumer loss. Revenue and the producer gain are transfers.
- **Quota versus tariff:** same price and quantity effects; the difference is who gets area C.
- **VER:** the rent goes to foreign exporters, so it is the worst for the importing nation.
- **Small versus large:** only a large country can gain from a tariff, via the terms of trade.
- **An export subsidy raises the home price**, it does not lower it.

So: Ask two questions for every restriction: who pays the higher price, and who pockets the gap?



IDEA

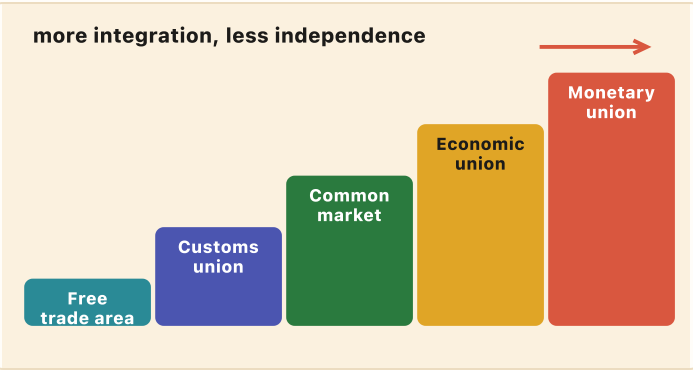
Capital restrictions: why and with what effect

Governments limit capital flows in or out to:

- reduce volatility of home asset prices from fast-moving "hot money";
- manage the exchange rate or defend a fixed rate;
- keep home interest rates low by stopping money from leaving;
- protect strategic industries from foreign ownership.

Effects: short-term breathing room (Malaysia in 1998), but capital is allocated less efficiently, the cost of capital rises, and future foreign investors stay wary.

So: Capital controls buy short-run stability at the cost of long-run efficiency and investor trust.



AT A GLANCE

Trading blocs: the integration ladder

Level	Adds	Example
Free trade area	No internal tariffs; own outside tariffs	USMCA
Customs union	+ common outside tariff	SACU (southern Africa)
Common market	+ free movement of labour and capital	Mercosur (its aim)
Economic union	+ shared institutions and policy	EU
Monetary union	+ one currency	Eurozone

So: Each level keeps everything below it and adds one more step of integration.



WORKED EXAMPLE

Trade diversion: a bad switch

What we know

- **Tariff** = 20: Home's tax on imported widgets
- **X** = 100: outsider, cheapest maker
- **Y** = 110: new free trade partner, tariff-free

What we want: is the switch good for Home as a nation?

1. Before: X 100 + 20 = 120; Y 110 + 20 = 130. Buy from X
   Same tax on both, so the cheaper maker wins. Tax: 20 a widget.
2. After: Y 110; X still 120. Switch to Y
   A tax break, not a lower cost, makes Y look cheaper.
3. Buyers save 120 - 110 = 10; the government loses its 20
   The tax it used to collect is gone.
4. Nation: +10 - 20 = -10 per widget
   Home now pays a foreigner 110 for what X sold for 100.

Trade diversion: Home loses 10 per widget.

So: **Diversion swaps the cheapest outside maker for a dearer member, and the nation loses.**



costly home output replaced:
trade creation

WORKED EXAMPLE

Trade creation: a good switch

What we know

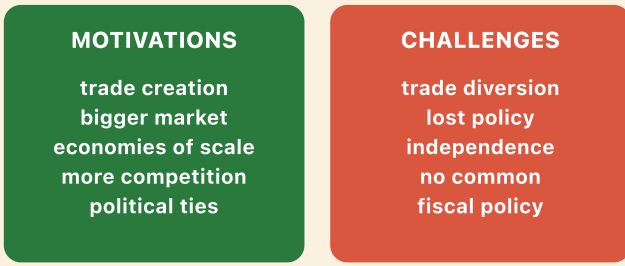
- **Home's cost** = 115: Home's own factories, per widget
- **Tariff** = 20: on all imports; Y's widgets go tariff-free after the deal
- **X, Y** = 100, 110: outsider's and partner's price

What we want: is the switch good for Home as a nation?

1. Before: X 120, Y 130 with tariff; home-made 115. Home makes its own
   No imports, so no tariff revenue.
2. After: Y comes in at 110, below 115. Home buys from Y
   The partner really is cheaper than home factories.
3. Nation: 115 - 110 = +5 per widget
   Workers are freed for other jobs; no tax revenue is lost.

Trade creation: Home gains 5 per widget.

So: **Creation replaces costly home output with cheaper member imports; only creation is a gain.**



IDEA

Why join a bloc, and what it costs

Motivations and advantages: trade creation, a larger market so firms reach economies of scale, sharper competition, more foreign investment, and closer political ties that reduce the risk of conflict.

Challenges: trade diversion; loss of national policy independence, worst in a monetary union, where a country gives up its own interest rate and exchange rate; and no common fiscal policy to transfer money to a member in trouble. Greece in 2010 to 2012 is the textbook case.

So: The deeper the integration, the bigger the market and the smaller each member's room to run its own policy.



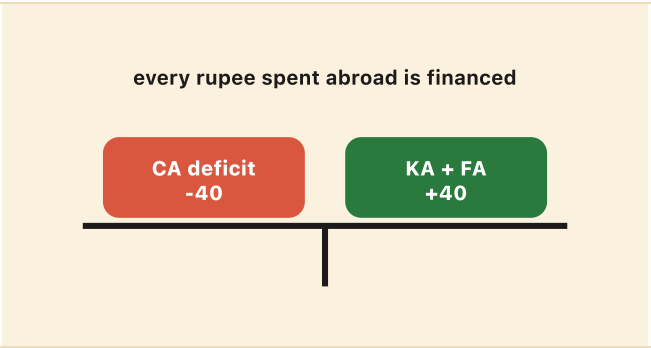
AT A GLANCE

The balance of payments: three accounts

The **balance of payments** records all transactions between a country's residents and the rest of the world.

Account	Records
Current	Goods and services trade; income (wages, interest, dividends); current transfers (remittances, aid)
Capital	Capital transfers (debt forgiveness); buying or selling non-produced, non-financial assets such as natural resource rights
Financial	Investment flows: direct, portfolio, loans, deposits, reserve assets

So: **Current = trade and income; capital = one-off capital transfers; financial = buying and selling assets.**



FORMULA

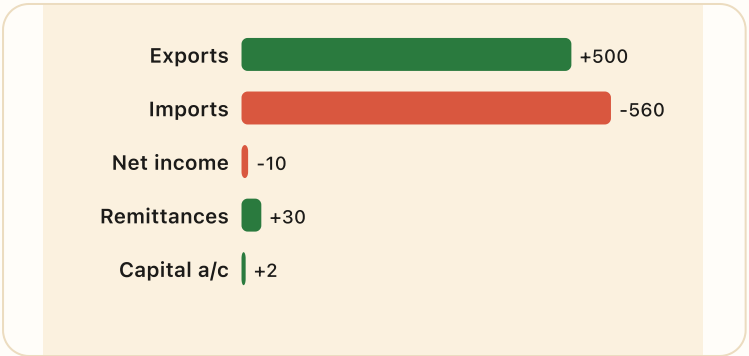
Why the balance of payments sums to zero

$$CA + KA + FA = 0$$

Symbol	In plain words	Example
CA	current account: trade, income and transfers	-40
KA	capital account: one-off transfers, such as debt forgiveness	0
FA	financial account: borrowing or selling assets abroad; + = money in	+40

Why zero: Chintu spends Rs 40 more than he earns. The 40 comes from a loan or selling his gold coin: spending gap CA -40, money in FA +40. Countries book every transaction twice the same way. (The IMF's BPM6 counts FA as net asset purchases, so there CA + KA = FA.)

So: A current account deficit must be financed by a capital plus financial account surplus of the same size.



WORKED EXAMPLE

Filling in the balance of payments

What we know

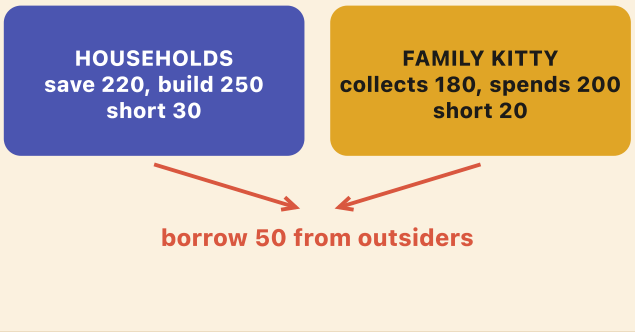
- **Exports, imports** = 500, 560: goods and services
- **Net income** = -10: earned abroad minus paid abroad
- **Transfers** = +30: remittances and aid received
- **KA** = +2: capital account (one-off capital transfers)

What we want: FA, the financial account (money in from borrowing or selling assets)

1. CA = 500 - 560 - 10 + 30 = -40
   Trade, income and transfers together: short 40.
2. CA + KA + FA = 0, so FA = -(CA + KA)
   Every payment is booked twice, so the accounts net to zero.
3. FA = -(-40 + 2) = +38
   KA covers 2 of the gap; borrowing or selling assets covers 38.

FA = +38: a net inflow of capital.

So: A current account deficit is paid for by borrowing or selling assets abroad.



STORY

A joint family that spends more than it earns

Think of a country as one big joint family.

- **The households (private sector)** save Rs 220, but spend Rs 250 building a new floor. Short by 30.
- **The family kitty run by the eldest (the government)** collects Rs 180 from everyone and spends Rs 200. Short by 20.

Together the family is short **50**. It must borrow 50 from outsiders or sell them some gold. Seen from outside, the family buys 50 more from the world than it sells: a **current account deficit of 50**.

So: **When households and government together spend more than they earn, the gap shows up as a current account deficit.**



spend more than you save: borrow abroad

FORMULA

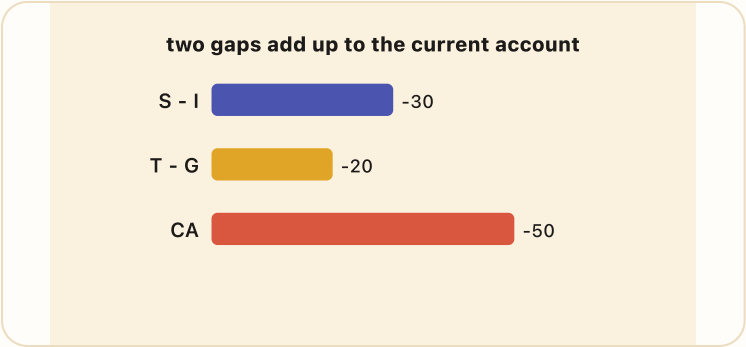
The current account and national saving

$$CA = (S - I) + (T - G)$$

Symbol	In plain words	Example
S	private saving	220
I	private investment (factories, homes)	250
T	taxes the government collects	180
G	government spending	200
CA	current account, about exports minus imports	-50

Where it comes from: income Y is spent as C (consumption) + I + G + (X - M), X exports, M imports. It is also used as C + S + T. Set them equal, cancel C: X - M = (S - I) + (T - G). Invest more than you save, or run a budget deficit, and CA goes negative.

So: **A current account deficit means the nation spends more than it saves; a budget deficit pushes it that way.**



WORKED EXAMPLE

Saving, investment and the deficit

What we know

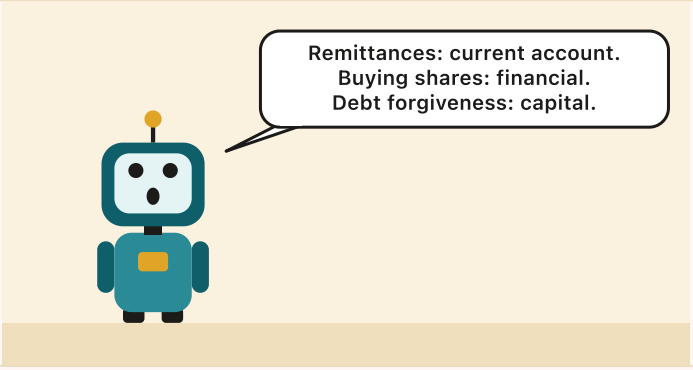
- **S, I** = 220, 250 billion: private saving, private investment
- **T, G** = 180, 200 billion: taxes collected, government spending

What we want: CA, the current account; then CA if the budget balances

1. Private gap: S - I = 220 - 250 = -30
   They invest 30 more than they save.
2. Budget balance: T - G = 180 - 200 = -20
   A budget deficit.
3. CA = -30 + (-20) = -50
   Both gaps must be borrowed from abroad.
4. If T = G: CA = -30 + 0 = -30
   Foreign borrowing shrinks by 20.

CA = -50 billion, a deficit.

So: Cutting a budget deficit shrinks the current account deficit, other things equal.

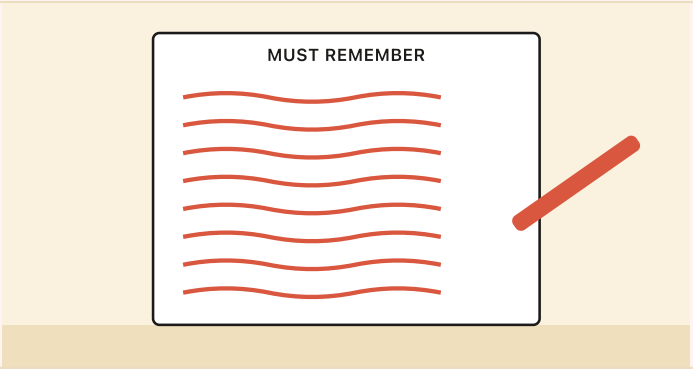


EXAM TRAP

Balance of payments traps

- **Remittances** from workers abroad are current transfers, in the current account, not the capital account.
- **Buying foreign shares or bonds** is a financial account item.
- **The capital account is small:** capital transfers and non-produced assets only. Investment flows sit in the financial account.
- **A current account deficit is not automatically bad:** borrowing to fund productive investment can pay for itself.
- **Sign conventions differ:** check whether FA is shown as net inflows or net asset purchases.

So: **Place each item by its nature: trade and income go current, one-off transfers go capital, assets go financial.**

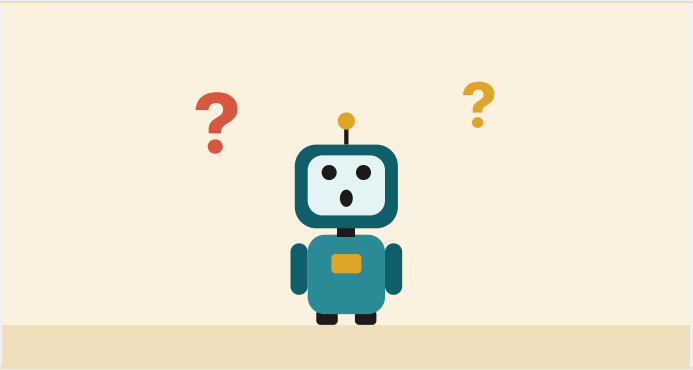


SHARMA JI'S RED PEN

8 things to remember

1. Gains: specialisation, scale, variety, competition, technology. Costs: jobs, inequality, adjustment.
2. Lower opportunity cost = comparative advantage; it sets trade.
3. Trade price must lie between the two OCs for both to gain.
4. Ricardian: technology. H-O: endowments; abundant factor gains.
5. Small-country tariff: nation loses the two triangles.
6. Quota rent goes to licence holders; under a VER it goes abroad.
7. FTA, customs union, common market, economic union, monetary union.
8. CA + KA + FA = 0; CA = (S - I) + (T - G).

So: Scribbled by Sharma ji on the international trade chapter.



EXAM QUESTION 1 OF 8

In one day, a worker in Country A makes 8 shirts or 2 phones. A worker in Country B makes 6 shirts or 3 phones. Which statement is correct?

A. Country A has a comparative advantage in phones.

B. Country B has a comparative advantage in phones.

C. Country B has an absolute advantage in both goods.

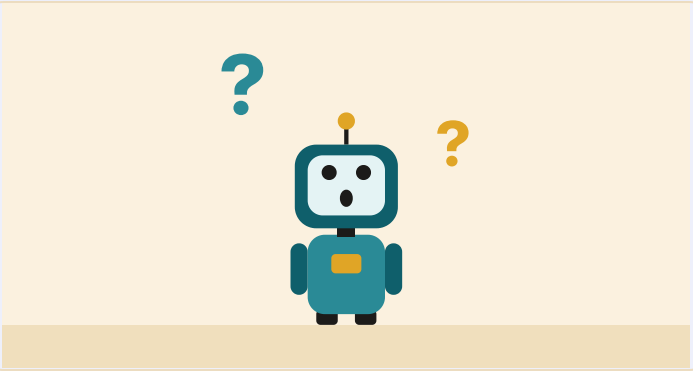
Answer on the next page.

So: **Pick an answer, then turn the page.**

ANSWER

In one day, a worker in Country A makes 8 shirts or 2 phones. A worker in Country B makes 6 shirts or 3 phones. Which statement is correct?

Answer: B. Opportunity cost of a phone: A gives up 8/2 = 4 shirts; B gives up 6/3 = 2 shirts. B's is lower, so B has the comparative advantage in phones. A is wrong because A's phone OC is the higher one (A's advantage is in shirts, OC 0.25 phone versus 0.5). C is wrong because A makes more shirts per worker (8 versus 6).



EXAM QUESTION 2 OF 8

A small country's world price is $10, where home demand is 120 and home supply is 40. A $2 tariff raises the home price to $12, where demand is 110 and supply is 50. The deadweight loss from the tariff is closest to:

A. $20

B. $120

C. $10

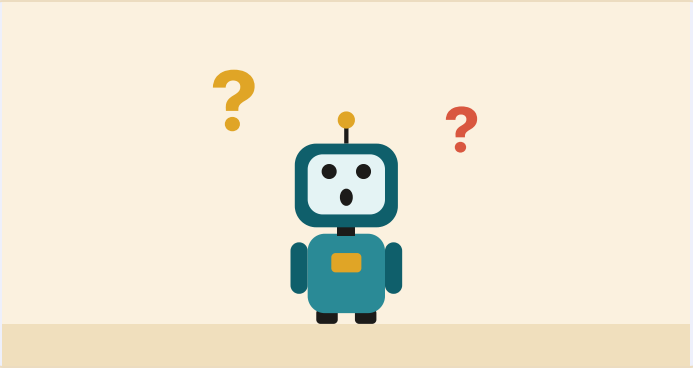
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

A small country's world price is $10, where home demand is 120 and home supply is 40. A $2 tariff raises the home price to $12, where demand is 110 and supply is 50. The deadweight loss from the tariff is closest to:

Answer: A. Deadweight loss is the two triangles: 0.5 x 2 x (50 - 40) + 0.5 x 2 x (120 - 110) = 10 + 10 = $20. B is the tariff revenue, 2 x (110 - 50) = 120, which is a transfer, not a loss. C counts only one of the two triangles.



EXAM QUESTION 3 OF 8

Under a voluntary export restraint, the quota rent is most likely captured by:

A. the importing country's government

B. the importing country's consumers

C. the foreign exporters

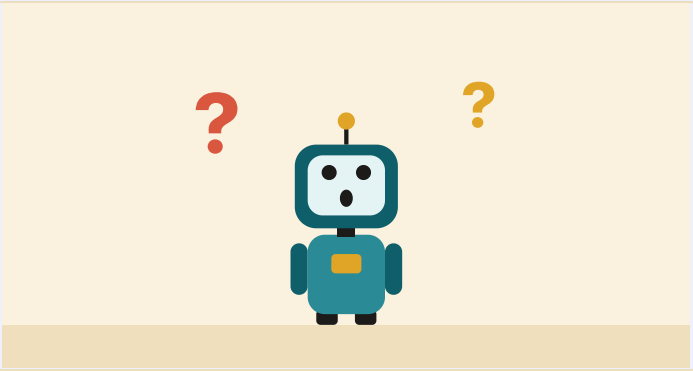
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

Under a voluntary export restraint, the quota rent is most likely captured by:

Answer: C. Under a VER the exporting country controls the right to sell, so foreign exporters buy at the world price and sell at the higher home price, keeping the rent. The importing government collects nothing, so A is wrong. Consumers pay the higher price, so they lose, and B is wrong.



EXAM QUESTION 4 OF 8

Members of a trading bloc remove internal trade barriers, share a common external tariff, allow free movement of labour and capital, coordinate economic policy and use one currency. The bloc is best described as a:

A. monetary union

B. common market

C. customs union

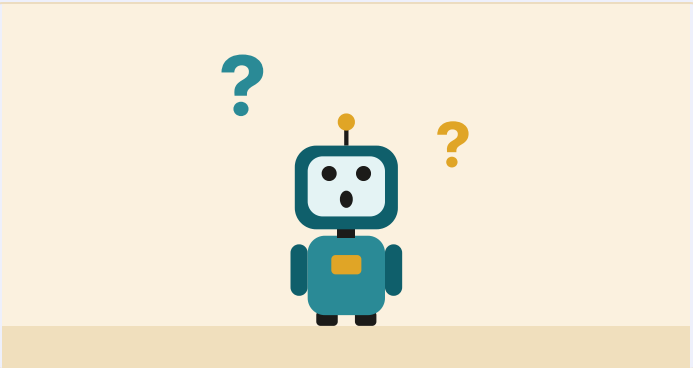
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So: Pick an answer, then turn the page.

ANSWER

Members of a trading bloc remove internal trade barriers, share a common external tariff, allow free movement of labour and capital, coordinate economic policy and use one currency. The bloc is best described as a:

Answer: A. A single currency is the extra step that defines a monetary union, the deepest level. A common market has free factor movement but no common currency or shared economic policy, so B is wrong. A customs union stops at a common external tariff, so C is wrong.



EXAM QUESTION 5 OF 8

After joining a free trade area, a country stops importing a good from the lowest-cost producer outside the area and imports it instead from a higher-cost member. This is best described as:

A. trade creation

B. economies of scale

C. trade diversion

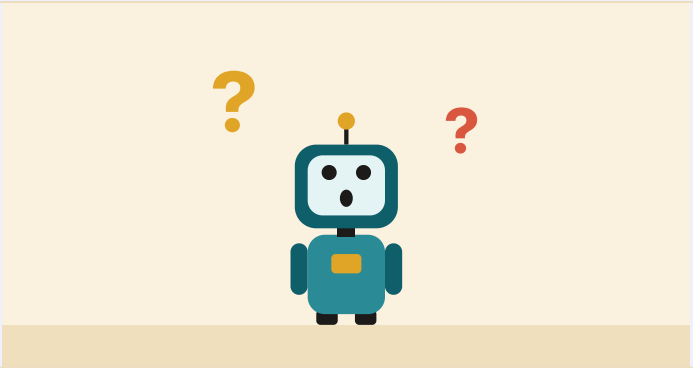
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

After joining a free trade area, a country stops importing a good from the lowest-cost producer outside the area and imports it instead from a higher-cost member. This is best described as:

Answer: C. Shifting from a lower-cost outsider to a higher-cost member because of tariff preferences is trade diversion, and it lowers efficiency. Trade creation is the reverse: costly home output or imports replaced by cheaper member imports, so A is wrong. Economies of scale are lower unit costs from bigger output, unrelated to the switch described, so B is wrong.



EXAM QUESTION 6 OF 8

According to the Heckscher-Ohlin model, when a labour-abundant country opens up to trade, the most likely result is that:

A. owners of capital gain the most

B. workers' wages rise

C. both workers and capital owners lose

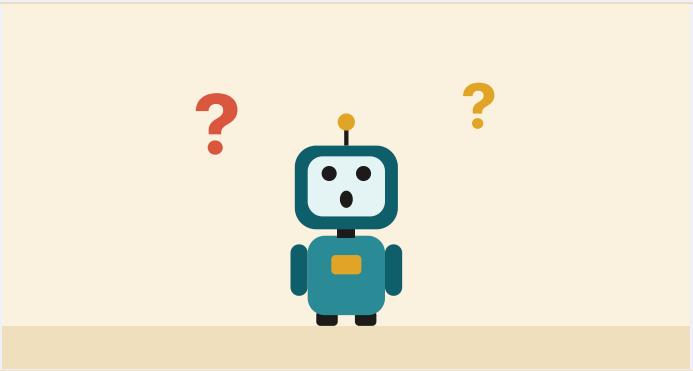
Answer on the next page.

So: **Pick an answer, then turn the page.**

ANSWER

According to the Heckscher-Ohlin model, when a labour-abundant country opens up to trade, the most likely result is that:

Answer: B. The country exports labour-intensive goods, raising demand for its abundant factor, labour, so wages rise. Capital is the scarce factor, so its return tends to fall, which rules out A. Trade raises total income, and the abundant factor gains, so C is wrong.



EXAM QUESTION 7 OF 8

A country has private saving of 300, private investment of 320, tax revenue of 150 and government spending of 170. Its current account balance is closest to:

A. -40

B. 0

C. -20

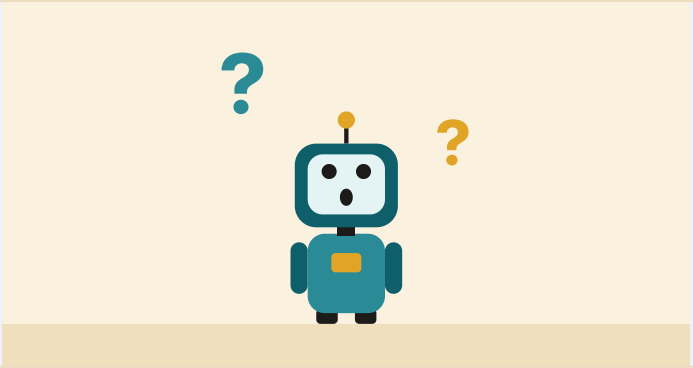
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

A country has private saving of 300, private investment of 320, tax revenue of 150 and government spending of 170. Its current account balance is closest to:

Answer: A. CA = (S - I) + (T - G) = (300 - 320) + (150 - 170) = -20 - 20 = -40. B assumes the two gaps cancel, but both are negative. C counts only one of the two gaps.



EXAM QUESTION 8 OF 8

Which is least likely an objective of government restrictions on capital flows?

A. Reducing the volatility of domestic asset prices

B. Protecting strategic industries from foreign ownership

C. Raising domestic interest rates to attract foreign capital

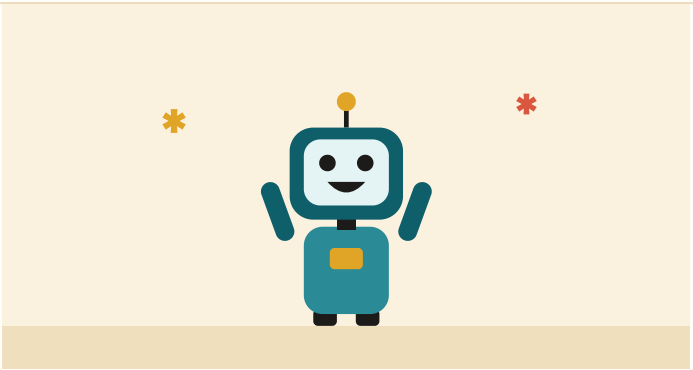
Answer on the next page.

So: **Pick an answer, then turn the page.**

ANSWER

Which is least likely an objective of government restrictions on capital flows?

Answer: C. Capital controls are used to keep domestic interest rates low by stopping capital from leaving, not to raise rates to pull capital in; restricting flows works against attracting foreign money. A and B are standard objectives, along with managing the exchange rate.



CHAPTER 6 DONE

Trade, sorted

You can compute opportunity costs and the gains from specialisation, tell Ricardo from Heckscher-Ohlin, draw and value the four tariff areas, say who keeps the quota rent, climb the trading-bloc ladder, spot trade diversion, and balance the balance of payments. Pinky has quietly bought an imported charger.

So: **Swipe on, or jump to another chapter from the Menu.**



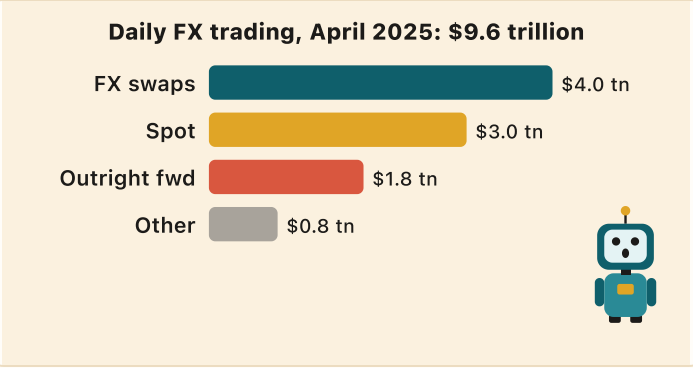
BEFORE YOU START

The cast, and words this chapter uses

A Mumbai investment firm. Sharma ji trusts nothing without data ("Data dikhao"); Pinky is sure of everything, especially when wrong.

- **Appreciate / depreciate:** a currency gains or loses value against another.
- **CPI:** consumer price index: the price level of a typical basket of goods.
- **Price elasticity of demand:** % change in quantity for a 1% change in price, used as a positive number.
- **Current account:** mainly a country's exports minus imports of goods and services.

So: **You can start this chapter cold. Everything it needs is on this card.**



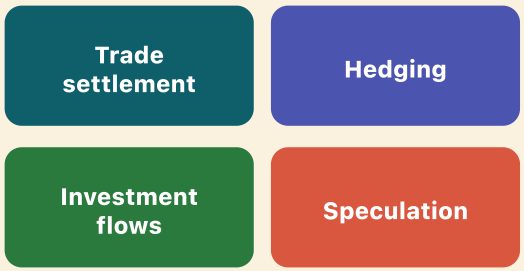
STORY

The biggest market nobody sees

Chintu's first morning at a Mumbai investment firm. Sharma ji asks where the world's biggest market is. Chintu guesses the NSE. Wrong.

The **foreign exchange (FX) market** is where currencies are swapped for each other. It has no building and no single exchange: banks and clients trade over the counter (OTC), by screen and phone, 24 hours a day on weekdays. In April 2025 it turned over about **$9.6 trillion a day**, and the US dollar was on one side of 89% of trades. BIS Triennial Survey 2025

So: The FX market is a global, decentralised OTC market, and the US dollar sits on one side of most trades.

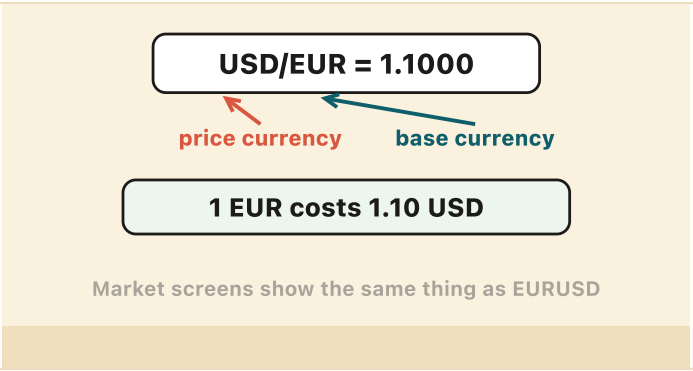


AT A GLANCE

Four jobs the FX market does

- **Trade settlement:** a Pune auto-parts exporter is paid in euros but pays wages in rupees, so it sells euros.
- **Hedging:** the same exporter locks today's rate for a payment due in 3 months, so a rupee rally cannot eat its margin.
- **Investment flows:** a US pension fund buying Indian bonds must first buy rupees.
- **Speculation:** a hedge fund buys yen because it thinks the yen will rise. Speculators also add liquidity.

So: **FX exists to settle trade, move investment across borders, transfer risk (hedging) and take risk (speculation).**



NOTATION, READ TWICE

What is the 1? The CFA A/B quote

The CFA curriculum writes a quote as **A/B**: the number of units of **A, the price currency**, needed to buy **one unit of B, the base currency**.

USD/EUR = 1.1000 means 1.10 US dollars per 1 euro. **INR/USD = 85.0000** means 85 rupees per 1 dollar.

Dealers write the same price as EUR/USD or EURUSD, base currency first: the opposite order. So never trust the slash. Ask: which currency is the 1?

So: **In A/B, B is the 1. If A/B rises, B buys more A, so B has appreciated.**



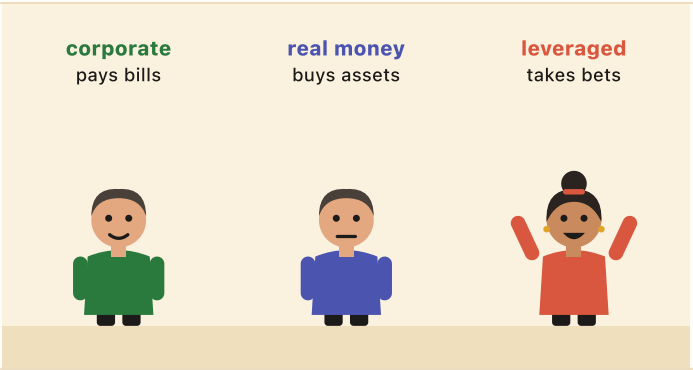
WHO TRADES, 1 OF 3

The sell side: dealer banks

The **sell side** is mostly large dealer banks. They quote two prices all day: a **bid** (the price at which they buy the base currency) and a higher **offer** or ask (the price at which they sell it).

What they want: to earn the bid-offer spread on huge volume, serve corporate and investor clients, and manage their own inventory of currencies. Dealers also trade with each other in the **interbank market** to offload positions.

So: **Sell side = dealers who make prices and earn the spread; everyone else is on the buy side.**

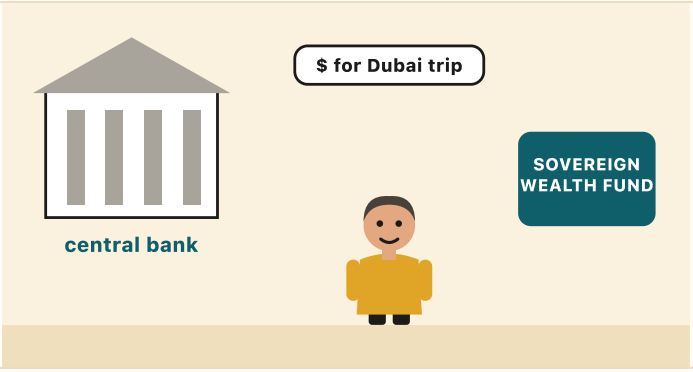


WHO TRADES, 2 OF 3

The buy side: firms and investors

Who	Examples	What they want
Corporates	exporters, importers, multinationals	pay and get paid; hedge to remove FX risk
Real money accounts	mutual funds, pension funds, insurers, endowments	buy foreign assets; little or no borrowing
Leveraged accounts	hedge funds, prop traders	profit from moves using borrowed money (Pinky's dream job)

So: **Corporates mostly hedge, real money invests, leveraged accounts speculate with borrowed money.**

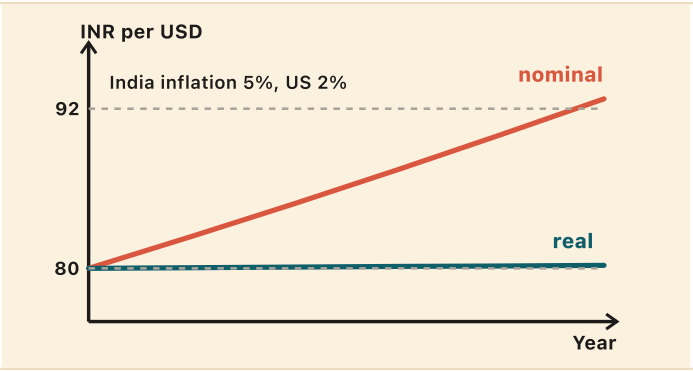


WHO TRADES, 3 OF 3

The buy side: states and households

Who	What they want
Governments	pay for imports and foreign debt
Central banks	manage reserves; buy or sell currency to steer or calm the rate (the RBI sells dollars when the rupee slides)
Sovereign wealth funds	invest national savings or oil money abroad for the long run (Norway, Abu Dhabi)
Retail	tourists, students paying fees abroad, remittances, small traders

So: **Central banks are the participants who can move the price on purpose; retail is tiny per trade but huge in number.**



PICTURE IT

Nominal versus real exchange rate

The **nominal exchange rate** is the quote you see on the screen: 85 rupees per dollar.

The **real exchange rate** adjusts that quote for the price levels in both countries. It measures how many Indian baskets of goods one US basket buys.

In the chart the rupee weakens 3% a year, from 80 to 92.7 per dollar. But Indian prices also rise 3% a year faster than US prices, so the real rate barely moves (80.2). An American tourist in Goa feels no richer.

So: Nominal is the screen price of money; real is the price of goods. Competitiveness depends on the real rate.

real(d/f) = S(d/f) x CPI_f / CPI_d

foreign prices up: real rate up

domestic prices up: real rate down



FORMULA

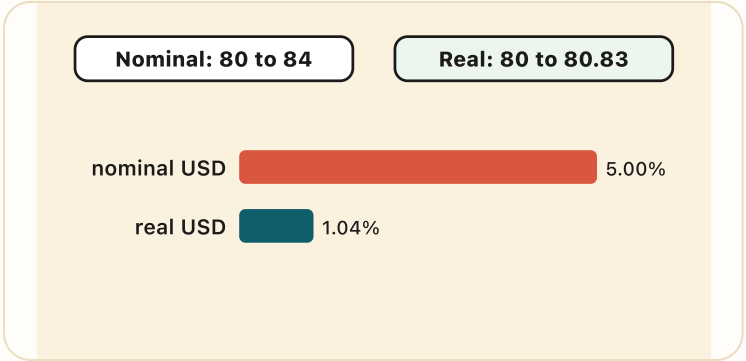
The real exchange rate

$$\text{real}\ (d/f) = S(d/f) \times CPI_f \div CPI_d$$

Symbol	In plain words	Example
d, f	d = domestic, your home country; f = foreign	d = India, f = US
S(d/f)	screen rate: home money paid for ONE unit of foreign money	INR/USD 84
CPI_f	foreign price level: cost of a typical US basket, as an index	102
CPI_d	home price level: same for an Indian basket	106

In plain words: how many Indian shopping baskets one US basket buys. US prices up: more. Indian prices up: fewer. A **rise** means the rupee **depreciated in real terms**.

So: **Real (d/f) rises if the nominal d/f rate rises, foreign prices rise, or domestic prices fall.**



WORKED EXAMPLE

Did the rupee really get weaker?

What we know

- **S(d/f)** = 80, now 84: INR/USD: ₹ for ONE $
- **CPI_d** = 100, now 106: Indian prices (d = India, home)
- **CPI_f** = 100, now 102: US prices (f = US, foreign)

What we want: the real INR/USD now, and the dollar's real rise.

1. Real now = 84.0000 × 102 ÷ 106 = 80.8302
   Screen rate × US prices ÷ Indian prices. Last year: 80.
2. Nominal: 84 ÷ 80 - 1 = 5.00%. Real: 80.8302 ÷ 80 - 1 = 1.04%
   The dollar is the 1 in INR/USD, so new over old for both.

Real INR/USD = 80.8302: the dollar is up only 1.04% in real terms. Indian inflation ate most of the rupee's fall.

So: Always deflate: a currency that falls by about the inflation gap has not made exports cheaper.

Last year: 1 US basket = 1.0000 Indian

Now: ₹8,568 ÷ ₹8,480 = 1.0104 Indian

real gain 1.04%, not 5.00%

PLAYED OUT IN MONEY

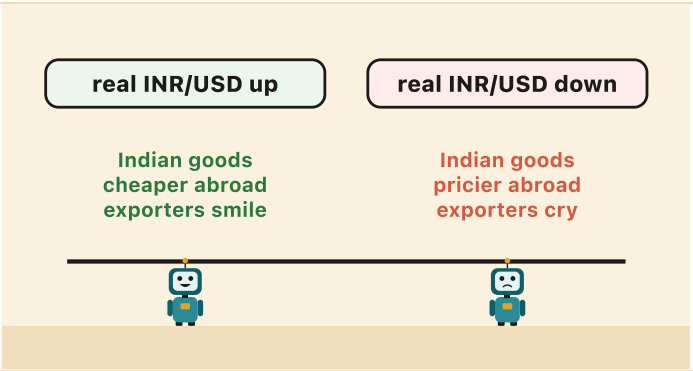
Same example, played out in shopping baskets

Say both baskets cost the same last year. US basket: $100 × 80 = **₹8,000**. Indian basket: **₹8,000**. One US basket = 1.0000 Indian basket.

Now: US basket $102 × 84 = **₹8,568**. Indian basket up 6% = **₹8,480**. One US basket = 8,568 ÷ 8,480 = **1.0104** Indian baskets.

An American's dollars buy only 1.04% more Indian goods, not 5.00%. The formula said the same: 80.8302 ÷ 80 = 1.0104.

So: The real rate is the nominal rate measured in baskets of goods, not in rupees.



INTERPRET IT

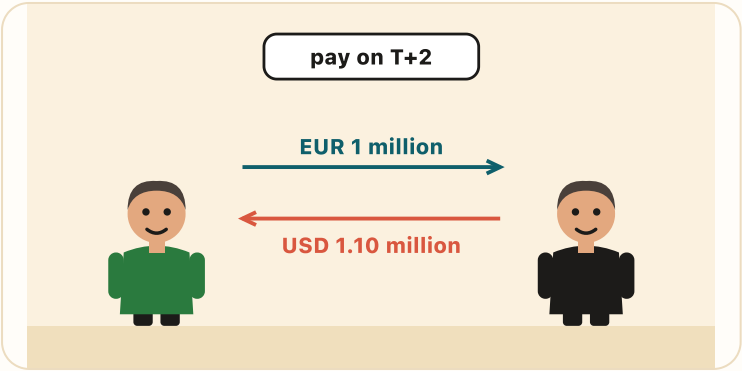
Rising real rate: who gains?

Quote it as domestic per foreign, real (d/f). For India that is real INR/USD.

- **Real (d/f) rises:** the rupee buys fewer US goods; Indian goods look cheap to Americans. The rupee has a **real depreciation** and Indian exports gain competitiveness.
- **Real (d/f) falls:** real appreciation of the rupee. Tirupur T-shirts look pricey in New York; imports look cheap in Mumbai.

The trap: in d/f form, a rise is bad news for the domestic currency but good news for competitiveness.

So: A higher real d/f rate means a weaker domestic currency in real terms and more competitive domestic exports.



DEALS, 1 OF 3

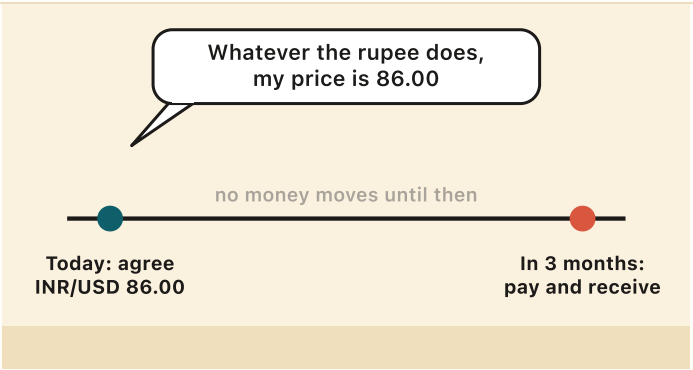
Spot: swap money now

A **spot** transaction exchanges two currencies at today's rate for immediate delivery. "Immediate" in FX usually means **T+2**, two business days after the trade date. A few pairs, such as USD/CAD, settle T+1.

Used for: paying an invoice, buying foreign shares, a tourist's travel money, a speculator's quick bet.

Example: at USD/EUR = 1.1000, buying EUR 1 million costs USD 1.10 million, paid two business days later.

So: Spot = today's price, settled in about two business days.



DEALS, 2 OF 3

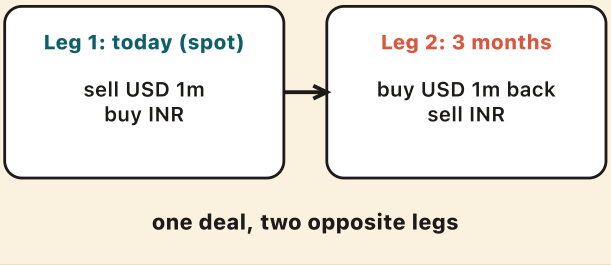
Forward: lock the rate now, swap later

A **forward contract** fixes today the rate for an exchange on a date beyond spot settlement, for example in 3 months.

A Mumbai electronics importer owes USD 500,000 in 3 months. If the rupee falls from 85 to 90 per dollar, the bill jumps by INR 2.5 million. She buys dollars forward at 86.00 and now knows her rupee cost today: INR 43 million.

Used for: hedging future payments and receipts. Speculators use them too.

So: A forward removes the uncertainty of a future exchange by fixing the rate now.



DEALS, 3 OF 3

FX swap: spot one way, forward back

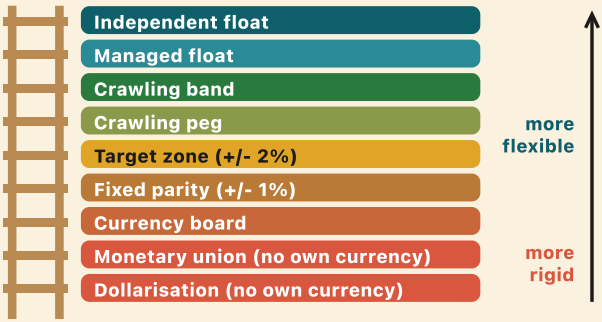
An **FX swap** combines a spot deal and a forward deal in the **opposite direction**, agreed together.

Used for:

- **Rolling a hedge:** a forward is about to mature but the exposure lasts longer, so the hedger swaps it out to a later date.
- **Funding:** a bank with dollars but needing rupees for 3 months swaps them, in effect borrowing rupees against dollars.

FX swaps are the most traded FX product by value.

So: FX swap = spot leg plus reverse forward leg; it shifts timing, not long-run exposure.



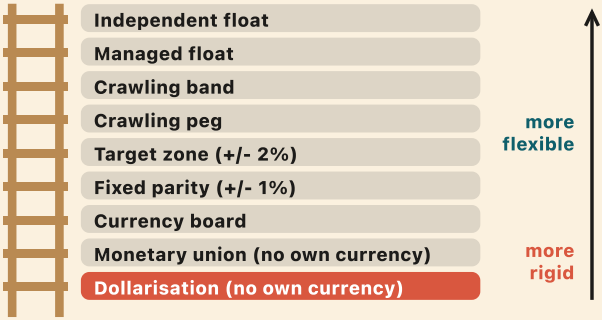
PICTURE IT

The regime ladder: rigid to flexible

An **exchange rate regime** is the set of rules a country follows for its currency's value. From most rigid to most flexible:

1. No separate legal tender: dollarisation, monetary union
2. Currency board
3. Fixed parity
4. Target zone
5. Crawling peg
6. Crawling band
7. Managed float
8. Independent float

So: Climbing the ladder buys monetary policy freedom and costs exchange rate stability.



REGIME 1A

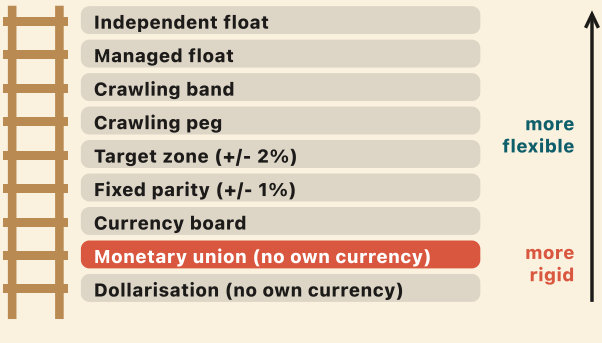
Dollarisation: use someone else's money

What: a country drops its own currency and uses another, usually the US dollar. Ecuador (since 2000), El Salvador and Panama use the dollar.

Plus: imports the credibility and low inflation of the issuer; no exchange rate risk with the US.

Minus: no own monetary policy and no say in the Fed's; no lender of last resort for local banks; the issuer keeps the profit from printing money (seigniorage). Policy is set for Texas, not Quito.

So: Dollarisation buys credibility by giving up monetary policy completely.



REGIME 1B

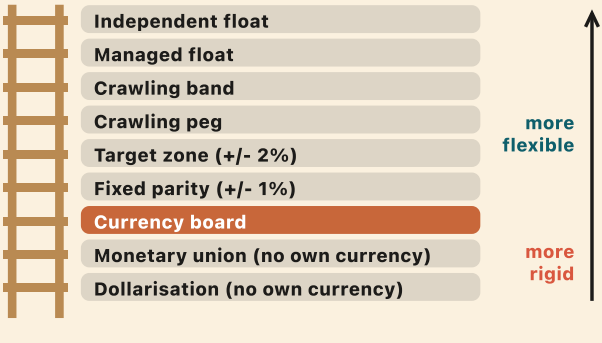
Monetary union: share one currency

What: several countries adopt one common currency run by one central bank. The euro area and the ECB are the classic case.

Plus: no exchange rate risk inside the union, lower transaction costs, deeper markets. Unlike dollarisation, members get a vote on policy.

Minus: one interest rate for all. When Greece was in crisis it could neither cut its own rates nor devalue; all adjustment fell on wages, jobs and budgets.

So: A monetary union shares a currency and a central bank; members lose their own rate and devaluation as shock absorbers.



REGIME 2

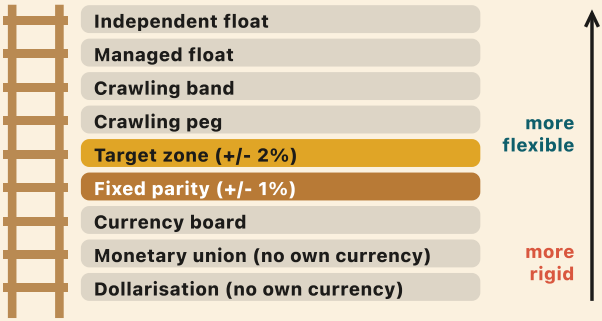
Currency board: a peg written into law

What: a legal commitment to swap local currency for a named foreign currency at a fixed rate, with the local money base fully backed by foreign currency reserves. Hong Kong: HKD 7.80 per USD, kept inside 7.75 to 7.85.

Plus: very credible; the board still earns interest on its reserves.

Minus: money supply and interest rates simply follow the reserves, so no independent monetary policy, and little room to act as lender of last resort.

So: A currency board is a hard peg with 100% reserve backing and almost no monetary discretion.



REGIMES 3 AND 4

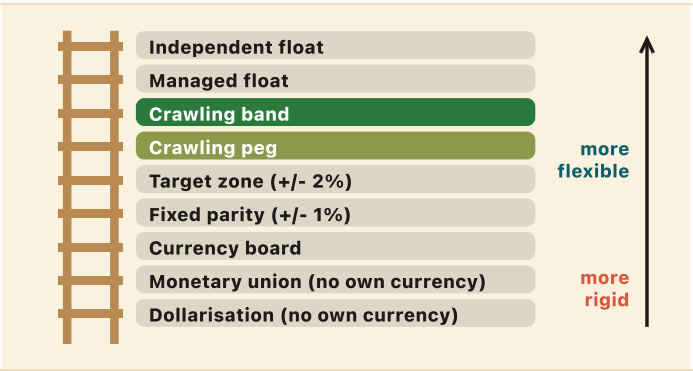
Fixed parity and target zone

Fixed parity: the rate is pegged to one currency or a basket and kept within about plus or minus 1%, using intervention and interest rates. No law forces 100% reserve backing. Saudi Arabia keeps 3.75 riyals per US dollar.

Target zone: the same, with a wider band, about plus or minus 2%. Denmark's krone moves in a band against the euro.

Trade-off: some stability and a little policy room, but the parity can be changed, which invites speculative attacks if reserves look thin.

So: Fixed parity and target zone are pegs without a legal backstop; the band width is the difference.



REGIMES 5 AND 6

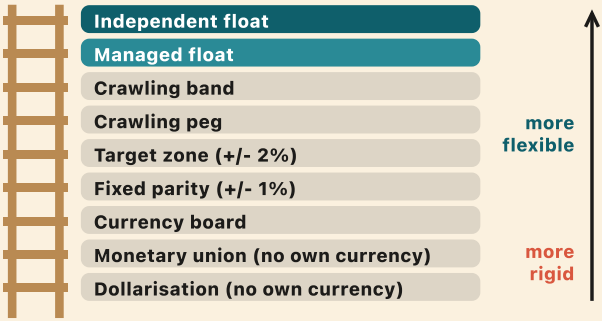
Crawling peg and crawling band

Crawling peg: the peg is reset in small steps. Passive: to offset inflation so competitiveness holds. Active: pre-announced steps to pull inflation expectations down. Used by high-inflation Latin American economies in the 1980s and 1990s.

Crawling band: a band around the peg that widens over time, a planned exit from a peg towards a float. Chile and Israel used bands in the 1990s.

Trade-off: more flexibility than a fixed peg, but every step announces policy and can still be attacked.

So: A crawl moves the peg in steps; a widening band is a staircase out of a peg.



REGIMES 7 AND 8

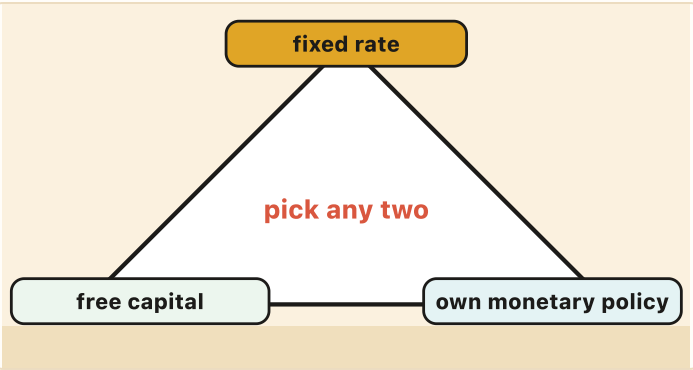
Managed float and independent float

Managed float: no announced target, but the central bank intervenes based on the balance of payments, inflation or jobs. The RBI says it targets no level, yet buys and sells dollars to smooth sharp rupee moves.

Independent float: the market sets the rate; monetary policy chases domestic goals such as inflation, and intervention is rare. USD, EUR, JPY, GBP.

Trade-off: full policy freedom and an automatic shock absorber, at the cost of volatility. Managed floats can drift into hidden pegs.

So: Floating regimes keep monetary independence and accept exchange rate volatility.



BIG PICTURE

The impossible trinity

No country can have all three at once:

- a fixed exchange rate
- free movement of capital
- independent monetary policy

Hong Kong: fixed rate plus free capital, so its rates follow the US. The US: free capital plus own policy, so the dollar floats. China has long kept capital controls to run its own policy while managing the yuan.

So: Every regime choice is a choice of which corner of the trinity to give up.



TRADE, 1 OF 6

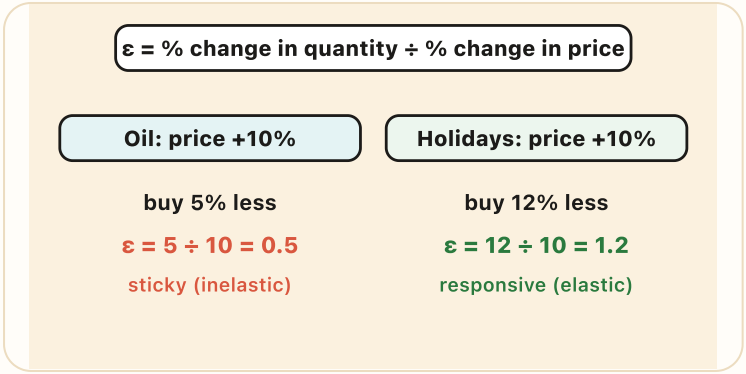
How a weaker currency changes trade

Say the rupee weakens 10%. Two things happen at once:

- **Exports:** our tea gets cheaper for Americans, so they buy more packets. More rupees come in. Good.
- **Imports:** each oil barrel now costs ₹8,800 instead of ₹8,000. We buy a few fewer barrels (good) but pay more for each one (bad).

So the trade balance improves only if buyers react **strongly enough** to the new prices. How strongly people react has a name: **elasticity** (next card).

So: A weaker currency helps trade only if export and import quantities react enough to the price change.



TRADE, 2 OF 6

Elasticity: how much people react to price

Elasticity (ε, Greek "epsilon") = how many % the quantity bought changes when the price changes by 1%.

- **Less than 1:** people barely react (oil, medicine). You need it, you buy it.
- **More than 1:** people react a lot (holidays, clothes). Price up, they walk away.

We always use it as a **positive** number. ε_X = how strongly foreigners react to the price of **our exports**. ε_M = how strongly **we** react to the price of **our imports**.

So: **Elasticity 0.5 means a 10% price change moves quantity 5%; 1.2 means it moves 12%.**

$$\omega X\ \varepsilon X + \omega M\ (\varepsilon M - 1) > 0$$

export gain change in import bill

above 0: weaker currency improves trade balance

below 0: it makes the trade balance worse

if exports = imports: $\varepsilon X + \varepsilon M > 1$

FORMULA

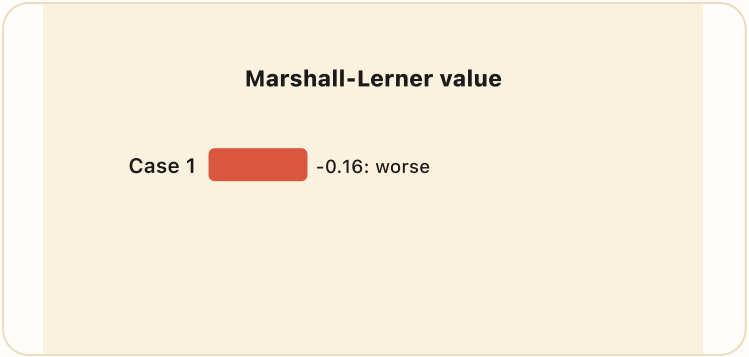
The Marshall-Lerner condition

$$\omega_X\ \varepsilon_X + \omega_M\ (\varepsilon_M - 1) > 0$$

Symbol	In plain words	Example
ω_X	exports' share of all trade: exports ÷ (exports + imports)	40 ÷ 100 = 0.40
ω_M	imports' share of all trade: imports ÷ (exports + imports)	60 ÷ 100 = 0.60
ε_X	how strongly foreigners react to our export prices	1.2
ε_M	how strongly we react to import prices	0.7

Why the "- 1": a 1% weaker rupee makes every import 1% dearer before anyone reacts. We then buy ε_M% less. If ε_M is below 1, that cut is too small and the import bill **rises**.

So: **In one line: extra export money minus extra import bill, each weighted by its size, must be above 0.**



WORKED EXAMPLE

Will the depreciation help? Case 1: sticky buyers

What we know

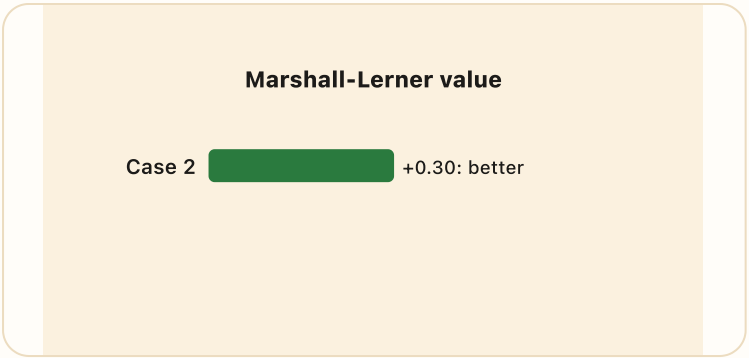
- **X** = 40 billion: exports
- **M** = 60 billion (a deficit of 20): imports
- ε_X = 0.5: foreigners' reaction to our prices
- ε_M = 0.4: our reaction to import prices

What we want: the currency falls; is $\omega_X \varepsilon_X + \omega_M (\varepsilon_M - 1)$ above 0 (better) or below 0 (worse)?

1. $\omega_X = 40 \div 100 = 0.40$; $\omega_M = 60 \div 100 = 0.60$
   Imports are the bigger slice, so they count for more.
2. $0.40 \times 0.5 + 0.60 \times (0.4 - 1) = 0.20 - 0.36 = -0.16$
   Export gain 0.20; import bill up 0.36 because we barely cut back.

-0.16 is below 0, so the trade balance worsens.

So: **Weak reactions, so habits barely change and only the higher import price shows up.**



WORKED EXAMPLE

Will the depreciation help? Case 2: responsive buyers

What we know

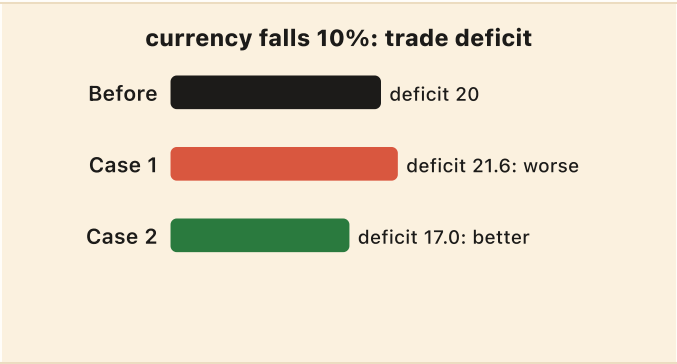
- **X** = 40 billion: exports
- **M** = 60 billion (a deficit of 20): imports
- ε_X = 1.2: foreigners' reaction to our prices
- ε_M = 0.7: our reaction to import prices

What we want: the currency falls; is $\omega_X \varepsilon_X + \omega_M (\varepsilon_M - 1)$ above 0 (better) or below 0 (worse)?

1. $\omega_X = 40 \div 100 = 0.40$; $\omega_M = 60 \div 100 = 0.60$
   Imports are the bigger slice, so they count for more.
2. $0.40 \times 1.2 + 0.60 \times (0.7 - 1) = 0.48 - 0.18 = +0.30$
   Export gain 0.48 beats the small rise in the import bill, 0.18.

+0.30 is above 0, so the trade balance improves.

So: **Foreigners react strongly to our cheaper goods, and that export gain beats the slightly bigger import bill.**



TRADE, 3 OF 6

Same example, played out in money

Same country: exports 40, imports 60, currency falls **10%**. All in home currency.

Case 1 (ε_X 0.5, ε_M 0.4)

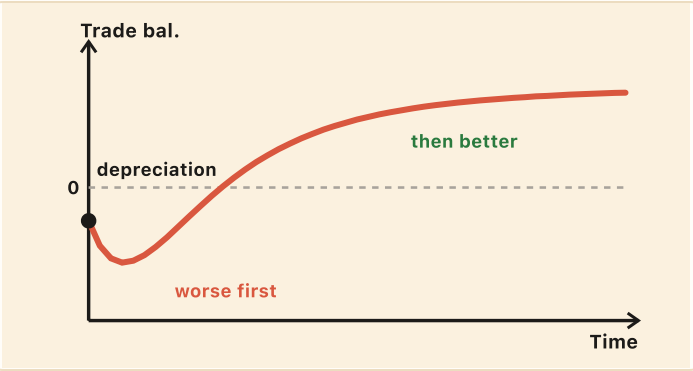
Exports: volume +5% (0.5 × 10%) → 40 + 2 = **42**
Imports: price +10%, volume -4% (0.4 × 10%), so bill up about 6% → 60 + 3.6 = **63.6**
Deficit: 20 → **21.6**. Worse by 1.6.

Case 2 (ε_X 1.2, ε_M 0.7)

Exports: volume +12% → 40 + 4.8 = **44.8**
Imports: price +10%, volume -7%, bill up about 3% → 60 + 1.8 = **61.8**
Deficit: 20 → **17.0**. Better by 3.0.

The link: change = formula value × total trade × 10%. Case 1: -0.16 × 100 × 10% = -1.6. Case 2: +0.30 × 100 × 10% = +3.0.

So: The Marshall-Lerner number is just the change in the trade balance per 1% depreciation, per unit of trade.



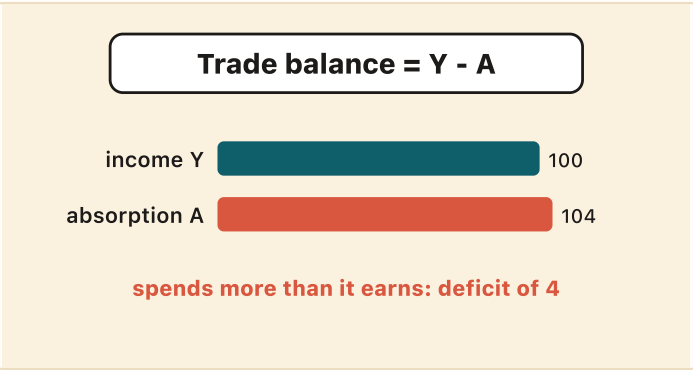
TRADE, 4 OF 6

The J-curve: worse before better

Right after a depreciation, the trade balance often **worsens**. Import orders were signed months ago at fixed volumes, and now each one costs more domestic currency. Buyers take time to switch suppliers.

Over time volumes adjust: foreigners buy more of the cheaper exports, locals buy fewer imports. Elasticities grow, Marshall-Lerner kicks in and the balance improves. Plotted against time, the path looks like a J.

So: Short-run elasticities are low, so the trade balance dips before it improves: the J-curve.



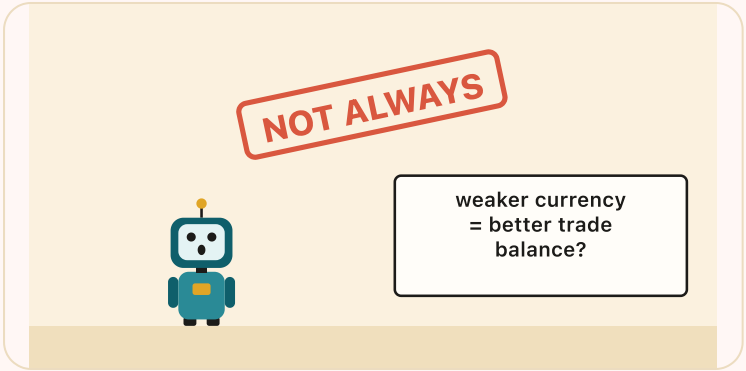
TRADE, 5 OF 6

The absorption approach

The **absorption approach** looks at the whole economy. **Absorption** A is total domestic spending: consumption + investment + government spending. Trade balance = income Y minus absorption A. Equivalently, saving minus investment.

So a depreciation improves the trade balance only if it raises income relative to spending. With spare capacity, it can switch demand to home-made goods and lift output. At full employment output cannot rise, so spending must fall instead.

So: A trade deficit means a country spends more than it earns; fixing it needs income up or absorption down.



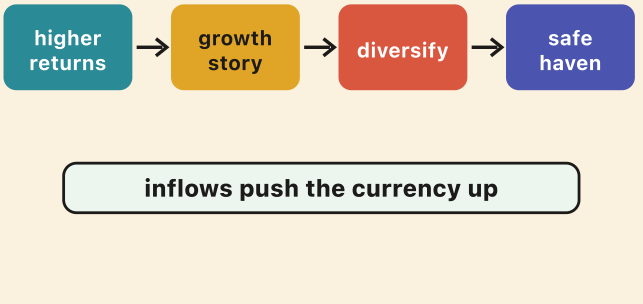
EXAM TRAP

Trade, 6 of 6: depreciation traps

Pinky: "Let the rupee fall 10% and the deficit vanishes." Sharma ji: "Data dikhao."

- **"Depreciation always improves the trade balance."** Wrong: only if Marshall-Lerner holds, and often after a J-curve dip.
- **Forgetting the weights.** With a deficit, $\omega_M > \omega_X$. The shortcut $\varepsilon_X + \varepsilon_M > 1$ is only exact for balanced trade.
- **Elasticity signs.** Plug elasticities in as positive numbers.
- **Absorption.** If the economy is at full employment, depreciation alone mainly raises prices.

So: Check elasticities, timing and spare capacity before believing a depreciation will cure a deficit.



CAPITAL, 1 OF 4

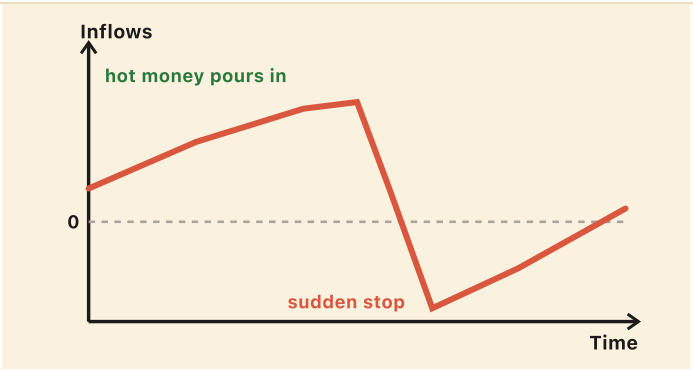
Why money crosses borders

Capital moves to where it expects the best **risk-adjusted return**: higher interest rates, faster growth, cheap assets, diversification, or safety in a storm.

- **FDI** (building a factory, buying a firm): slow, sticky.
- **Portfolio flows** (shares, bonds): quick in, quick out.

A country with a current account deficit must finance it with a financial account surplus: foreigners buying its assets or lending to it. Strong inflows bid up the currency.

So: Current account deficits are financed by capital inflows; those inflows lift the currency while they last.



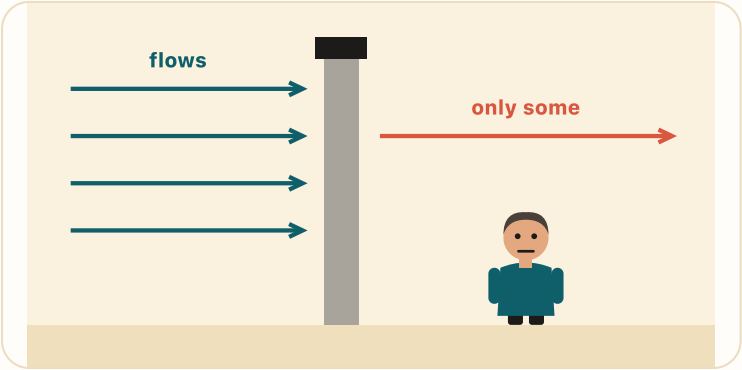
CAPITAL, 2 OF 4

Hot money and the sudden stop

Hot money is short-term portfolio money chasing yield. It arrives fast and leaves faster.

A **sudden stop** is an abrupt halt or reversal of inflows. The currency slumps, reserves drain, rates spike and the current account deficit must close almost overnight. In 2013 a hint that the US Fed would slow bond buying (the taper tantrum) pulled money out of emerging markets; the rupee fell past 68 per dollar, a record low then.

So: Hot money flatters a currency on the way in and wrecks it on the way out.



CAPITAL, 3 OF 4

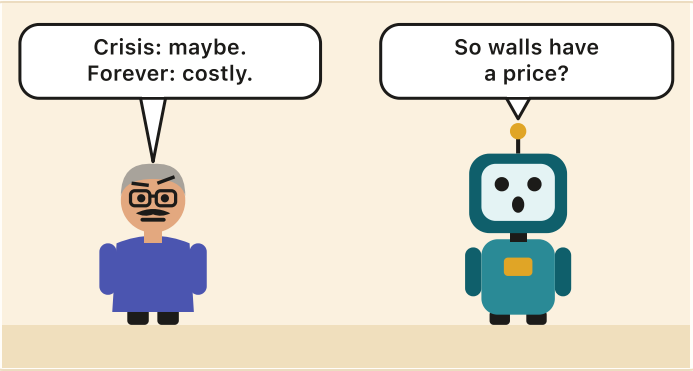
Why governments use capital controls

Capital controls are limits on moving money in or out: taxes, caps, approvals or bans. Common objectives:

- reduce volatility of domestic asset prices and the currency
- keep a fixed exchange rate while running own monetary policy (the trinity)
- keep domestic interest rates low
- protect strategic industries from foreign ownership

India caps residents' outward remittances under the LRS at USD 250,000 a year and limits foreign ownership in some sectors.

So: Capital controls trade efficiency for stability, and are most defensible during a crisis.



CAPITAL, 4 OF 4

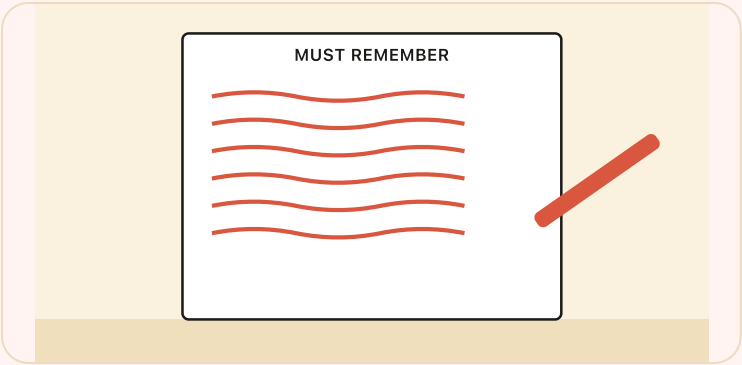
The cost of controls

Controls are not free:

- capital goes to less productive uses, so growth can suffer
- foreign investors demand a higher return, or stay away
- people find ways around them (mis-invoicing trade, informal channels)
- they can delay reforms the economy needs

Malaysia imposed outflow controls in 1998 and pegged the ringgit at 3.80 per dollar to buy time during the Asian crisis. Even the IMF now accepts temporary controls in a crisis.

So: Controls can buy time in a crisis but distort capital allocation if kept for long.

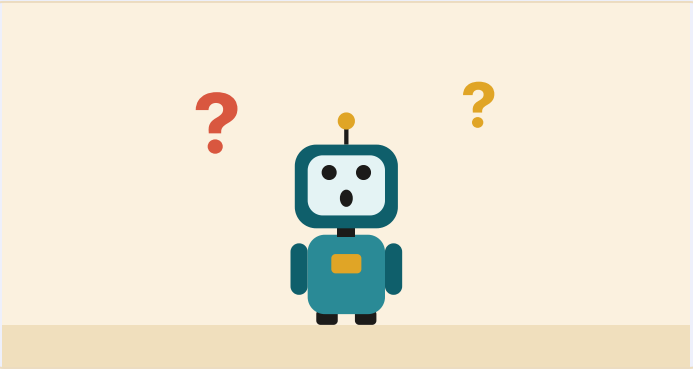


SHARMA JI'S RED PEN

6 things to remember

1. A/B = units of A per 1 B. USD/EUR 1.1000 = 1.10 dollars per euro. Dealers write EURUSD. What is the 1?
2. Real (d/f) = S(d/f) × CPI_f/CPI_d. Up = more competitive domestic exports.
3. Sell side: dealer banks. Spot T+2; forward: rate now, swap later; FX swap: spot plus opposite forward.
4. Least to most flexible: no own currency, currency board, fixed parity, target zone, crawls, managed float, free float.
5. Marshall-Lerner: $\omega_X\varepsilon_X + \omega_M(\varepsilon_M - 1) > 0$. J-curve: worse, then better. Absorption: TB = income minus spending.
6. Hot money leaves fast. Controls aim at stability, pegs, low rates, strategic firms.

So: Scribbled by Sharma ji on the chapter summary.



EXAM QUESTION 1 OF 8

If INR/USD moves from 85.0000 to 86.0000, which statement is correct under the CFA convention?

A. The rupee has appreciated against the dollar

B. The dollar has appreciated against the rupee

C. The quote is indirect from India's point of view

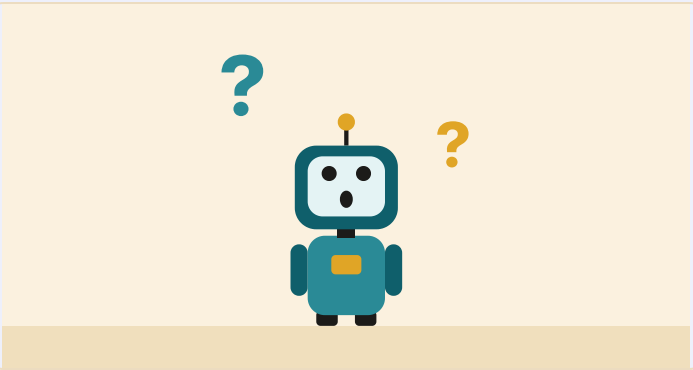
Answer on the next page.

So: **Pick an answer, then turn the page.**

ANSWER

If INR/USD moves from 85.0000 to 86.0000, which statement is correct under the CFA convention?

Answer: B. INR/USD gives rupees per 1 dollar, so the dollar (base) is the 1. More rupees per dollar means the dollar appreciated. A is the reverse. C is wrong: rupees per dollar has the domestic currency as the price currency, which is a direct quote for an Indian.



EXAM QUESTION 2 OF 8

From Brazil's view, the nominal BRL/USD rate rises from 5.0000 to 5.5000 over a year while Brazil's CPI rises from 100 to 110 and US CPI from 100 to 104. The real BRL/USD rate is closest to:

A. 5.2000

B. 5.8173

C. 5.5000

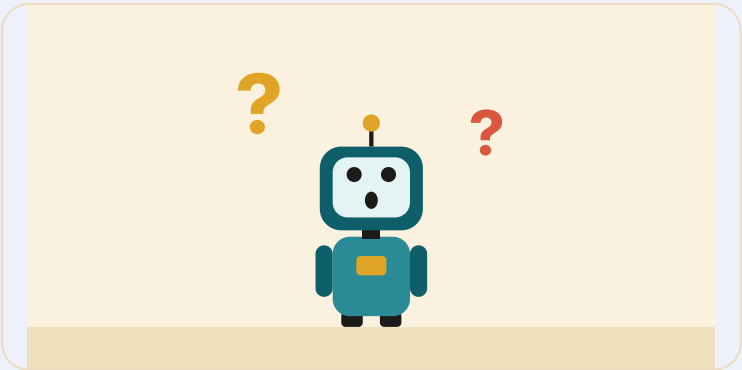
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

From Brazil's view, the nominal BRL/USD rate rises from 5.0000 to 5.5000 over a year while Brazil's CPI rises from 100 to 110 and US CPI from 100 to 104. The real BRL/USD rate is closest to:

Answer: A. Real (d/f) = S(d/f) × CPI_f/CPI_d = 5.5000 × 104/110 = 5.2000. B inverts the price ratio (domestic over foreign). C ignores inflation entirely: that is the nominal rate.



EXAM QUESTION 3 OF 8

Which participant is most likely to be on the sell side of the FX market?

A. A pension fund buying foreign bonds

B. A hedge fund betting on the yen

C. A large dealing bank quoting bid and offer prices

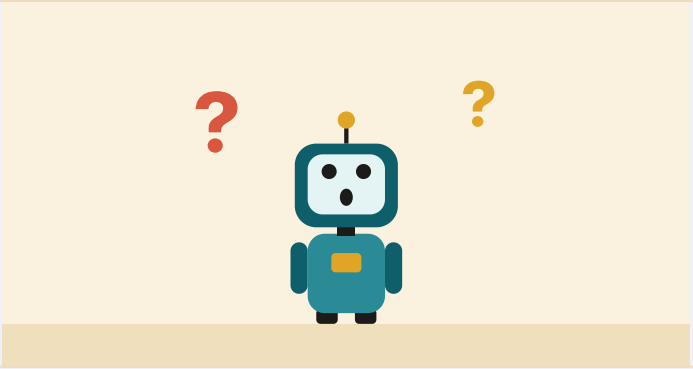
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

Which participant is most likely to be on the sell side of the FX market?

Answer: C. The sell side makes prices: large dealer banks quoting two-way prices and earning the spread. A is a real money account and B a leveraged account; both are buy side.



EXAM QUESTION 4 OF 8

Which exchange rate regime gives a country the least monetary policy independence?

A. A target zone

B. A currency board

C. A managed float

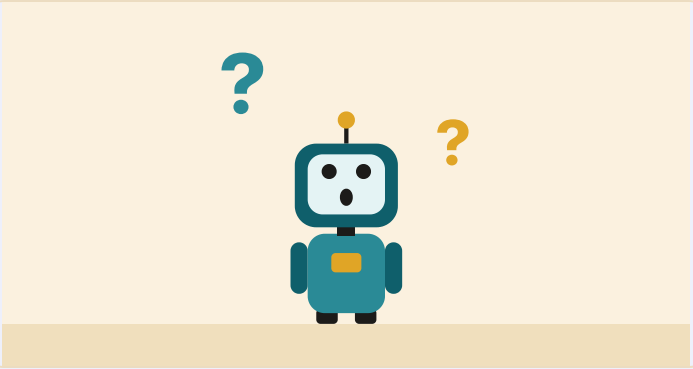
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

Which exchange rate regime gives a country the least monetary policy independence?

Answer: B. A currency board legally fixes the rate and fully backs the money base with foreign reserves, so money supply follows reserves. A target zone allows a band of about plus or minus 2% and some discretion. A managed float has no fixed target at all.



EXAM QUESTION 5 OF 8

A country's exports are 30 and imports 70. Price elasticities of demand are 0.9 for exports and 0.5 for imports. After a depreciation, its trade balance will most likely:

A. improve, because 0.9 + 0.5 > 1

B. stay unchanged, because the effects cancel

C. worsen, because the Marshall-Lerner condition fails

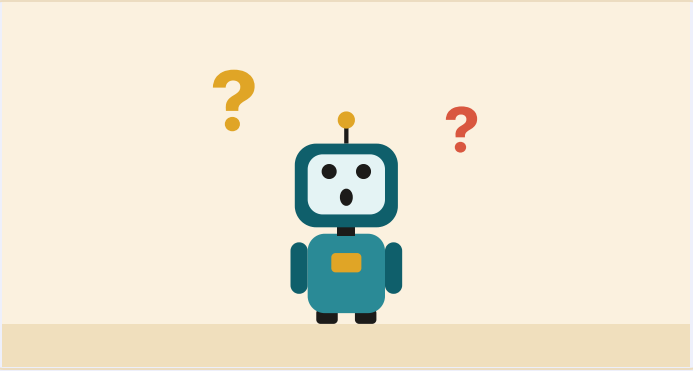
Answer on the next page.

So: **Pick an answer, then turn the page.**

ANSWER

A country's exports are 30 and imports 70. Price elasticities of demand are 0.9 for exports and 0.5 for imports. After a depreciation, its trade balance will most likely:

Answer: C. $\omega_X = 0.30$, $\omega_M = 0.70$. $0.30 \times 0.9 + 0.70 \times (0.5 - 1) = 0.27 - 0.35 = -0.08 < 0$, so the balance worsens. A uses the balanced-trade shortcut, which ignores the large import weight. B would need a value of exactly zero.



EXAM QUESTION 6 OF 8

The J-curve describes a trade balance that, after a depreciation:

A. first worsens and then improves

B. improves at once and then fades

C. worsens permanently because imports are inelastic

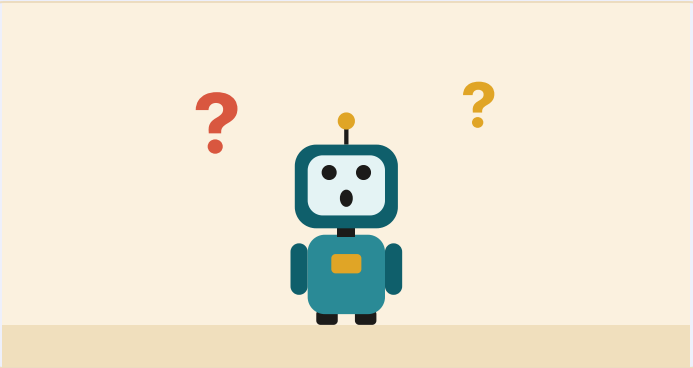
Answer on the next page.

So: **Pick an answer, then turn the page.**

ANSWER

The J-curve describes a trade balance that, after a depreciation:

Answer: A. Short-run volumes are locked in by existing contracts while import prices rise, so the balance dips; volumes adjust later and it improves. B reverses the order. C confuses short-run inelasticity with a permanent state.



EXAM QUESTION 7 OF 8

Under the absorption approach, a depreciation in an economy at full employment improves the trade balance only if:

A. domestic spending falls relative to income

B. export prices rise in foreign currency

C. the central bank cuts interest rates

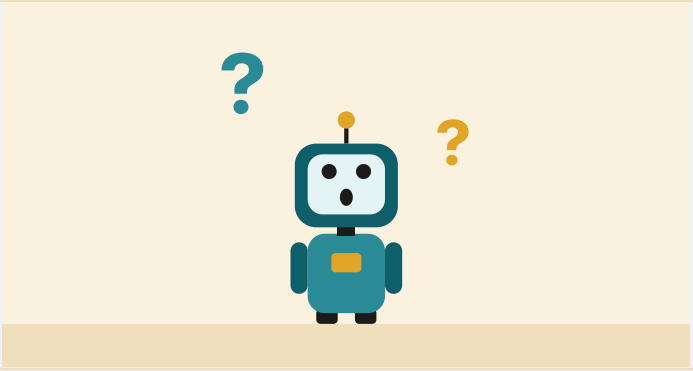
Answer on the next page.

So: **Pick an answer, then turn the page.**

ANSWER

Under the absorption approach, a depreciation in an economy at full employment improves the trade balance only if:

Answer: A. Trade balance = income minus absorption. At full employment income cannot rise, so absorption must fall. B would make exports less competitive. C would raise spending, which pushes absorption up and worsens the balance.



EXAM QUESTION 8 OF 8

Which is least likely an objective of capital controls?

A. Keeping a fixed exchange rate while running independent monetary policy

B. Reducing volatility of domestic asset prices

C. Raising the return foreign investors demand

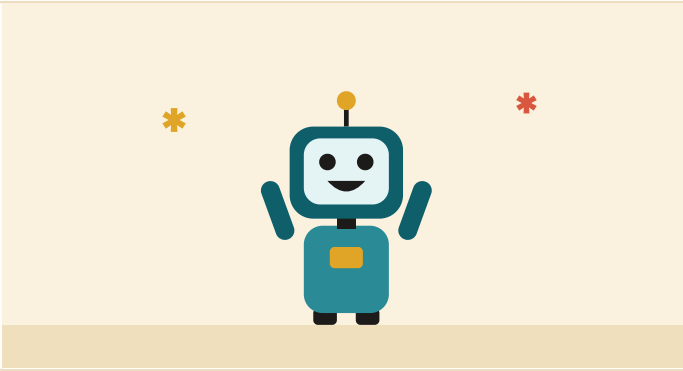
Answer on the next page.

So: **Pick an answer, then turn the page.**

ANSWER

Which is least likely an objective of capital controls?

Answer: C. Controls are used to defend pegs while keeping monetary independence (A) and to calm asset prices (B). A higher return demanded by investors is a cost of controls, not an aim.

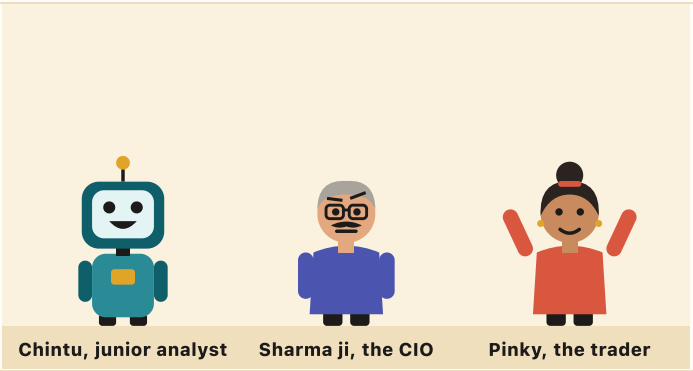


CHAPTER 7 DONE

Capital flows and FX, done

You can read an A/B quote, deflate a nominal rate into a real one, name who trades FX and why, place nine regimes on the ladder, test a depreciation with Marshall-Lerner, sketch the J-curve, use the absorption identity, and explain hot money and capital controls. Sharma ji nods once. That is a lot.

So: **Swipe on, or jump to another chapter from the Menu.**



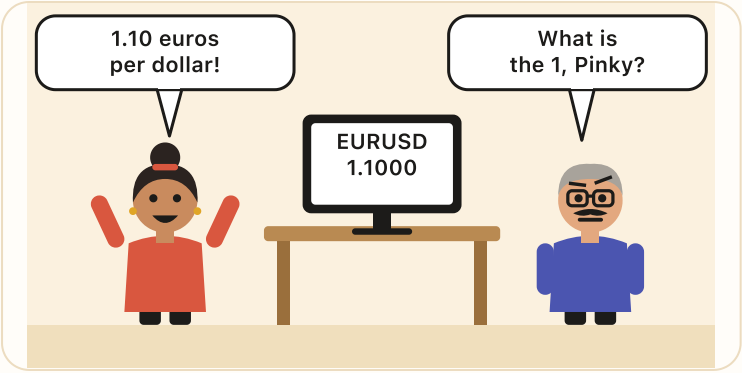
BEFORE YOU START

The cast, and words this chapter uses

A Mumbai investment firm. Sharma ji trusts nothing without data ("Data dikhao"); Pinky is sure of everything, especially when wrong.

- **Appreciate / depreciate:** a currency gains or loses value against another.
- **Pip:** the smallest quoted step: 0.0001 for most pairs, 0.01 for yen pairs.
- **Annual interest rate:** a rate quoted per year; for 90 days you earn only 90/360 of it.
- **Arbitrage:** a riskless profit from mismatched prices, with no money of your own at stake.

So: **You can start this chapter cold. Everything it needs is on this card.**



STORY

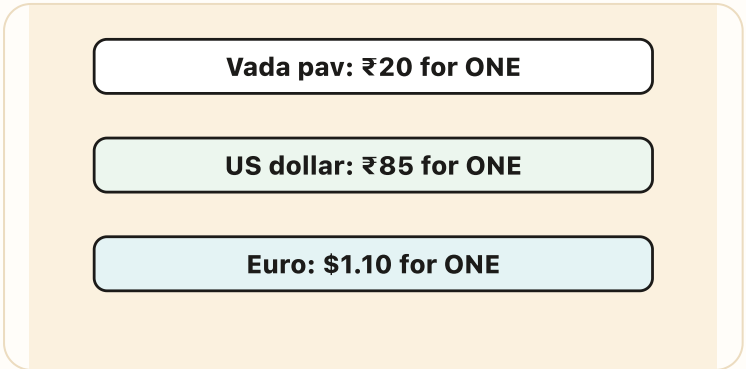
Pinky reads the screen backwards

Pinky's screen shows **EURUSD 1.1000**. She shouts: "1.10 euros per dollar, buy euros, they are cheap!"

Sharma ji does not look up. "What is the 1, Pinky?"

The 1 is the euro. One euro costs 1.10 dollars. Pinky had the whole trade upside down. In this chapter every number hangs on that one question, so we ask it on every card.

So: Before any FX calculation, find the currency that is the 1.



STORY

A currency quote is just a price tag

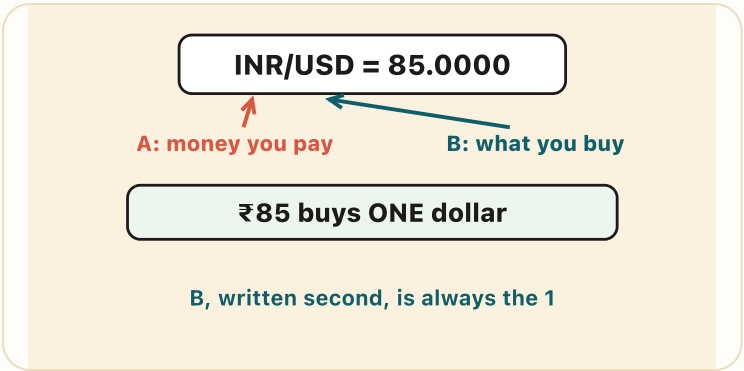
The stall board says **vada pav ₹20**. Nobody is confused: you pay 20 rupees, you get ONE vada pav.

A currency quote is the same price tag. The only twist: the thing you buy is another currency.

- ₹85 for one dollar: you pay 85 rupees, you get ONE dollar. The dollar is the vada pav here.
- $1.10 for one euro: you pay 1.10 dollars, you get ONE euro.

Every quote answers one question: how much of one money do I hand over to get ONE unit of the other?

So: A quote is a price tag: so much of one money for ONE unit of another.



FORMULA

Reading A/B: what A and B mean

`A/B = how many A you pay for ONE B`

Symbol	In plain words	Example
A	the money you **pay with**. CFA name: price currency	INR in INR/USD
B	the thing you **buy**, always ONE unit. CFA name: base currency	USD in INR/USD
the number	how many A one B costs	85: ₹85 buys $1

The bit people find weird: the 1 is the currency written **second**. Read the slash as "per one": USD/EUR = 1.1000 is dollars per one euro. If the number goes up, ONE B costs more A: B got stronger, A weaker.

So: A/B = how many A you pay for ONE B. B is the 1.



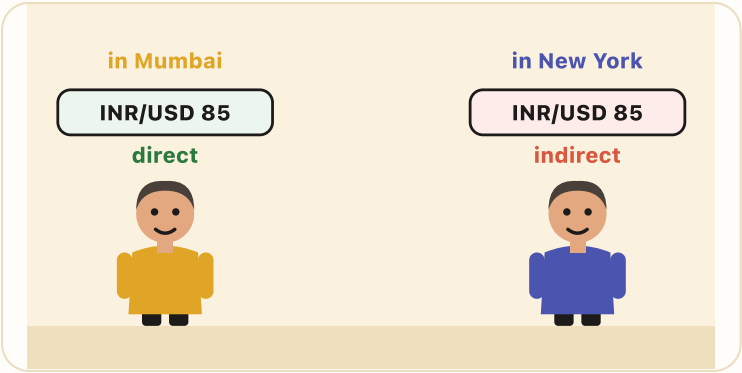
NOTATION

CFA order versus dealer order

Meaning	CFA (A/B)	Dealer screen
1 EUR = 1.1000 USD	USD/EUR	EUR/USD, EURUSD
1 USD = 85.0000 INR	INR/USD	USD/INR, USDINR
1 USD = 150.0000 JPY	JPY/USD	USD/JPY, USDJPY
1 GBP = 1.2500 USD	USD/GBP	GBP/USD, GBPUSD

The CFA writes the money you pay first and the thing you buy (the 1) second. Dealers do the opposite: the 1 first. Exam questions use the CFA order; real screens use the dealer order.

So: **Same number, opposite slash: always translate a quote into "1 of this = x of that" before you calculate.**



NOTATION

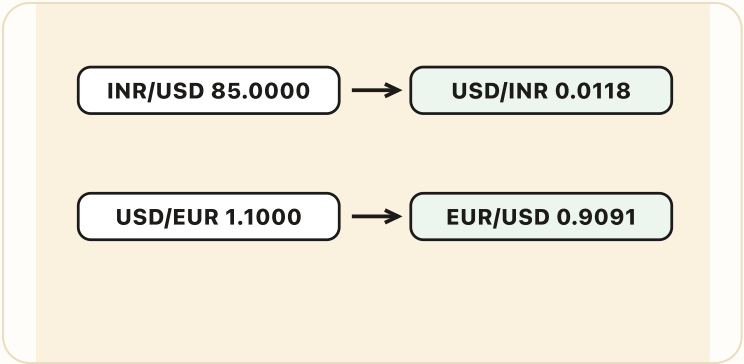
Direct and indirect quotes

Direct and indirect depend on **whose home** you stand in. Domestic currency = your home money (the rupee, for an Indian).

- **Direct quote:** your home money is A, the money you pay. For an Indian, INR/USD = 85.0000: ₹85 for one dollar. Reads like a shop price.
- **Indirect quote:** your home money is B, the thing being bought. For an Indian, USD/INR = 0.0118: 0.0118 dollars for one rupee.

So the same INR/USD quote is direct in Mumbai and indirect in New York.

So: Direct = home money is the money you pay (A); indirect = home money is the 1 (B).



WORKED EXAMPLE

Flip a quote

What we know

- **INR/USD** = 85.0000: rupees you pay for ONE dollar
- **USD/EUR** = 1.1000: dollars you pay for ONE euro

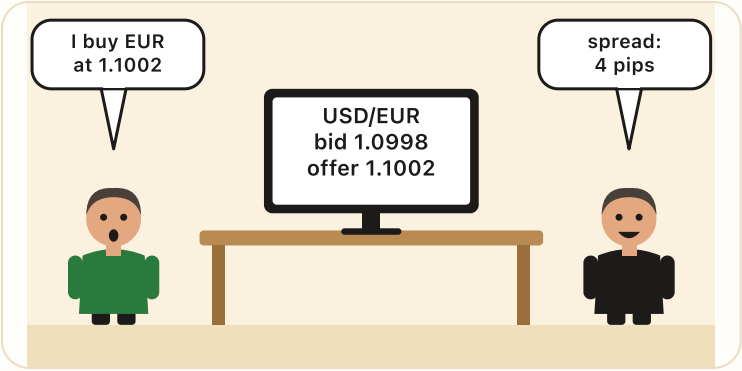
What we want: USD/INR (dollars for ONE rupee) and EUR/USD (euros for ONE dollar).

1. USD/INR = 1 ÷ 85.0000 = 0.0118
   ₹85 buys one dollar, so ₹1 buys one 85th of a dollar: about 1.2 US cents.
2. EUR/USD = 1 ÷ 1.1000 = 0.9091
   $1.10 buys one euro, so $1 buys 1 ÷ 1.10 of a euro.

USD/INR = 0.0118 dollars per rupee; EUR/USD = 0.9091 euros per dollar.

Sense check: a rupee is worth far less than a dollar, so one rupee buys only a small slice of a dollar.

So: **To flip A/B into B/A (swap who pays and what is bought), divide 1 by the quote.**



NOTATION

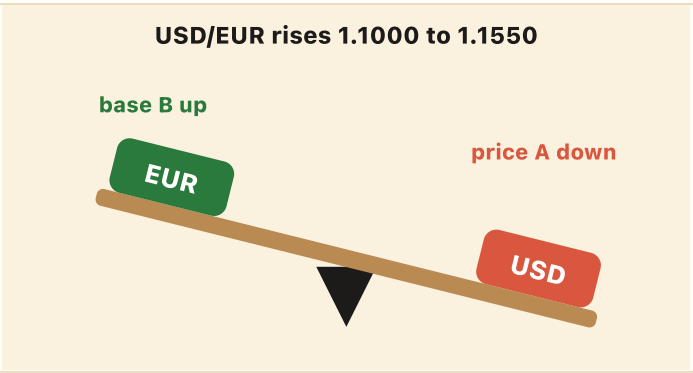
Bid and offer: the dealer always wins

Like the airport money counter, a dealer shows two prices for ONE unit of B. Here USD/EUR: dollars per one euro.

- **Bid 1.0998:** what the dealer **pays you** for one euro. You sell euros here.
- **Offer (ask) 1.1002:** what **you pay** the dealer for one euro. You buy euros here.

You always get the worse side; the 4-pip gap (the spread) is the dealer's income. Flipping a two-way quote swaps the sides: EUR/USD bid = 1 ÷ 1.1002 = 0.9089, offer = 1 ÷ 1.0998 = 0.9093.

So: You buy the 1 (B) at the offer and sell it at the bid; the offer is always the higher number.



PICTURE IT

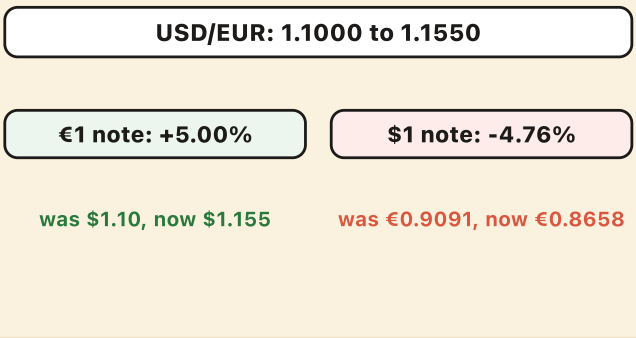
The currency see-saw

USD/EUR = dollars you pay for ONE euro. It rises from 1.1000 to 1.1550: one euro now costs more dollars.

- The euro (B, the 1) got dearer: it **appreciated**.
- Each dollar (A, the money paid) now buys less euro: it **depreciated**.

If the quote falls, flip the see-saw. The quote is the price of B, so B always moves the same way as the number.

So: A/B up: B up, A down. A/B down: B down, A up.



STORY

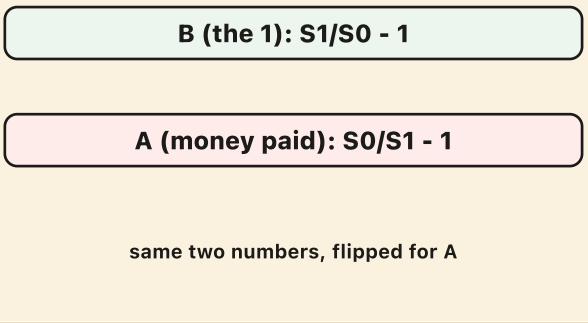
Hold one note of each

Pinky keeps a 1-euro note and a 1-dollar note in her wallet. USD/EUR moves from 1.1000 to 1.1550.

- **Euro note:** it fetched $1.10, now $1.155. Gain 0.055 on 1.10 = **+5.00%**.
- **Dollar note:** it fetched 1 ÷ 1.10 = €0.9091, now 1 ÷ 1.155 = €0.8658. Loss = **-4.76%**.

One move, two different sizes. Each currency's change is measured by what ONE note of it buys, so the dollar uses the flipped quote.

So: **Measure each currency by what one unit of it buys; the two changes are never equal and opposite.**



FORMULA

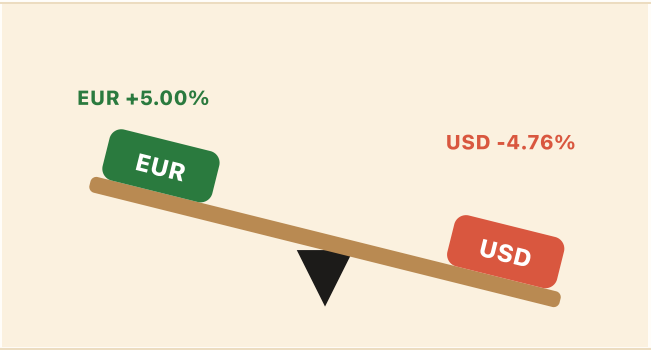
Percentage change in each currency

$$\text{B (the 1)}: S_1 \div S_0 - 1$$
$$\text{A (money paid)}: S_0 \div S_1 - 1$$

Symbol	In plain words	Example
S_0	the old A/B quote: A paid for ONE B, before	1.1000 USD/EUR
S_1	the new A/B quote, after	1.1550
B	base currency: the 1, the thing bought	EUR: +5.00%
A	price currency: the money you pay with	USD: -4.76%

Why A is old over new: one unit of A buys 1 ÷ S of B. Its change is $(1 \div S_1) \div (1 \div S_0) - 1$, which tidies to $S_0 \div S_1 - 1$. That is Pinky's dollar note. So first ask: is my currency the 1 (B) or the money paid (A)?

So: B, the 1: new over old. A, the money paid: old over new.



WORKED EXAMPLE

The euro gains, the dollar loses

What we know

- S_0 = 1.1000: old USD/EUR: dollars paid for ONE euro
- S_1 = 1.1550: new USD/EUR
- **B / A** = EUR / USD: the 1 (bought) / the money paid

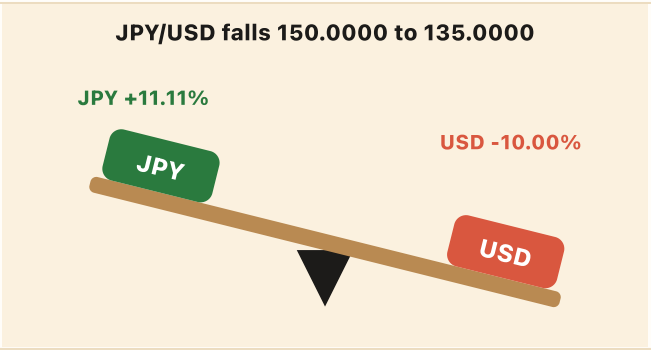
What we want: the % change in the euro and in the dollar.

1. Euro: 1.1550 ÷ 1.1000 - 1 = +5.00%
   The euro is the 1 (B), so new over old: one euro now fetches 5% more dollars.
2. Dollar: 1.1000 ÷ 1.1550 - 1 = -4.76%
   The dollar is the money paid (A), so old over new: one dollar now fetches fewer euros.

Euro appreciated 5.00%; dollar depreciated 4.76%.

Sense check: quote up, so the 1 (euro) up and the dollar down, by less.

So: The 1 moves with the quote; the money paid moves against it, by a different size.



WORKED EXAMPLE

The yen rally

What we know

- S_0 = 150.0000: old JPY/USD: yen paid for ONE dollar
- S_1 = 135.0000: new JPY/USD
- **B / A** = USD / JPY: the 1 (bought) / the money paid

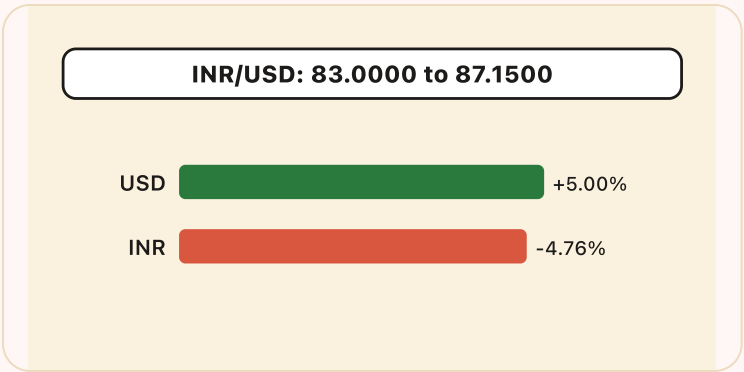
What we want: the % change in the dollar and in the yen.

1. Dollar: 135 ÷ 150 - 1 = -10.00%
   The dollar is the 1 (B), so new over old: one dollar now fetches 10% fewer yen.
2. Yen: 150 ÷ 135 - 1 = +11.11%
   The yen is the money paid (A), so old over new: one yen now fetches more dollars.

Dollar depreciated 10.00%; yen appreciated 11.11%.

Sense check: a dollar costs fewer yen now: dollar down, yen up.

So: **A falling quote means the 1 fell and the money paid rose.**



EXAM TRAP

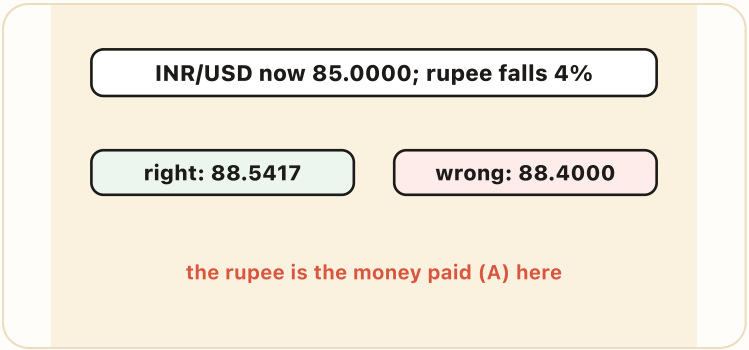
Not equal and opposite

INR/USD (rupees for ONE dollar) rises from 83.0000 to 87.1500. The dollar rose 5.00%. Did the rupee fall 5%? **No.**

In money: ₹100 bought 100 ÷ 83 = $1.2048; now 100 ÷ 87.15 = $1.1474. That is -4.76%, i.e. 83 ÷ 87.15 - 1.

- Read the question: "the rupee against the dollar" asks about the rupee, not the quote.
- Never subtract quotes (87.15 - 83); always divide.

So: Compute each currency's change separately: new/old for the 1, old/new for the money paid.



WORKED EXAMPLE

Working backwards: the rupee falls 4%

What we know

- **S_0** = 85.0000: INR/USD today: rupees for ONE dollar
- **A** = INR: the money paid, falls 4%
- **B** = USD: the 1, the thing bought

What we want: S_1, the new INR/USD.

1. Rupee change = $S_0 \div S_1 - 1 = -0.04$

   The rupee is A, the money paid, so its change is old over new.

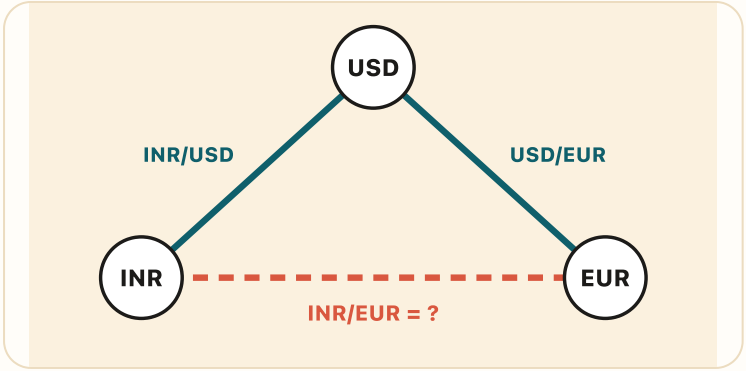
2. $85.0000 \div S_1 = 0.96$, so $S_1 = 85.0000 \div 0.96 = 88.5417$

   ₹1 now buys 4% fewer dollars, so one dollar costs 85 ÷ 0.96 rupees.

New INR/USD = 88.5417, not 85 × 1.04 = 88.4000 (that would be the dollar rising 4%).

Sense check: a weaker rupee means more rupees per dollar, and 88.5417 is above 85.

So: When the money paid (A) moves by x%, divide the old quote by (1 + x), never multiply.



PICTURE IT

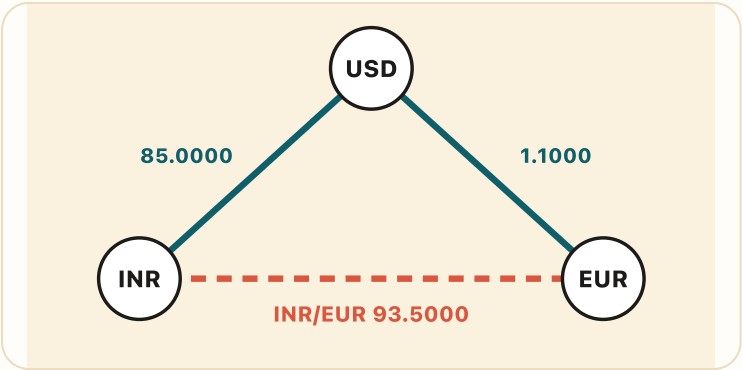
Cross rates: the triangle

A **cross rate** is the rate between two currencies worked out through a third, usually the US dollar.

In money: the bank quotes rupees and euros only against the dollar. One euro costs $1.10. Each of those dollars costs ₹85. So one euro costs 1.10 × 85 = ₹93.50. The dollar was just a stepping stone.

In symbols, cancel the middle currency like a fraction: **INR/USD × USD/EUR = INR/EUR**. If it does not cancel, flip one quote first.

So: Walk through the middle currency, or cancel it like a fraction; both give the same number.



WORKED EXAMPLE

Cross rate 1: rupees per euro

A Mumbai importer pays in euros.

What we know

- **INR/USD** = 85.0000: rupees paid for ONE dollar
- **USD/EUR** = 1.1000: dollars paid for ONE euro

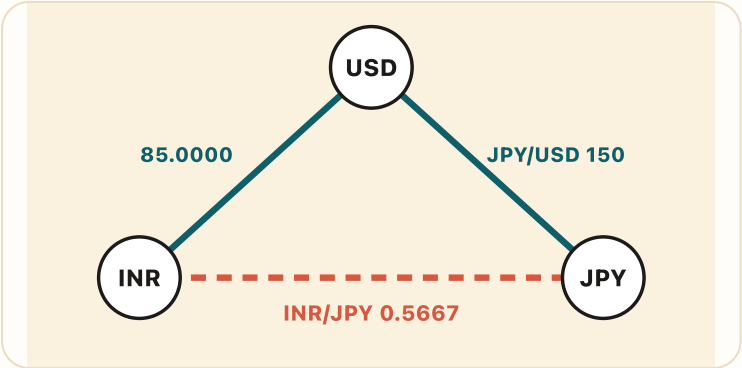
What we want: INR/EUR: rupees paid for ONE euro.

1. One euro = $1.10
   That is what USD/EUR = 1.1000 says.
2. $1.10 × ₹85 each = ₹93.50, so INR/EUR = 93.5000
   Each dollar costs 85 rupees. In symbols: INR/USD × USD/EUR, and the USD cancels.

INR/EUR = 93.5000 rupees for ONE euro.

Sense check: a euro is worth more than a dollar, so it must cost more than 85 rupees.

So: When the middle currency cancels directly, simply multiply.



WORKED EXAMPLE

Cross rate 2: divide when the 1 is shared

What we know

- **INR/USD** = 85.0000: rupees paid for ONE dollar
- **JPY/USD** = 150.0000: yen paid for ONE dollar

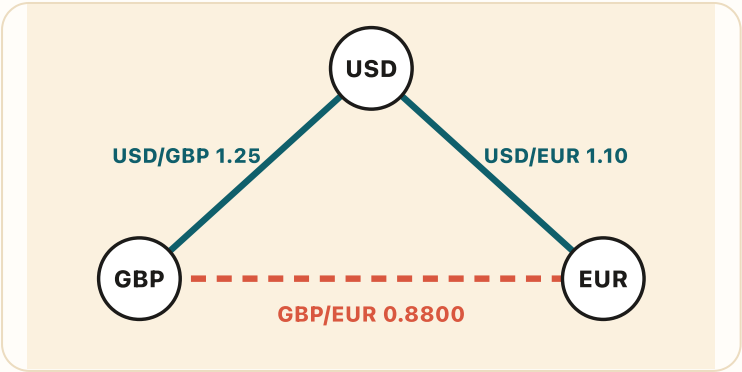
What we want: INR/JPY: rupees paid for ONE yen.

1. 150 yen = $1 = ₹85
   Both quotes price ONE dollar, so the dollar links them.
2. One yen = 85 ÷ 150 = ₹0.5667
   Share ₹85 across 150 yen. In symbols: INR/USD ÷ JPY/USD, because USD is the 1 in both.

INR/JPY = 0.5667 rupees per yen (flipped: JPY/INR = 1.7647).

Sense check: a yen is a small coin, worth well under one rupee.

So: Two quotes with the same 1 (B): divide one by the other.



WORKED EXAMPLE

Cross rate 3: pounds per euro

What we know

- **USD/GBP** = 1.2500: dollars paid for ONE pound
- **USD/EUR** = 1.1000: dollars paid for ONE euro

What we want: GBP/EUR: pounds paid for ONE euro.

1. One euro = $1.10; one pound = $1.25
   Both quotes give a price in dollars (A is USD in both).
2. One euro = 1.10 ÷ 1.25 = 0.8800 pounds
   How many $1.25 pounds make $1.10? Same as GBP/USD (1 ÷ 1.25 = 0.8000) × USD/EUR.

GBP/EUR = 0.8800 pounds per euro (EUR/GBP = 1.1364).

Sense check: the pound ($1.25) is worth more than the euro ($1.10), so a euro costs less than one pound.

So: Two quotes with the same money paid (A): divide, in the order that leaves the pair you want.

$$\text{INR}/\cancel{\text{USD}} \times \cancel{\text{USD}}/\text{EUR} = \text{INR}/\text{EUR}$$

cancel like fractions, then check the order

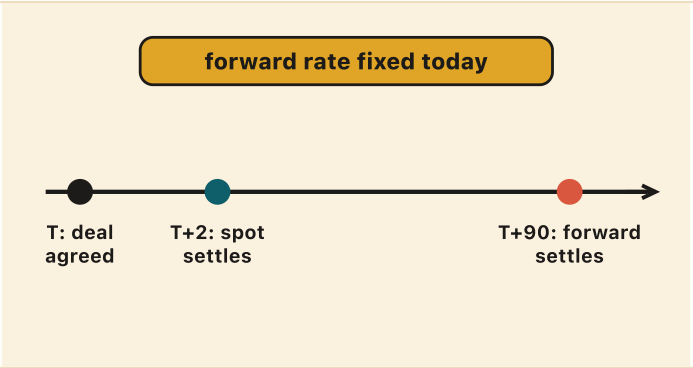


EXAM TRAP

Cross rate traps

- **Multiplying blindly.** INR/USD × JPY/USD gives nonsense units (INR·JPY per USD²). Check the cancellation first.
- **Answer upside down.** The question asks GBP/EUR (pounds for one euro) and you give EUR/GBP. Distractors are often the exact inverse.
- **Dealer order.** "EURUSD" on a screen is USD/EUR in CFA notation.
- **Sense check.** The stronger currency costs more units of the weaker one.

So: **Write the units, cancel the middle currency, and compare your answer with its inverse before choosing.**



PICTURE IT

Spot and forward on one timeline

A **spot** deal: today's price, money swapped about two business days later (T+2).

A **forward** deal: the price is fixed today, but money is swapped later, for example in 90 days. Like booking a train ticket today at today's fare for a trip next month.

The forward rate is not a guess about the future. It comes from today's spot rate and the two interest rates, as the next cards show.

So: **Forward = price agreed now, money swapped later.**

F = S + points / 10,000

yen pairs: F = S + points / 100

+ points: B at a premium; - points: discount

FORMULA

Forward points

$$F = S + \text{points} \div 10{,}000$$
$$\text{(yen pairs: } F = S + \text{points} \div 100\text{)}$$

Symbol	In plain words	Example
S	spot rate: A paid for ONE B, swapped in 2 days	USD/EUR 1.1000
F	forward rate: A paid for ONE B, fixed today, swapped later	1.1045
points	dealer shorthand for F - S, counted in the quote's last decimal	+45 = +0.0045

Why 10,000: most quotes have 4 decimals, so one point is 0.0001. Yen quotes have 2 decimals, so one point is 0.01: divide by 100.

Plus points: ONE B costs more for later delivery, a forward **premium** on B. Minus points: a forward **discount**.

So: Scale the points to the quote's last decimal, then add them to spot, keeping the sign.

USD/EUR 1.1000 + 45 pts = 1.1045

JPY/USD 150.0000 - 85 pts = 149.1500

USD/GBP 1.2500 - 31 pts = 1.2469

WORKED EXAMPLE

Points into forward rates

What we know

- **USD/EUR** = spot 1.1000, points +45: $ per ONE euro
- **JPY/USD** = spot 150.0000, points -85: yen per ONE $
- **USD/GBP** = spot 1.2500, points -31: $ per ONE pound

What we want: each 3-month forward rate F.

1. USD/EUR: 1.1000 + 45 ÷ 10,000 = 1.1045
   4-decimal quote, so one point = 0.0001: +0.0045.
2. JPY/USD: 150.0000 - 85 ÷ 100 = 149.1500
   Yen quote, 2 decimals, so one point = 0.01: -0.85.
3. USD/GBP: 1.2500 - 31 ÷ 10,000 = 1.2469
   Minus points: the forward is below spot.

Forwards: 1.1045, 149.1500, 1.2469. Euro at a premium; dollar (vs yen) and pound at a discount.

So: The sign of the points says whether the 1 (B) is at a premium (+) or a discount (-).

premium on B = F/S - 1

F > S
B at a premium

F < S
B at a discount

A is always on the other side

FORMULA

Forward premium or discount

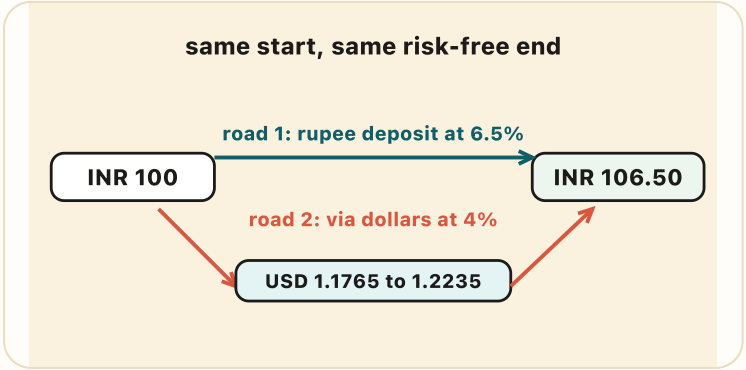
Forward premium on B = F ÷ S − 1

Symbol	In plain words	Example
F	forward rate: A paid for ONE B on the later date	USD/EUR 1.1045
S	spot rate: A paid for ONE B now	1.1000
B	the 1, the thing bought	EUR: 1.1045 ÷ 1.1000 - 1 = +0.409%

- **Above 0:** ONE B costs more for later delivery: B is at a **forward premium**.
- **Below 0:** B is at a **forward discount**.

A is always on the other side: if the euro is at a premium against the dollar, the dollar is at a discount against the euro.

So: Forward premium or discount = F ÷ S - 1, and it belongs to the 1 (B).



STORY

Two roads for Chintu's ₹100

Chintu has ₹100 to park for one year. Rupee FD: 6.50%. Dollar deposit: 4.00%. Today ₹85 buys one dollar.

- **Road 1, stay home:** ₹100 grows to **₹106.50**.
- **Road 2, go abroad:** ₹100 ÷ 85 = $1.1765. At 4% it grows to $1.2235. He locks TODAY the rate for changing it back in a year (a forward).

Which locked rate makes both roads pay the same? ₹106.50 ÷ $1.2235 = **₹87.0433 per dollar**.

Any other rate and one road is free money with zero risk; traders would pile in until it closed. So the market forward must be 87.0433. That is **covered interest rate parity**.

So: **The forward is the rate that makes "stay home" and "go abroad and lock the way back" pay exactly the same.**

$$F = S \times (1 + iA \times t) / (1 + iB \times t)$$

iA = rate on A, the money you pay with: on top

iB = rate on B, the 1: below

t = days / 360

higher-rate currency: forward discount

FORMULA

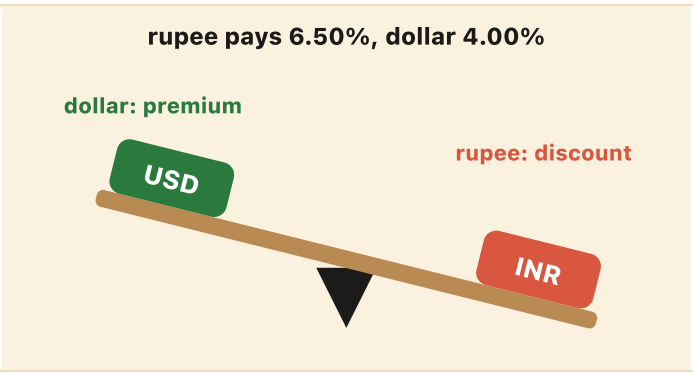
Covered interest rate parity

$$F = S \times (1 + i_A t) \div (1 + i_B t)$$

Symbol	In plain words	Example
A, B	in the quote A/B: A = the money you pay with, B = the thing bought (the 1)	INR/USD: A = ₹, B = $
S, F	A/B quotes, A paid for ONE B: S = spot (today), F = forward (later, price fixed today)	85.0000, 87.0433
i_A, i_B	yearly interest rate on A (money paid) and on B (the 1)	₹ 6.50%, $ 4.00%
t	days ÷ 360: slice of a year (CFA year = 360 days)	1 year = 1; 90 days = 0.25

Chintu's two roads: ₹1 grows to $1 + i_A \times t$ at home, \$1 to $1 + i_B \times t$ abroad. $85 \times 1.065 \div 1.04 = 87.0433$.

So: Rate of A on top, rate of B below, both times days/360.



THE WEIRD BIT

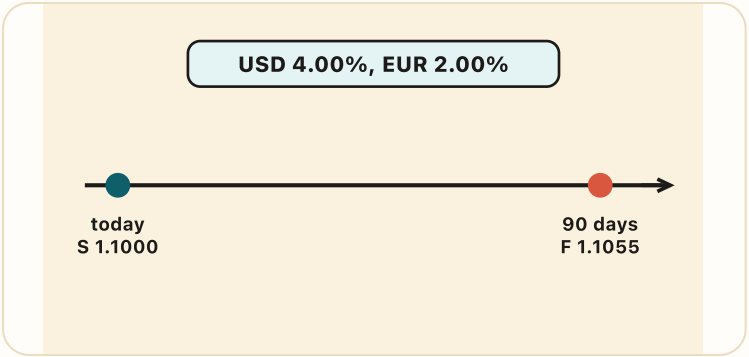
High interest, cheap forward

Most people expect the currency that pays **more** interest to be **dearer** in the forward. It is the opposite.

The rupee pays 2.5% more interest than the dollar. If Chintu could also change back at today's 85, Road 2 would lose and everyone would pile into rupee FDs for free. So the forward gives that extra interest back: one dollar costs 87.0433 rupees in a year instead of 85, 2.40% more.

The higher-rate currency (rupee) trades at a **forward discount**; the lower-rate one (dollar) at a **forward premium**.

So: Higher interest rate today, weaker rate in the forward: the two cancel, so no road is a free lunch.



WORKED EXAMPLE

CIP: a 3-month USD/EUR forward

What we know

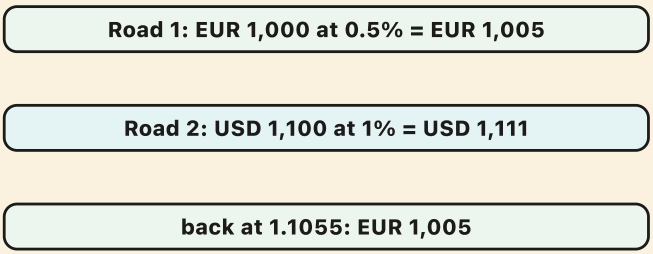
- **S** = 1.1000: USD/EUR spot: dollars for ONE euro
- $\mathbf{i_A}$ = 4.00% a year: dollar rate (A = USD, the money paid)
- $\mathbf{i_B}$ = 2.00% a year: euro rate (B = EUR, the 1)
- **t** = 0.25: 90 days ÷ 360

What we want: F, the 90-day forward USD/EUR.

1. Top: 1 + 0.04 × 0.25 = 1.0100. Bottom: 1 + 0.02 × 0.25 = 1.0050
   In 90 days $1 grows to $1.0100 and €1 to €1.0050: a quarter of each yearly rate.
2. F = 1.1000 × 1.0100 ÷ 1.0050 = 1.1055, points +55
   Dollars grow faster, so later on a euro costs more dollars.

F(USD/EUR) = 1.1055. The euro, the lower-rate currency, is at a forward premium.

So: **The higher-rate dollar trades at a forward discount: forward euros cost more dollars than spot euros.**



PLAYED OUT IN MONEY

Same forward, played out in euros

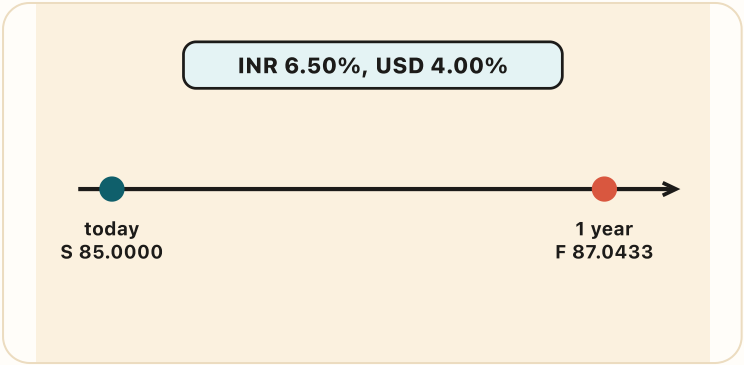
Pinky's Frankfurt cousin has EUR 1,000 for 90 days. Same rates as the last card.

Road 1, stay in euros: 2% × 0.25 = 0.5% → **EUR 1,005.00**.

Road 2, via dollars: 1,000 × 1.1000 = USD 1,100. At 4% × 0.25 = 1% → USD 1,111. Change back at the forward: 1.1055 dollars buy one euro, so divide: 1,111 ÷ 1.1055 ≈ **EUR 1,005.00**.

Same end. The dollar's extra 0.5% of interest is eaten exactly by the dearer euro on the way back.

So: Covered interest parity: a hedged foreign deposit earns exactly what a home deposit earns.



WORKED EXAMPLE

CIP: a 1-year INR/USD forward

What we know

- **S** = 85.0000: INR/USD spot: rupees for ONE dollar
- i_A = 6.50%: rupee rate (A = INR, the money paid)
- i_B = 4.00%: dollar rate (B = USD, the 1)
- **t** = 1: 360 days ÷ 360

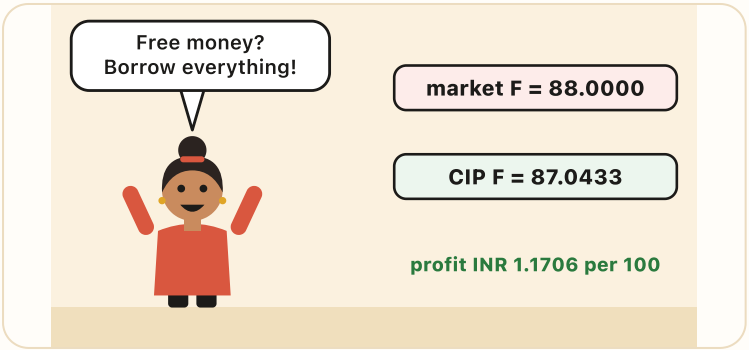
What we want: F, the 1-year forward INR/USD.

1. F = 85.0000 × 1.0650 ÷ 1.0400 = 87.0433
   ₹1 grows to ₹1.065; $1 grows to $1.04. Full-year rates: t = 1.
2. Dollar premium = 87.0433 ÷ 85 - 1 = 2.40%
   One dollar (B) costs 2.40% more for delivery in a year.

F(INR/USD) = 87.0433: rupee at a forward discount, dollar at a 2.40% premium.

Sense check: the same 87.0433 Chintu found with his two roads.

So: **Higher-rate rupee, cheaper rupee in the forward market.**



WORKED EXAMPLE

When the forward is wrong: arbitrage

What we know

- **S** = 85.0000; rupee 6.50%, dollar 4.00%: INR/USD spot
- **F** = 88.0000, fair (CIP) is 87.0433: dealer's 1-year forward

What we want: a riskless profit with no money of her own.

1. Borrow ₹100 at 6.50%: owe ₹106.50 in a year
   The bank's money, not Pinky's own.
2. ₹100 ÷ 85 = $1.1765; at 4% → $1.2235
   Road 2, the dollar road.
3. Sell $1.2235 forward at 88 → ₹107.6706
   The dealer pays ₹88 per dollar, more than the fair 87.0433.
4. Repay ₹106.50; keep ₹1.1706
   Profit locked on day one, whatever the rupee does.

Riskless profit: ₹1.1706 per ₹100 borrowed.

So: **Arbitrageurs selling the overpriced forward push it back to the parity rate.**

$$(F - S)/S \approx (iA - iB) \times t$$

exact: divide by $(1 + iB \times t)$

rate gap in, forward premium out

FORMULA

The approximate forward premium

$$(F - S) \div S \approx (i_A - i_B) \times t$$

Symbol	In plain words	Example
F, S	forward and spot: A paid for ONE B	USD/EUR 1.1055, 1.1000
i_A	yearly rate on A, the money paid	dollar 4%
i_B	yearly rate on B, the 1	euro 2%
t	days ÷ 360	90 days = 0.25

- 3-month USD/EUR (4% vs 2%): approx 2% × 0.25 = 0.50%; exact 0.498%.
- 1-year INR/USD (6.5% vs 4%): approx 2.50%; exact 2.5% ÷ 1.04 = 2.404%.

So: **Premium on B ≈ rate of A minus rate of B, times days/360.**

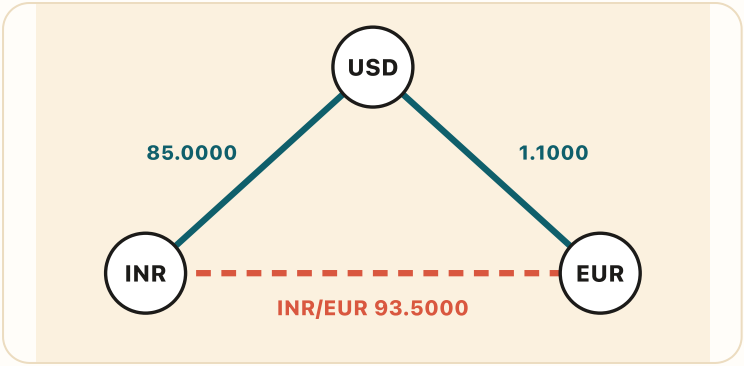


EXAM TRAP

Forward and parity traps

- **Rates upside down.** The rate of A (the money paid) goes on top. Swap them and the premium flips sign.
- **Annual rates for short periods.** Use 4% × 90/360, not 4%.
- **"Higher rate means stronger forward."** Wrong: the higher-rate currency trades at a forward discount.
- **Forward as forecast.** CIP is an arbitrage price, not a prediction of the future spot rate.
- **Points scale.** Divide by 100 for yen pairs.

So: **Rate of A on top, days over 360, and the high-yield currency is at a discount.**



WORKED EXAMPLE

Exam-style, part 1: build the cross spot

A Pune machine-tool maker must pay EUR 200,000 in 180 days and wants to lock the rupee cost today. Step 1: the INR/EUR spot.

What we know

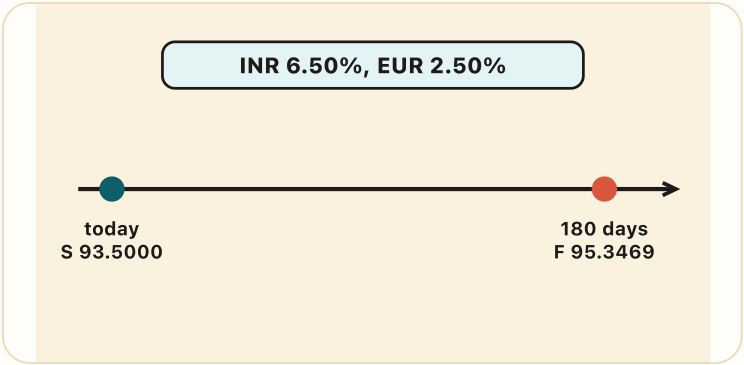
- **INR/USD** = 85.0000: rupees for ONE dollar
- **USD/EUR** = 1.1000: dollars for ONE euro
- **rates** = used in part 2: rupee 6.50%, euro 2.50% a year

What we want: INR/EUR: rupees for ONE euro, today.

1. One euro = $1.10 = 1.10 × ₹85 = ₹93.50
   The dollar is a stepping stone: INR/USD × USD/EUR, the USD cancels.

INR/EUR spot = 93.5000. In INR/EUR, the rupee is A (money paid) and the euro is B (the 1).

So: In multi-step questions, build the cross spot first, then apply parity to that pair.



WORKED EXAMPLE

Exam-style, part 2: the 180-day forward

What we know

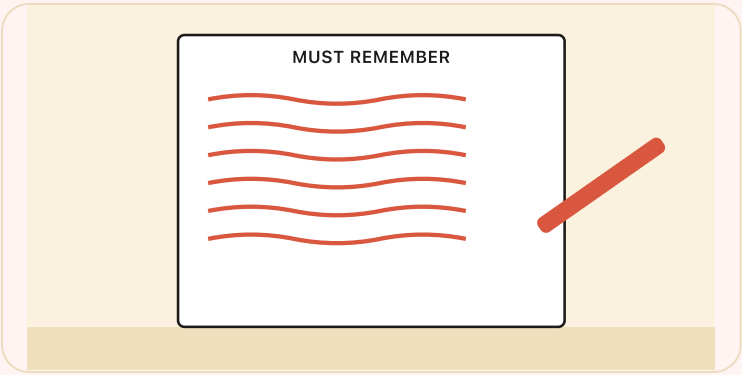
- **S** = 93.5000: INR/EUR spot, ₹ for ONE euro
- $\mathbf{i_A}$, $\mathbf{i_B}$ = 6.50%, 2.50%: rates on ₹ (A, paid) and € (B, the 1)
- **t** = 0.5: 180 days ÷ 360

What we want: F, and the rupee bill for EUR 200,000.

1. 1 + 0.065 × 0.5 = 1.0325; 1 + 0.025 × 0.5 = 1.0125
   Half a year: half of each yearly rate.
2. F = 93.5000 × 1.0325 ÷ 1.0125 = 95.3469
   Rupees grow faster, so a euro costs more rupees later.

F = 95.3469. Locked bill = 200,000 euros × 95.3469 = ₹19,069,380 (euro premium 1.98%).

So: **Cross first, parity second, then multiply by the amount.**

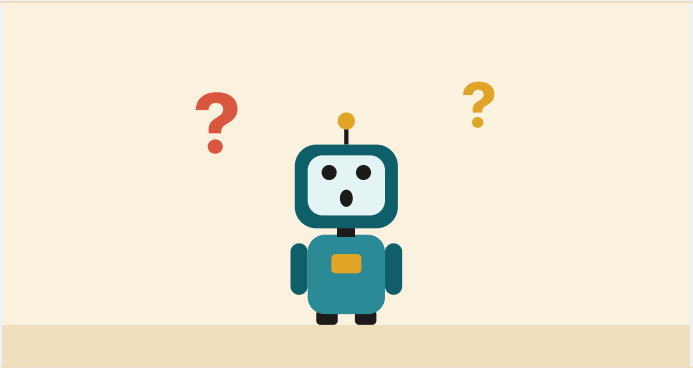


SHARMA JI'S RED PEN

6 things to remember

1. A/B = how many A you pay for ONE B. A = money you pay with (price currency); B = the thing bought, the 1 (base). INR/USD 85 = ₹85 buys $1. Dealers write USDINR.
2. Direct quote: home money is A, the money paid. Flip a quote with 1 ÷ S.
3. B (the 1) changes by S1/S0 - 1; A (money paid) by S0/S1 - 1. Not equal and opposite.
4. Cross rates: cancel the middle currency; invert first if it does not cancel.
5. Points / 10,000 (yen pairs / 100). F > S: B (the 1) at a forward premium.
6. $F = S \times (1 + i_A t)/(1 + i_B t)$: i_A = rate of the money paid, i_B = rate of the 1, t = days/360. Higher-rate currency at a forward discount.

So: **Scribbled by Sharma ji on the chapter summary.**



EXAM QUESTION 1 OF 8

USD/GBP moves from 1.2500 to 1.2000. The percentage change in the US dollar against the pound is closest to:

A. -4.00%

B. +4.17%

C. +4.00%

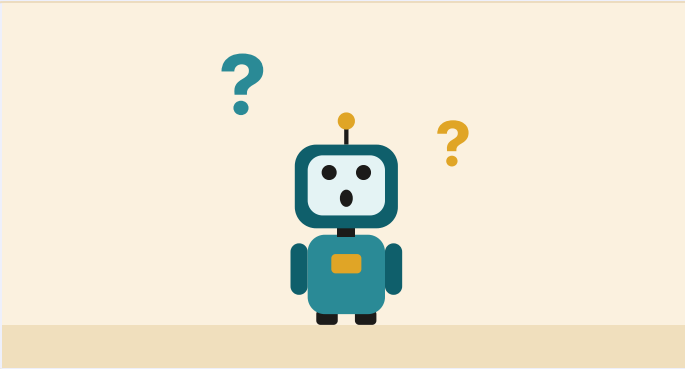
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

USD/GBP moves from 1.2500 to 1.2000. The percentage change in the US dollar against the pound is closest to:

Answer: B. USD is the price currency, so its change is S0/S1 - 1 = 1.25/1.20 - 1 = +4.17%: the dollar appreciated. A is the pound's change (base: 1.20/1.25 - 1). C assumes the two moves are equal and opposite.



EXAM QUESTION 2 OF 8

For an investor based in Japan, which quote is direct?

A. JPY/USD = 150.0000

B. USD/JPY = 0.0067

C. EUR/USD = 0.9091

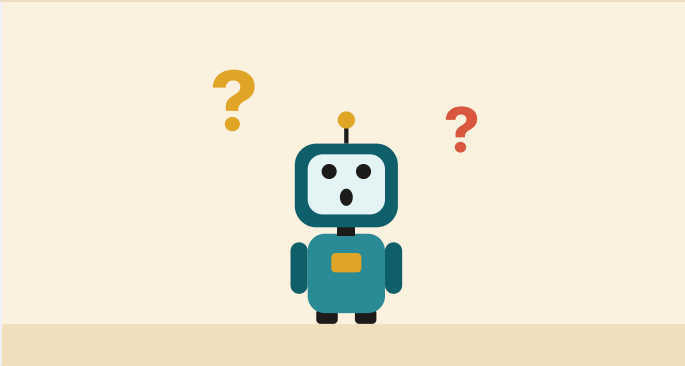
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

For an investor based in Japan, which quote is direct?

Answer: A. A direct quote has the domestic currency as the price currency: yen per dollar. B has the yen as the base, so it is indirect. C does not involve the yen at all.



EXAM QUESTION 3 OF 8

INR/USD = 85.0000 and EUR/USD = 0.9200. The INR/EUR cross rate is closest to:

A. 78.2000

B. 0.0108

C. 92.3913

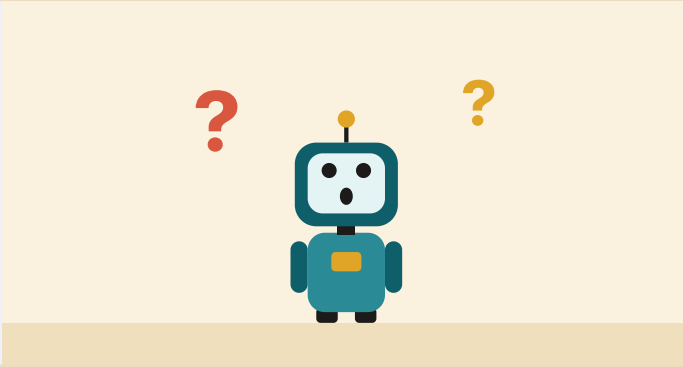
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

INR/USD = 85.0000 and EUR/USD = 0.9200. The INR/EUR cross rate is closest to:

Answer: C. Both quotes have USD as base, so divide: INR/EUR = (INR/USD)/(EUR/USD) = 85/0.92 = 92.3913. A multiplies, which leaves units of INR × EUR per USD². B is EUR/INR, the inverse of the answer.



EXAM QUESTION 4 OF 8

JPY/USD spot is 150.0000 and the 6-month forward points are +120. The forward rate is:

A. 150.0120

B. 151.2000

C. 148.8000

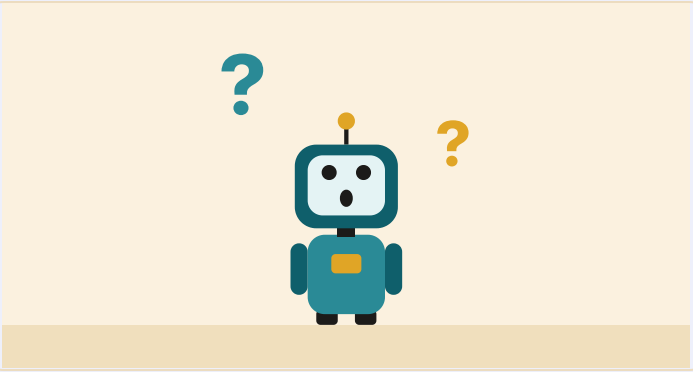
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

JPY/USD spot is 150.0000 and the 6-month forward points are +120. The forward rate is:

Answer: B. Yen pairs quote to 2 decimals, so divide points by 100: 150.0000 + 1.20 = 151.2000. A scales by 10,000, the rule for 4-decimal pairs. C subtracts positive points.



EXAM QUESTION 5 OF 8

USD/GBP spot is 1.3000, the 90-day USD rate is 3.00% and the GBP rate is 5.00% (Actual/360). The 90-day forward USD/GBP is closest to:

A. 1.3065

B. 1.2936

C. 1.2752

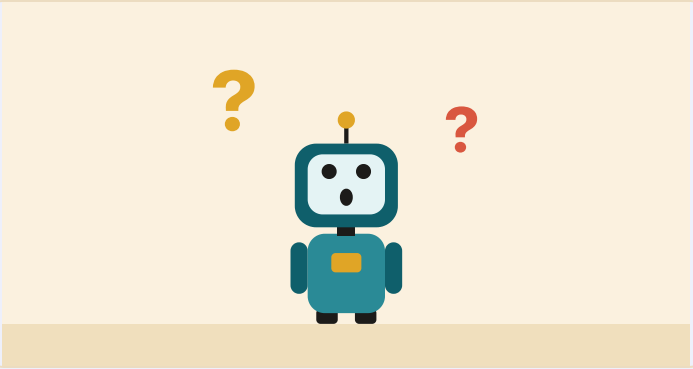
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

USD/GBP spot is 1.3000, the 90-day USD rate is 3.00% and the GBP rate is 5.00% (Actual/360). The 90-day forward USD/GBP is closest to:

Answer: B. F = 1.3000 × (1 + 0.03 × 0.25)/(1 + 0.05 × 0.25) = 1.2936: the higher-rate pound is at a forward discount. A puts the base currency's rate on top. C uses full-year rates for a 90-day contract.



EXAM QUESTION 6 OF 8

Under covered interest rate parity, the currency with the higher interest rate will:

A. trade at a forward premium

B. appreciate in the spot market over time

C. trade at a forward discount

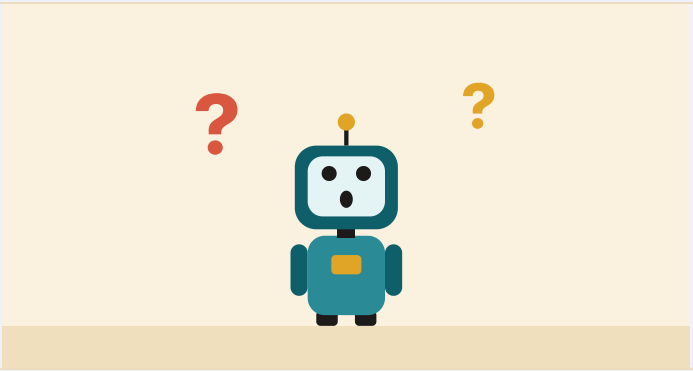
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

Under covered interest rate parity, the currency with the higher interest rate will:

Answer: C. The forward must offset the interest gap, or a hedged deposit would beat a home deposit risk-free. So the higher-rate currency is at a forward discount. A has it backwards. B is a forecast about future spot rates, which CIP does not make.



EXAM QUESTION 7 OF 8

INR/USD spot 85.0000, 1-year rates INR 6.00%, USD 4.00% (use t = 1). A dealer quotes the 1-year forward at 86.0000. An arbitrageur should:

A. borrow USD, buy INR spot, invest in INR and sell INR forward (buy USD forward)

B. borrow INR, buy USD spot, invest in USD and sell USD forward

C. do nothing, because 86.0000 is close to spot

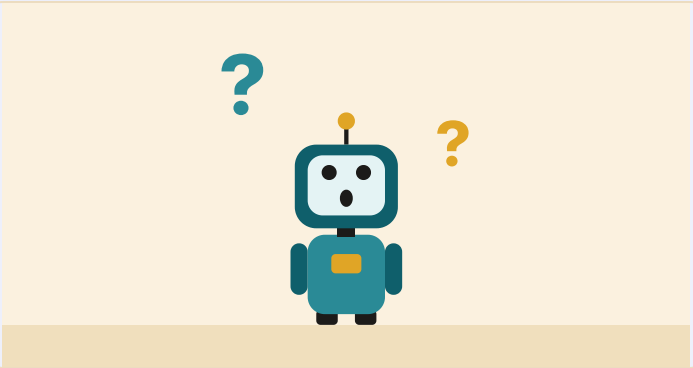
Answer on the next page.

So: **Pick an answer, then turn the page.**

ANSWER

INR/USD spot 85.0000, 1-year rates INR 6.00%, USD 4.00% (use t = 1). A dealer quotes the 1-year forward at 86.0000. An arbitrageur should:

Answer: A. Parity forward = 85 × 1.06/1.04 = 86.6346. The quote 86.0000 is too low, so dollars are cheap forward: buy them forward. Borrow USD 1, get INR 85, grow to INR 90.10, buy USD at 86: USD 1.0477, repay USD 1.04, profit USD 0.0077. B would be right if the forward were above parity. C ignores a riskless profit.



EXAM QUESTION 8 OF 8

USD/EUR spot 1.1000, USD rate 4%, EUR rate 2%. The 180-day forward premium on the euro is closest to (approximate formula):

A. 2.00%

B. 1.00%

C. 0.50%

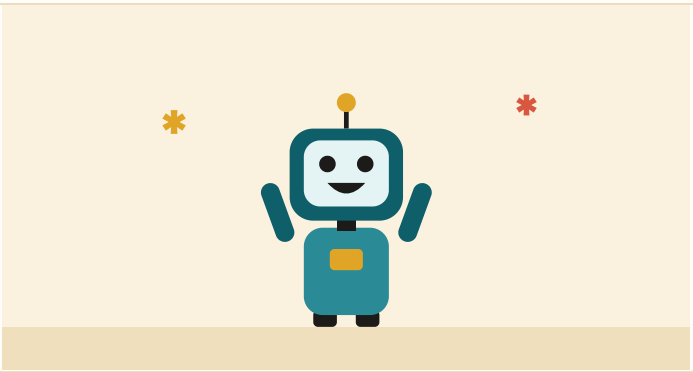
Answer on the next page.

So: Pick an answer, then turn the page.

ANSWER

USD/EUR spot 1.1000, USD rate 4%, EUR rate 2%. The 180-day forward premium on the euro is closest to (approximate formula):

Answer: B. Premium on base ≈ $(i_{USD} - i_{EUR}) \times 180/360 = 2\% \times 0.5 = 1.00\%$ (exact 0.990%). A forgets to scale the annual gap to half a year. C scales as if it were 90 days.



CHAPTER 8 DONE

Exchange rate calculations, done

You can find the 1 in any quote, translate dealer notation, flip and cross quotes, measure each currency's move correctly, turn points into forwards, price a forward with covered interest parity and spot an arbitrage. Pinky now asks "what is the 1?" before every trade.

So: **Swipe on, or jump to another chapter from the Menu.**